



DFSMSrmm Implementation

(Course Code SS21)

Student Notebook

ERC 2.0

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Course Description

DFSMSrmm Implementation

Duration: 4 days

Purpose

This course is designed to teach the student how to implement the Removable Media Manager (DFSMSrmm) functional component of DFSMS/MVS and optional feature of z/OS. DFSMSrmm provides life cycle management of tape data which includes scheduling of tape volumes to and from storage locations (vaults).

Audience

People responsible for implementing life cycle management for tape data using DFSMSrmm (a component of z/OS DFSMS). Personnel implementing DFRMM may also find the course beneficial.

Prerequisites

Those attending this course should understand basic MVS allocation and sequential processing such as can be obtained in *Storage Management Fundamentals* (J3701/SS06). The student should also be able to code MVS job control language statements (JCL) for tape data sets. Knowledge of JCL may be obtained by attending *MVS JCL and Utilities* (H3838).

Objectives

After completing this course, you should be able to:

- Select DFSMSrmm options suitable for your installation
- Create a control data set and journal for DFSMSrmm
- Define DFSMSrmm tape policies (Vital Record Specifications - VRS) for retention and movement.
- Create procedures for:
 - Starting DFSMSrmm
 - Initializing tapes
 - Performing control data set backup

- Use the DFSMSrmm utilities to:
 - Produce volume inventory and movement reports
 - Perform control data set reorganization
 - Perform control data set recovery
- Use the **SYS1.SAMPLIB** samples to produce additional reports.
- Identify access control requirements for using DFSMSrmm functions.

Agenda

Day 1

Welcome
Unit 1 - Introduction to DFSMSrmm
Exercise 1 - DFSMSrmm ISPF Application
Unit 2 - DFSMSrmm Day-to-Day Procedures
Exercise 2 - DFSMSrmm Inventory Management and Reporting Facilities

Day 2

Unit 3 - DFSMSrmm Implementation Considerations
Exercise 3 - DFSMSrmm Implementation Wizard
Unit 4 - Vital Record Specifications

Day 3

Exercise 4 - Using VRS for Retention Management
Unit 5 - Advanced DFSMSrmm VRS Management
Exercise 5 - Using VRS for Volume Movement

Day 4

Unit 6 - Miscellaneous DFSMSrmm Topics
Exercise 6 - Using VRS Chains for Retention Management
Unit 7 - Tape Volume Pools and Tape Libraries

Unit 1. Introduction to DFSMSrmm

What This Unit Is About

This unit describes the terminology used in the DFSMSrmm functional component of z/OS DFSMS, an optional feature of z/OS, and discusses DFSMSrmm usage categories.

What You Should Be Able to Do

After completing this unit, you should be able to:

- List the components of the DFSMSrmm control data set
- Identify the users of DFSMSrmm
- Understand the terms used for tape volumes and their storage locations

How You Will Check Your Progress

Accountability: Checkpoint questions

References

SC26-7404	<i>DFSMSrmm Guide and Reference</i>
SC26-7405	<i>DFSMSrmm Implementation and Customization Guide</i>
SC26-7406	<i>DFSMSrmm Reporting</i>

Objectives

When you have finished this unit, you should be able to:

- Understand DFSMSrmm terminology
- Identify different types of users for command driven interactions with DFSMSrmm

Notes:

Before we go into the details of DFSMSrmm (DFSMS removable media manager), it is necessary to understand the terminology used in DFSMSrmm publications.

Each enterprise has its own definition of job responsibilities. For DFSMSrmm, IBM has defined five user job categories. We will see which categories of users can perform which tasks.

1.1 DFSMSrmm Terminology

DFSMS/MVS Tape Support

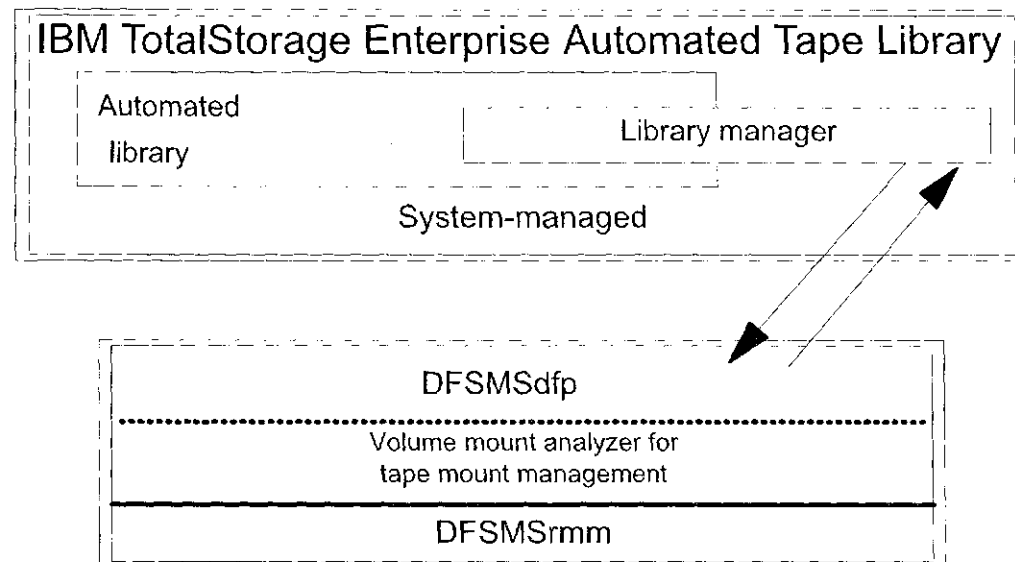


Figure 1-2. DFSMS/MVS Tape Support

SS212.0

Notes:

DFSMSrmm is part of an overall solution to tape management issues that has been incorporated into z/OS DFSMS. The DFSMSdfp component provides for a system-managed tape environment that uses IBM 3494 tape library data servers while DFSMSrmm provides the life cycle management of tape volumes and their content. The volume mount analyzer assists in analyzing and understanding the existing tape environment. Using the volume mount analyzer and implementing system-managed tape are covered in another course *Analyzing and Managing DFSMS Tape Data*, ES590.

We will first look at the terms used in the DFSMSrmm documentation:

- SC26-7404** *DFSMSrmm Guide and Reference*
- SC26-7405** *DFSMSrmm Implementation and Customization Guide*
- SC26-7406** *DFSMSrmm Reporting*

What Is DFSMSrmm?

The DFSMSrmm (removable media manager) is:

- ✓ A tape management system
- ✓ A storage manager for

- ☆ **Tapes and cartridges**

- Physical volumes
- Logical volumes
- Stacked volumes

- ☆ **Media libraries**



- ☆ **Removable media**

All dates are stored in four digits.

It is the basis for the future of removable media management.

Figure 1-3. What is DFSMSrmm?

SS212.0

Notes:

DFSMSrmm provides life cycle management of tape resident data. It is a *tape management system*. DFSMSrmm also manages other removable media, such as optical disks, that you define to it.

Currently, it provides automated management of tape data. While it is removable, DFSMSrmm does not provide automated life cycle management for data (objects) residing on optical disk.

DFSMSrmm Terminology (1 of 4)

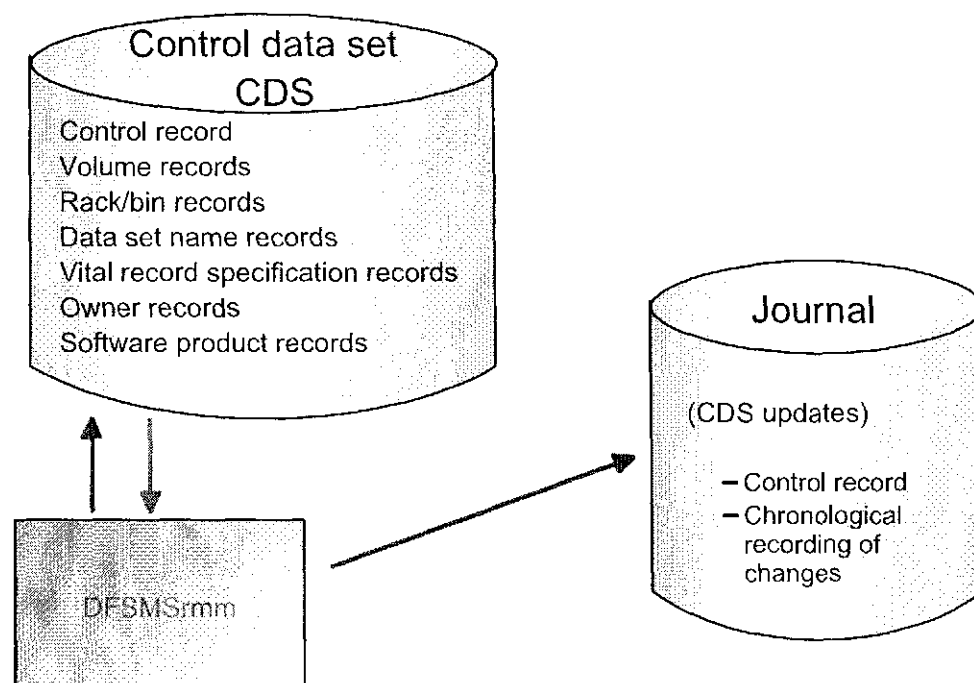


Figure 1-4. DFSMSrmm Terminology (1 of 4)

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Notes:

DFSMSrmm uses a single VSAM key sequenced data set (KSDS) to control tape activity. This data set is referred to as the DFSMSrmm control data set or **CDS**. The CDS is a very important *catalog* of tape information. This information is reflected in the following record types:

Control	Identifies the KSDS as a DFSMSrmm control data set.
Rack	A location that may contain a tape volume.
Volume	Identifies a specific tape volume (cartridge or reel).
Data set	Identifies a data set residing on one or more of the volumes.
Owner	Identifies the owner of a non-scratch volume.
VRS	Vital Record Specification: Retention and movement policy for data sets or volumes.

Software product Identifies product names and the volumes associated with the product.

CDS backup procedures help to protect against the loss of the control data set's contents. DFSMSrmm journals changes that have occurred since the last point-in-time back up of the CDS. This provides forward recovery from the last backup copy.

DFSMSrmm Terminology (2 of 4)

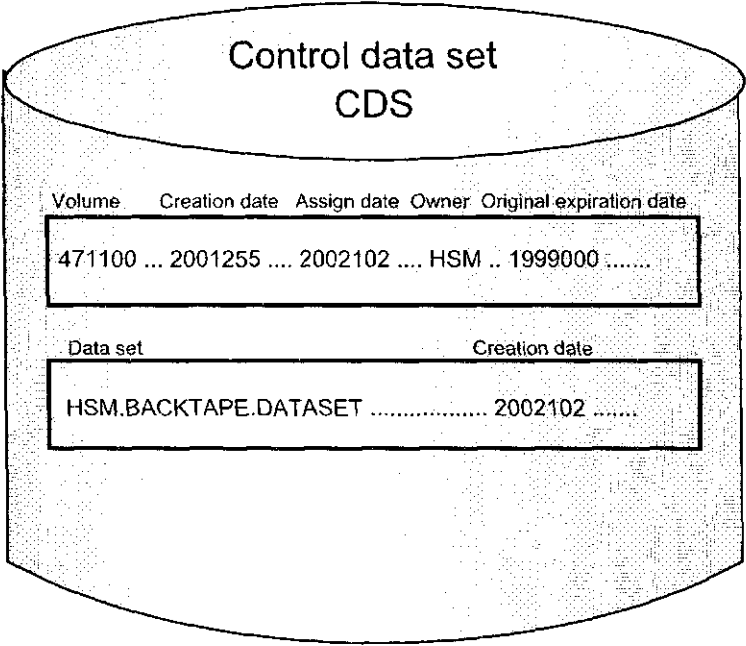


Figure 1-5. DFSMSrmm Terminology (2 of 4)

SS212.0

Notes:

Creation date	The creation date for a data set is the date on which a data set is written to tape. The creation date can also be the date on which a data set was read if it was created before DFSMSrmm went into use. The creation date is updated each time a data set is replaced without being extended. The creation date for volumes and other resources defined to DFSMSrmm is the date the resource is defined to DFSMSrmm or the date specified on the command as the creation date.
Assign date	<u>The date on which the volume was assigned to a user or returned to scratch status.</u>
Original expiration date	The date supplied via EXPDT or RETPD when a data set is created.
Owner	In DFSMSrmm, a person defined as a DFSMSrmm user owning volumes. An owner is defined to DFSMSrmm through an owner ID.

DFSMSrmm Terminology (3 of 4)

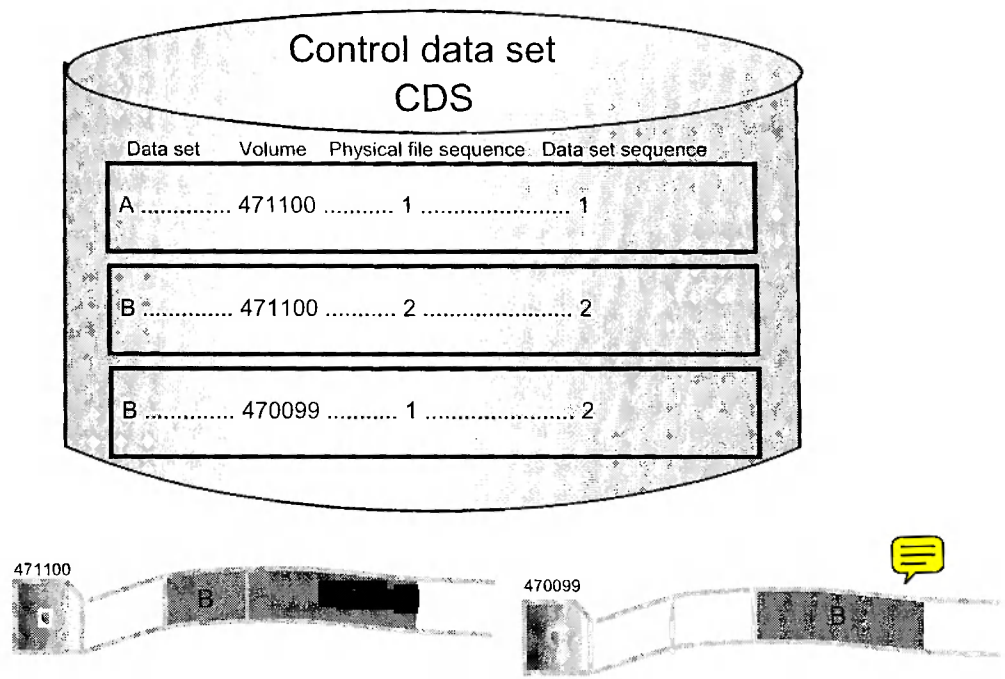


Figure 1-6. DFSMSrmm Terminology (3 of 4)

SS212.0

Notes:

- Physical file sequence number** Relative position of a file on a volume
- Data set sequence number** Logical position of a data set relative to others for a multifile single volume or a multifile/multivolume set

DFSMSrmm Terminology (4 of 4)

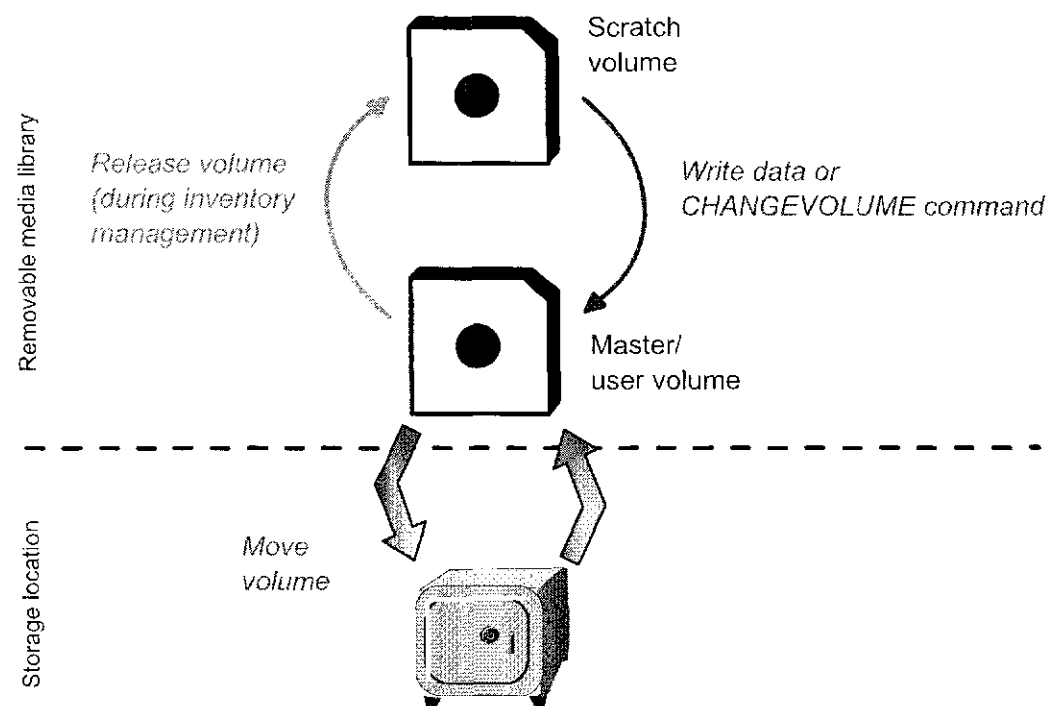


Figure 1-7. DFSMSrmm Terminology (4 of 4)

SS212.0

Notes:

The status of a volume defined in DFSMSrmm is subject to change on occasion. Writing data on a scratch volume will change its status automatically to master. Once all data on a master volume has expired, the volume is released during inventory management. The release of a volume means that its status changes to *scratch*.

Inventory management is the processing performed by the DFSMSrmm utility **EDGHSKP** to manage the tape inventory. This includes:

- Vital record processing, which determines which data sets to retain and what volume moves are required, based on policies defined to DFSMSrmm.
- Expiration processing, which identifies volumes ready to be released and returned to scratch.
- Storage location management processing, which assigns shelf locations to volumes being moved to storage locations.
- DFSMSrmm control data set functions, which back up the control data set and the journal, and finally reset the journal.
- Report extract processing, which creates an extract data set for report processing use.

Change of Volume Status

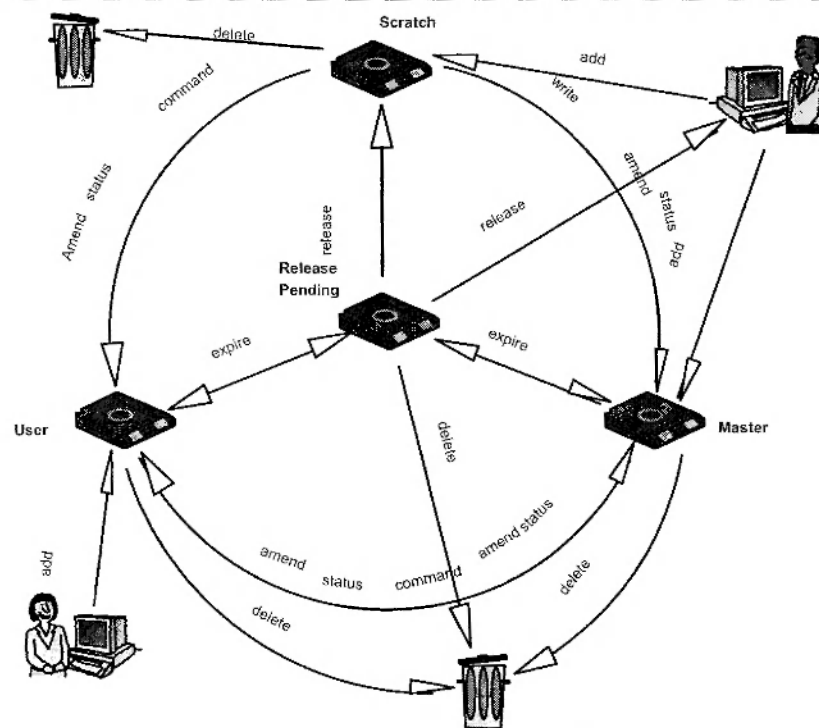


Figure 1-8. Change of Volume Status

SS212.0

Notes:

- Master** A private volume containing data that is protected against overwrite processing based on the DFSMSrmm EDGRMMxx parmlib MASTEROVERWRITE operand.
- Scratch** The status of a tape volume that is available for general use, because the data on it has expired. You request a scratch volume when you omit the volume serial number on a request for a tape volume mount.
- User (volume)** A volume assigned to a user, which can contain any data and can be rewritten as many times as the user wishes until the volume expires.

Volume status is changed automatically by DFSMSrmm in most cases. You can use the CHANGEVOLUME subcommand to amend the status manually.

Note: CHANGEVOLUME does not allow you to change the status of a volume from *master* or *user* to *scratch*. To release a volume manually, use the DELETEVOLUME subcommand with the RELEASE operand. This will set the availability of the volume to *pending release*. During the next inventory management run the status of the volume will change to *scratch* if the only pending release action is *return to scratch*.

Libraries and Locations (1 of 4)

A removable media library contains all the tape and optical volumes that are available for immediate use. A removable media library can include the following types of libraries:

- **System-managed tape libraries**

Location(*library_name*)

IBM TotalStorage Enterprise Automated Tape Library

IBM TotalStorage Virtual Tape Server

- **Non-system-managed tape libraries**

Location(*SHELF, installation_defined*)

Traditional tape libraries

Other robots

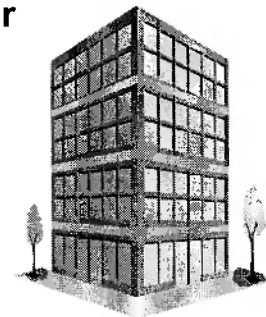


Figure 1-9. Libraries and Locations (1 of 4)

SS212.0

Notes:

All models of the IBM 3495 tape library dataserver had been withdrawn from marketing by April 14, 1998. DFSMSrmm does support those currently available.

System-managed tape libraries are assigned a name by which they are known to DFSMS. This name is used by DFSMSrmm as a location name. All other tape volumes in the computing center are considered to be in a SHELF location. SHELF includes non-IBM automated tape libraries.

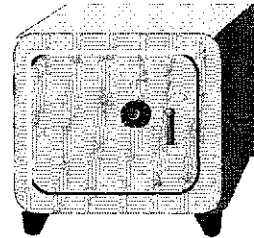
A tape volume in the computing center is assigned either to a SHELF location or as a volume in an IBM system-managed tape library. DFSMSrmm assigns the location to which a volume is added: that is to SHELF or the name of the system-managed library, as the volume's HOME location.

Libraries and Locations (2 of 4)

Storage locations can be:

- Built-in storage locations

LOCAL
DISTANT
REMOTE



- Installation defined storage locations

Any eight-character name for each storage location is acceptable.

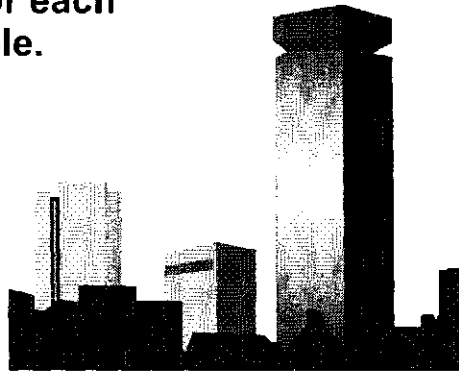
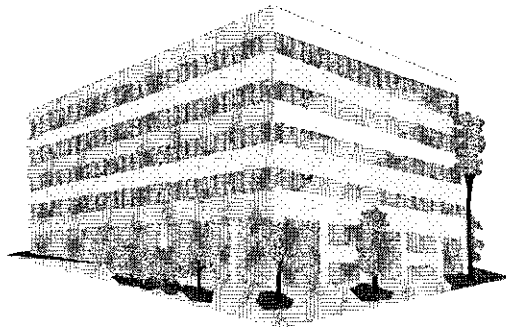


Figure 1-10. Libraries and Locations (2 of 4)

SS212.0

Notes:

DFSMSrmm managed volumes may also be stored outside a system-managed tape library or off the shelf in a storage location. In addition to the built-in three storage locations of LOCAL, DISTANT, and REMOTE, any number of installation defined storage locations can be used.

Movement of Volumes

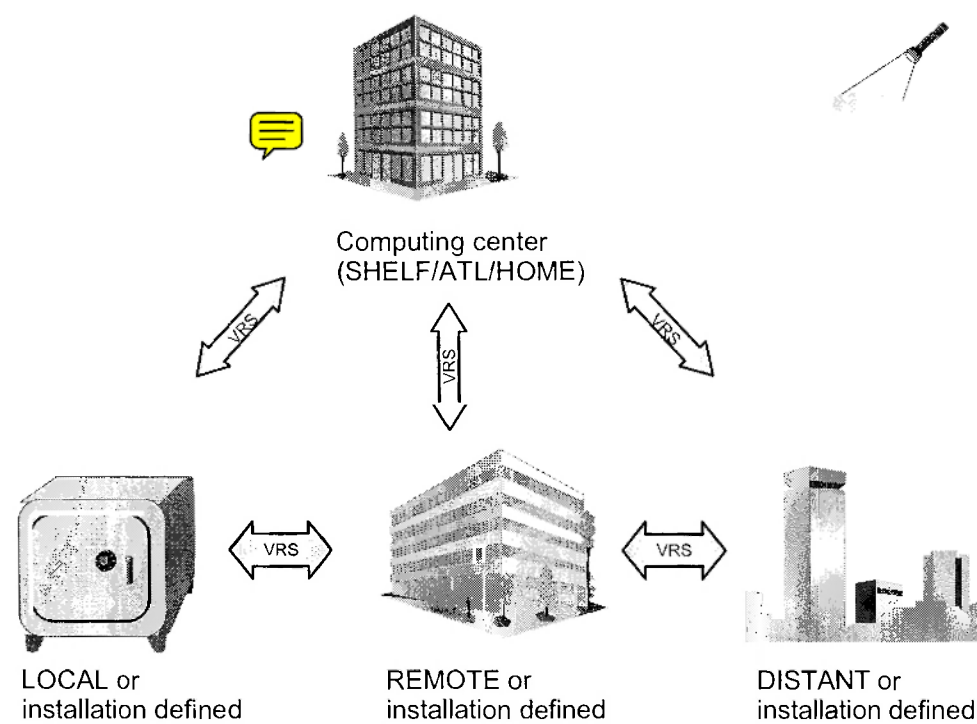


Figure 1-11. Movement of Volumes

SS212.0

Notes:

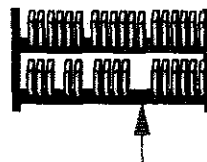
To schedule moves of volumes, we need to define vital record specifications (VRS). A volume may be moved around from one storage location to another through the use of chains of VRSs. Normally, return to scratch for a volume will happen after its return to its home location.

Libraries and Locations (3 of 4)

Computing center *rack*

TYPE(S) *scratch pool*

TYPE(R) *rack pool*



Loan location

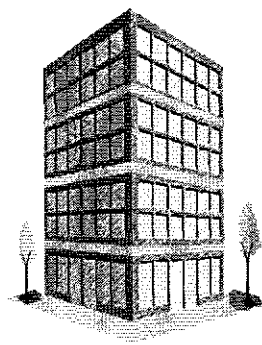


Figure 1-12. Libraries and Locations (3 of 4)

SS212.0

Notes:

For tape volumes in the computing center DFSMSrmm uses a six-position identifier called a **rack** to represent the shelf location of a tape volume. DFSMSrmm defaults to using the volume serial identification as the **rack** location. However, it also allows the volume serial to be different from the **rack** ID.

The DFSMSrmm base design is that a system will use a common pool of tapes for all users. To support special situations, DFSMSrmm has two types of tape volume pools, *scratch* and *rack*:

- | | |
|----------------|---|
| Scratch | Volumes in scratch status may be used to satisfy non-specific requests for scratch tapes. |
| Rack | Volumes are available for scratch mount requests only if the pool is committed by user exit EDGUX100 or an SMS storage group. An uncommitted pool might be used for external tapes such as those that enter and leave an installation on an ad hoc basis. It is also used for environments where tapes are dedicated to an application that calls for all its tapes by specific volser. |

For volumes in an IBM system-managed tape library, the rack and the volume serial must match.

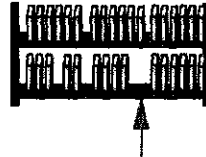
Loan location

The location where the volume resides other than in the removable media library or a storage location. A loan location can be specified by using one to eight characters to allow the tracking of a volume that has been manually removed from normal locations managed by DFSMSrmm. This allows for a more descriptive specification than a general *out of the area* classification. For example, a tape sent to IBM service containing trace data for diagnostic data could be assigned the loan location, **IBMSERV**.

Libraries and Locations (4 of 4)

Acceptable names for bins depend on storage locations.

Built-in storage locations
Names of *bins* are built of six digits.



Installation defined storage locations
Names of *bins* are built of six alphanumeric or national characters in any combination.

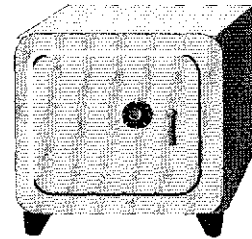
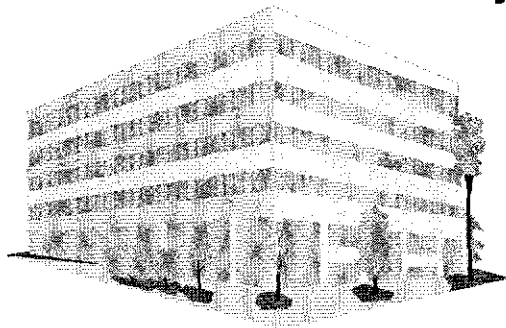


Figure 1-13. Libraries and Locations (4 of 4)

SS212.0

Notes:

Storage locations may be arranged so that specific slots or bins are set aside to store volumes. Bins can be defined for specific media names if necessary. DFSMSrmm also supports locations where volume location management is not required.

Installation defined names have the same naming conventions as MVS simple names. They can be one to eight characters in length and must start with a non-numeric value.

Supported Hardware

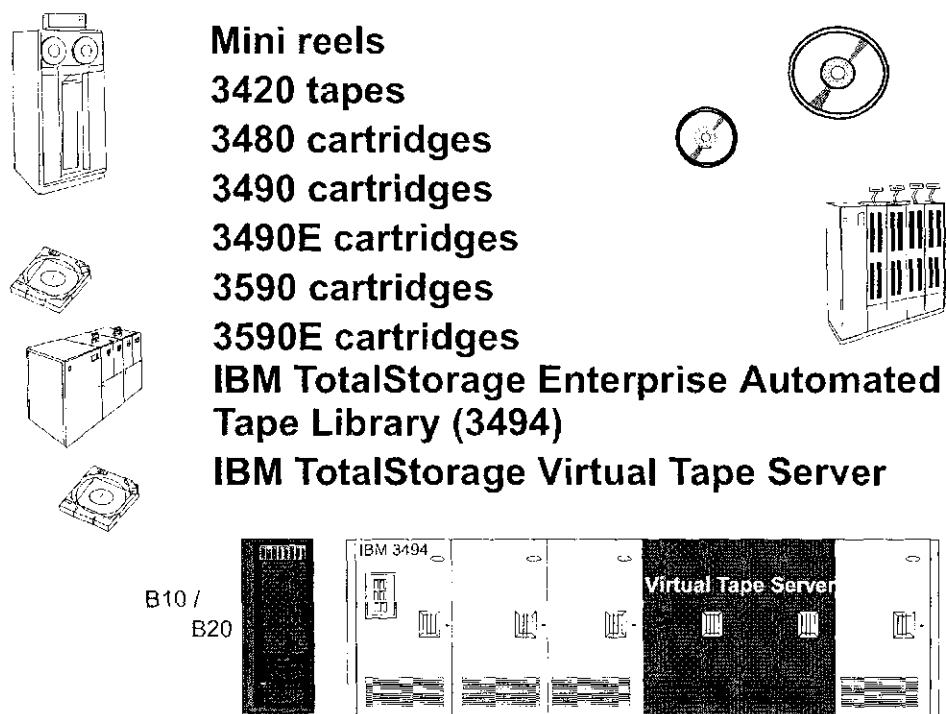


Figure 1-14. Supported Hardware

SS212 0

Notes:

DFSMSrmm is a tape management system. It does not manage tape devices. For this reason, you will not find the device type in any field of the volume record.

The volume record contains fields showing information on

- Media type
- Recording format
- Volume type (physical, logical or stacked)

The volume type is always physical if you do not use volumes in a virtual tape server.

Media type and recording format are automatically recorded by DFSMSrmm when the volume is used for the first time.

Media Types

S/390 Tape Cartridge



Cartridge Type	CST	CST	ECCST	3590	3590	3590	3590	3590	3590
				Standard	Standard	Standard	Extended	Extended	Extended
Tracks	18	36	36	128	256	384	128	256	384
Native	200 MB	400 MB	800 MB	10 GB	20 GB	30 GB	20 GB	40 GB	60 GB
Compacted	600 MB	1.2 GB	2.4 GB	30 GB	60 GB	90 GB	60 GB	120 GB	180 GB
ATL Media Label	1 or blank	1 or blank	E	J	J	J	K	K	K
OAM Media Type	01	01	02	03	03	03	04	04	04
SMS Tape Category	MEDIA 1	MEDIA 1	MEDIA 2	MEDIA 3	MEDIA 3	MEDIA 3	MEDIA 4	MEDIA 4	MEDIA 4

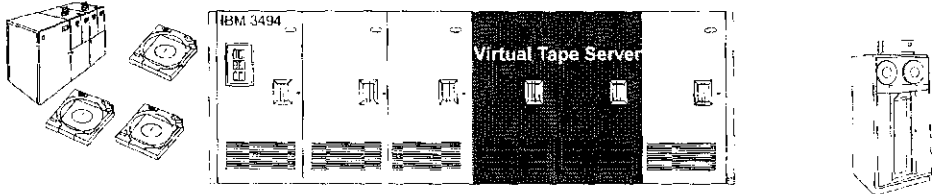


Figure 1-15. Media Types SS212.0

Notes:

Note that some cartridge types can be operated by using different hardware device types. That is why these media show different capacities in the table.

Drives and Supported Media Types

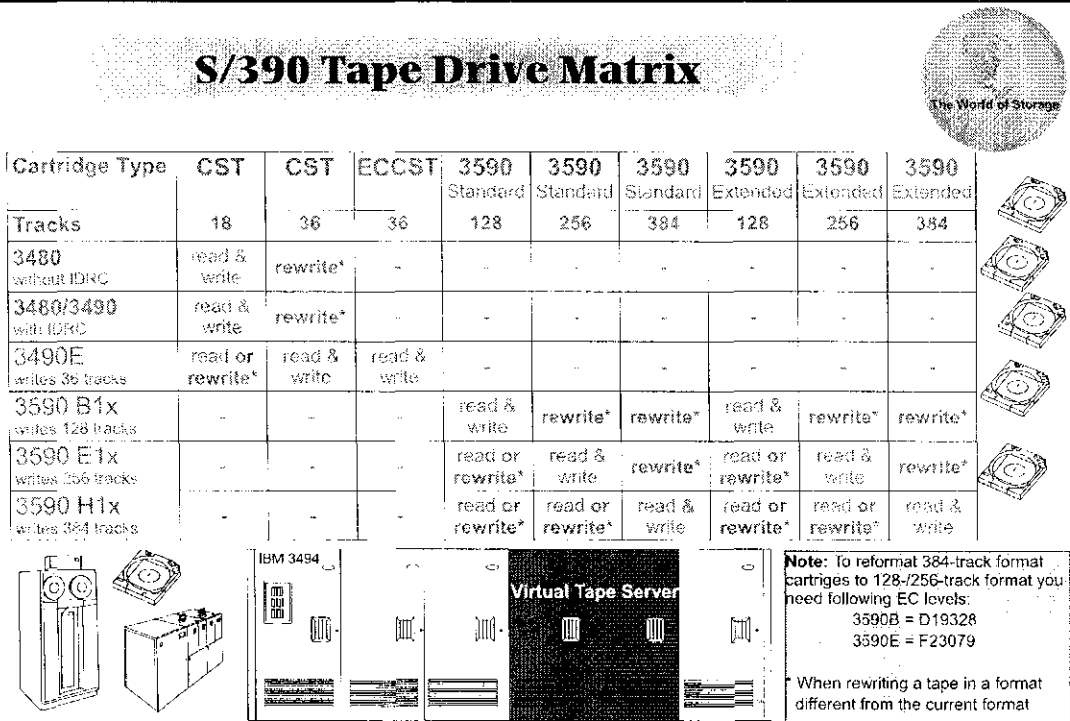
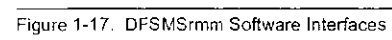


Figure 1-16. Drives and Supported Media Types SS212.0

Notes:

Cartridges can be used on several devices having different capabilities such as HPCT on 3590 B1x and 3590E1x. Note that cartridges that are initially written on 3590B1x can be read or rewritten on 3590E1x, but that you cannot append data unless you use the original device type.

This restriction also applies to CST and EHPCT.



Notes:

Subsystem interface (SSI) The means by which system routines request services of the master subsystem, a job entry subsystem, or other subsystems defined to the subsystem interface.

Reject during Mount Verification

A private volume is mounted in response to a scratch request

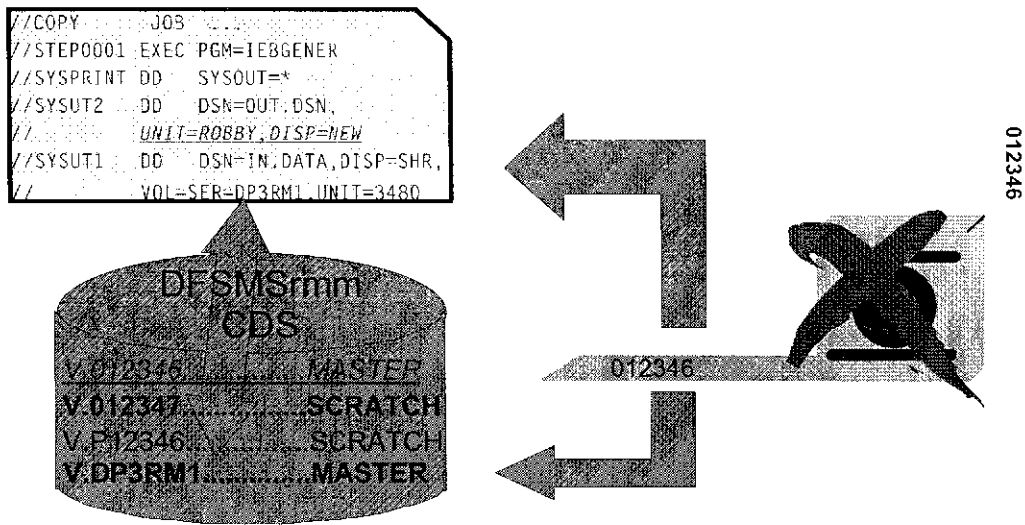


Figure 1-18. Reject during Mount Verification

SS212.0

Notes:

DFSMSrmm protects tapes with master or user status from being used in response of a scratch mount request, thus hindering users from destroying unexpired data.

Reject during Open Processing

An attempt is made to use a specific scratch volume

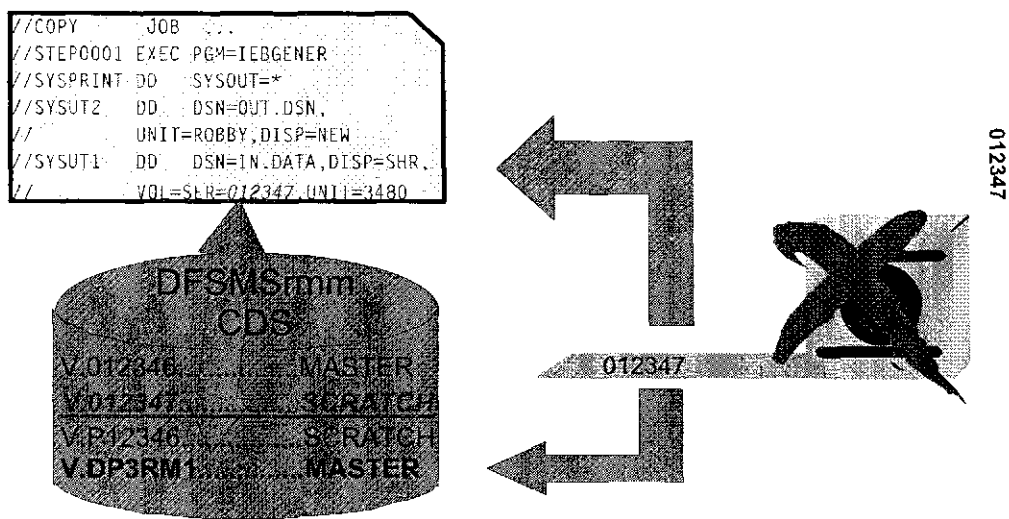


Figure 1-19. Reject during Open Processing

SS212.0

Notes:

DFSMSrmm protects the contents of volumes with scratch status against any use. The only way to re-access data on volumes that were inadvertently released, is to **change the status manually to master or user** by a person with proper authorization, such as the librarian.

Modify Mount Message

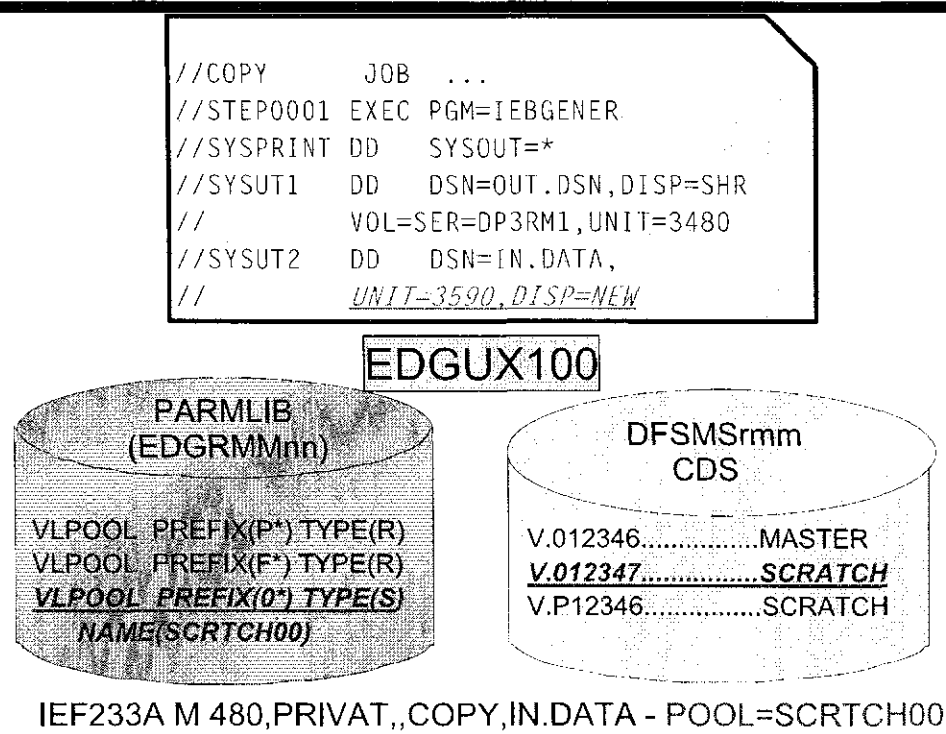


Figure 1-20. Modify Mount Message

SS212.0

Notes:

During mount processing DFSMSrmm searches for an acceptable scratch pool (for scratch requests) which can be displayed in a modified mount message.

If you plan to use several scratch pools, you have two options for pooling management:

- The sample EDGUX100 exit provides a working solution for application type scratch pools based on job names and data set names
- You can use DFSMS automatic class selection (ACS) routines processing to set a storage group name, which overrides EDGUX100 processing.

High Speed Locate

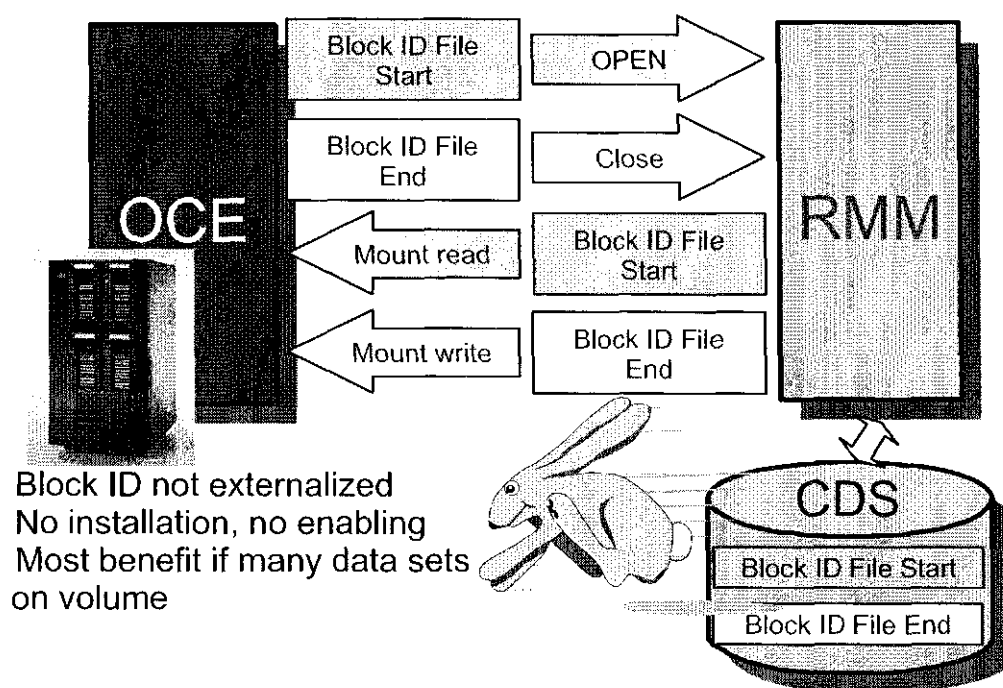


Figure 1-21. High Speed Locate

SS212.0

Notes:

DFSMSrmm records in the control data set the starting and ending tape block IDs for each data set when it is created. With this technique, DFSMSrmm enables the use of tape block IDs for applications that do not support them.

The benefit of using this function is that it enables high speed positioning when there are a large number of data sets in a high capacity tape cartridge.

To gain benefit from this function, all you need is OS/390 Version 2 Release 10 and DFSMSrmm. The system and DFSMSrmm will automatically record and retrieve block IDs, and then use these for high speed data set positioning whenever it is possible.

Hardware Exploitation

DFSMSrmm records the following information for media that is written on 3590 devices:

Capacity

Percent used

Volume level

Data set level

DFSMS/MVS 1.4 +

OS/390 DFSMS 2.10 +

DFSMSrmm considers hardware compaction

Details direct from hardware

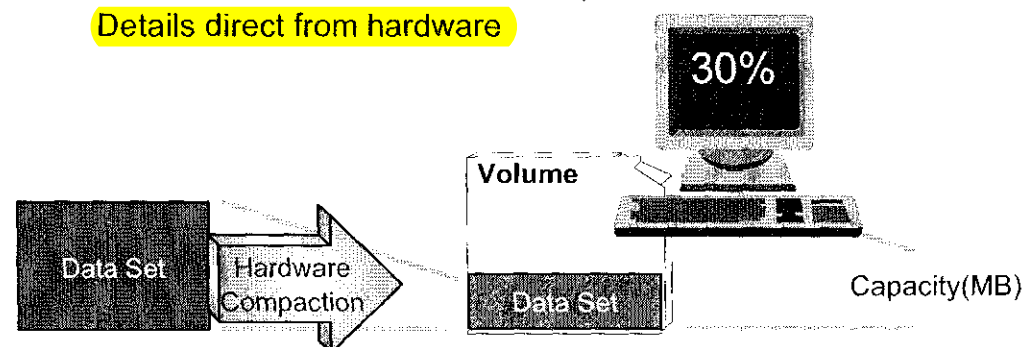


Figure 1-22. Hardware Exploitation

SS212.0

Notes:

At the volume and data-set levels, DFSMSrmm supports information on cartridge capacity and on percentage of the capacity that has been used, for tapes written on 3590 devices.

This does not work for virtual volumes, because from the perspective of a host, it is 3490E devices that are used.

1.2 DFSMSrmm Users

Who Can Use DFSMSrmm and How?

Figure 1-23. Who Can Use DFSMSrmm and How?

SS212.0

Notes:

Applications use tapes to store and retrieve data. Storage administrators use tapes to store backup and inactive copies of application data residing on disk. System programmers receive software products and maintenance on tape. Operators handle the mechanics of mounting and demounting tapes, while librarians manage the tape inventory.

To read or write data on tape, you need only the proper RACF authorization for the data set to open. For this reason, anybody can use DFSMSrmm implicitly, even without knowing it.

But to use or update information in RMM without performing tape I/O—for example by using TSO subcommands or the ISPF interface of DFSMSrmm to list some volume or data-set information, users need proper authorization based on profiles located in the general resource class, FACILITY, in RACF.

Who Can Use DFSMSrmm and How?

Examples for DFSMSrmm users are:

- General user
- Tape librarian
- Storage administrator
- System programmer
- Operator
- Inventory manager

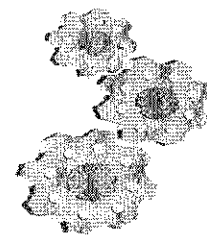


Figure 1-24. Who Can Use DFSMSrmm and How?

SS212.0

Notes:

DFSMSrmm considers any user not defined in one of the other categories to be a general user. A user may have a TSO ID or may use tape services via a batch job. Either a job has an explicit userid assigned to it, or the system derives one.

The DFSMSrmm Primary Option Menu

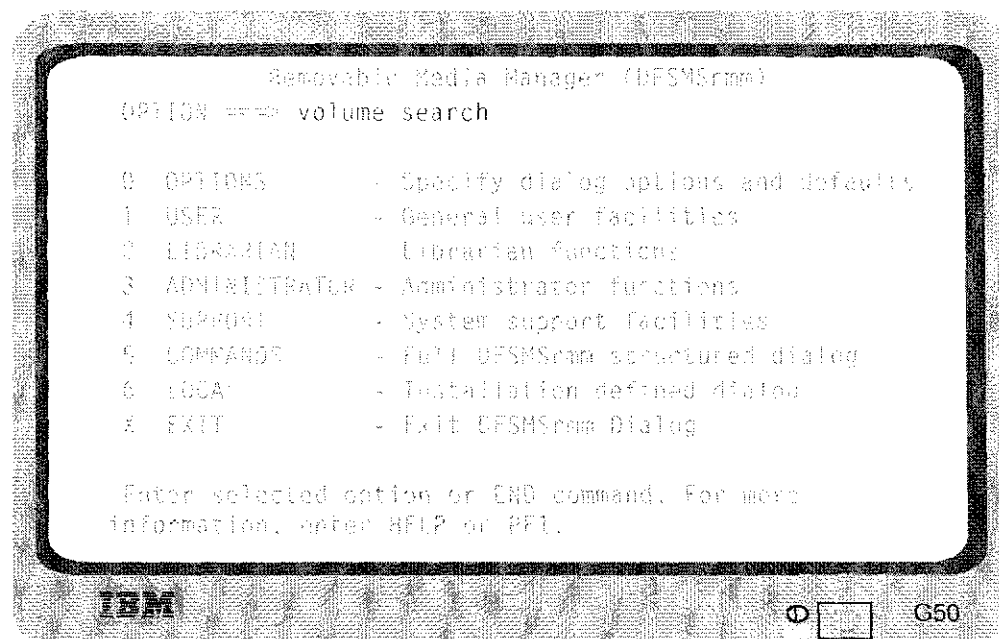


Figure 1-25. The DFSMSrmm Primary Option Menu

SS212.0

Notes:

TSO users communicate with DFSMSrmm by using the *RMM* TSO command. The subcommands of *RMM* are regarded as the DFSMSrmm commands and are documented in the *DFSMSrmm Guide and Reference*. To facilitate the use of DFSMSrmm, an ISPF dialog is provided.

Additionally users can enter fast path commands like *volume search*. For a complete list of fast path commands see *DFSMSrmm Guide and Reference, Appendix A*.

Volume Search Panel

DFSMSrmm Volume Search Page 1 of 2

Volume SC* May be generic. Leave blank for all volumes.

Owner * Owned by a specific user. Default is your user ID.

Media class mm Limit to a specific media class.

Pool mm or to a particular pool.

Status MASTER, USED, SC=DD, RELEASE, VIO, ENTRY, INIT or All

Start mmmm volumes assigned since YYY/DDD

Retaining mmmm volumes retained up to YYY/DDD

Limit 99 Limit search to first mm volumes

IBM G50

Figure 1-26. Volume Search Panel

SS212.0

Notes:

You can search volumes by using specific or generic volume specifications.

If you enter an asterisk in the OWNER field, volumes with any owner will be included in the resulting list. This is very important when searching for scratch volumes, which always have an empty OWNER field. If you leave the OWNER field blank, the search will use your TSO user ID by default.

Limit defaults to 10.

There are additional search criteria on page 2.

You will find more information about the DFSMSrmm ISPF Interface in *Appendix A* and in the *DFSMSrmm Guide and Reference*.

General User

General user functions

- Manually request a scratch volume
- Change information about an owned volume or data set
- Update information about your owner ID
- Manually release an owned volume
- Create lists of resources and display information about most resources defined to DFSMSrmm



Figure 1-27. General User

SS212.0

Notes:

The general user should be able to use the DFSMSrmm CDS like a catalog and to work on volumes owned by himself.

Read Access-General User

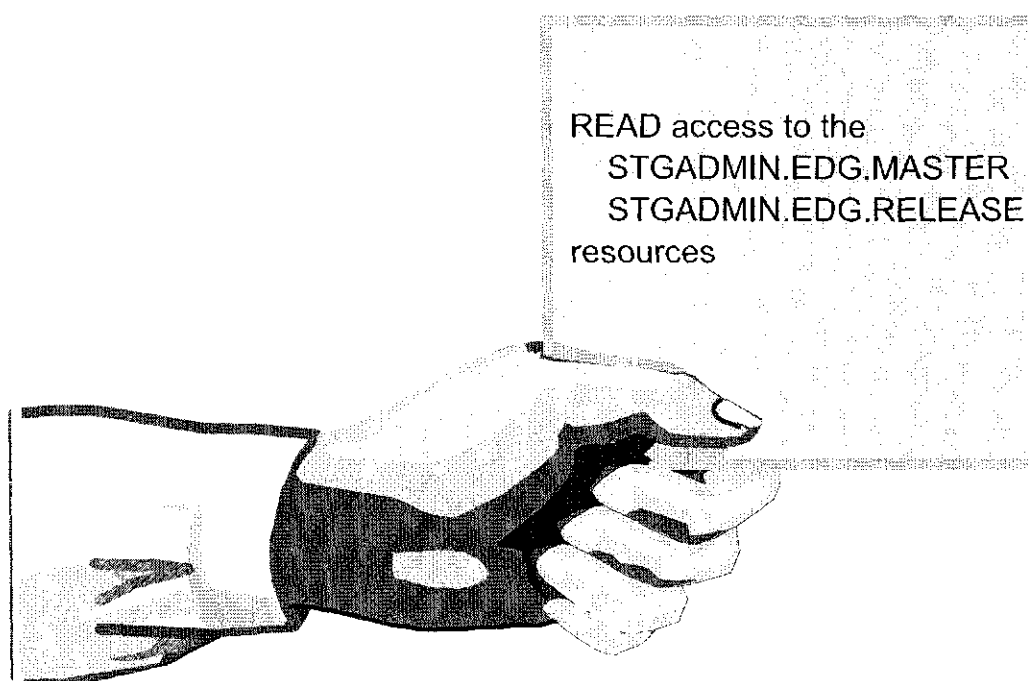


Figure 1-28. Read Access-General User

SS212.0

Notes:

DFSMSrmm uses resource facility classes defined in a security product such as RACF. To perform the functions DFSMSrmm assumes for a general user, read access is suggested to STGADMIN.EDG.MASTER and STGADMIN.EDG.RELEASE facility class profile. Read access to the latter resource is needed to release even owned volumes.

Tape Librarian

Tape librarian functions

- Define new volumes
- Add shelf locations to the removable media library and storage locations
- Add, change, and delete information about resources defined to DFSMSrmm
- Manually release any volume
- Manage vital record specification
- Confirm volume movements and actions



Figure 1-29. Tape Librarian

SS212.0

Notes:

There are many functions performed by a tape librarian.

Control and Update Access-Tape Librarian

Tape librarian

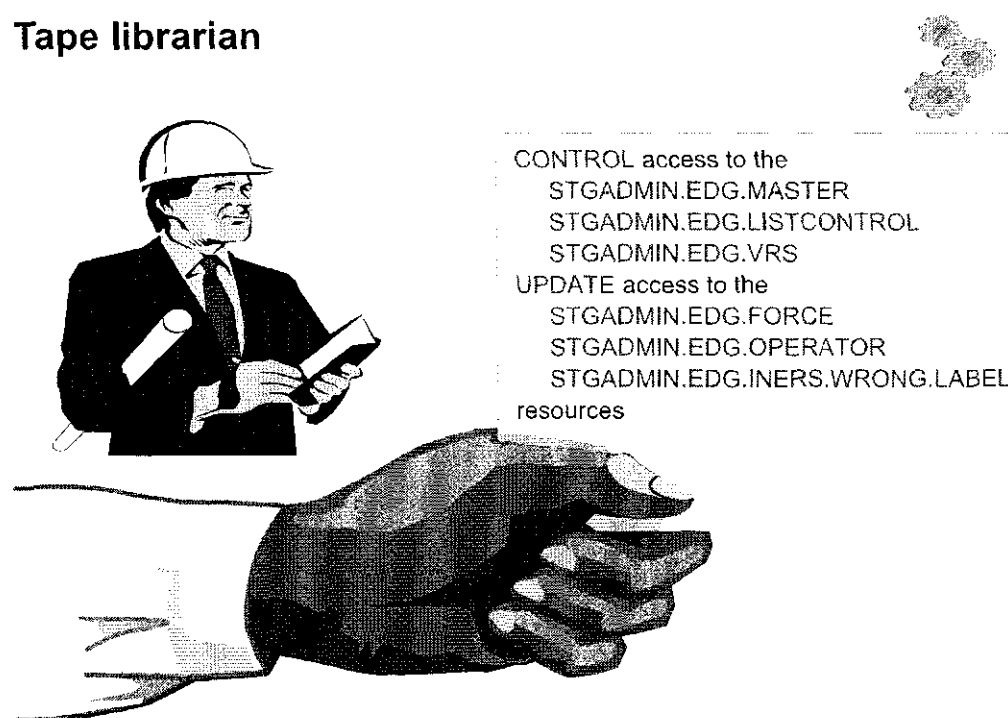


Figure 1-30. Control and Update Access-Tape Librarian

SS212.0

Notes:

Notice that DFSMSrmm does not rely on read for access control. A librarian needs control access to the same resource to which the general user has read access (STGADMIN.EDG.MASTER).

Storage Administrator

Storage administrator functions

- Define retention and storage policies for data sets and volumes
- Change information about any volume they own, their owner ID, and any vital record specification
- Manually request a scratch volume, and release a volume they own
- Delete a vital record specification
- Create lists of resources and display information about resources defined to DFSMSrmm



Figure 1-31. Storage Administrator

SS212.0

Notes:

Storage administrators, with the librarian, may be involved in establishing life cycle management policies for tape resident data. The slide shows an example of what might be performed. Your storage administrators might also help the librarian to release volumes other than their own.

Read, Control, and Update Access

Storage administrator

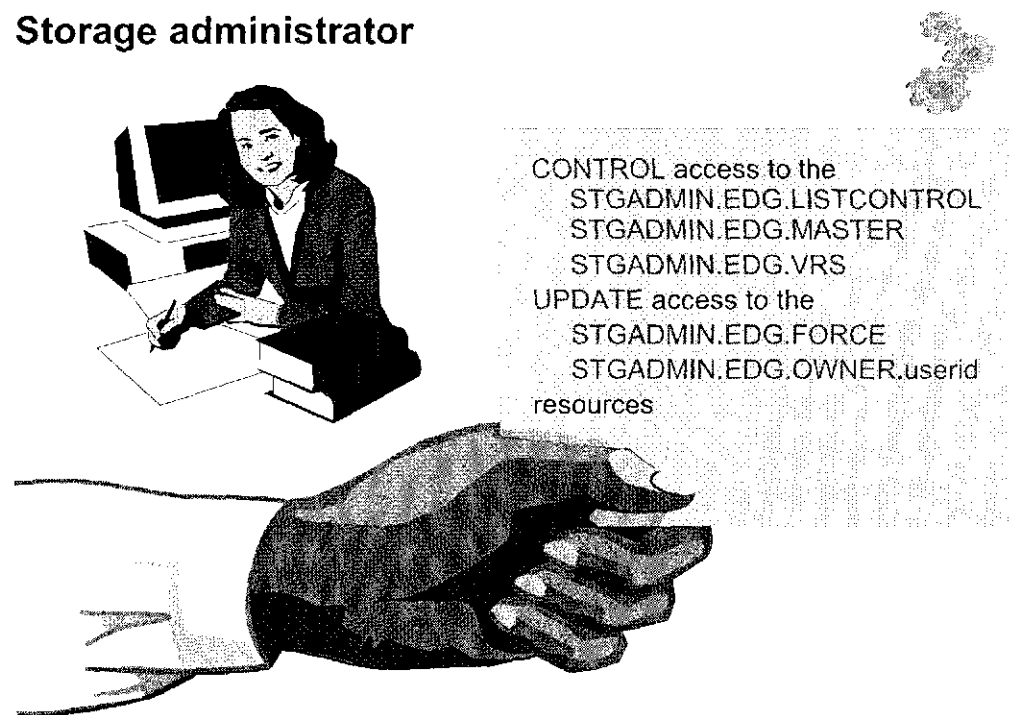


Figure 1-32. Read, Control, and Update Access

SS212.0

Notes:

To perform the functions shown, the access levels you see in the slide are needed. If a storage administrator is required to release volumes, CONTROL access is needed to STGADMIN.EDG.MASTER.

System Programmer

System programmer functions

- Display parmlib options and control data set control information
- Add, change, and delete information about any volume owner
- Manually request a scratch volume, and release a volume they own
- Create a list of software products and display information about any resource defined to DFSMSrmm



Figure 1-33. System Programmer

SS212.0

Notes:

Since system programmers receive and install software products, they should be allowed to create the software product information in DFSMSrmm. They are also generally involved in setting up the DFSMSrmm address space and the system options.

Control Access

System programmer

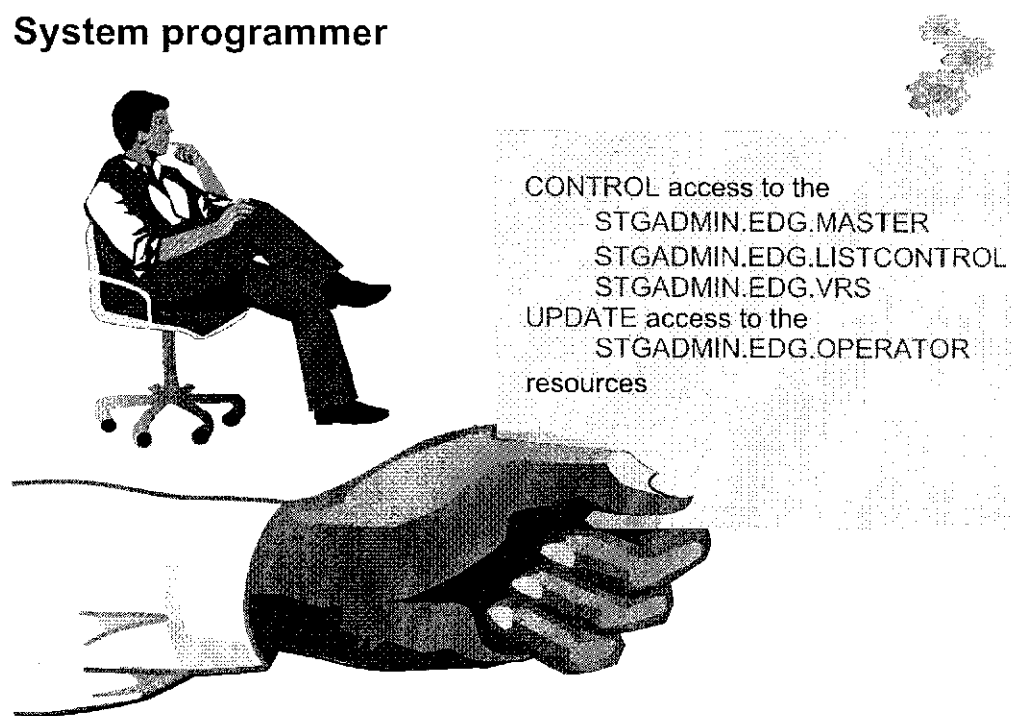


Figure 1-34. Control Access

SS212.0

Notes:

CONTROL access to MASTER is needed in order to create software product entries in DFSMSrmm.

Operator

Operator functions

- Fetch and mount tapes from specific pools or shelf locations, as specified in mount messages
- Manually erase and initialize tapes
- Create lists of scratch tapes available for use

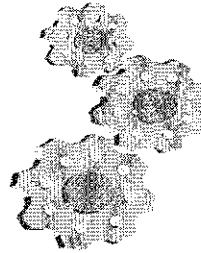


Figure 1-35. Operator

SS212.0

Notes:

The primary role of an operator is to mount and demount tapes. They need to know which ones can be mounted as scratch tapes. Operators are also required to initialize new tapes or reinitialize old ones. The chapter, *DFSMSrmm Operator Procedures* in the *DFSMSrmm Guide and Reference* provides operator information.

Read, Update and Alter Access-Operator

READ access to the
STGADMIN.EDG.MASTER
UPDATE access to
STGADMIN.EDG.OPERATOR
ALTER access to
STGADMIN.EDG.RESET.SSI
resources

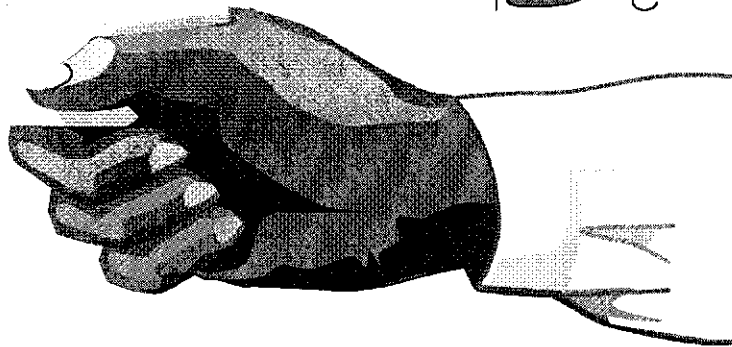


Figure 1-36. Read, Update and Alter Access-Operator

SS212.0

Notes:

READ access to STGADMIN.EDG.MASTER and UPDATE access to STGADMIN.EDG.OPERATOR profiles provide the basic operator functions. Having ALTER access to STGADMIN.EDG.RESET.SSI provides access to the program task *EDGRESET*, which is required when removing the DFSMSrmm address space from the system. We will discuss starting and stopping DFSMSrmm in more detail in the next unit.

Inventory Manager

Operator functions

- Create an extract of the control data set
- Select volumes to be retained and moved as vital records or disaster recovery
- Perform expiration processing
- Perform storage location management processing
- Back up the control data set



Figure 1-37. Inventory Manager

SS212.0

Notes:

DFSMSrmm dynamically records and validates tape processing. However, the analysis of the inventory is not automatically run, because it requires the control data set for its own use. While the inventory processing is running, it can have an effect on tape performance. For example, tape activity is restricted during the process of backing up the control data set. Therefore, it is the inventory manager that schedules and executes the processing of the DFSMSrmm control data set—that is, of the tape inventory. This is the process whereby tapes are returned to scratch status, while others are identified as those for shipping to a storage location. In general terms, we refer to this process as housekeeping.

Read Access - Inventory Manager

Inventory manager

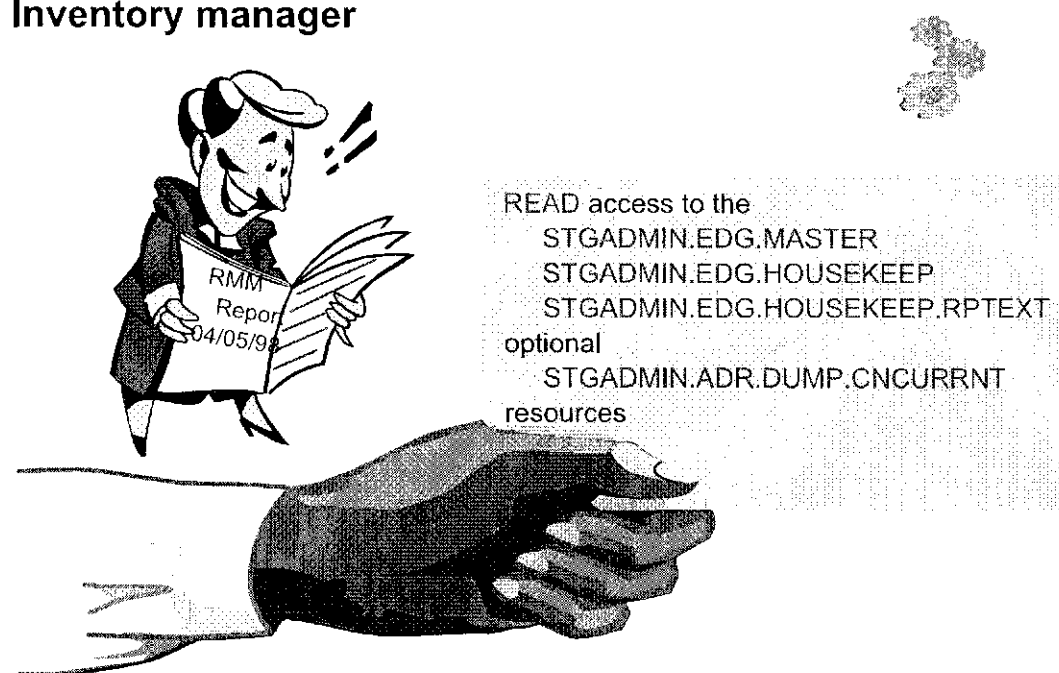


Figure 1-38. Read Access - Inventory Manager

SS212.0

Notes:

This facility class profile controls the use of the DFSMSrmm housekeeping utility program.

Checkpoint

Indicate whether each of the following statements is true or false.

1. During open of any tape data set, DFSMSdfp will receive a block ID for high speed locate from DFSMSrmm.
2. Installation defined storage locations are supported by DFSMSrmm as long as the name follows MVS simple name standards (one to eight positions in length starting with a non-numeric) and the name is not reserved for use by DFSMSrmm (such as ALL, SHELF, and HOME).
3. Any TSO userid can perform DFSMSrmm general user functions.
4. DFSMSrmm uses facility class resources to determine the authority of a user to perform requested functions.

Figure 1-39. Checkpoint

SS212.0

Notes:

Write down your answers (true or false) here:

- 1.
- 2.
- 3.
- 4.

Unit Summary

Now that you have completed this unit, you should be able to:

- Understand DFSMSrmm terminology
- Identify different types of users for command driven interactions with DFSMSrmm

Figure 1-40. Unit Summary

SS212.0

Notes:

Unit 2. DFSMSrmm Day-to-Day Procedures

What This Unit Is About

This unit describes some of the DFSMSrmm procedures that operators and administrators repeatedly perform.

What You Should Be Able to Do

After completing this unit, you should be able to:

- Use operator commands to control DFSMSrmm
- Understand the need for daily housekeeping
- Use the different reporting facilities

How You Will Check Your Progress

Accountability: Checkpoint questions

References

SC26-7404	<i>DFSMSrmm Guide and Reference</i>
SC26-7405	<i>DFSMSrmm Implementation and Customization Guide</i>
SC26-7406	<i>DFSMSrmm Reporting</i>

Objectives

The objectives for this module are as follows:

- To learn the operator commands that control DFSMSrmm
- To gain a basic understanding of DFSMSrmm inventory management
- To learn how to use the DFSMSrmm reporting facilities

Figure 2-1. Objectives

SS212.0

Notes:

Before going into implementation of DFSMSrmm in detail, let's have a look at some of the daily operations that operators and administrators must carry out.

2.1 Using Operator Commands to Control DFSMSrmm

z/OS DFSMSdfp without Using DFSMSrmm

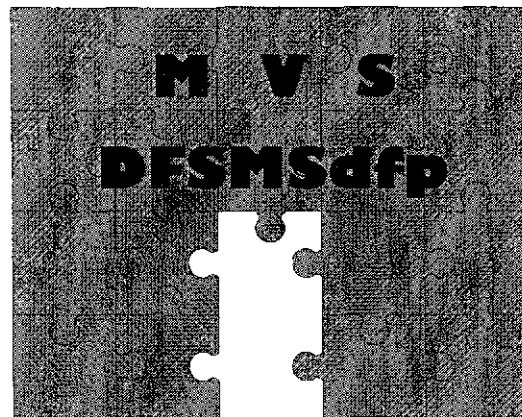


Figure 2-2. z/OS DFSMSdfp without Using DFSMSrmm

SS212.0

Notes:

Without a tape management system, we would be missing a large part of the inventory information normally associated with tape data. MVS records only the unit and volume serial numbers for cataloged tape data sets. There is no information about the data set, such as expiration or size. DFSMS system-managed tape offers no more information, because it actually manages tape volumes rather than the data sets.

Defining DFSMSrmm to z/OS

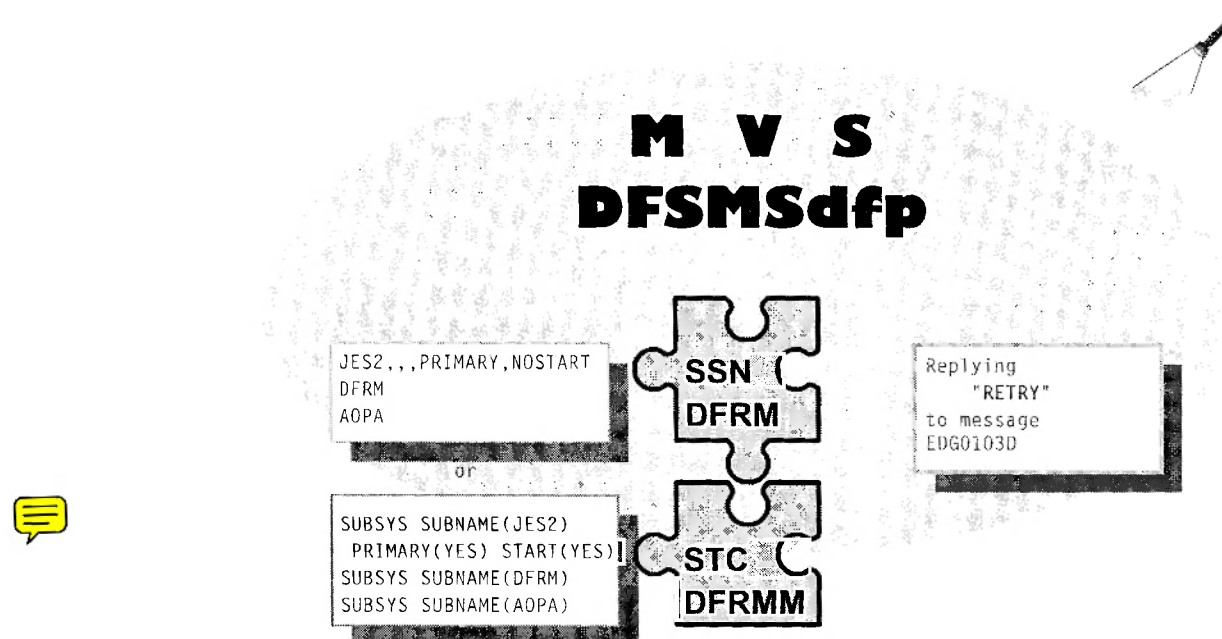


Figure 2-3. Defining DFSMSrmm to z/OS

SS212.0

Notes:

In most cases, you will want to become familiar with DFSMSrmm. By using the **manual operation mode** (by typing `OPTION OPMODE(MANUAL)` in EDGRMMxx), you can start the address space without affecting tape activities. By identifying DFSMSrmm to MVS, without automatically starting the subsystem interface, you can start DFSMSrmm in manual mode. When DFSMSrmm is running in manual mode, you can use RMM TSO subcommands, the DFSMSrmm ISPF dialog, and inventory management functions for all types of media. When in manual mode, DFSMSrmm neither records volume usage nor validates volumes.

Enabling DFSMSrmm's Subsystem Interface

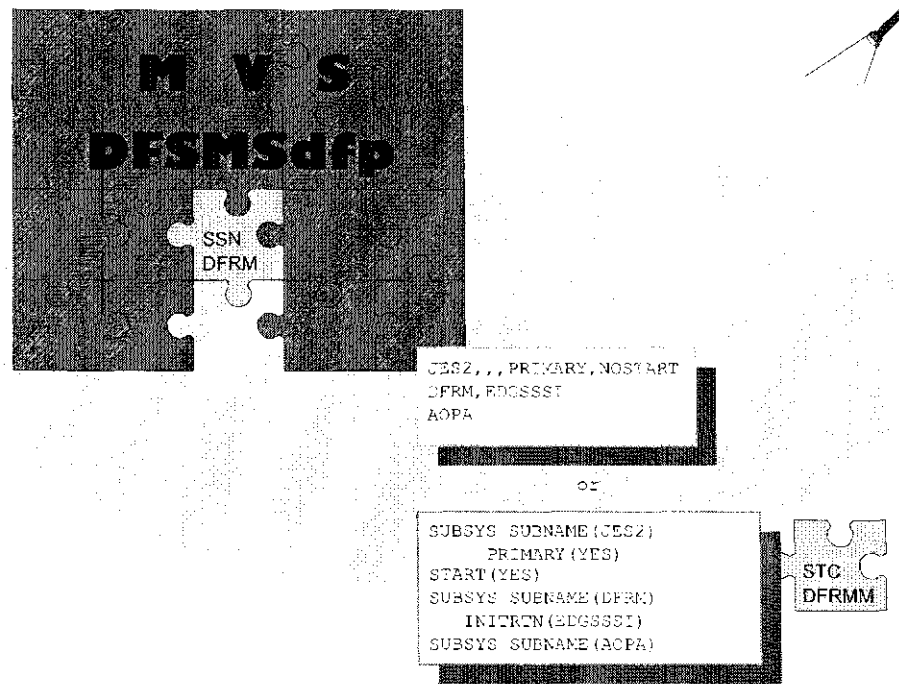


Figure 2-4. Enabling DFSMSrmm's Subsystem Interface

SS212.0

Notes:

Once you are ready to have DFSMSrmm record tape activity, it is important that tape activity not occur unless DFSMSrmm is active. By having the subsystem interface initialized during IPL means that tape activity is prevented until DFSMSrmm is either started or the subsystem interface is reset.

Starting DFSMSrmm

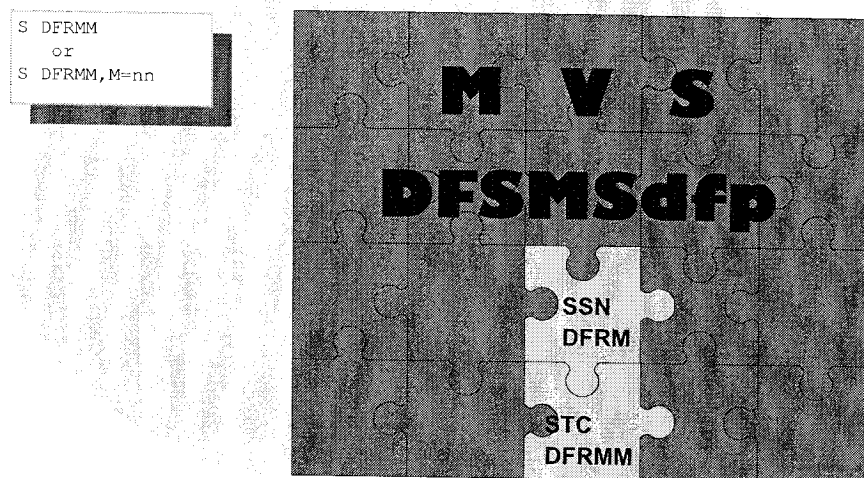


Figure 2-5. Starting DFSMSrmm

SS212.0

Notes:

With DFSMSrmm identified as a subsystem, we can start the address space. There is a sample procedure, EDGDFRMM, in SYS1.SAMPLIB that can be copied into SYS1.PROCLIB as DFRMM. If your DFSMSrmm parameters are in the system parmlib (SYS1.PARMLIB) and contain the data set names of your control data set and journal, you can use the procedure without modification. However, it is highly recommended that you enable problem determination aid tracing and logging. If you do, you need to add your data definition (DD) statements for the log data sets to the procedure.

Starting DFSMSrmm

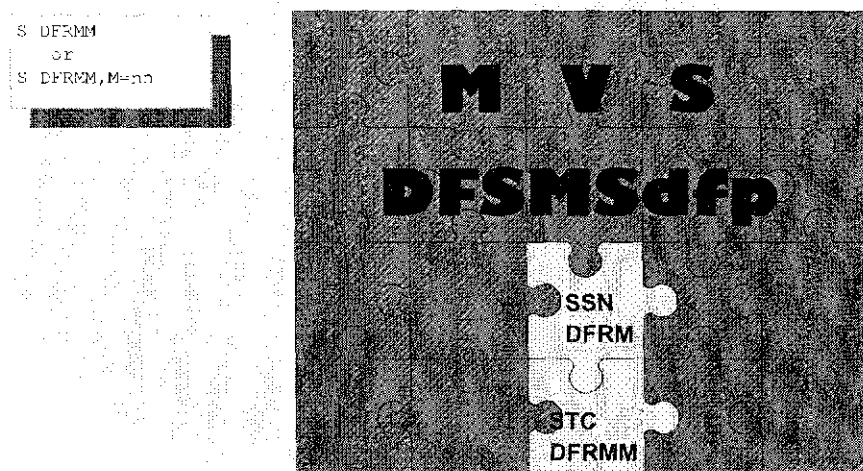


Figure 2-5. Starting DFSMSrmm

SS212.0

Notes:

With DFSMSrmm identified as a subsystem, we can start the address space. There is a sample procedure, EDGDFRMM, in SYS1.SAMPLIB that can be copied into SYS1.PROCLIB as DFRMM. If your DFSMSrmm parameters are in the system parmlib (SYS1.PARMLIB) and contain the data set names of your control data set and journal, you can use the procedure without modification. However, it is highly recommended that you enable problem determination aid tracing and logging. If you do, you need to add your data definition (DD) statements for the log data sets to the procedure.

Stopping DFSMSrmm

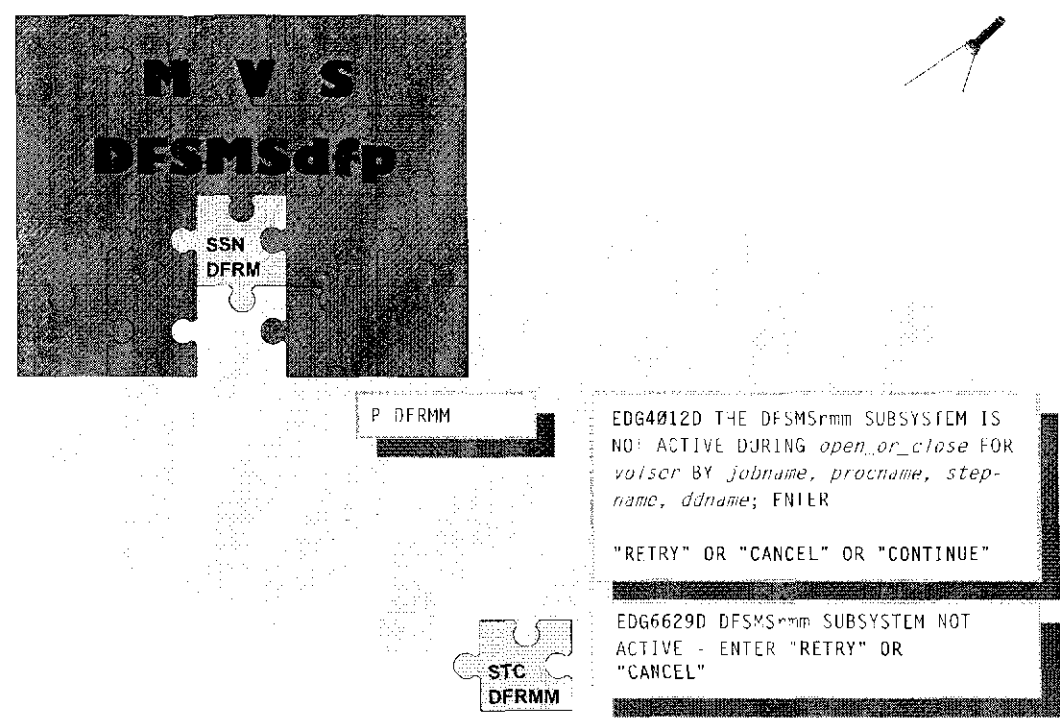


Figure 2-6. Stopping DFSMSrmm

SS212.0

Notes:

After stopping DFSMSrmm when it is in *record*, *warning*, or *protect* mode, the subsystem interface prevents tape activities. If a tape data set is opened or closed, an error message is issued. For OPEN processing, DFSMSrmm prevents the data set from being opened until DFSMSrmm is restarted. However, *continue* will allow the data set to be closed, but the control data set will not reflect this event.

The second message is an example of a message that appears when DFSMSrmm is stopped while EDGINERS was executing and initializing tapes. To avoid losing control data set updates, the reply should be *cancel*.

Disabling the DFSMSrmm Subsystem Interface

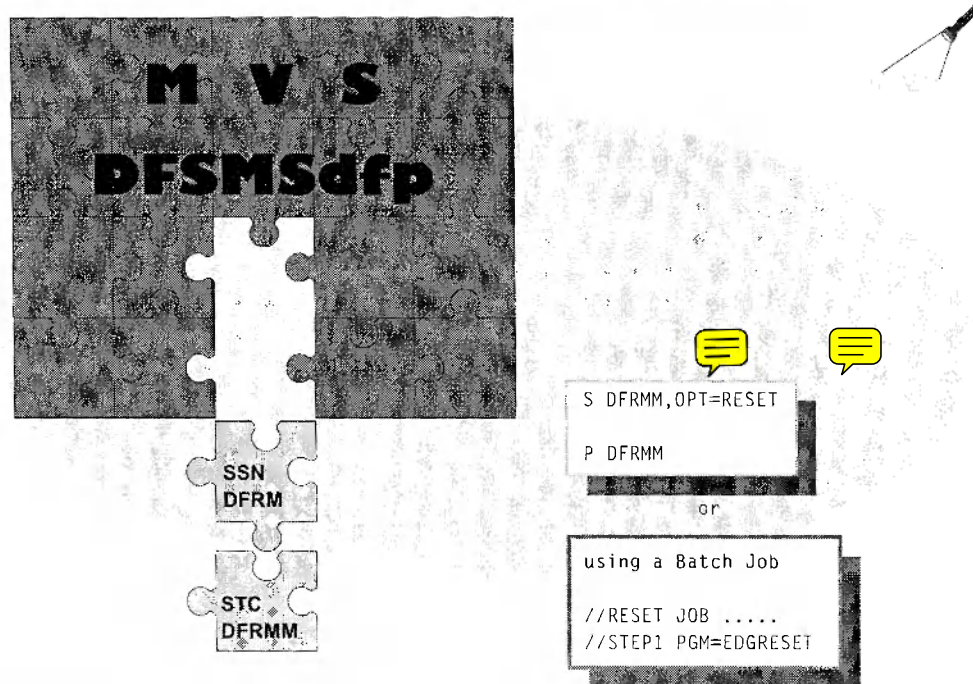


Figure 2-7. Disabling the DFSMSrmm Subsystem Interface

SS212.0

Notes:

During a conversion, it can happen that the conversion of the current tape management system database into the DFSMSrmm control data set did not work properly. During the analysis of what needs to be done to correct the situation, it may be necessary to disable DFSMSrmm. An authorized user can do this by issuing the commands shown. After the stop has completed, the DFSMSrmm subsystem interface is disabled and tape activity can continue. The DFSMSrmm control data set that was in use will no longer remain current.

Restarting the DFSMSrmm Subsystem

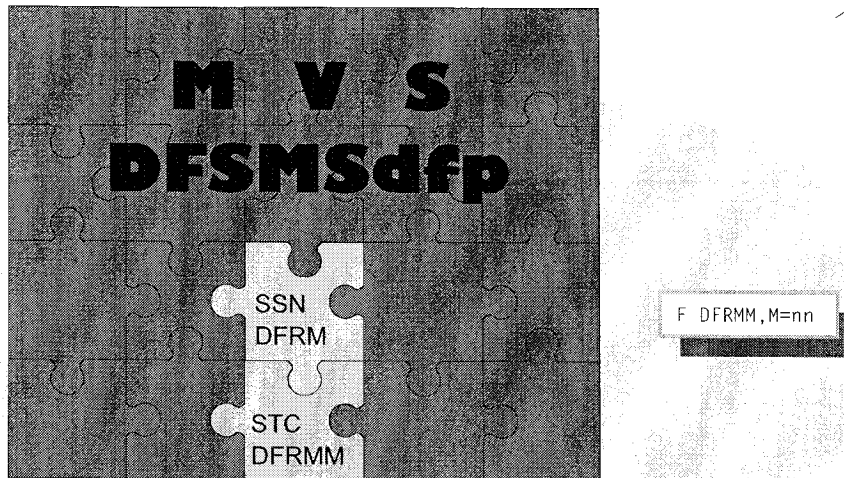


Figure 2-8. Restarting the DFSMSrmm Subsystem

SS212.0

Notes:

As the implementation progresses, it will be necessary to change the mode of operation for DFSMSrmm. To do this, you change the OPMODE value in an EDGRMMxx member, and then use the MODIFY command to restart DFSMSrmm.

The subsystem temporarily stops and re-initializes itself with the new OPMODE. If there is tape activity when the DFSMSrmm subsystem is inactive, DFSMSrmm issues a RETRY message to the operator for data sets being opened or closed, and a "subsystem not active" message for TSO subcommand activity. However, the time during reinitialization is very short, and the chance of a request during that time is small.

During this restart, DFSMSrmm issues the following message:

```
EDG0103D DFSMSrmm SUBSYSTEM INTERFACE IS INACTIVE -
      ENTER "IGNORE", "CANCEL" OR "RETRY"
```

After replying *retry* to this message, DFSMSrmm initializes its subsystem interface.

Use this command whenever you update DFSMSrmm parameters and want to implement the change.

Quiescing DFSMSrmm

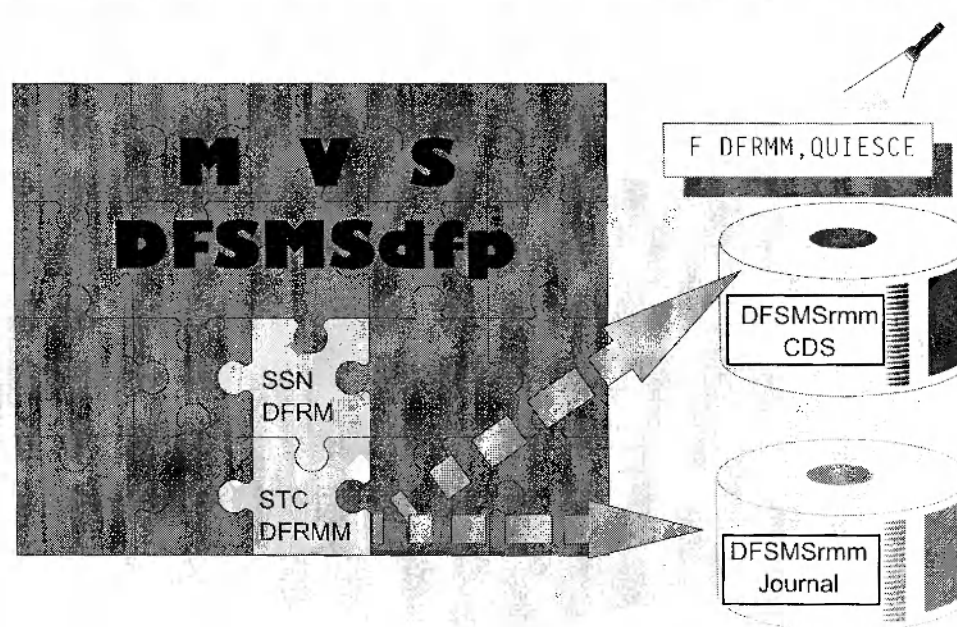


Figure 2-9. Quiescing DFSMSrmm

SS212.0

Notes:

When performing maintenance or recovery on the DFSMSrmm control data set or journal, you must de-allocate them. The following command allows DFSMSrmm to suspend activities and de-allocate the control data set and journal:

MODIFY DFRMM,QUIESCE

After performing any necessary activities, restart DFSMSrmm:

MODIFY DFRMM,M=xx

where xx is an appropriate parmlib member ID.

Commands that Control DFSMSrmm

Starting DFSMSrmm

- S DFRMM
- S DFRMM,M=01

Restarting DFSMSrmm

- F DFRMM,M=02
- F DFRMM,REFRESH EXITS

Stopping DFSMSrmm

- P DFRMM

Disabling DFSMSrmm

- S DFRMM,OPT=RESET
- P DFRMM

Quiescing DFSMSrmm

- F DFRMM,QUIESCE

Display activities

- F DFRMM, QUERY ACTIVE

Figure 2-10. Commands that Control DFSMSrmm

SS212.0

Notes:

Starting DFSMSrmm:

The first start command uses the default parameters coded in the procedure.

The second start uses the member EDGRMM01 in the parmlib specified in the JCL for procedure DFRMM.

Restarting DFSMSrmm:

The first MODIFY changes the DFSMSrmm environment based on the parameters in parmlib member, EDGRMM02.

The second MODIFY causes the DFSMSrmm exits to be reloaded. If DFSMSrmm is running in a shared environment, you must repeat this step on each system.

Stopping DFSMSrmm:

While the command appears to stop DFSMSrmm, DFSMSrmm does not necessarily go away. If DFSMSrmm is running in any mode other than manual when the command is issued, the interface is still present and will prevent subsequent tape activity.

2.2 Inventory Management

Recording when DFSMSrmm Is Active

- **Automatic recording**
(Open/Close/End-of-volume)
- **Manual updating**
 - ISPF dialog
 - TSO RMM subcommands
 - Application program interface
 - Set of utilities
- **Creation of SMF records**
 - Security records
 - Audit trail records

Figure 2-11. Recording when DFSMSrmm Is Active

SS212.0

Notes:

DFSMSrmm automatically records information during open, close and end-of-volume processing.

Records are also updated when requested by an authorized user via the TSO RMM subcommands or the ISPF dialog as well as when the DFSMSrmm utilities are being executed.

Based on the options specified in EDGRMMxx, DFSMSrmm will also record user SMF records for CDS updates and classified data set OPENs.

Volume Status

Scratch

A term that means the volume is free and available for use.

Master

A private volume containing data that is available for write processing based on the DFSMSrmm option **MASTEROVERWRITE** in parmlib member **EDGRMMxx**.

User

A volume assigned to a user, which can contain any data and be rewritten as many times as the user wishes until the volume expires.

Figure 2-12. Volume Status

SS212.0

Notes:

Scratch tapes are available to be written in response to a non-specific request. DFSMSrmm does not allow them to be mounted in response to a specific request regardless of whether the request is to read or to write. If a volume returns to scratch it still contains data that was formerly written on (unless being erased). So in most cases you could save such data from being overwritten by changing the volume status back to *master* or *user*. A scratch tape never has an owner.

Volumes containing data are considered *master* volumes. The EDGRMMxx **MASTEROVERWRITE** OPTION determines what write processing can occur to a volume in master status. A master volume will have an owner.

A volume in user status is assigned to an owner, and it can contain data. However, the volume can be rewritten as needed until it expires.

DFSMSrmm Management of Tape Volumes

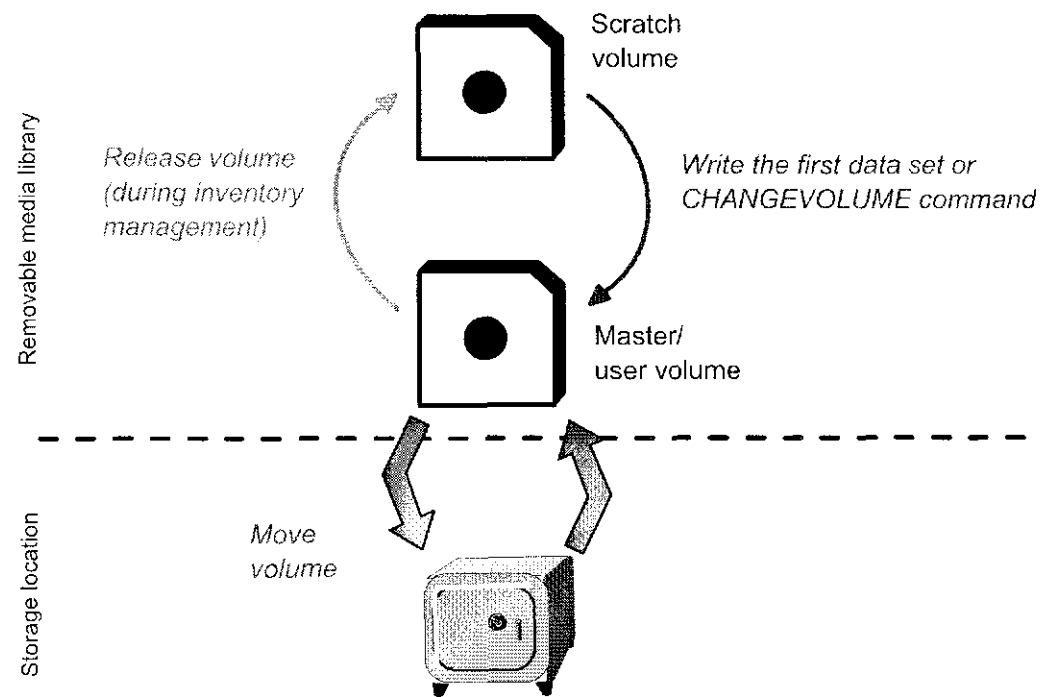


Figure 2-13. DFSMSrmm Management of Tape Volumes

SS212.0

Notes:

The status of volumes in DFSMSrmm changes from scratch to master or user and vice versa.

Change of status from master/user to scratch is called release of a volume. A release cannot be done by command. You always need a housekeeping run to complete the release process of a volume.

Change of Volume Status

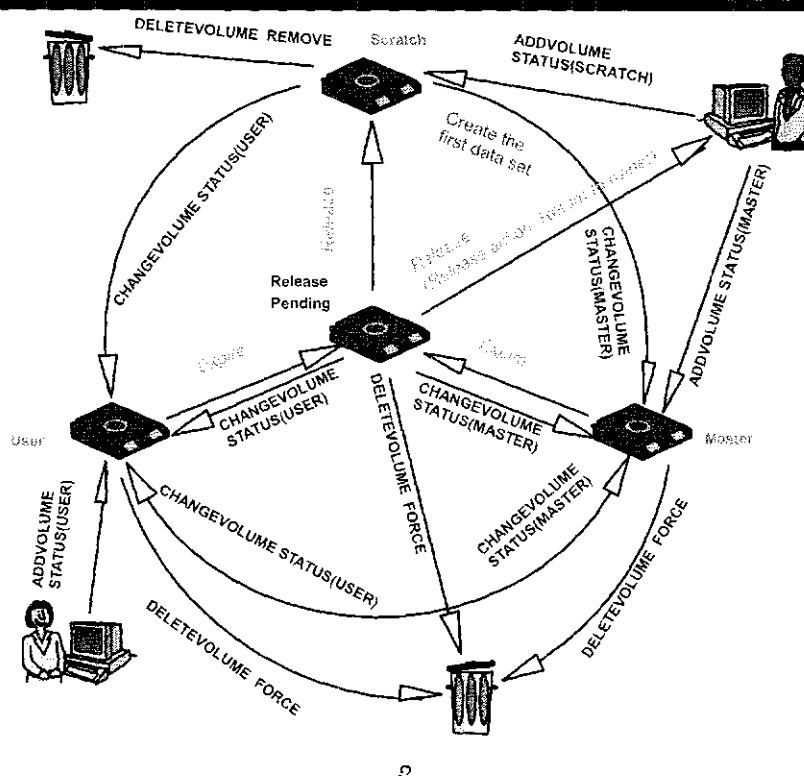


Figure 2-14. Change of Volume Status

SS212.0

Notes:

The librarian can use the following command to add volumes of any kind of status to DFSMSrmm at any time:

`ADDVOLUME volser STATUS(SCRATCH|MASTER|USER) ...`

To change the status of a volume, an authorized user (a librarian or an owner) can use this command:

`CHANGEVOLUME volser STATUS(MASTER|USER) ...`

An authorized user can use the following command to cause the expiration of a master or a user volume:

`DELETEVOLUME volser RELEASE`

Volumes expire automatically during inventory management if their expiration date is reached or exceeded and if they are no longer retained by VRS.

During expiration (without regard to whether this is caused by a command or happens automatically), availability will be set to pending release while the status remains master or user.

A release is carried out during the next housekeeping for volumes being marked as pending release if the only release action is *return to scratch*.

Scheduling of DFSMSrmm Utilities

The following actions are scheduled daily:

- Process vital records
- Perform expiration processing
- Create an extract data set
- Back up the DFSMSrmm control data set
- Back up the DFSMSrmm journal
- Check DFSMSrmm CDS space utilization

Figure 2-15. Scheduling of DFSMSrmm Utilities

SS212.0

Notes:

The inventory management program EDGHSKP performs the following functions:

VRSEL	Vital records processing
EXPROC	Expiration processing
DSTORE	Storage location management
BACKUP	Control data set and journal backup and reset of the journal
RPTEXT	Creation of an extract data set for reporting

These functions can be run individually or in the same step. If two or more are run in the same day, such as VRSEL, EXPROC, BACKUP, and RPTEXT, it is more efficient to run them in the same step.

A check of CDS space utilization can be done only by using access method services command, LISCTC ENT(...) ALL.

Scheduling of DFSMSrmm Utilities (Cont)

Perform storage location management	Weekly
Create volume movement and inventory reports	Weekly
Create security and audit reports	Monthly
Initialize and erase volumes	Weekly
Reorganize DFSMSrmm control data set	Once a year

Figure 2-16. Scheduling of DFSMSrmm Utilities (Cont)

SS212.0

Notes:

In our example, tapes are sent to the storage locations weekly. Since initializing tapes may require tape drives and operator intervention (mounts), this example schedules that operation weekly. For this example, weekly audit reports are adequate.

To reorganize the control data set, you can use the utility, EDGBKUP. Refer to the *DFSMSrmm Implementation and Customization Guide* for details. The control data set must be de-allocated by DFSMSrmm for this process.

Processing of Vital Records

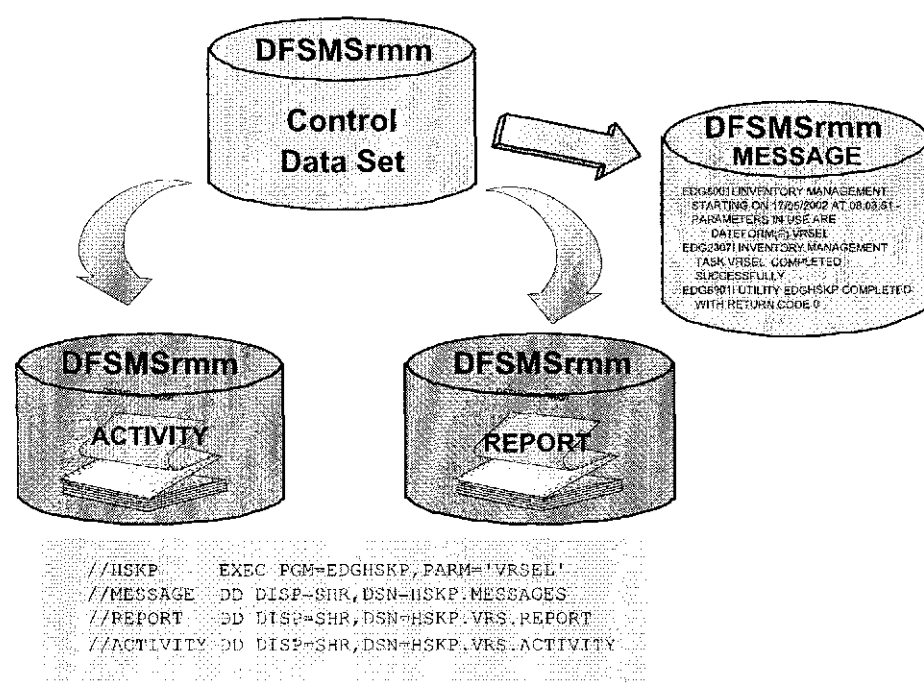


Figure 2-17. Processing Vital Records

SS212.0

Notes:

VRSEL retains volumes based upon matching data sets and volumes to VRS policies. You can ensure that your policies have been read by coding **OPTION VRSMIN(n,FAIL)** in EDGRMMxx where n is the minimum number of VRS policies DFSMSrmm should read during processing. If you want processing to continue, you can replace FAIL with INFO or WARNING. The WARNING option will set a return code of four for the run. Volumes whose VRS policy is for movement as well as retention will be flagged for further processing (DSTORE).

You can execute a production run of vital records processing by using the following:

```

//HSKP      EXEC PGM=EDGHSKP, PARM='VRSEL'
//MESSAGE   DD DISP=SHR, DSN=HSKP.MESSAGES
//REPORT    DD DISP=SHR, DSN=HSKP.VRS.REPORT
  
```

to check the vital record specifications that match data sets and to see which versions of the data sets are being retained.

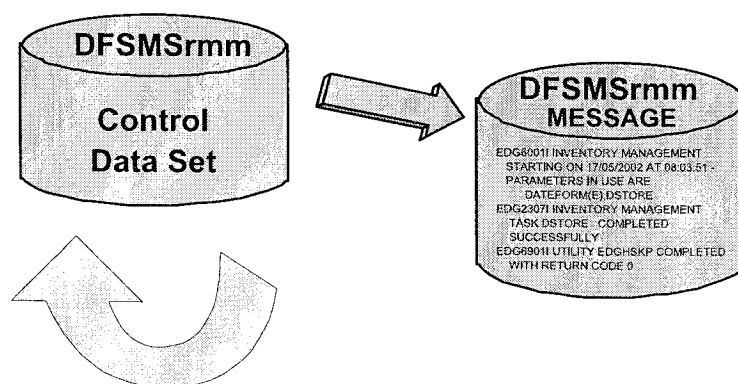
If VRS changes have been made, run a VERIFY step:

```
//HSKP EXEC PGM=EDGHSKP, PARM='VRSEL, VERIFY'  
//MESSAGE DD DISP=SHR, DSN=HSKP.MESSAGES  
//REPORT DD DISP=SHR, DSN=HSKP.VRS.REPORT  
//ACTIVITY DD DISP=SHR, DSN=HSKP.ACTIVITY
```

Member EDGJACTP in SYS1.SAMPLIB will produce reports from the *activity* file.

VRSEL will not run until a VERIFY run has completed successfully, unless **OPTION**
VRCHANGE(INFO) has been chosen in EDGRMMxx.

Performing Storage Location Management



```
//STEP1    PGM=EDGHSKP, PARM='DSTORE'
//SYSPRINT DD SYSOUT=*
//MESSAGE  DD DSN=RMM.MASTER.MESSAGE,
//          DISP=SHR
```

Figure 2-18. Performing Storage Location Management

SS212.0

Notes:

The DSTORE phase of housekeeping processing assigns a shelf location, known as the bin number. This assignment is done to volumes moving to and among shelf-managed storage locations and the process sets a destination storage location.

Performing Expiration Processing

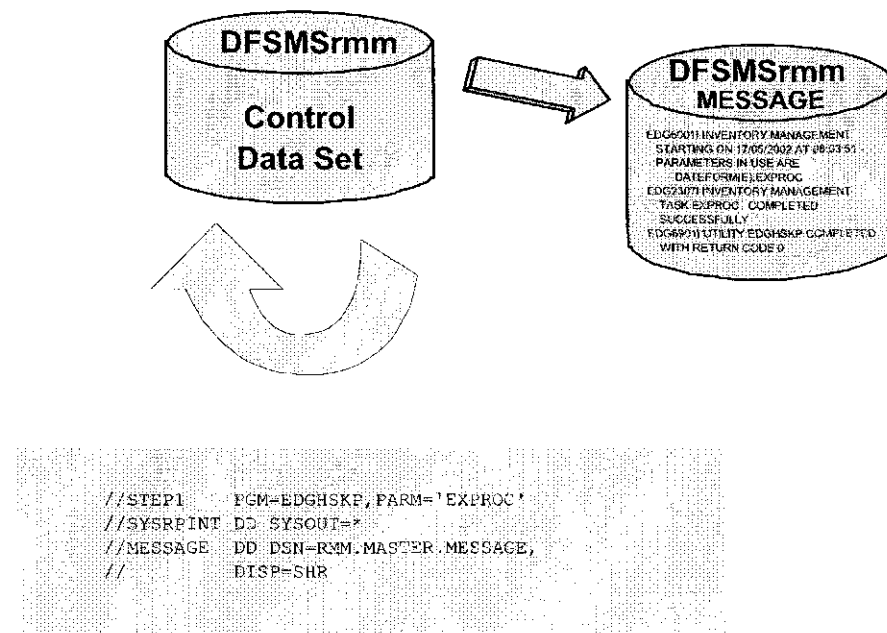


Figure 2-19. Performing Expiration Processing

SS212.0

Notes:

DFSMSrmm checks the expiration date for volumes not managed by a vital record specification. If the expiration date has been reached, DFSMSrmm changes the volume availability to *pending release*. DFSMSrmm checks for any actions that should be taken before the volume can be released. DFSMSrmm defers most release actions for later processing. However, DFSMSrmm does process the "notify owner" release action immediately if an electronic address is provided for the owner and the DFSMSrmm system option NOTIFY is in use.

When a volume reaches its expiration date, and is not being retained by a vital record specification, it is not immediately returned to scratch or to its owner. Two runs of expiration processing are needed to return a volume. The first run of expiration processing sets the volume availability to pending release. No other information is changed in case the volume is reclaimed before it is returned to scratch status. Expiration processing must be run a second time to complete the return.

Note: You can also return a volume to scratch during only one run of EXPROC using vital record specification options. However, this might not work in every case.

Creating an Extract Data Set

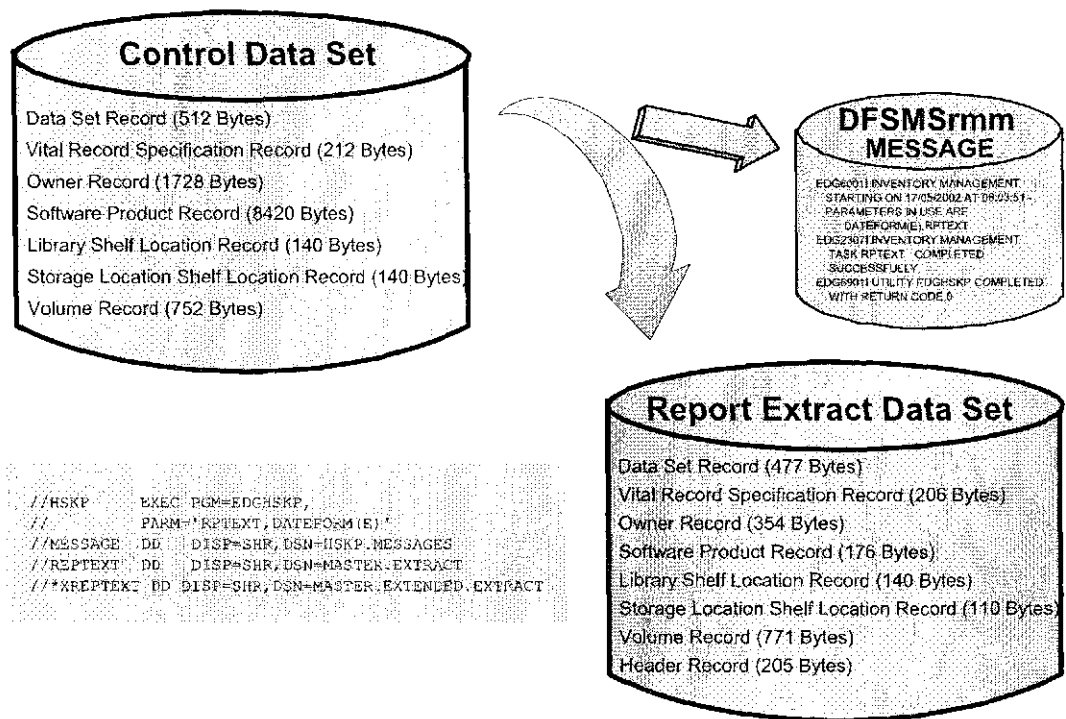


Figure 2-20. Creating an Extract Data Set SS212.0

Notes:

Recall that DSTORE assigned a destination and shelf location (where applicable). The EDGRPTD program produces inventory and movement reports. However, it requires an extracted version of the control data set.

Choose your preferred form for the date by coding **DATEFORM(x)** where x can be:

- A** American - mm/dd/yyyy
- E** European - dd/mm/yyyy
- I** ISO - yyyy/mm/dd
- J** Julian - yyyy/ddd
- d** Default - Installation's default in EDGRMMxx, initially Julian

The record size values shown in the figure are subject to change. Please refer to the section titled *Assessing DASD Space and Placement for the Extract Data Set* in the *DFSMSrmm Implementation and Customization Guide*, SC26-4932 for storage estimates.

Backing Up the Control Data Set

While DFSMSrmm active

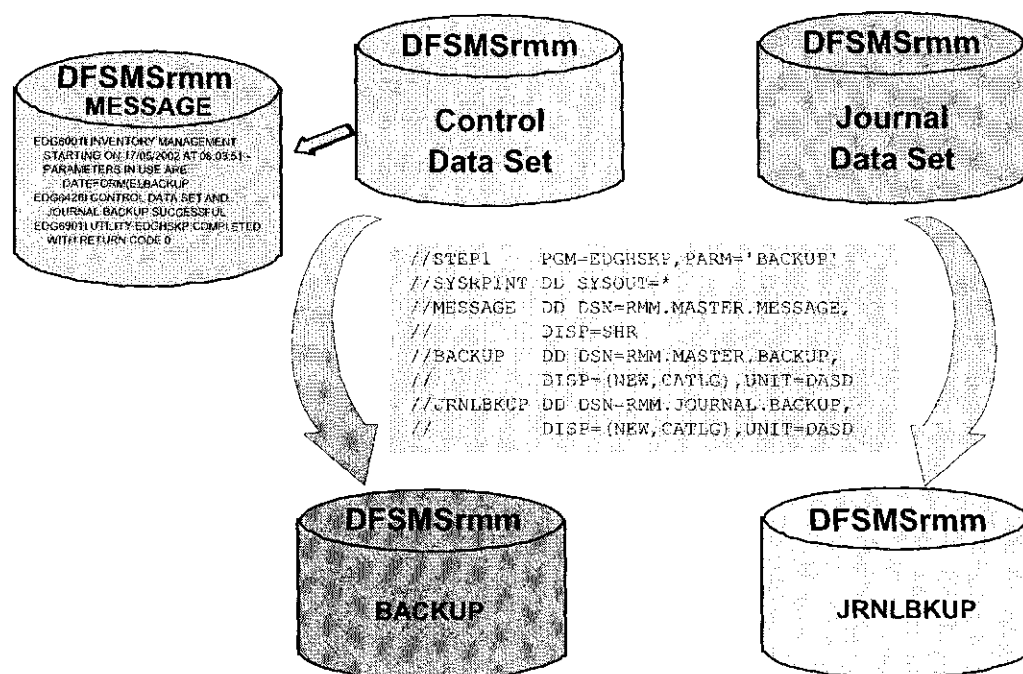


Figure 2-21. Backing Up the Control Data Set

SS212.0

Notes:

The simple way of backing up the DFSMSrmm control data set and journal makes internal use of IDCAMS REPRO to copy the data. The figure shows some sample JCL to do this. It is recommended, however, that you use generation data sets and retain more than one backup version. Note that only DASD data sets are supported for output. When both data sets are successfully backed up, the journal data set is nulled.

IDCAMS enqueues exclusively the input data sets that will make the CDS and journal unavailable for DFSMSrmm during backup. This means that DFSMSrmm is unable to process open/close/end-of-volume or command requests. Our next example shows how to avoid these disadvantages.

Backing Up the Control Data Set (Cont)

While DFSMSrmm active

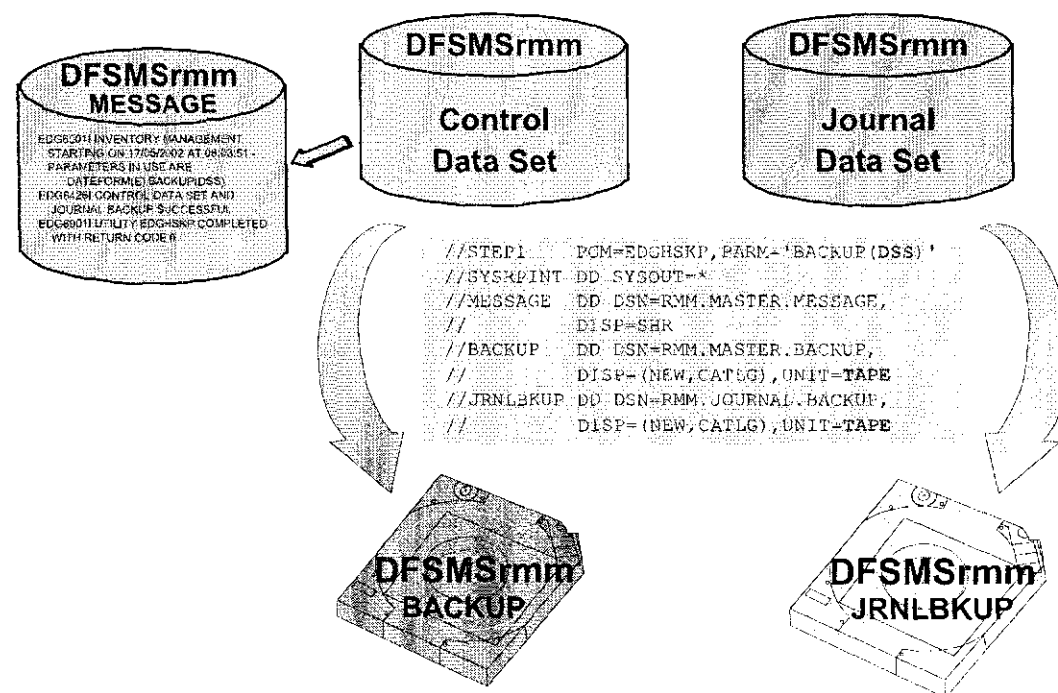


Figure 2-22. Backing Up the Control Data Set (Cont)

SS212.0

Notes:

Requesting `PARM='BACKUP(DSS)'` will internally use DFSMSdss with the keyword `CONCURRENT`, which will perform the backup as a fast point-in-time backup by using different hardware features depending on availability. In the following sample code, the use of concurrent copy by DFSMSdss is requested:

```

//EDGHSKP EXEC PGM=EDGHSKP, PARM='BACKUP(DSS)'
//MESSAGE DD DISP=SHR, DSN=RMM.MESSAGE
//BACKUP   DD DISP=(,CATLG), UNIT=TAPE, DSN=BACKUP.CDS(+1),
//          LABEL=(,SL)
//JRNLBKUP DD DISP=(,CATLG), UNIT=TAPE, DSN=BACKUP.JRNL(+1),
//          LABEL=(2,SL), VOL=REF=*.BACKUP
//DSSOPT   DD *
//          CONCURRENT OPTIMIZE(1) VALIDATE
//          /*
  
```

This process does not hinder overall tape processing, even if, as can sometimes happen, the time to write the backups to tape takes several minutes.

2.3 Reporting Facilities of DFSMSrmm

Available Reports with EDGRPTD

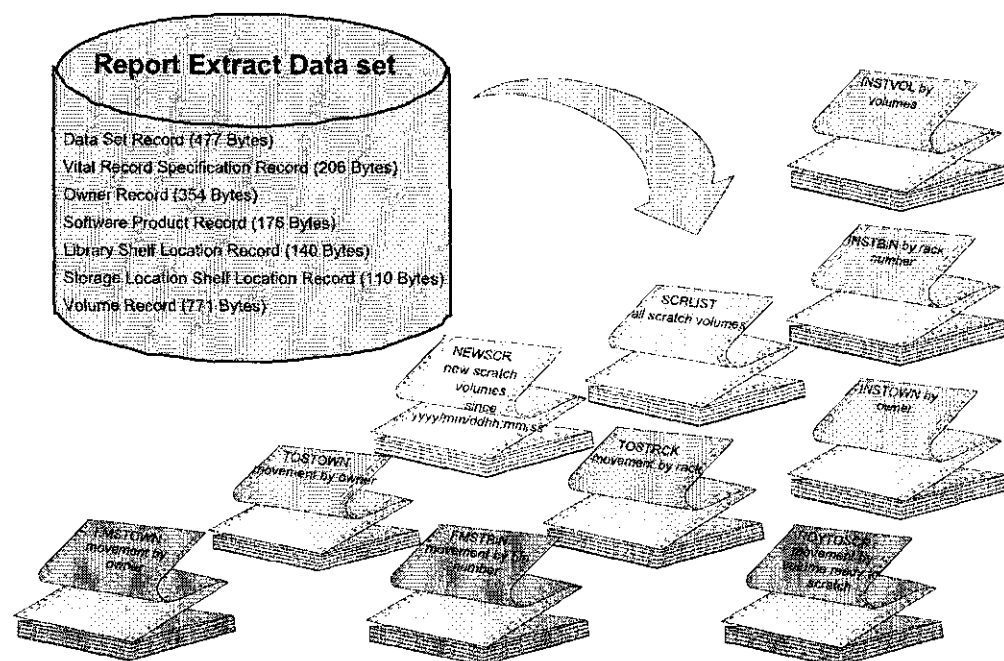


Figure 2-23. Available Reports with EDGRPTD

SS212.0

Notes:

The extract file created by the **RPTTEXT** processing phase is used by EDGRPTD to produce volume inventory and movement reports, as the following code sample shows:

```
//RPTD      EXEC PGM=EDGRPTD,
//          PARM='SEC('INTERNAL USE ONLY'),DATEFORM(I),LINECOUNT(54)'
//REPTTEXT DD DISP=SHR,DSN=RMMTST.PR0914X.REPTTEXT
//SYSPRINT DD SYSOUT=*
//INSTVOL  DD DISP=SHR,DSN=RMMTST.REPORT.INSTVOL
//INSTBIN  DD DISP=SHR,DSN=RMMTST.REPORT.INSTBIN
//INSTOWN  DD DISP=SHR,DSN=RMMTST.REPORT.INSTOWN
//TOSTRCK  DD DISP=SHR,DSN=RMMTST.REPORT.TOSTRCK
//TOSTOWN  DD DISP=SHR,DSN=RMMTST.REPORT.TOSTOWN
//FMSTBIN  DD DISP=SHR,DSN=RMMTST.REPORT.FMSTBIN
//FMSTOWN  DD DISP=SHR,DSN=RMMTST.REPORT.FMSTOWN
//RDTOSCR  DD DISP=SHR,DSN=RMMTST.REPORT.RDTOSCR
//NEWSOCR  DD DISP=SHR,DSN=RMMTST.REPORT.NEWSOCR
//SCRDATE  DD DISP=SHR,DSN=RMMTST.INPUT.SCRDATE
```

```
//SCRLIST DD DISP=SHR,DSN=RMMTST.REPORT.SCRLIST
//SYSOUT DD DISP=SHR,DSN=RMMTST.REPORT.DFSORT
```

To select a report, simply specify the appropriate DD statement.

The reports are:

INSTVOL	Output file for the report that contains the inventory of volumes by location sorted by volume serial number.
INSTBIN	Output file for the report that contains the inventory of volumes by location sorted by rack or bin number. The storage location report is sorted by bin number. All other reports are sorted by rack number.
INSTOWN	Output file for the report that contains the inventory of volumes by location sorted by owner.
RDYTOSCR	Output file for movement reports sorted in ascending order by rack report column, which can be either a bin or a rack number. The report only contains volumes that are ready to become scratch tapes when their move HOME is confirmed. Note: Volumes listed in the Ready-To-Scratch report are excluded from either the TOSTRCK or the FMSTBIN report.
TOSTRCK	Output file for movement reports sorted by rack number: Volumes to be moved from SHELF to storage locations Volumes to be moved between system-managed libraries Volumes to be moved from system-managed libraries to storage locations
FMSTBIN	Output file for movement reports sorted by bin number: Volumes to be moved between storage locations Volumes to be moved from storage locations to SHELF Volumes to be moved from storage locations to system-managed tape libraries
TOSTOWN	Output file for movement reports sorted by owner: Volumes to be moved from SHELF to storage location Volumes to be moved from system-managed libraries to storage locations Volumes to be moved between system-managed libraries
FMSTOWN	Output file for movement reports sorted by owner: Volumes to be moved from storage locations to SHELF Volumes to be moved from storage locations to system-managed libraries Volumes to be moved between storage locations
NEWSCR	Output file for a list of scratch volumes sorted by volume serial number: Scratch volumes returned to scratch status since the date specified in the SCRDATE input file.
SCRDATE	Input/output file for a list of new scratch volumes:

Sequential data set containing one record (FB, record length: 80). Each time a scratch list report is produced, DFSMSrmm updates the SCRDATE file with the highest scratch date and time for a volume. DFSMSrmm uses the date and time to determine which volumes to include in the new scratch list report. DFSMSrmm includes all scratch volumes with a newer assigned date and time in the new scratch listing.

SCRLIST Output file for a complete list of scratch volumes sorted by volume serial number.

EDGRPTD Scratch Reporting

Processes RMM extract file

With movement and inventory reports

New DD statements for **EDGRPTD**

SCRDATE

Data set which contains just a date/-time record

Example:

2000/01/0517:15:00

Used as input to select volumes for new scratch report

Used as output to save highest scratch date/-time for next run

NEWSCR

Output file for the report of volumes scratched since SCRDATE date and time

SCRLIST

Output file for the report of all scratch volumes

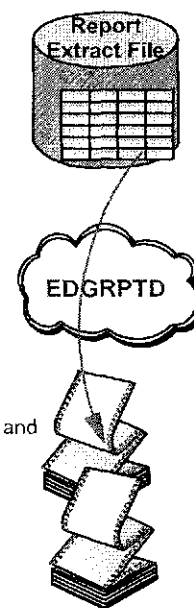


Figure 2-24. EDGRPTD Scratch Reporting

SS212.0

Notes:

Support for scratch lists was released with OS/390 DFSMS 2.10. You can choose between a list containing all scratch volumes and a list of new scratch volumes. New volumes are selected based on the date specified in the data set allocated using DD card SCRDATE.

EDGRPTD Scratch Reporting (Example)

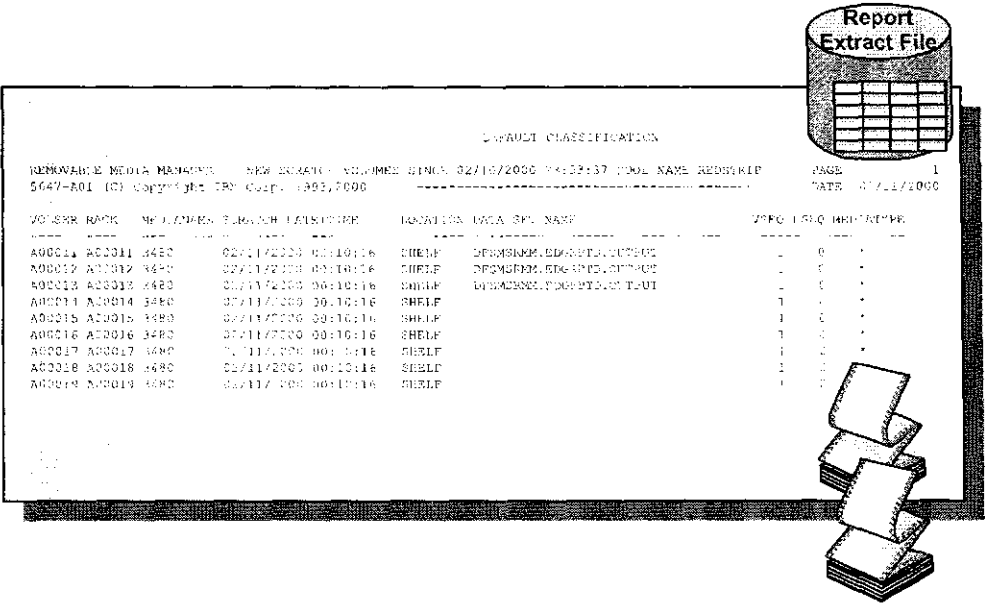


Figure 2-25. EDGRPTD Scratch Reporting (Example) SS212.0

Notes:

Scratch volumes are sorted by volume serial numbers.

Extended Report Extract Data Set

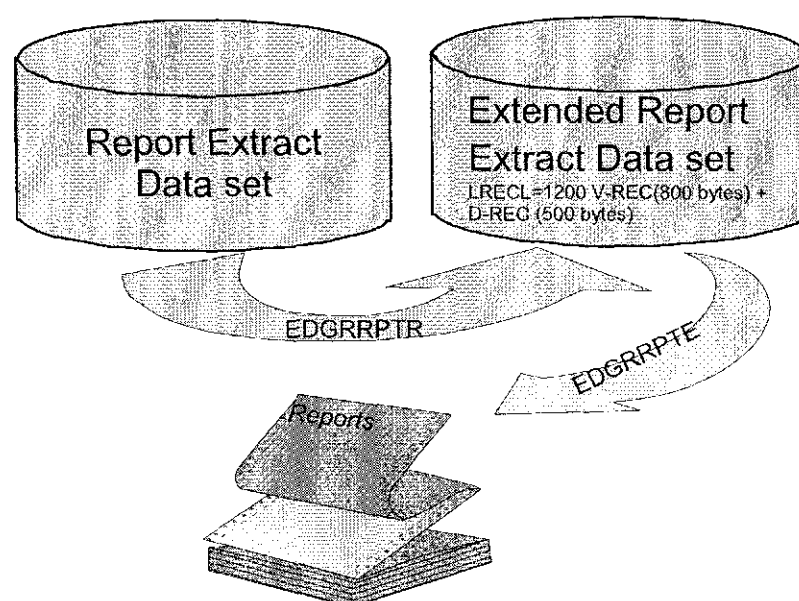


Figure 2-26. Extended Report Extract Data Set

SS212.0

Notes:

The DFSMSrmm report utility EDGRPTD does not provide any data set reports. IBM supplies some sample reports for DFSMSrmm extended reporting. One set (which we will present next) is created from an extended report extract file. When extended reporting came up, IBM provided only a REXX CLIST, EDGRRPTR, which would create the extended extract file. Nowadays, an extended report extract file can be created by REPTEXT during housekeeping.

Extended Reporting

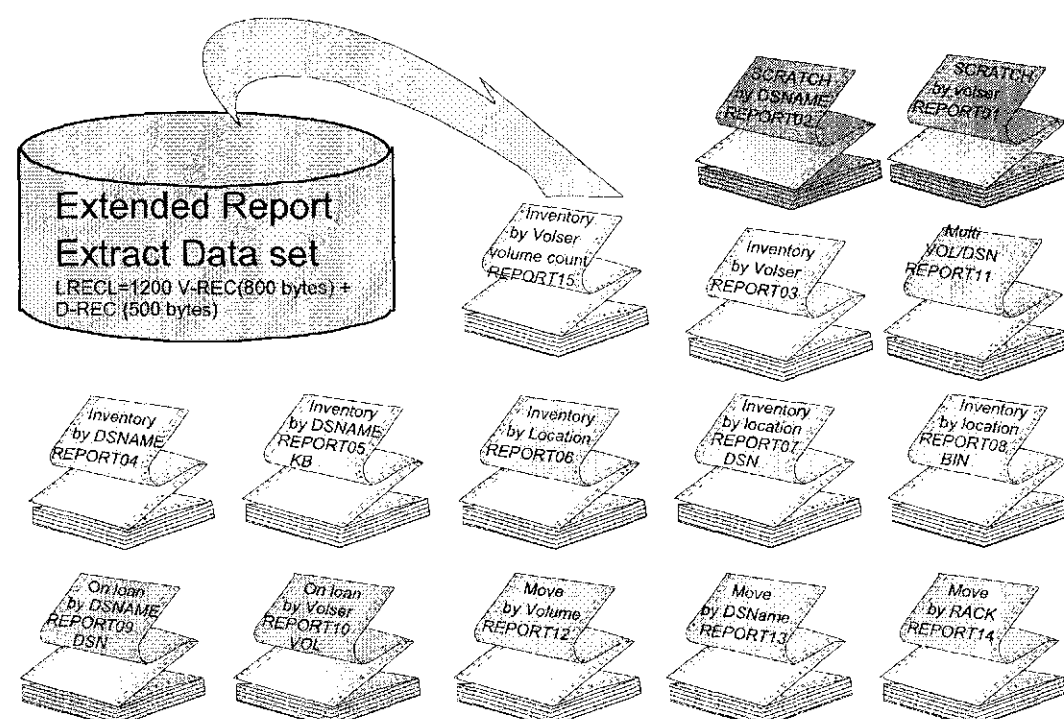


Figure 2-27. Extended Reporting

SS212.0

Notes:

The following reports are available:

REPORT01	Pull list for scratch tapes sorted by volume serial number
REPORT02	Pull list for scratch tapes sorted by data set name
REPORT03	Inventory list by volume serial number
REPORT04	Inventory list by data set name
REPORT05	Inventory of data sets including number of KB used
REPORT06	Inventory of volume serial number by location
REPORT07	Inventory of data set names by location
REPORT08	Inventory of BIN numbers by location
REPORT09	List of all data set names at loan locations
REPORT10	List of all volume serial numbers at loan locations
REPORT11	List of all multiple volume, multiple data sets
REPORT12	Movement report including the first data set name on the volume.
REPORT13	Movement report sorted by storage location bin number
REPORT14	Movement report sorted by volume serial number
REPORT15	Inventory list sorted by volume serial number including volume count

Reporting with OW47651

OW47651 introduced changes to RPTEXT processing

- Run RPTEXT at any time, independent of inventory management
- Presence of XREPTTEXT DD requests extended records

```
//EXTRACT EXEC PGM=EDGHSKP, PARM='RPTEXT, DATEFORM=1'
//MESSAGE DD DISP=SHR, DSN=MY.MESSAGE
//XREPTTEXT DD DISP=SHR, DSN=MY.EXTRACT
```

- Extended Records contain volume and data set information concatenated

- New file serialization to limit multiple use

For MESSAGE and extract files

SYSZRMM RPTEXT.dsname.volume

SYSZRMM HSKP.dsname.volume

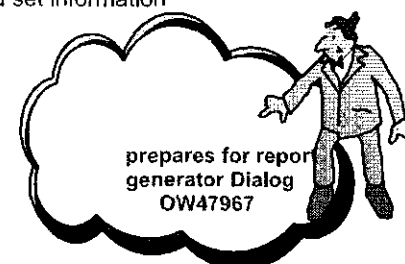


Figure 2-28. Reporting with OW47651

SS212.0

Notes:

For extended reporting, a different format is needed for report extracts. Starting with APAR OW47651, you can specify one of the DD statements, REPTTEXT or XREPTTEXT. The job will fail if you specify both DD cards.

In addition RPTEXT can run in parallel with other EDGHSKP jobs. Each of them uses different output data sets for messages and report extracts.

- EDGJRPT sample JCL is simplified
 - Single step job
 - Reports enabled via DD statement
- EDGRRPTE updated
 - Performs sorting as required
 - Checks for DD to produce a report
 - Uses extended records
 - CCARD DD enables simple custom

- New EXEC does not support old parameters
- Both EXECs can process old or new extended extract records
- Use new EDGJRPT sample job, or, run existing EDGRRPTE exec

SS212.0

CCARD DD provides the opportunity to customize the reports. See *DFSMSrmm Reporting* for details.

Reporting with DFSORT

DFSORT R14 makes it easier with *symbols*

```
//STEP1 EXEC PGM=ICETOOL
//TOOLMSG DD SYSOUT=*
//DFSMSG DD SYSOUT=*
//EXTRACT DD DSN=RMM.MASTER.EXTRACT,DISP=SHR
//SYMNAMES DD DSN=SYS1.MACLIB(EDGEXTSY),DISP=SHR
//TEMP DD DISP=(,DELETE),UNIT=SYSDA,DSN=&&T,SPACE=(TRK,(5,5),,RLSE)
//RPT1CNTL DD *
INCLUDE COND=(RVTYPE,EQ,RVTYPEID,
RVSTATUS,EQ,RVSCR,
RVASDATE,GT,2000/031)
SORT FIELDS=(RVDSNAM1,A)
//TOOLIN DD *
SORT FROM(EXTRACT) TO(TEMP) USING(RPT1)
DISPLAY .....
HEADER('DSN') ON(RVDSNAM1)-
HEADER('VOLSER') ON(RVVOLSER)-
HEADER('SG') ON(RVSGNAME)-
HEADER('DATE') ON(RVASDATE)-
HEADER('TIME') ON(RVASTIME)
//REPORT DD .....
```

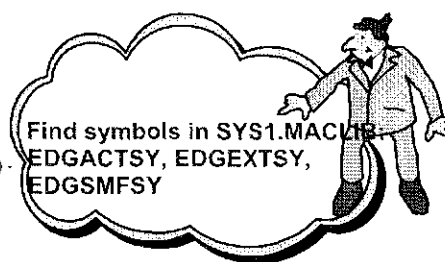


Figure 2-30. Reporting with DFSORT

SS212.0

Notes:

All batch reporting in DFSMSrmm is making use of a standard sort program such as DFSORT. Thus, if you have the required training, you can produce any report you want, based on the information contained in the report extract data set.

However, working with a sort program is sometimes very hard because you have to use offset numbers to specify your requests. Using symbols with DFSORT R14 makes it more comfortable to specify what you want to be selected in your report.

Reporting with DFSORT

DFSORT and ICETOOL samples are provided

Some are converted to use symbols

Examples:

- » EDGJBCAV Create RMM ADDVOLUME subcommands from a list of barcode scanned volumes
- » EDGJCOMB Perform an audit of the tape library using a list of barcode scanned volumes
- » EDGJCVB Create RMM CHANGEVOLUME subcommands for volumes in storage locations that can be used as input to other jobs
- » EDGJNSCR Create a report of volumes that have returned to scratch status
- » EDGJRACK Create a report of rack prefixes
- » EDGJROWN Create a report of owners sorted by name and by department number
- » EDGJVLTM Create a report of volumes moving to storage locations
- » EDGJACTP Report from the ACTIVITY file

See the EDGJxxxx samples in SAMPLIB

Notes:

There are some sample jobs contained in SYS1.SAMPLIB that can be useful in some situations.

Report Generator

Dialog extension to create reports

- * Select report input records
- * Specify record selection and sorting
- * Specify report contents
- * Generate and run JCL
- * Print or view report

Dialog manages

- * types of report based on input
- * report definitions
- * IBM provided, installation and user level reporting

Use with any records mapped by ASMH macro

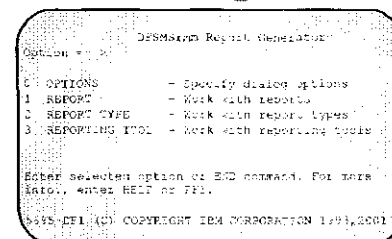


Figure 2-32. Report Generator

SS212.0

Notes:

To allow any customer to develop customized reports with little effort and to make it easy to administer those special reports, now you can use report generator.

You can even include your own mapping macros to describe records that should be used for reporting. Thus, you can extend report generator to generate reports for different sources of information that are distinct from DFSMSrmm.

Report Generator Options

Customize libraries

- Sysprog updates EDGRMAIN defaults
Including optional installation lib
- No need for pre-allocation

Report definition library contains

- EDGGTOOL - Report tool definitions
- EDGGRTD - Report type definitions
- Members - Report definitions

JCL library

Generated JCL for reports

```
DFSMSrmm Report Options
Command ==>
Report definition libraries:
User . . . . . 'WG01MW.REPORT.LIB'
Installation . . . . . 'SYS1.SAMPLIB'
Product . . . . . 'SYS1.SAMPLIB'
User report JCL library . 'WG01MW.REPORT.JCL'
DFSMSrmm allocates user libraries if they do
not exist.
```

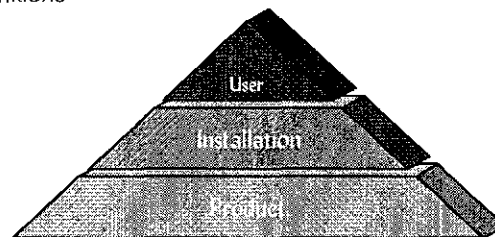


Figure 2-33. Report Generator Options

SS212.0

Notes:

Reporting can be founded on

- IBM samples
- Installation- or user-level customization

Each user can specify report definition libraries and a user-report JCL library.

Work with Reports

Press ENTER !
or
Select

DFSMSrmm Report Definition Search

Command ===>

Report name . . . May be generic. Leave blank for all reports.

User id Leave blank for all user ids.

Libraries (enter S): Select one or more library.

User Default are all defined libraries.

Installation

Product

The following line commands will be available when the list is displayed:

A - Add a new report definition D - Delete a report definition

G - Generate and save the JCL E - Edit and submit the JCL

N - Copy a report definition S - Display or change the report def

T - Select a reporting tool

A list of all report definitions is displayed

Figure 2-34. Work with Reports

SS212.0

Notes:

Make your selection from

- All reports (specify nothing)
- Reports by a specific user
- Reports from a specific library

Work with Reports: List of Report Definitions

DFSMSrmm Report Definitions

Row 1 to 4 of 4
Scroll left> CSK

Command ==>

The following line commands are valid: A,D,G,J,N,S, and T

S Name	Report title	Report type	User id
DROPPED	Data Sets dropped from VRS	ACTIV. IT file records	WOODMW
S REPORT01	Scratch tapes by volume serial	Extended Records	WOODMW
REPORT02	Scratch tapes by Data Set Name	Extended Records	WOODMW
REPORT03	Inventory by Volser	Inventory of Volumes	WOODMW

***** Bottom of data *****

Use line commands to process the definitions

- A - Add a new report definition
- D - Delete a report definition
- G - Generate and save a report JCL
- J - Edit and submit a report JCL
- N - Copy a report definition
- S - Display or change a report definition
- T - Select a reporting tool

Figure 2-35. Work with Reports: List of Report Definitions SS212.0

Notes:

Use line commands to work with the reports presented in the list.

At any time you can

- Add a new report definition
- Use an existing definition to create a report
- Display or change an existing definition
- Select a reporting tool

Work with Reports: Report Definition Details

DFSMSrmm Report Definition - REPORT01 Row 1 to 11 of 162
Command ==> Scroll ==> CSR

Report title . . . Scratch tapes by volume serial
Report footer . . .
Reporting tool . : EDGRGICE Report width: 173

Select the field name to specify a criteria

S	CO	SC	Field name	Column header text	CW	Len	Typ
S	1	1A	XVOLUME	Volser	6	6	C
	2		XDDNAME	Data set name	44	44	C
	3		XVOLUME	Vol-Seq.	8	4	C
	4		XDLABNO	ESN-Seq.	8	4	C
	5		XDCRDATE	Create dat	10	10	C
	6		XVEXEDTO	Org. Exptd	10	10	C
	7		XVOPFN	C	1	1	C
	8		XDAREND	A	1	1	C
	9		XVLABEL	Lab	3	3	C
	10		XVMDTY	media ty	8	8	C
	11		XVMDRRC	rec Fmt.	8	8	C

Figure 2-36. Work with Reports: Report Definition Details

SS212.0

Notes:

Each field name and, column header text is displayed along with display properties of the field.

Some of the fields on a line can be edited according to the table on the next page. You can specify a report title as well as a report footer. The reporting tool EDGRGICE is shipped with report generator and must be specified for reports based on the extended report extract file.

Typing the line command **S** brings up more details about the selection criteria that must be met in order for a record to be selected.

Using the Report Definition Screen

S	Select to view Criteria Details
CO	Enter the Report Column Number Column Numbers Recalculated
SO	Specify Sort Order and Sequence 1A - First sort Field, Ascending
Field Name	This is the field name extracted from the Input Record Mapping Macro
Column Header	Header Text Extracted From Macro Modify to Suit - CW is Recalculated
CW	Calculated Report Column Width You can override the value
Len	Field Length
Typ	Field Data Type

Figure 2-37. Using the Report Definition Screen

SS212.0

Notes:

The table gives an overview of the contents of the columns in the report definition screen. You can specify

- The report column number
- The sort order of the list
- The column header
- The report column width

to tailor the list.

Report Selection Criteria

DFSMSrmm Report Criteria - REPORT01				Row 1 of 4 of 4
Command ==>				Scroll ==> CSR
Report title : Scratch Lapes by volume serial				
The following line commands are valid: B,D,I,N,F,R, and T				
Comparison operators: EQ =, NE <>, GT >, GE >=, LT <, LE <=, IN, BW				
Conjunction: AND, OR, AND(,)AND				
S	Field name	Op	Compare value(s)	Conj Len Typ
	RXTYPE	EQ	X	1 C
	XVSTATUS	EQ	SCRATCH	8 C
	XDDNSSEQ	EQ	' 1'	4 C
	XVOLSER			6 C
***** Bottom of data *****				

Specify selection operators and comparison data

- EQ - Equal, NE - Not equal, GT - Greater than
- GE - Greater than or equal, LT - Less than, LE - Less than or equal to
- IN - In (Specify a list separated by a comma)
- BW - Between (Specify two inclusive values)

Figure 2-38. Report Selection Criteria SS212.0

Notes:

By using report criteria you can specify conditions that must be met to include a record in the report output.

Report Example

Scratch tapes by volume serial									
- 1 -									
2001/07/13 07:30:20									
Volser	Data set name	Vol-Seq.	DSN-Seq.	Create dat.	Org. Exptd.	O	A	Tab	
AL0702	DSNC	1	0	05/23/2001		X	N	SL	
AL0703	DSNC	1	0	05/23/2001		X	N	SL	
AL0704	DSNC	1	0	05/23/2001		X	N	SL	
DSV001	DSNA	1	0	05/23/2001		X	N	SL	
DSV002	DSNA	1	0	05/23/2001		X	N	SL	

Figure 2-39. Report Example SS212.0

Notes:

This is the look of the sample report entitled 'Scratch Tapes by Volume Serial,' which includes the following information:

- The data set name
- The volume sequence number
- The data set sequence number
- The creation date of the data set
- The original expiration date
- The data set *open* indicator
- The data set abend indicator
- The label type
- The media type (invisible)
- The recording format (invisible)

Report Types

```

                                DFSMSrmm Report Types
Command --->                                Row 1 to 3 of 17
                                                Scroll ---> CSR

The following line commands are valid: A,C,D,R, and S

S Name      Description
-----
S EDGACTRC  ACTIVITY file records
  Macro library . . : 'SYS1.MACLIB'
  Applicable macros : EDGACTRC
  Input data set. . : 'RMM.ACTIVITY'
-----
EDGRDTEXT  Inventory of Data Sets
  Macro library . . : 'SYS1.MACLIB'
  Applicable macros : EDGRDTEXT
  Input data set. . : 'RMM.EXTRACT'
-----
EDGRKEXT   Inventory of VRs
  Macro library . . : 'SYS1.MACLIB'
  Applicable macros : EDGRKEXT
  Input data set. . : 'RMM.EXTRACT'
-----
```

Figure 2-40. Report Types SS212.0

Notes:

If you work with report types instead of reports, you would select option 2 in the primary option menu of report generator.

This part of the application shows information available by record type.

You can define your own report types based on existing macro definitions.

Typing the line command S allows you to specify new report type criteria (and to see the current settings if they are available).

To create a new report based on a report type, first select the type of record which should be used as a model (this would need a line command **R** instead).

Report Type Example: Activity File Records

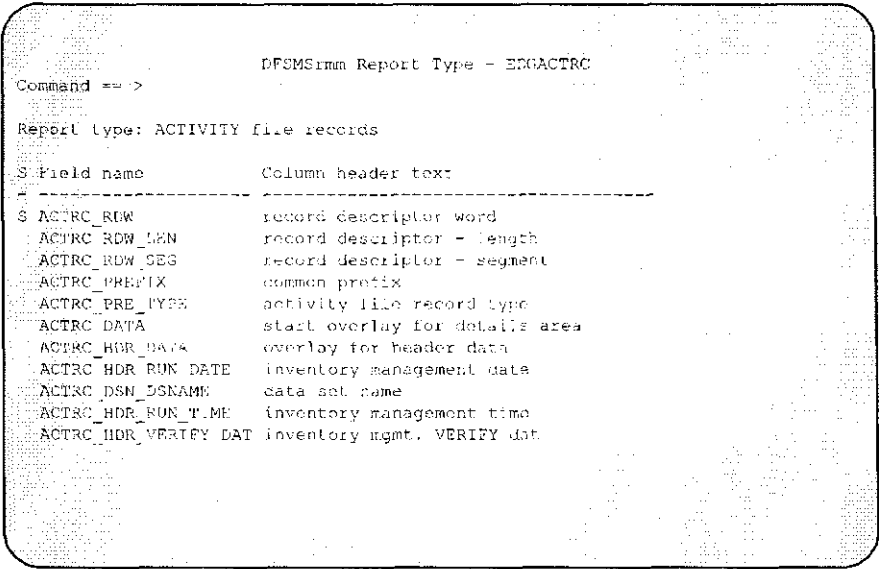


Figure 2-41. Report Type Example: Activity File Records SS212.0

Notes:

After having chosen a report type to specify new report type criteria, you can choose one or more fields in which to specify selection criteria for records to be included in the report (line command S).

Report Type Criteria

```
DFSMSrmm Report Type Criteria - DDGACTRC Row 1 to 2 of 2
Command ==> Scroll ==> CSR

Report Type: ACTIVITY file records

The following line commands are valid: B,B,I,N,P,R, and F
Comparison operators: EQ =, NE <>, GT >, GE >=, LT <, LE <=, IN, BW
Conjunctions: AND, OR, AND(, )AND

S Field name      Op Compare value(s)      Conj Len Typ
-----
ACTRC PRE_TYPE    EQ 5                      1 C
ACTRC RDW         4 N
***** Bottom of data *****
```

Specify selection operators and comparison data

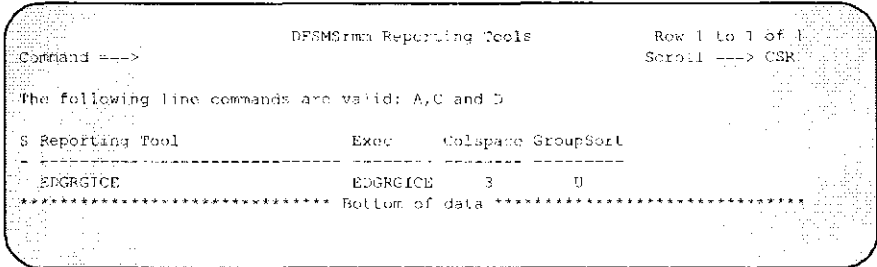
- EQ - Equal, NE - Not equal, GT - Greater than
- GE - Greater than or equal, LT - Less than, LE - Less than or equal to
- IN - In (Specify a list separated by a comma)
- BW - Between (Specify two inclusive values)

Figure 2-42. Report Type Criteria SS212.0

Notes:

Now specify criteria to select records from input. In our example, we want only data set records from the output of a VRSEL run with VERIFY which is recorded in output data set associated with DD card ACTIVITY.

Report Tools



Rexx EXEC interface

- * Report definition
 - Input passed as variables
- * JCL generated using ISPF file tailoring skeleton
- * Using EDGRGICE as a sample
 - Produces DFSORT ICETOOL statements

Figure 2-43. Report Tools

SS212.0

Notes:

A reporting tool is a REXX EXEC that builds control statements to create reports that use a reporting utility, such as DFSORT ICETOOL. The EXEC processes a report definition and uses an ISPF skeleton to generate JCL, which runs the report job.

You can add, change, or delete reporting tools in report generator.

Using Report Generator

IBM ships RMM starter set

- * Existing RMM mapping macros
- * Report types for RMM records
- * Sample report definitions

Installation creates customized report definitions

- * Use RMM types
- * Start from shipped samples
- * Populates installation report library
- * Defines types for non-RMM reporting

User creates reports as required

- * Reuses installation and product report definitions
- * Customizes or creates reports
- * Submits jobs as required



Figure 2-44. Using Report Generator

SS212.0

Notes:

Report Generator is a powerful tool which can be used according to your needs.

Using ISPF

Powerful search capability
Volumes, data sets, and much more

Command ==>

DFSMsrm Volume Search

Volume May be generic. Leave blank for all volumes

Owner Owned by a specific user. Default is your userid

Media name Limit to a single media name

Pool prefix or to a particular pool

Status Select volume status. Default is ALL

Since Volumes assigned since YYYY/DD

Retention Volumes retained up to YYYY/DD

The following line commands will be available when the list is displayed:

C - Change volume information E - Eject volume

I - Data set information L - List volume chain

O - Owner information R - Release volume

V - Volume information

Retention Volumes retained up to YYYY/DD

Limit Limit search to first nnn volumes

Home Limit to volumes with this home location name

Location Limit to volumes in a single location

In contains Stashed volumes

Volume type (LOGICAL , PHYSICAL or STACKED)

Media type Tape media type (* , OCT, ECCTC, HPC or ERCT)

Label Tape label standard (for example HL)

Current version Label version number(for example 1)

Required version Label version number(for example 4)

Density Tape recording density

Format Tape recording format (* , 16TRACK, 16TRACK, 120TRACK or 256TRACK)

Compaction Limit to volumes containing compacted data

Attributes Tape special attributes (HOME or EDOCDAT)

Destination Limit by destination

Required Limit by required location

Command ==>

DFSMsrm Volumes (Page 1 of 2) Row 29 to 41 of 1,006

Use the RIGHT command to view other data columns Scroll ==> PAGE

The following line commands are valid: C,I,O,R,E,L and V

Serial	Owner	Assigned date	Expiration date	Status	Location	Inaction	Exit	Data sets
003025	IBM	1998/307	1999/365	MASTER	SHLFP	REMOTE	Y	3
003245	IBM	1998/240	1999/365	MASTER	SHLFP	REMOTE	Y	3
003398	IBM	1998/120	1999/365	MASTER	SHLFP	REMOTE	Y	3
003891	IBM	1999/309	1999/365	MASTER	SHLFP	REMOTE	Y	3
005662	IBM	1998/132	1999/365	MASTER	SHLFP	REMOTE	Y	3
004389	IBM	1998/042	1999/365	MASTER	SHLFP	REMOTE	Y	3
004564	IBM	1998/022	1999/365	MASTER	SHLFP	REMOTE	Y	3
004679	IBM	1998/086	1999/365	MASTER	SHLFP	REMOTE	Y	3
004995	IBM	1998/705	1999/365	MASTER	SHLFP	REMOTE	Y	3
005382	IBM	1998/190	1999/365	MASTER	SHLFP	REMOTE	Y	3
005677	STANJON	2001/182	2001/202	MASTER	SHLFP	N		
005678	STANJON	2001/182	2001/202	MASTER	SHLFP	N		
005683	STANJON	2001/162	2001/202	MASTER	SHLFP	N		

Figure 2-45. Using ISPF SS212.0

Notes:

You can use the standard ISPF interface to create on-screen reports.

Using ISPF (Cont.)

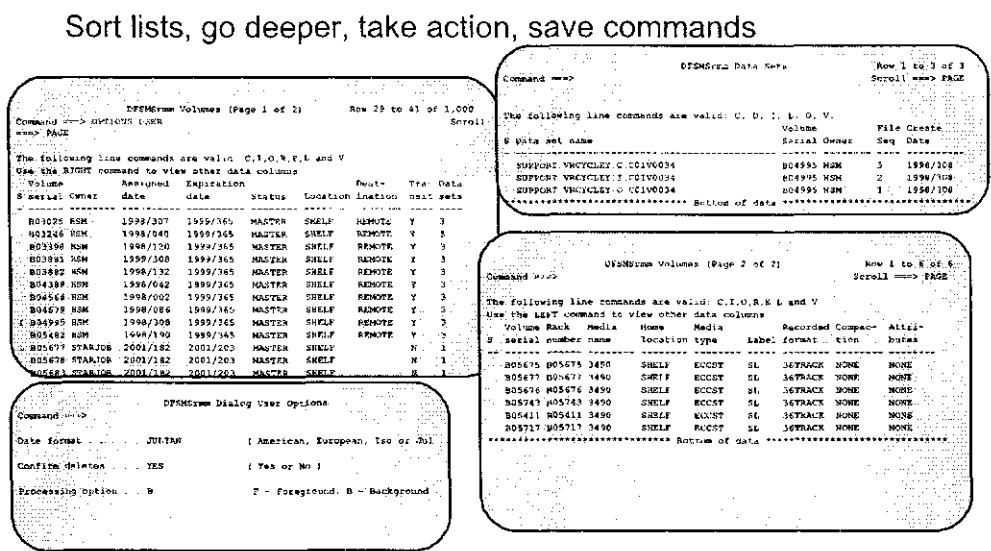


Figure 2-46. Using ISPF (Cont.) SS212.0

Notes:

You can sort the resulting lists, go deeper using line commands, or take any actions supported by line commands. You can also save the commands normally used to create the lists in the foreground in a data set that might be used to run the same commands in the batch TSO environment.

Using TSO

Powerful information retrieval and usage

SEARCH and LIST combined with CLIST

A summary list of all data sets on exported volumes,
and a list of all the data set details

```
RMM SV VOL(*) OWNER(*) LIMIT(*) CONTAINER(*) NOLIST-  
CLIST('RMM SD VOLUME(*) LIMIT(*)')  
EXEC EXEC.RMM
```

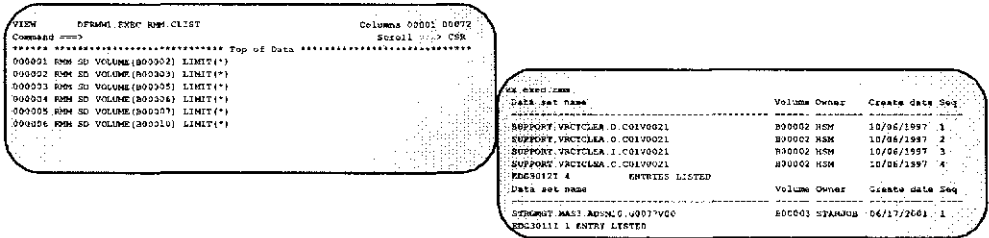


Figure 2-47. Using TSO SS212.0

Notes:

By using TSO RMM subcommands to search items based on specific criteria, you can receive simple lists or even generate commands to be executed later for all items that meet the search criteria by using the CLIST parameter.

Checkpoint

Indicate whether each of the following statements is true or false.

1. The status of a MASTER or a USER volume can be changed to scratch without running inventory management.
2. When you are backing up the DFSMSrmm CDS and journal with EDGHSKP, you can use tape for output when DFSMSdss is used, but not for access method services (AMS).
3. If the DFSMSrmm subsystem interface has been initialized, DFSMSrmm must be started or the subsystem interface must be reset before tape activity can take place.
4. The DFSMSrmm utility EDGRPTD cannot be used to produce a list of scratch volumes.
5. The sample reports shipped with report generator can use the report extract data set as well as the extended report extract data set for input.

Figure 2-48. Checkpoint

SS212.0

Notes:

Write down your answers here:

- 1.
- 2.
- 3.
- 4.
- 5.

Unit Summary

Now that you have completed this unit, you should be able to:

- Be able to use operator commands to control DFSMSrmm
- Have a basic understanding of DFSMSrmm inventory management
- Be able to use the DFSMSrmm reporting facilities
 - EDGRPTD
 - ISPF interface
 - Report generator
 - RMM TSO subcommands

Unit 3. DFSMSrmm Implementation Considerations

What This Unit Is About

This unit describes the steps required to prepare for the activation of DFSMSrmm.

What You Should Be Able to Do

After completing this unit, you should be able to:

- Define a DFSMSrmm control data set
- Define a DFSMSrmm journal data set
- Create DFSMSrmm proclib members
- Modify MVS parmlib members to support DFSMSrmm
- Create the DFSMSrmm parmlib member
- Identify the DFSMSrmm exits that may be needed for your environment
- Identify DFSMSrmm considerations when creating DFSMShsm tapes

How You Will Check Your Progress

Accountability: Checkpoint questions

References

- | | |
|-----------|--|
| SC26-7404 | <i>DFSMSrmm Guide and Reference</i> |
| SC26-7405 | <i>DFSMSrmm Implementation and Customization Guide</i> |

Objectives

After completing this unit, you should be able to:

- Define the DFSMSrmm control data set
- Define the DFSMSrmm journal
- Create DFSMSrmm proclib members
- Modify MVS parmlib members to support DFSMSrmm
- Create the DFSMSrmm parmlib member
- Identify DFSMSrmm exits that may be needed for your environment
- Identify DFSMSrmm considerations when creating DFSMShsm tapes

Notes:

3.1 DFSMSrmm Data Sets, Procedures, and PARMLIB Entries

DFSMSrmm Customization Wizard

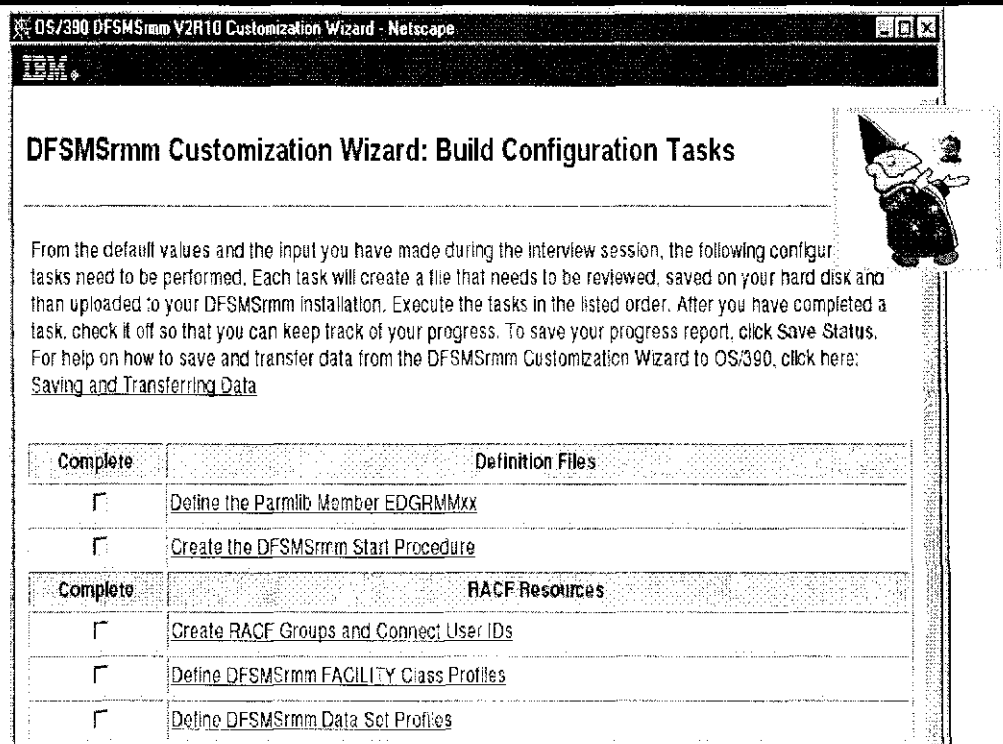


Figure 3-2. DFSMSrmm Customization Wizard

SS212.0

Notes:

The wizard was made available in July 2001 for OS/390 R10 and z/OS.

It can be helpful in setting up DFSMSrmm. It creates:

- A parmliib member EDGRMMxx
- JCL to allocate data sets
- JCL to setup RACF security
- Started procedures

Transfer the results to a PC or to a z/OS for use. Use these URLs:

- <http://www-1.ibm.com/servers/eserver/zseries/zos/wizards/dfsms/rmminstall>
- <http://www.ibm.com/servers/eserver/zseries/zos/wizards/>
- <http://www-1.ibm.com/servers/s390/os390/bkserv/wizards.html>

DFSMSrmm Control Data Set Size

Control Data Set Content	DASD Space
Control record	1KB
Data sets	416KB for every 1000
Shelf locations in the library that do not contain volumes	140KB for every 1000
Shelf locations in storage locations	140KB for every 1000
Owners	38KB per 1000 volumes
Software products, average five volumes per product	420KB for every 1000
Volumes	1MB for every 1000
Vital record specifications	212KB for every 1000

Figure 3-3. DFSMSrmm Control Data Set Size SS212.0

Notes:

The size of the DFSMSrmm control data set depends primarily on the number of volumes and data sets it has to manage. The estimates given are per 1000 of each kind of record except for the base control record itself.

DFSMSrmm Control Data Set Allocation

The DFSMSrmm control data set is a VSAM key-sequenced data set (KSDS) that contains the complete inventory of the removable media library. This is a sample job to allocate DASD space for the control data set:

```
//ALLOCCDS JOB ...
//STEP1 EXEC PGM=IDCAMS
//SYSPRINT DD SYSOUT=*
//SYSIN DD *
    DEFINE CLUSTER (NAME(RMM.CONTROL.DSET) -
        FREESPACE(20 20) -
        KEY(56 0) -
        REUSE -
        RECSZ(512 9216) -
        SHR(3 3) -
        STORCLAS(GSPACE) -
        VOLUMES(SMS052)) -
    DATA (NAME(RMM.CONTROL.DSET.DATA) -
        KILOBYTES(14381 1438) -
        CISZ(10240)) -
    INDEX (NAME(RMM.CONTROL.DSET.INDEX) -
        KILOBYTES(1438 143) -
        CISZ(2048))
/*
```

Figure 3-4. DFSMSrmm Control Data Set Allocation

SS212.0

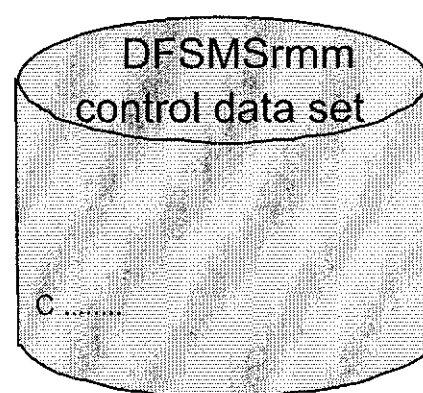
Notes:

Access method services (IDCAMS) is used to create the DFSMSrmm control data set. It is a VSAM key sequenced data set with its 56 byte key starting in the first position of the record. In our example, we are selecting a specific volume since we need to insure that it is placed such that it is protected via hardware (array DASD for example) and will be able to perform suitably (For example, a cache enabled, DASD fast write volume).

Note that the CDS size is limited to four GB unless an EF (extended format) VSAM data set with EA (extended addressability) is used. Support for VSAM extended addressability was released with APAR OW47639. To define an EA DFSMSrmm CDS you must include the DATA CLAS keyword in the AMS DEFINE command and reference a data class that fits your needs.

To convert a non-EF DFSMSrmm CDS you have to allocate a new EF/EA CDS and copy the contents of the old CDS into the new one by using AMS REPRO. There is no other way to perform this task.

Initializing the Control Data Set



```
//STEP1    PGM=EDGUTIL,  
//          PARM='CREATE'  
//SYSRPINT DD SYSOUT=*  
//MASTER  DD DISP=SHR,  
//          DSN=RMM.MASTER.DSET  
//SYSIN    DD *  
CONTROL  
/*
```

Figure 3-5. Initializing the Control Data Set

SS212.0

Notes:

The DFSMSrmm utility, EDGUTIL, must be used to create the control record in the control data set.

You can specify one to eight alphanumeric characters as a CDSID, which DFSMSrmm can use to identify the CDS by name during startup. Using the CDSID is a kind of branding; You should not change it later on, because changing it can affect inventory management.

DFSMSrmm Journal Size Calculation

Journal Content	DASD Space
Changes by users	1.5KB for each change made
Data sets	1.5KB for each data set retained by a vital record specification
Data sets no longer retained by VRS	1.5KB for each data set no longer retained by VRS
Non-scratch mounts	6.7KB for each mount
Scratch Mount	8.3KB for each mount
Vital Record Specification	1.3KB for each vital record specification created

Expiring volumes	1.5KB for each expiring volume
Volumes	1.5KB for each volume retained by a vital record specification
Volumes no longer retained by VRS	1.5 KB for each volume that has not reached its expiration date and is no longer retained by VRS
Volumes returned to scratch	3.0KB for each volume returned to scratch
Volume checked in/out	2.6KB for each volume in or out the library
Volumes to and from storage locations	3.5KB for each volume moved to or from a storage location

Figure 3-6. DFSMSrmm Journal Size Calculation

SS212.0

Notes:

Changes that occur to the control data set are logged in a journal. One of the inventory management functions is to back up the control data set and null the journal. Thus, the journal should be large enough to accommodate all the changes that occur between the runs that back up the CDS and reset the journal.

The DFSMSrmm Journal

The journal contains a record of all changes made to the DFSMSrmm control data set since the last backup

This is a sample job to allocate DASD space for the journal:

```
//ALLOC      JOB ...
//STEP1      EXEC PGM=IEFBR14
//SYSPRINT   DD   SYSOUT=*
//JOURNAL    DD   DISP=(NEW,CATLG),DSN=RMM.JOURNAL.DSET,
//           VOL=SER=RMM002,UNIT=SYSDA,STORCLAS=GSPACE,
//           AVGREC=U,SPACE=(4096,(2000))
```

Figure 3-7. DFSMSrmm Journal

SS212.0

Notes:

DFSMSrmm sets LRECL and BLKSIZE on startup. It should be placed such that an outage that affects the control data set does not affect the journal. This may mean placing the journal on a volume connected to a controller other than the CDS.

The DFSMSrmm Message File

The DFSMSrmm message file lists the messages the DFSMSrmm subsystem issues during inventory management.

This is a sample job to allocate DASD space for the message file:

```
//ALLOC    JOB ...
//STEP1    EXEC PGM=IEFBR14
//SYSPRINT DD  SYSOUT=*
//MESSAGE  DD  DISP=(NEW,CATLG),DSN=RMM.MESSAGE.DSFT,
//          VOL=SFR=RMM002,UNIT=SYSDA,
//          LRECL=133,BLKSIZE=0,DSORG=PS,RECFM=FBA,
//          SPACE=(TRK,1,1))
```

Figure 3-8. DFSMSrmm Message File

SS212.0

Notes:

The housekeeping program, EDGHSKP, produces messages. It is mandatory to use a pre allocated and cataloged data set, because it is used by both the EDGHSKP utility and the DFSMSrmm subsystem address space.

EDGHSKP needs all data sets that are allocated by one of the following DD names, to be pre-allocated and cataloged:

ACTIVITY
MESSAGE
REPORT
REPTXT
XREPTXT

Recommendation: Copy these files to GDGs during housekeeping. You might find that having multiple versions of these files is helpful when it comes to diagnosing problems or creating reports.

New Procedures

These members are shipped in SYS1.SAMPLIB to create procedures in the procedure library:

Member name	Procedure can be used
EDGDFRMM	To start the DFSMSrmm subsystem address space
EDGXPROC	To replenish scratch volumes in an automated tape library data server
EDGBETT	To automate control data set backup and clearing of the journal after message EDG2107E has occurred
EDGPDO	To copy the problem determination aid log data sets to tape
EDGLABEL	To initialize or erase tapes

Figure 3-9. New Procedures

SS212.0

Notes:

DFSMSrmm supplies some sample procedures. Others are recommended. Let's look at the suggested procedures:

EDGDFRMM	A sample to start DFSMSrmm is can be found in SYS1.SAMPLIB in the member <i>EDGDFRMM</i> . In keeping with the DFSMSrmm manual, this should be copied over a procedure name DFRMM.
EDGXPROC	This member can also be found in SYS1.SAMPLIB. It is used in conjunction with IBM system-managed 3494 automated tape libraries. This procedure is named in EDGRMMxx.
EDGBETT	You can set up a procedure that is automatically started when the journal reaches its threshold. This procedure must be named in EDGRMMxx.
EDGPDO	A pair of problem-determination logs can be created for DFSMSrmm. Once one is filled, DFSMSrmm switches to the other. A procedure to save the contents of the full data set is recommended.
EDGLABEL	The member EDGLABEL in SYS1.SAMPLIB can be copied over as LABEL. It can be used for initializing and erasing tapes using the DFSMSrmm utility EDGINERS.

Sample Procedure for DFSMSrmm

Create a procedure in SYS1.PROCLIB to start the DFSMSrmm subsystem address space.

SYS1.SAMPLIB(EDGDFRMM) contains a sample:

```
//DFRMM      PROC M=00,OPT=MAIN
//IEFPROC    PGM=EDG&OPT.,PARM='&M',
//           TIME=1440,REGION=40M
//PARMLIB    DDNAME=IEFRDER
//IEFRDER    DD  DISP=SHR,DSN=SYS1.PARMLIB
//MASTER     DD  DISP=SHR,DSN=RMM.CONTROL.DSET
//JOURNAL     DD  DISP=SHR,DSN=RMM.JOURNAL.DSET
//EDGPDOX    DD  DISP=SHR,DSN=RMM.TRACE.PDOX
//EDGPDOY    DD  DISP=SHR,DSN=RMM.TRACE.PDOY
```

Figure 3-10. Sample Procedure for DFSMSrmm

SS212.0

Notes:

The sample procedure contains the statements for the control data set and the journal. However, these can be specified in the EDGRMMxx parmlib member. If they are specified both here and in EDGRMMxx, the DD statements in the procedure are ignored.

The statement for parmlib identifies the library to be searched for the EDGRMMnn member during DFSMSrmm startup. If you specify the parmlib data set in the DFSMSrmm procedure, the data set remains allocated while DFSMSrmm is active. IEFRDER enables operators to specify a different parmlib data set name when they are starting the DFSMSrmm procedure.

Recommendation: Do not specify any of these four DD statements. Let DFSMSrmm dynamically allocate the PARMLIB to SYS1.PARMLIB, or use the concatenated parmlib support.

To provide detailed information for IBM in case of software problems, enable the problem determination aid facility that uses the two data sets allocated by EDGPDOX/EDGPDOY. This works in the same way as the PDA shipped with DFSMSshm. Combine PDA with a procedure such as EDGPDO to copy information to tape after data set close and switch.

Procedure for DFSMSrmm

Specify the MASTER DD and JOURNAL DD information in the parmlib member EDGRMMnn.



Sample JCL:

```
//DFRMM      PROC M=00,OPT=MAIN
//IEFPROC    PGM=EDG&OPT.,PARM='&M',
//           TIME=1440,REGION=40M
//EDGPDOX    DD  DISP=SHR,DSN=RMM.TRACE.PDOX
//EDGPDOY    DD  DISP=SHR,DSN=RMM.TRACE.PDOY
```

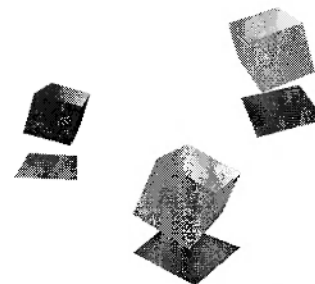


Figure 3-11. Procedure for DFSMSrmm

SS212.0

Notes:

In this example, we only have left in the sample procedure the DD statements for the data sets that are used by the problem determination aid facility. These need to be pre-allocated on the same volume.

Also, the DD statements for the CDS and the journal have been removed and must appear in the EDGRMMnn parmlib member. Also, DFSMSrmm will dynamically allocate SYS1.PARMLIB or any other library in the parmlib concatenation to find the EDGRMMnn member.

Product Setup

Member name	Purpose
IEFSSNxx	To define DFSMSrmm to MVS
IKJTSOxx	To authorize RMM TSO command and utilities
GRSRNLxx	To define how global resource serialization handles the reserve on the DFSMSrmm control data set
IGDDFPKG or IFAPRDxx/IDAPRDxx	To ensure that DFSMSrmm is enabled
SMFPRMxx	To ensure that DFSMSrmm SMF records are generated
EDGRMMxx	To define the installation options for DFSMSrmm

Figure 3-12. Product Setup

SS212.0

Notes:

In addition to the primary DFSMSrmm options in EDGRMMxx, updates in other members of PARMLIB are required. For example, DFSMSrmm runs as a started task so we need to update the sub system name parameters (IEFSSNxx).

IEFSSNxx

Define DFSMSrmm to MVS:

Add an entry to IEFSSNxx to define the DFSMSrmm subsystem to MVS

Example:

JES2,,,PRIMARY,START	JES2
DFRM	Name of the DFSMSrmm subsystem
AOPA	Netview

Example with subsystem active:

JES2,,,PRIMARY,START	JES2
DFRM,EDGSSSI	Name of the DFSMSrmm subsystem
AOPA	Netview

Figure 3-13. IEFSSNxx

SS212.0

Notes:

This syntax applies to MVS 5.1 or earlier. Adding DFRM allows it to be started via an operator command. If any recording is desired, specify the subsystem interface initialization program, **EDGSSSI**. Once the interface is initialized, DFSMSrmm must be started to allow tape activity, or the interface must be reset.

IEFSSNxx (Syntax with MVS 5.2 or Higher)

Define DFSMSrmm to MVS (5.2 or higher):

Add an entry to IEFSSNxx to define the DFSMSrmm subsystem to MVS.

Example:

```
SUBSYSTEM SUBNAME(JES2) /*Primary subsystem start */
    PRIMARY(YES) START(YES)
SUBSYSTEM SUBNAME(DFRM) /* DFSMSrmm subsystem */
SUBSYSTEM SUBNAME(AOPA) /* Netview */
```

Example with subsystem active:

```
SUBSYSTEM SUBNAME(JES2) /*Primary subsystem start */
    PRIMARY(YES) START(YES)
SUBSYSTEM SUBNAME(DFRM) /* DFSMSrmm subsystem */
    INITRTN(EDGSSSI) /* active */
SUBSYSTEM SUBNAME(AOPA) /* Netview */
```

Figure 3-14. IEFSSNxx (Syntax with MVS 5.2 and Higher)

SS212.0

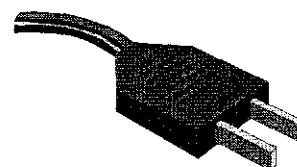
Notes:

This slide shows MVS 5.2 examples. DFSMSrmm should be started before Netview so that write to operator messages are modified by DFSMSrmm before they get to Netview.

Manual Start of the Subsystem Interface

To start the DFSMSrmm procedure, use one of the following commands:

```
S DFRMM
S DFRMM,M=99
```



Message:

```
EDG0103E DFSMSrmm SUBSYSTEM INTERFACE
IS INACTIVE -
      ENTER "IGNORE", "CANCEL" OR "RETRY"
```

Enter RETRY to activate the subsystem interface

Figure 3-15. Manual Start of the Subsystem Interface

SS212.0

Notes:

If the subsystem interface was not started automatically during IPL, we have the situation described above. Normally, you should reply RETRY to activate the subsystem interface.

IKJTSOxx

Authorize RMM commands:

```
AUTHCMD NAMES(RMM)
```

Call DFSMSrmm through TSO:

```
AUHTSF NAMES(EDGHSKP  
              EDGUTIL  
              EDGRPTD  
              EDGAUD)
```

```
AUTHPGM NAMES(EDGHSKP  
              EDGUTIL  
              EDGRPDT  
              EDGAUD)
```

Figure 3-16. IKJTSOxx

SS212.0

Notes:

To authorize the DFSMSrmm command RMM and utilities that you can use within TSO or call through the TSO service facility, make the specifications as shown in the figure.

GRSRNLxx (Optional)

Convert the SYSTEMS enqueue to a local SYSTEM enqueue:

```
RNLDEF RNL(EXCL) TYPE(SPECIFIC)
QNAME(SYSZRMM)
RNAME(MASTER.RESERVE)
```

Convert the RESERVE to a SYSTEMS enqueue:

```
RNLDEF RNL(CON) TYPE(SPECIFIC)
QNAME(SYSZRMM)
RNAME(MASTER.RESERVE)
```

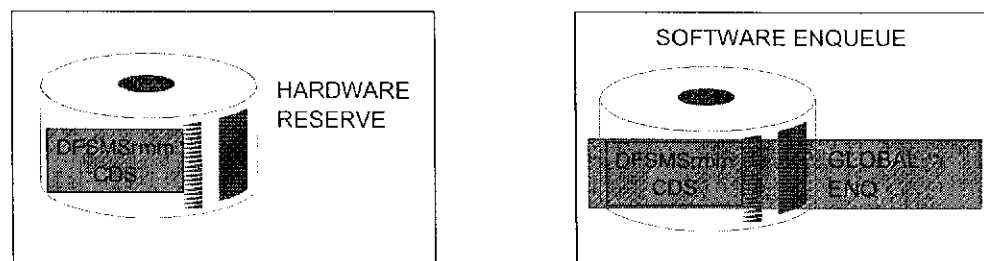


Figure 3-17. GRSRNLxx (Optional)

SS212.0

Notes:

This is required only if you are using a global resource serialization ring in a multiple host environment:

If there is little else on the volume containing the control data set, convert the systems enqueue to local, while leaving the hardware reserve/release in effect.

Otherwise, convert the reserve to a SYSTEMS enqueue.

GRSRNLxx Symbolic Names

DFSMSrmm Resource Symbolic Names in GRSRNLxx

Major Name	Minor Name	Resource
SYSZRMM	MASTER.RESERVE	DFSMSrmm control data set serialization SCOPE=SYSTEMS
SYSZRMM	RMM.ACTIVE	Ensure only one system run per MVS image SCOPE=SYSTEM
SYSZRMM	RMM.BUFFER.CONTROL	Buffer management SCOPE=STEP
SYSZRMM	EDGINERS.volser	Serialize volume labeling SCOPE=SYSTEMS
SYSZRMM	SHUTDOWN	DFSMSrmm shutdown processing SCOPE=SYSTEM

Figure 3-18. GRSRNLxx Symbolic Names

SS212.0

Notes:

Here are some of the resources used by DFSMSrmm to make sure, that processes in one or more DFSMSrmm address spaces will be done in a well-ordered sequence and that some processes will be unique in a task, in a step, on a system or on a complex.

With exception of the CDS serialization there remains nothing to be done during DFSMSrmm setup.

SCHEDxx (Optional)

Updating SCHEDxx:

```
PPT  PGMNAME(EDGMAIN)
      NODSI
```

```
PPT  PGMNAME(EDGRESET)
      NODSI
```

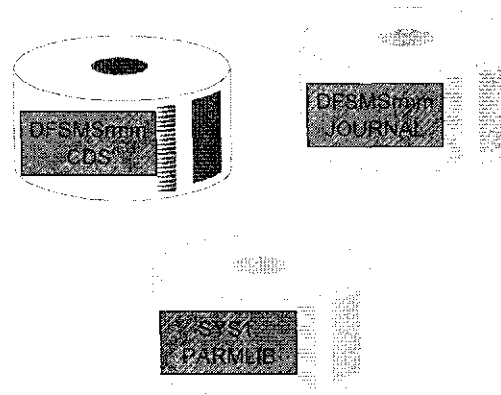


Figure 3-19. SCHEDxx (Optional)

SS212.0

Notes:

If you update SCHEDxx, the DFSMSrmm started procedure does not enqueue on any of the data sets it allocates. If you specify NODSI on a PPT entry for an DFSMSrmm program, you must also take steps like making the journal unmovable, to prevent the DFSMSrmm data sets from being moved while DFSMSrmm is active. For example, DFSMSdss DEFRAG could move the journal while DFSMSrmm is active.

If you want to ensure that the parmlib data set named in the DFSMSrmm started procedure does not cause enqueue contention while DFSMSrmm is running.

Rather than use NODSI, specify the CDS and Journal in the EDGRMMxx parmlib member. DFSMSrmm can be QUIESCED which removes the enqueue on them, but only if allocated via parmlib.

Recommendation: Do not perform this step.

Licensing (MVS)

Ensure that IGDDFPKG was updated during installation to reflect that DFSMSrmm is licensed for your use as a component of DFSMS/MVS.

DFSMS_OFFERING=FULL
or
DFSMS_OFFERING=5
or
DFSMS_OFFERING=4

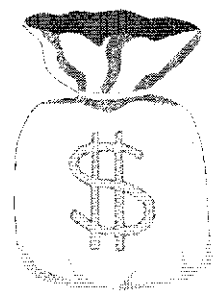


Figure 3-20. Licensing (MVS)

SS212.0

Notes:

This is only valid for MVS V 5 (or earlier) systems. An offering level of 4 or higher is required for DFSMSrmm.

Note:

Valid offerings FULL DFSMSdfp + DFSMSdss + DFSMShsm + DFSMSrmm

- 5 DFSMSdfp + DFSMSdss + DFSMSrmm
- 4 DFSMSdfp + DFSMSrmm
- 3 DFSMSdfp + DFSMSdss + DFSMShsm
- 2 DFSMSdfp + DFSMSdss
- 1 DFSMSdfp

If IGDDPFGK is not found in the SYS1.PARMLIB, or if syntax errors are detected during processing, the default value of 1 will be used for the offering.

Licensing (OS/390 and z/OS)

Ensure that IFAPRDxx was updated during installation to reflect that DFSMSrmm is licensed for your use as a component of OS/390 DFSMS or z/OS DFSMS.

```
PRODUCT OWNER('IBM CORP')  
NAME(OS/390)  
ID(5647-A01)  
VERSION(*) RELEASE(*) MOD(*)  
FEATURENAME(DFSMSRMM)  
STATE(ENABLED)
```

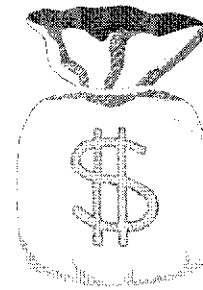


Figure 3-21. Licensing (OS/390 and z/OS)

SS212.0

Notes:

This is the information to be provided on OS/390 and z/OS systems for enabling DFSMSrmm.

EDGRMMxx

The parmlib member EDGRMMxx offers the following statements to initialize DFSMSrmm:

LOCDEF - Defining Storage Locations

MNTMSG - Defining Mount and Fetch Messages

OPTION - Defining System Options

REJECT - Defining Tapes not available on Systems

SECCLS - Defining Security Classes

VLPOOL - Defining Pools

Notes:

Listed alphabetically are the statements that make up the EDGRMMxx parameters for DFSMSrmm. The primary statement is the OPTION as it defines the system options. We will look at the parameters for most statement.

EDGRMMxx OPTION

```
Panel: Help
-----
EDGRMMxx System Options Display
Command: >

Panel suffix: WT
Operating mode: PROTECT
Data set:
  Control: RMHC2.LAPLE.DASER
  Journal:
  RDD id: RDD1
  Catalog: RDD1
  Protection:
  Default: 5
  Maximum: 10
  Catalog: 12
  PDA: 20
  Block count: 0
  Block size: 0
  Log: 0
  Miscellaneous:
    RAMP support: AUTOMATIC
    MAXIMUM limit: 0
    IRL check: NO
    Updated: YES
    System notify: YES
    DSR: YES
    Maximal overwrite: LAST
    Maximal change: KIML
    Association: YES
    Disposition to name: YES
    Disposition message ID: ETR1044
    ERGAT: NO
    SMS ACS: NO
    TRCAT purge: RELEASE
    Reuse bin: CONFIRMOUT
  SN3P2B:
    Purge: NO
    Update exits: YES
    Update commands: YES
    Update scratch: YES
    Date: 2000/01/01
  Journal threshold: 10
  CNTs:
    System id: 00000001
    Audit: 048
    Security: 048
  Report options:
    Lines per page: 54
    Data format: 00000000
  Auto-start procedures:
    Scratch procedure: RMHC2P00
    Backup procedure:
  TRS:
    Job name: 12
    Minimum count: 1
    Minimum action: 1
    Change: INFO
    VMSL processing: NO
    Retain by: VOLUME
    Move by: VOLUME
  Date: 2000/01/01
  Date: 2000/01/01
```

Figure 3-23. EDGRMMxx Option SS212.0

Notes:

The option statement is used to set the system options for DFSMSRmm. There are many. We will cover some of them on the following pages.

You can have a look at the current settings at any time using TSO subcommand LISTCONTROL OPTION or using the DFSMSRmm ISPF interface if you enter fast path command CONTROL SYSTEM (see above).

DFSMSrmm DSNNAME and JRNLNAME

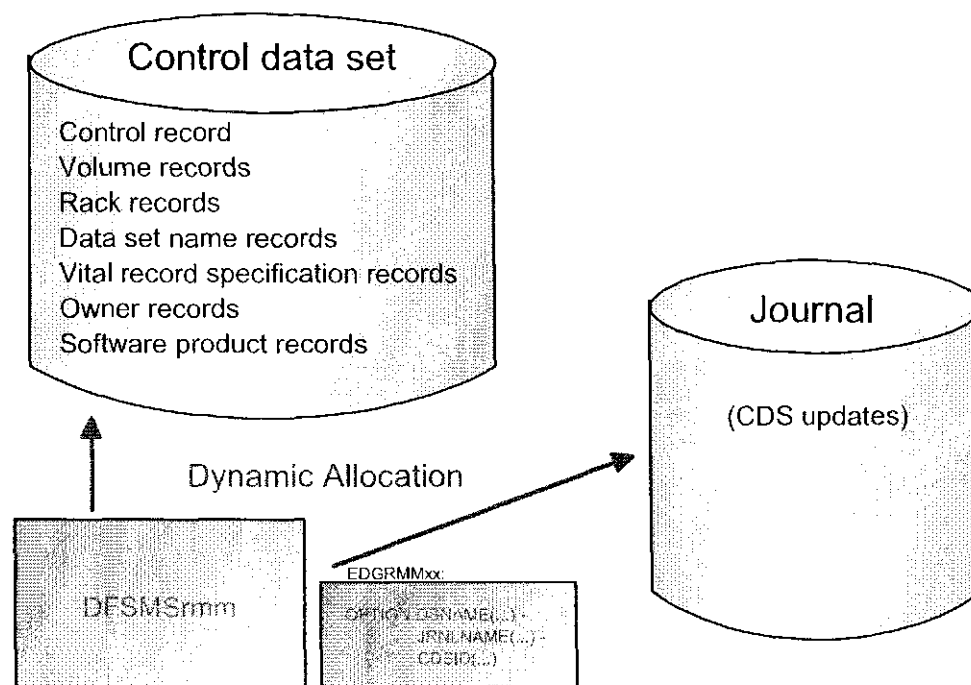


Figure 3-24. DFSMSrmm DSNNAME and JRNLNAME

SS212.0

Notes:

You can specify the name of the DFSMSrmm control data set and the journal data set. If they are not specified here, they must be specified in the procedure used to start DFSMSrmm. If in both locations specified, this name is used.

This is the recommended way to allocate CDS and journal data set (see Unit 2).

Optionally a one to eight alphanumeric character string can be used as an identifier of the control data set. When you start DFSMSrmm, the CDSID is compared to the ID stored in the control record.

If the IDs match, startup continues. If the control data set does not have an ID, DFSMSrmm creates the ID in the control record from the CDSID. If the IDs do not match, startup fails and DFSMSrmm issues a message to select another pamlib member.

If you do not specify the matching CDSID (if already present in the control record), DFSMSrmm cannot use that control data set.

EDGUTIL can be used to set the CDSID in the control data set.

Automatic Reaction when Journal Fills Up

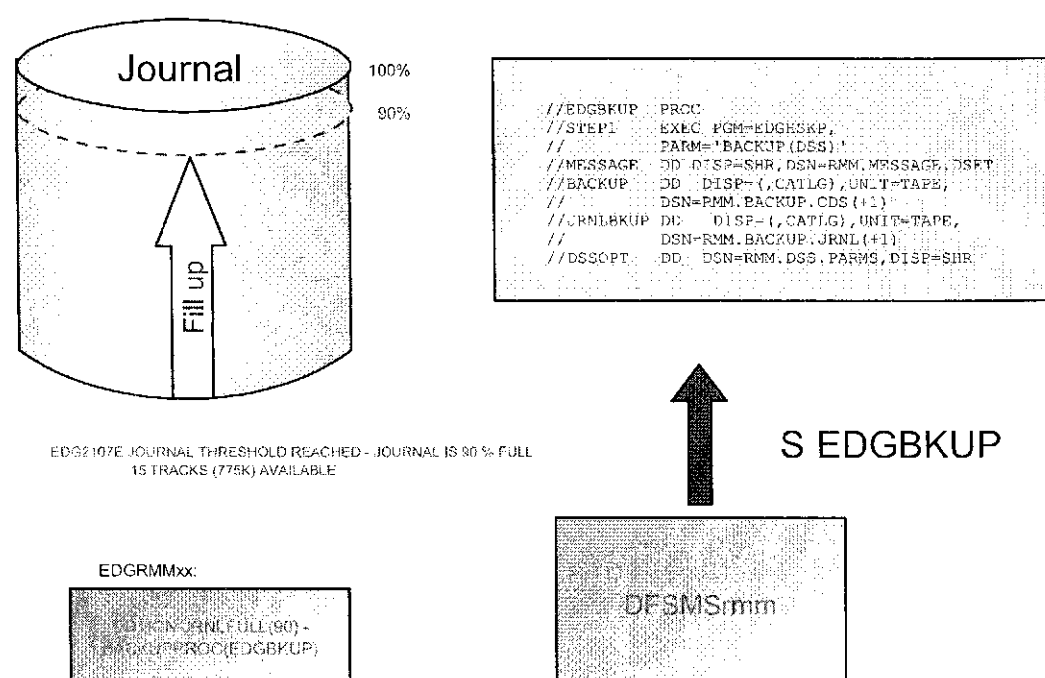


Figure 3-25. Automatic Reaction when Journal Fills Up

SS212.0

Notes:

A percentage value can be selected so that when the journal reaches the specified percent used, DFSMSrmm issues message EDG2107E. If this is not specified, 75% is used. If a BACKUPPROC is specified, it is started automatically by DFSMSrmm when the threshold is reached. There is no default for BACKUPPROC.

Use only a single system to trigger backup using the journal threshold. If you have multiple systems and the journal threshold is reached, each system can start the backup procedure at the same time if you use the same threshold number. To avoid this situation, you can define the backup procedure on only one system or specify different thresholds on each system. For example, you could specify a threshold of 75 on your main system and thresholds of 76 and 77 for your other systems. You can also disable threshold processing on a system by specifying a zero value.

Procedure for Backup when Journal Fills Up

EDGBKUP

Name of the procedure that you want to be started automatically when the journal percentage full threshold is reached. DFSMSrmm issues message **EDG2107E**.

Sample JCL:

```
//EDGBKUP  PROC
//STEP1    EXEC PGM=EDGHSKP,
//          PARM='BACKUP(DSS)'
//MESSAGE  DD  DISP=SHR,DSN=RMM.MESSAGE.DSET
//BACKUP   DD  DISP=(,CATLG),UNIT=TAPE,
//          DSN=RMM.BACKUP.CDS(+1)
//JRNLBKUP DD  DISP=(,CATLG),UNIT=TAPE,
//          DSN=RMM.BACKUP.JRNL(+1)
//DSSOPT   DD  DSN=RMM.DSS.PARMS,DISP=SHR
```

Figure 3-26. Procedure for Backup when Journal Fills Up

SS212.0

Notes:

It is recommended that generation data groups (GDGs) be defined to manage the control data set backup and journal backup copies. To be in sequence with regular backups done during inventory management, you should use the same target data sets (or GDGs) as specified in your inventory job. In our example, we use DFSMSdss rather than IDCAMS (Access Method Services). The data set **RMM.DSS.PARMS** would contain the parameters to be by DFSMSdss. For example:

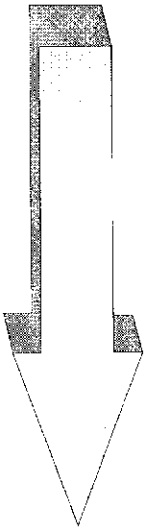
```
CONCURRENT OPTIMIZE(1) VALIDATE
```

DFSMSdss supports the use of concurrent copy, **CONCURRENT** parameter. Concurrent copy allows updates to be made while the backup is occurring. This feature has specific DASD storage control hardware requirements. The use of DFSMSdss also allows the backup up to be written directly to tape.

Modes of Operation

OPMODE

- M Manual mode
You can use RMM TSO subcommands, the DFSMSrmm ISPF dialog, and inventory management functions for all types of media.
- R Record-only mode
Records information about tape volumes used on the system, including details about volume owners and data set names.
- W Warning mode
DFSMSrmm issues error messages instead of rejecting tapes
- P Protect mode
DFSMSrmm is fully operational



EDGRMMxx:



Figure 3-27. Modes of Operation

SS212.0

Notes:

DFSMSrmm has 4 modes of operation:

- M** Manual mode. When DFSMSrmm is running in manual mode, you can use RMM TSO subcommands, the DFSMSrmm ISPF dialog, and inventory management functions for all types of media. DFSMSrmm does not record volume usage or validate volumes when in manual mode.
- R** In addition to what manual mode supports, DFSMSrmm records information about tape volumes used on the system, including details about volume owners and data set names. DFSMSrmm does not validate or reject volumes when in record-only mode.
- W** In addition to the functions available in Manual and Record modes, warning mode DFSMSrmm validates magnetic tape volumes as you use them. If DFSMSrmm discovers errors, DFSMSrmm issues error messages instead of rejecting tapes.
- P** In protect mode, DFSMSrmm is fully operational. It performs mount validation and, when necessary, rejects mounts due to validation errors. It also satisfies all non-specific requests for tape. This is the default.

OPMODE and Other OPTIONS

Mode \ Option	Manual	Record-only	Warning	Protect
EXPDTCHECK				
UNCATALOG				
TPRACF				
SMSTAPE UPDATE(...)				
SMSTAPE PURGE(...)				

-  : Option will not be honored in this OPMODE
-  : Option will be honored in this OPMODE

Figure 3-28. OPMODE and Other OPTIONS

SS212.0

Notes:

The **OPTION** values for **EXPDTCHECK**, **UNCATALOG**, **TPRACF** and **SMSTAPE**, are honored depending on chosen mode as indicated in the figure. **EXPDTCHECK** will be covered in the **VLPOOL** statement later in this unit.

During conversion to DFSMSrmm, other opmodes than protect mode will be used while the old tape management system is still operational and productive. Normally, updates of inventories like ICF catalogs, tape volume catalogs or RACF data base by DFSMSrmm are not wanted before conversion is finished. So care must be taken with the options marked in the figure as to be honored.

If an option is not honored, fixed settings are used by DFSMSrmm which will normally guarantee proper behavior of the system.

UNCATALOG, TPRACF and SMSTAPE

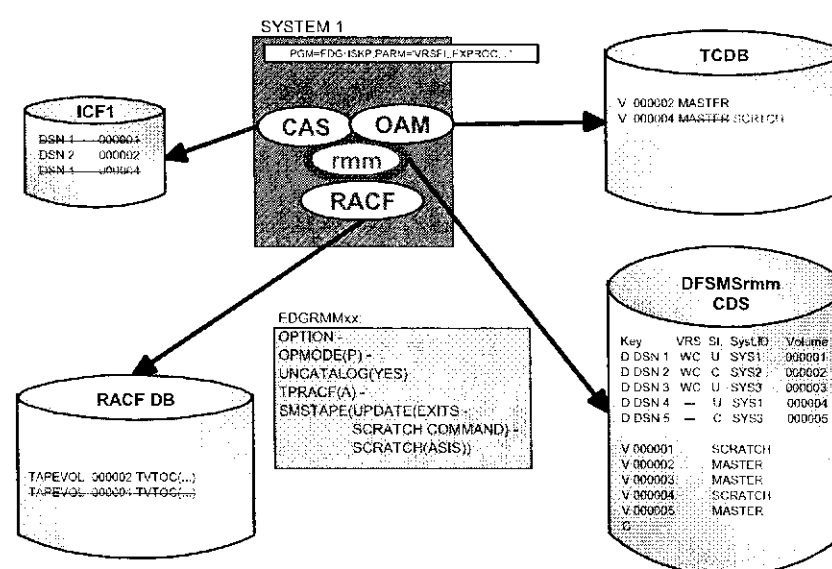


Figure 3-29. UNCATALOG, TPRACF and SMSTAPE

SS212.0

Notes:

If RACF tape security functions are enabled (TAPEVOL and/or TAPEDSN), **TPRACF** specifies DFSMSrmm part in interfacing with RACF. TPRACF(A) will ensure that whenever a scratch volume will be written with new data, a profile in class TAPEVOL exits and when a volume returns to scratch, an existing profile will be deleted. If not specified, TPRACF defaults to no interfacing with RACF.

Recall the **UNCATALOG** specifications are honored (if coded) for all **OPMODE** values. UNCATALOG(YES) will ensure uncatalog processing of all data sets on a volume returning to scratch. If not specified, UNCATALOG defaults to YES if DFSMSrmm runs in OPMODE(P). In all other opmodes the default is NO.

SMSTAPE controls **updates** of the TCDB by DFSMSrmm and ensures that the volume status in DFSMSrmm will override the OAM volume status during entry processing.

PURGE sets up the rules how DFSMSrmm affects the TCDB records during eject processing. For more details see Unit 7.

SMSTAPE(UPDATE(...)) defaults as shown in the visual when DFSMSrmm is running in protect mode.

Option MASTEROVERWRITE

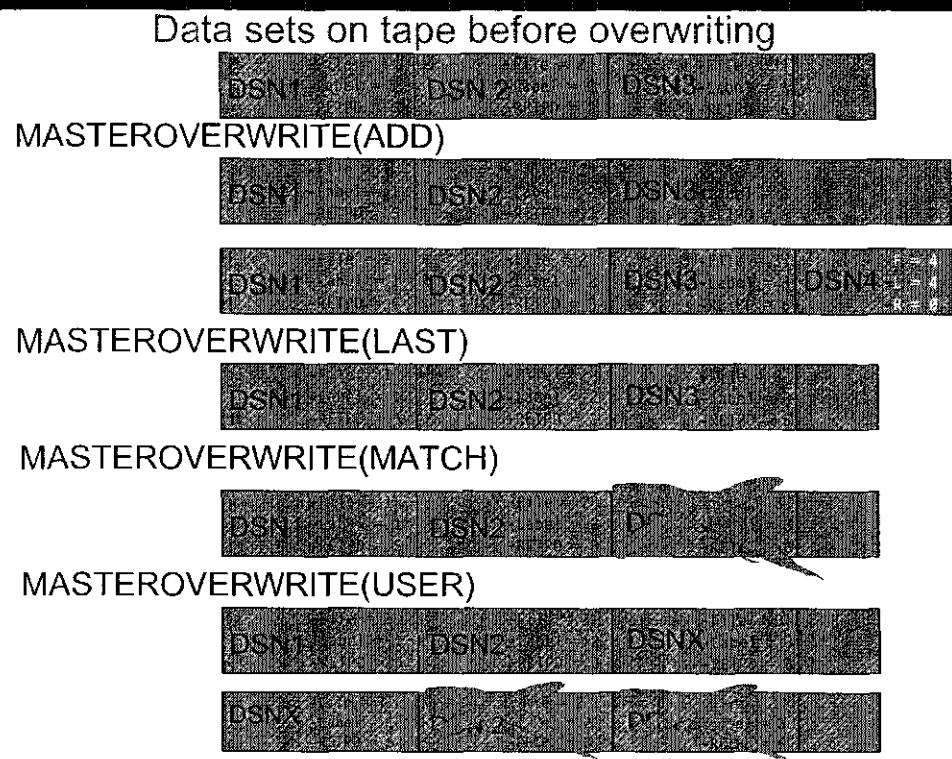


Figure 3-30. Option MASTEROVERWRITE

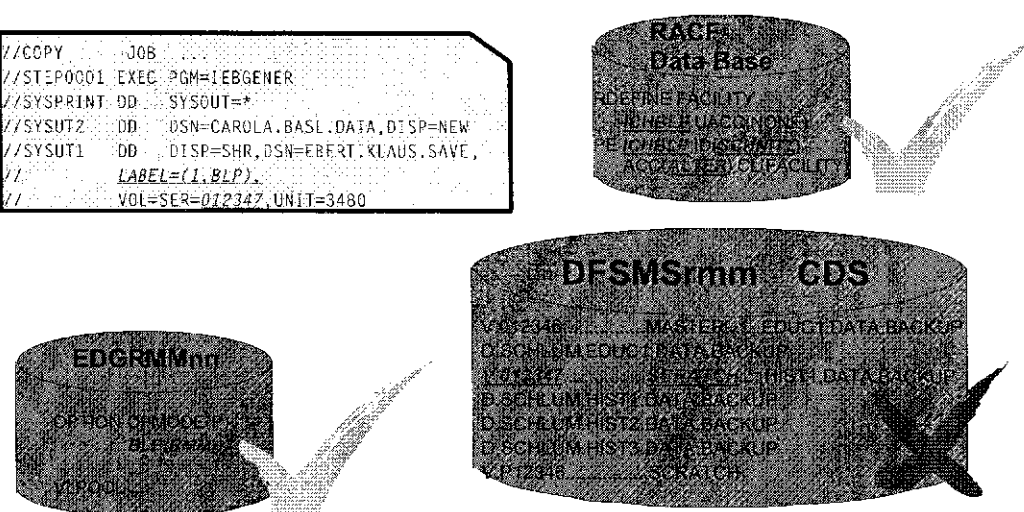
SS212.0

Notes:

For specific requests to overwrite data on a master volume, DFSMSrmm allows the overwriting of data based on criteria set by the EDGRMMxx MASTEROVERWRITE operand. DFSMSrmm checks that the data set name used matches the one DFSMSrmm has recorded. For generation data group data sets, DFSMSrmm removes the GDG suffix before checking the data set name.

If not specified, MASTEROVERWRITE defaults to LAST.

```
//CCPY      JOB
//STEP0001 EXEC PGM=IEBGNER
//SYSPRINT DD  SYSOUT=*
//SYSUT2   DD  DSN=CAROLA.BASL.DATA,DISP=NEW
//SYSUT1   DD  DISP=SHR,DSN=EBERT.KLAUS.SAVE,
//          LABEL=(1,B1P),
//          VOL=SER=012347,UNIT=3480
```



SS212.0

Volumes to be read must be in USER or MASTER status. This applies to either setting of BLP option. In our example, the volume is rejected by DFSMSrmm.

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Option BLP - Write to Master/Scratch Volume

Bypass label processing (BLP) is being used to write to a scratch or master volume unless you request DFSMSrmm to restrict BLP processing through EDGRMMxx in the parmlib

```
//COPY      JOB      ...
//STEP0001 EXEC PGM=IEBGENER
//SYSPRINT DD  SYSOUT=A
//SYSUT1 DD   DSN=CAROLA.BASE.DATA,DISP=SHR
//SYSUT2 DD   DISP=NEW,DSN=DTAUS,
//          LABEL=(1,BLP),
//          VOL=SE 012347,UNIT=3480
```

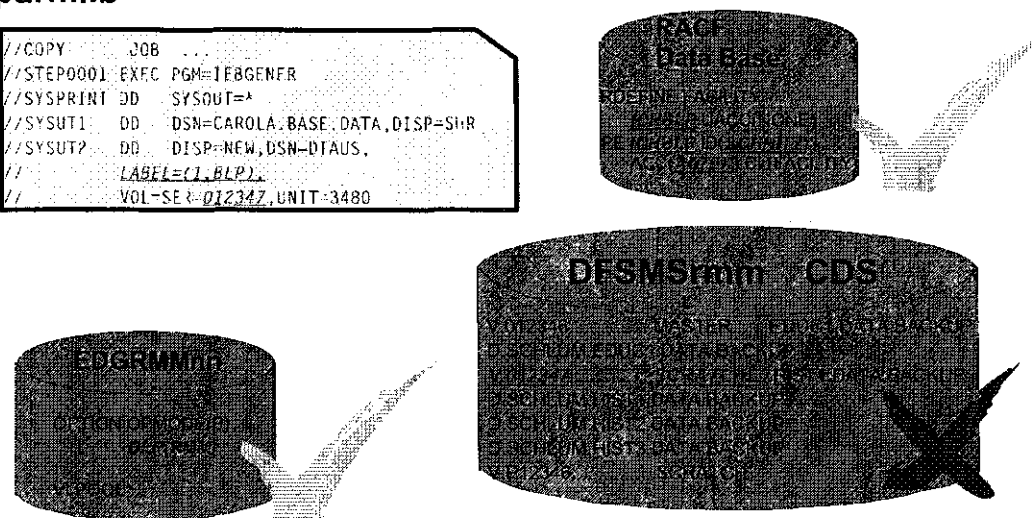


Figure 3-32. Option BLP - Write to Master/Scratch Volume

SS212.0

Notes:

This is for output while the previous example was for input. If you are using DFSMSrmm default settings BLP(RMM), a volume to be written must be in user status.

If DFSMSrmm rules are not desired (to rely on MVS and RACF controls instead), BLP(NORMMM) can be specified in the EDGRMMxx PARMLIB member. When you use BLP(NORMMM), BLP can be used

- for reading and writing of user and master volumes and
- for writing of scratch tapes.

Default Retention and Maximum Retention

OPTION**RETPD**

0 - 9999 Specifies the default retention period

MAXRETPD Maximum retention period a user can request for data sets on tape volumes

0 - 9999 The value is added to the current date to determine the maximum allowed retention date

NOLIMIT If you intend to use the dates 99365 or 99366 which means never to expire

Figure 3-33. Default Retention and Maximum Retention

SS212.0

Notes:

A minimum retention can be specified for data sets that are created and no EXPDT or RETPD is found in the JFCB.

MAXRETPD can be used to limit the effect of EXPDT or RETPD values. However, if the installation relies on 99365 and/or 99366 to mean never expire, the MAXRETPD should be specified as **NOLIMIT**.

MAXHOLD, IPLDATE, ACCOUNTING, MSG

OPTION

MAXHOLD

10 - 500 maximum number of activities
DFSMSrmm performs before the
reserve is released and reacquired

IPLDATE

Y to request IPL date checking
N to bypass IPL date checking

ACCOUNTING

JOB from JOB statement
STEP from EXEC (or JOB) statement

MSG

M mixed case message text
U upper case message text

Figure 3-34. MAXHOLD, IPLDATE, ACCOUNTING, MSG

SS212.0

Notes:

A value of 100 (the default) is recommended for MAXHOLD. Larger values increase lock out times for other users and increase the amount of storage used by DFSMSrmm.

If you specify **IPLDATE(Y)**, DFSMSrmm issues an operator message during startup only if the date of the last run of expiration processing did not occur within two days of the current date. The operator is prompted to enter the current date and day of the week. Initialization does not proceed until DFSMSrmm receives a valid reply. The IPLDATE operand helps prevent you from running expiration processing with an incorrect system date that can result in expiration of unexpired volumes.

Specifying **J** for **ACCOUNTING** has DFSMSrmm record the job accounting in volume records (this is the default). When you are using **S**, the step accounting information will be recorded. If no step accounting is present, the account information from the job is used.

There are messages that can be customized. They are contained in EDGMTAB. **MSG** is used to specify this value when you want messages to be displayed exactly as they appear in EDGMTAB (**M**ixed case message text) or to have them translated to **U**pper case.

Problem Determination Aid Facility

OPTION	
PDA	
NONE	specify to prevent DFSMSrmm from enabling the PDA facility
OFF	disable the trace facility
ON	enable the trace facility
PDABLKCT	number of blocks (2-255)
PDABLKSZ	DASD blocksize in kb (1-31)
PDALOG	
OFF	disable output to the PDA trace data set
ON	enable output to the PDA trace data set

Figure 3-35. Problem Determination Aid Facility

SS212.0

Notes:

These options enable/disable the **Problem Determination Aid** facility. The PDABLKCT is for in storage blocks. PDA is enabled by default.

Procedure for Copying PDA Data Sets

EDGPDO

to copy the Problem Determination Aid (PDA) log data sets to tape

Sample JCL that copies the inactive PDA log data set:

```
//EDGPDO    PROC LOG=PDOY
//STEP1     EXEC PGM=IEBGENER
//SYSPRINT  DD  SYSOUT=A
//SYSIN     DD  DUMMY
//SYSUT1    DD  DISP=SHR,DSN=RMM.TRACE.&LOG
//SYSUT2    DD  DISP=(NEW,CATALOG,DELETE,
// UNIT=TAPE,DSN=RMM.TRACE.&LOG..COPY(+1),
// DCB=*.SYSUT1
```

Figure 3-36. Procedure for Copying PDA Data Sets

SS212.0

Notes:

When the switch of logs occurs, the two data sets on the volume are renamed simultaneously. This means that the Y log data set is the one that just become full while the X data set is the one currently being updated. The procedure copies the content of the Y log data set to a generation data set.

DATEFORM, LINECOUNT, NOTIFY

OPTION

DATEFORM

A	American	mm/dd/yyyy	12/15/2001
E	European	dd/mm/yyyy	15/12/2001
I	ISO	yyyy/mm/dd	2001/12/15
J	Julian	yyyy/ddd	2001/349

LINECOUNT

10 - 255 default number of lines per page in reports

NOTIFY

N	Notification performed by librarian
Y	DFSMSrmm notifies the OWNER by sending an electronic message using the TSO TRANSMIT command. The OWNER information in the DFSMSrmm control data set must include a valid user ID and node name for the message to be sent.

Figure 3-37. DATEFORM, LINECOUNT, NOTIFY SS212.0

Notes:

- Four date formats for messages and reports are available.
- Specifies the default number of lines per page for reports, including heading and trailer lines. This can be overridden when executing DFSMSrmm report utility.
- To use electronic notification, the **OWNER** record must contain a node and userid.

Notify Messages

Add volume:

```
FROM: DFSMRM -- STUTVS                               Date and Time 09/02/97 09:37:44
-----
Subject: Volume DPMK1 has been added for software product 5655-086
A volume has been added on 1997/002 at 09:37:44 to a DFSMRM
software product which you own:

Product Number 5655-086 Level 1 V01R01M00
Name          DPMK1
Description   Tape management system

Volume  Rank  Feature Code  Description
-----  ---  -
DPMK1  000001  1234          DPMK1 initial action
```

Release volume:

```
FROM: DFSMRM -- STUTVS                               Date and Time 09/02/97 09:15:52
-----
Subject: DFSMRM volume expiration
Volume DPMK1 assigned to owner CUPROM on 1997/002 at 09:37:44
is now pending release.
If you wish the volume to be retained, please take immediate action.
You can use the dialog functions of the RMM CHANGVOLUME TSO command.
```

Update the text of a message or a note in the DFSMSrmm message table EDGMTAB to customize notification messages and notes

Figure 3-38. Notify Messages SS212.0

Notes:

The OPTION statement has a NOTIFY parameter that allows users to be notified by DFSMSrmm electronically. The owner record must contain a valid node and userid. Notification takes place when a volume whose pending action is set to notify goes into pending release status. It is also used to notify the owner of a program product record when a volume has been added to that product.

DFSMSrmm provides a series of messages in the DFSMSrmm message module, EDGMTAB. You can modify the notification messages to provide specific information to users. Examples are given in your *z/OS DFSMSrmm Implementation and Customization Guide*, SC26-7405.

Security Classification (SECCLS)

SECCLS

DESCRIPTION describes the security class

ERASE

- Y** indicates volumes must be erased
- N** indicates that they do not have to be erased

MASK specifies a data set name mask
In addition to standard data set naming conventions, you can use the following masking characters: *, .**, ** or %.

MESSAGE/MSG

- Y** issue operator confirmation messages
EDG4008A
- N** suppress the message

Figure 3-39. Security Classification (SECCLS)

SS212.0

Notes:

Data sets that do not match any of the masks specified in SECCLS definitions are assigned no security classification. DFSMSrmm uses a number 0 to indicate no security classification. Whenever you create a tape data set, DFSMSrmm uses the SECCLS masks to classify data sets and volumes.

The ERASE(Y) is useful when tapes need to be erased after they return to scratch status. Using SECCLS and this parameter allows the classified data to be created on a tape from a global scratch pool.

The Message option of SECCLS might be useful to trigger the creation of an external security label.

Security Classification (SECCLS)

SECCLS

NAME	Name for the security class
NUMBER	Indicate relative levels of importance (1-255)
SMF	
Y	write an SMF security record each time a tape data set in this class is opened
N	do not write it

Figure 3-40. Security Classifications (SECCLS)

SS212.0

Notes:

You must assign a security level and name. The higher the number, the higher the level of security. If you wish an audit trail of classified data set access, code **SMF(Y)**. You must also have a user SMF record number selected in the **SMFSEC(nnn) OPTION** statement.

Defining Pools of Tape Volumes

VLPOOL

DESCRIPTION describes the pool (between 1 and 32 characters)

EXPDTCHECK

- N** DFSMSrmm does not check or validate the expiration date of data sets on a tape
- Y** DFSMSrmm ensures that all unexpired tape data sets are not overwritten
- O** DFSMSrmm takes no action

MEDIANAME Name for all volumes in the pool

NAME Pool name that DFSMSrmm uses to update operator messages and tape drive displays for non-specific tape mount requests (can be a 1 to 8 character name, but the first character must be an alphanumeric or national character)

Figure 3-41. Defining Pools of Tape Volumes SS212.0

Notes:

Each VLPOOL command defines a pool. Each generic shelf location defined by PREFIX must be unique. If you do not specify any VLPOOL commands, or do not specify a VLPOOL command for POOL(*), DFSMSrmm automatically creates a default pool definition that includes all the shelf locations and volumes in them. The defaults for this pool are:

```
VLPOOL RACF(N) TYPE(S) EXPDTCHECK(O) MEDIANAME(3480) -
DESCRIPTION('DEFAULT POOL') PREFIX(*)
```

IBM recommends using EXPDTCHECK(N) when running a DFSMSrmm-managed scratch tape pool. DFSMSrmm lets you re-use tapes only when it is certain the information on them is no longer required. DFSMSrmm can easily automate re-use of tapes if they are still expiration protected, for example, when they are re-used as scratch tapes.

The MEDIANAME need not be any MVS generated name. It is meant to describe shelf locations for a unique media type, not a hardware tape drive.

Defining Pools of Tape Volumes

VLPOOL

PREFIX	Specifies a generic shelf location that is used in operator messages, TSO sub-commands, and the DFSMSrmm ISPF dialog
RACF	
Y	you want DFSMSrmm to create RACF tape profiles
N	you do not want DFSMSrmm to create tape profiles
SYSID	associates the scratch pool you are defining with a particular system
TYPE	
R	a rack pool
S	a scratch pool

Figure 3-42. Defining Pools of Tape Volumes

SS212.0

Notes:

The prefix can be from one to five positions in length (six is a specific volume). DFSMSrmm supports two types of pools, SCRATCH and RACK. Non-specific requests for a scratch tape cannot come from a RACK pool. A SCRATCH pool is used to satisfy these requests. The volume remains in the same DFSMSrmm scratch pool in that it occupies the same shelf space regardless of status (SCRATCH, MASTER, ETC).

Using SYSID restricts a SCRATCH pool to a particular system (SYSID must match the SYSID specified in the OPTION statement). DFSMSrmm enforces a match on SYSID when validating non-specific tape mounts. Only scratch volumes from a pool associated with a specific system can satisfy non-specific mount requests for that system. DFSMSrmm rejects volumes from other pools on that system.

If TPRACF is enabled (Y or P specified), the RACF(N) can be used to prevent DFSMSrmm from creating a profile.

Exit EDGUX100

Updating installation-wide exits

DFSMSrmm samples are provided in SYS1.SAMPLIB.

EDGUX100 can be used to

- Ignore foreign or duplicate volumes
- Use JCL special expiration dates to manage volumes by setting vital record specification management values to retain data sets
- Set any expiration date, include a zero value, for a data set
- Obtain information to create sticky Labels Request
- Record only the first file on a multi-file volume
- Select scratch pool by Jobname and/or dataset name

Figure 3-43. Exit EDGUX100

SS212.0

Notes:

In addition to the code listed above, the EDGUX100 in SYS1.SAMPLIB also contains sample code for selecting a SCRATCH pool based on job and/or data set name.

Refer to Chapter 9, *Using DFSMSrmm Installation Exits* in the *DFSMSrmm Implementation and Customization Guide*, SC26-4932.

Exit EDGUX200

EDGUX200 can be used to

- Return a volume to scratch status in an external inventory
- Prevent a volume from returning to scratch status
- Request that DFSMSrmm ignores/removes the data set information recorded for a volume

Notes:

The EDGUX200 sample in SYS1.PARMLIB provides an example of the proper entry and exit sequences. It does not provide sample coding for the uses shown.

Related Exits

Installation-wide exits for which DFSMSrmm samples are provided in SMPSTS:

- CBRUXCUA
Change use attribute
- CBRUXEJC
Cartridge eject
- CBRUXENT
Cartridge entry
- CBRUXVNL
Volume not in library
- EDGDFHSM
Programming interface to EDGTVEXT
- IGXMSGEX
Programming interface to EDGMSGEX

Figure 3-45. Related Exits

SS212.0

Notes:

Additional sample exits can be found in SYS1.SMPSTS. DFSMSrmm supplies these exits so that it can interface with IBM system-managed tape libraries (CBRUXxxx EXITS) as well as DFSMSshsm (EDGDFHSM or EDGTVEXT). The system uses DFSMSdfp MSGDISP to update the display screens on tape drives. DFSMSrmm supplies assembler source code for IGXMSGEX, the MSGDISP installation exit, that you can modify to include your own functions or to merge with another product's supplied exit code.

Refer to Chapter 8, *Using DFSMSrmm Programming Interfaces*, in the *DFSMSrmm Implementation and Customization Guide*, SC26-4932.

Starting with OS/390 DFSMS 2.10, DFSMSrmm no longer uses DFP installation exits for tape processing. Instead, DFSMSrmm processing is at installation exit points. By removing DFSMSrmm's code from the installation exits, there is less risk for malfunction by improper modifications or accidental deletion. Processing of tapes depends on DFSMSrmm and exits together.

3.2 DFSMSrmm with DFSMShsm and Tivoli Storage Manager

Running DFSMSrmm with DFSMShsm

Update ARCCMDxx to retain DFSMShsm tapes using expiration dates:

```
SETSYS TAPESECURITY (EXPIRATION RACF)
DEFINE DUMPCLASS (class
    AUTOREUSE
    TAPEEXPIRATIONDATE (DATE)
    RETENTIONPERIOD (day) )
```

Update ARCCMDxx to make DFSMShsm use a global scratch pool:

```
SETSYS SELECTVOLUME (SCRATCH)
SETSYS TAPEDELETION (SCRATCH)
```

Authorize DFSMShsm to the following resources in RACF class FACILITY:

Resource	Access
STGADMIN.EDG.MASTER	UPDATE
STGADMIN.EDG.RELEASE	READ
STGADMIN.EDG.OWNER.hsmid	UPDATE

Figure 3-46. Running DFSMSrmm with DFSMShsm SS212.0

Notes:

Since DFSMShsm performs its own tape management for backup, migration and dump tapes, its has its own criteria for when they expire. To interface with a tape management system DFSMShsm executes a tape volume deletion exit, **EDGTVEXT**. It is recommended that DFSMShsm obtain its tapes from a global scratch pool managed by DFSMSrmm. Even though a retention can be specified for DUMP tapes, DFSMShsm performs the expiration processing. DFSMSrmm should retain all DFSMShsm tapes until it returns them via the EDGTVEXT exit which is committed by SETSYS TAPESECURITY(EXPIRATION).

Refer to the chapter titled *Implementing DFSMShsm Tape Environments* in *DFSMShsm Implementation and Customization Guide*, GC35-0385.

In the event of a conflict between DFSMShsm and a tape management system, the security interface can be used. If SETSYS TAPESECURITY(RACF) is specified for DFSMShsm, it protects tapes it has in its inventory via the RACF TAPEVOL functions. DFSMShsm removes the protection when the volume is removed from its control.

Since DFSMShsm is requesting an early release (in DFSMSrmm terms), it needs to be authorized to the proper DFSMSrmm facility classes.

Running DFSMSrmm with DFSMShsm (2)

Using ARCTVEXT for communication with another tape management system during conversion to DFSMSrmm:

```
SETSYS EXITON(TV) in ARCCMDxx
```

How should DFSMSrmm react on a release request for volumes communicated via EDGTVEXT and EDGDFHSM?

OPTION TVEXTPURGE ... in EDGRMMxx

RELEASE	DFSMSrmm releases volume according to the release action set for the volume.
EXPIRE	Expiration date of volumes to be purged is being set to the current date.
NONE	DFSMSrmm takes no action for the volumes to be purged.

Figure 3-47. Running DFSMSrmm with DFSMShsm (2)

SS212.0

Notes:

During conversion to DFSMSrmm we still need communication between DFSMShsm and the old tape management system. After conversion is finished, you should remove

SETSYS EXITON(TV)

from ARCCMDxx.

DFSMShsm acts as an external data manager and decides by itself when to return a volume to scratch. Normally you should specify TVEXTPURGE(RELEASE) in EDGRMMxx to perform return to scratch according to the release action set of the volume. This is also the default.

If you want to combine expiration with VRSs including UNTILEXPIRED operand, specify TVEXTPURGE(EXPIRE). You might want to specify a grace period for return to scratch by using EXTRADAYS(...) with the matching VRS.

If you want DFSMSrmm not to release any volume in reaction of the notify communicated via EDGTVEXT or EDGDFHSM, specify TVEXTPURGE(NONE). Use of this operand should be likely very rare.

TSM Specifications

Specify permanent tape retention by entering the following TSM command:

update devclass cartridge expiration=99365

Retain TSM tapes using vital record processing:

RMM ADDVRS DSNAME('tsmprefix.') COUNT(99999)
CYCLES**

Set up tape installation-wide exit to identify the deletion exit to the server by specifying in the server option file:

DELetionexit EDGTVEXT

Notes:

The deletion exit must be reentrant and is run in the 24-bit addressing mode

Checkpoint

Indicate whether each of the following statements is true or false.

1. You must specify the DFSMSrmm control data set and the journal in the procedure used to start DFSMSrmm.
2. During conversion to DFSMSrmm from another tape management system you have the choice of several modes to run DFSMSrmm in parallel with the old tape management system, but you must not choose protect mode.
3. When you choose option BLP(RMM), scratch tapes may not be processed for input.
4. In a multihost environment, you must use a global resource serialization method to serialize the use of the DFSMSrmm CDS.
5. Which MASTEROVERWRITE option must be selected to allow a tape in master status to be overwritten by a data set not matching the existing data set name on the volume:
 - a. ADD
 - b. LAST
 - c. MATCH
 - d. USER
 - e. SCRATCH
6. DFSMSrmm does not make any changes, other than in its own control data set, when running in PROTECT mode.
7. DFSMSrmm provides a method of identifying tapes that contain classified data and marking the volume record to indicate that the tape must be erased before it can be returned to scratch status.

Figure 3-49. Checkpoint

SS212.0

Notes:

Write down your answers here:

- 1.
- 2.
- 3.
- 4.
- 5.
- 6.
- 7.

Unit Summary

Before activating DFSMSrmm, you need to:

- Define the DFSMSrmm control data set
- Define the DFSMSrmm journal
- Create DFSMSrmm proclib members
- Modify MVS parmlib members to support DFSMSrmm
- Create the DFSMSrmm parmlib member
- Identify DFSMSrmm exits that may be needed for your environment
- Identify DFSMSrmm considerations when creating DFSMShsm tapes

Notes:

Unit 4. Vital Record Specifications

What This Unit Is About

This unit describes the creation and use of vital record specifications.

What You Should Be Able to Do

After completing this unit, you should be able to:

- Define VRS policies for retention and movement
- Identify VRS retention options for retaining data sets
- Use DFSMSrmm VRS masking facility for retaining data sets
- Identify updates of control data set during inventory management

How You Will Check Your Progress

Accountability: Checkpoint questions

References

SC26-7404	<i>DFSMSrmm Guide and Reference</i>
SC26-7405	<i>DFSMSrmm Implementation and Customization Guide</i>
SC26-7406	<i>DFSMSrmm Reporting</i>
SC24-7331	<i>DFSMSdfp Storage Administration Reference</i>

Objectives

After completing this unit, you should be able to:

- Define VRS policies for retention and movement
- Identify VRS retention options for retaining data sets
- Use DFSMSrmm VRS masking facility for retaining data sets
- Identify updates of the control data set during VRSEL

Notes:

4.1 Vital Record Specifications

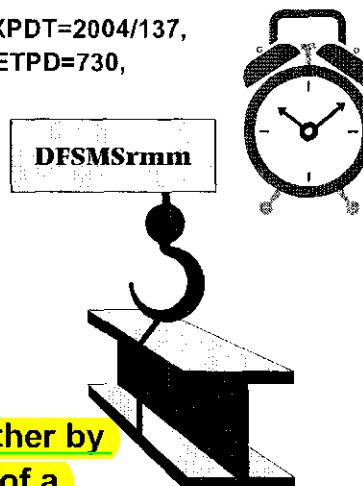
What Kind of Retention?

Job control language

```
// LABEL=EXPDT=2004/137, or // EXPDT=2004/137,  
// LABEL=RETPD=730, or // RETPD=730,
```

Vital record specification

Data set name and job name or
Volume serial number or
Management class (SMS) or
Management value (EDGUX100 user exit)



Expiration date and VRS are used together by DFSMSrmm to determine the retention of a volume.

Figure 4-2. What Kind of Retention

SS212.0

Notes:

DFSMSrmm does not use special dates. If a date is used (RETPD or EXPDT) and not overridden by an exit or DFSMS management class, DFSMSrmm retains the data set until the expiration date is reached. Data sets and volumes can also be retained through the use of VRS policies.

VRS Basics

There are three different types of vital record definitions available with the ADDVRS subcommand. Vital record specifications can use

- The data set name as a matching criteria

RMM ADDVRS DSNAME(...) JOBNAME(...) ...

- The volume serial number as a matching criteria

RMM ADDVRS VOLUME(...

- A name to provide a continuation of a VRS rule within a chain of vital record specifications

RMM ADDVRS NAME(...

Figure 4-3. VRS Basics

SS212.0

Notes:

The primary vehicle for retaining tape data is through the use of the DSNAME form of VRS definitions. Retention can be specified via cycles, days, or days since last reference. These three choices are mutually exclusive. If you do not request any of them while you are adding in a vital record specification, DFSMSrmm defaults to cycles.

A volume can also be retained by volser or a volser mask.

A NAME VRS is used for the chaining of vital record specifications. To compose a more complex rule including several vital record specifications, you must use a NAME VRS to be added after a DSNAME VRS. Further NAME VRSs may follow. Name VRSs must always be used to perform volume moves to more than one storage location.

Types of Retention Policies

Data set name by cycles

RMM ADDVRS DSNAME('SCHLUM.*.DATA') CYCLES COUNT(5)

RMM ADDVRS DSNAME('D99005') CYCLES COUNT(5)

Data set name by elapsed days

RMM ADDVRS DSNAME('SCHLUM.BACKUP.**') DAYS COUNT(30)

Data set name by days last referenced

RMM ADDVRS DSNAME('SCHLUM.BACKUP.DATA') -
LASTREFERENCEDAYS COUNT(7)

RMM ADDVRS DSNAME('D98007') LASTREFERENCEDAYS -
COUNT(7)

Data set name until specific date

RMM ADDVRS DSNAME('SCHLUM.**') DELETEDATE(2002/225)

Figure 4-4. Types of Retention Policies

SS212.0

Notes:

Vital record specifications, VRSSs, can be defined for volumes. We will concentrate on those directed at retaining data sets. For more complexity, a third type of vital record specification can be created. NAME VRSSs are used for multiple rules to retain or move volumes. Here we see examples of the data-set-name and volume-VRS definitions.

Types of Retention Policies

Data set name while cataloged

RMM ADDVRS DSNAME('SCHLUM.BACKUP.DATA') -
WHILECATALOGED

RMM ADDVRS DSNAME('M99000') WHILECATALOGED

Data set name until expired

RMM ADDVRS DSNAME('SCHLUM.BACKUP.DATA') -
UNTILEXPIRED

Data set name and job name

RMM ADDVRS DSNAME('**') JOBNAME(CHECK)

Data set open status

RMM ADDVRS DSNAME('ABEND') CYCLES COUNT(1)

RMM ADDVRS DSNAME('OPEN') LASTREFERENCEDAYS -
COUNT(2)

Figure 4-5. Types of Retention Policies

SS212.0

Notes:

WHILECATALOG will retain data sets as long as they are cataloged.

UNTILEXPIRED retains a data set until the volume expiration date is reached or until the data set is no longer retained (because of a VRS rule).

JOBNAME can be specified as an additional criteria for DSNAME VRS. For data sets to match the vital record specification, both criteria must be met.

Special data set status can be addressed by protected data set names of ABEND and OPEN. These vital record specifications match if the ABEND or OPEN flags in the DSN record in DFSMSrmm CDS are set, regardless of the real name of the data set.

Don't get confused about the example that includes DSNAME('M99000'). This is a valid data set name for a VRS but for a single level name (a name that does not contain a dot). Definitions like that are not intended to match real data set names. They are connected to a data set through the use of a management class (assigned to the data set by the management class ACS routine of SMS) or a management value (assigned to the data set by installation exit EDGUX100).

Using the Management Class to Retain Volumes

Define the management class names in the SMS configuration. Contents of the management class construct will be ignored (except when a data set is allocated to a system-managed tape volume, in which case the RETENTION LIMIT field is always honored).

M99000 Refers specifically to the
 expiration date 99000
 (LABEL=EXPDT=99000)

Figure 4-6. Using the Management Class to Retain Volumes

SS212.0

Notes:

DFSMS supports using a management class for system-managed tape data sets in order to limit retention specifications. DFSMSrmm will record the management class name that is assigned to the data set. This connection between a data set and a management class can be used for vital records processing.

Starting with OS/390, DFSMSrmm control via management class was extended to all tape data sets.

Assign a Management Class to a Data Set

```

Update ACS Routine
PROC 1 AMGMTCLAS
/*****
/* DEFINE FILTER FOR KEEP WHILE CATALOGED (99000) */
*****/
FILTLIST SPECIAL_DATE9 INCLUDE('1999000')
/*****
/* DEFINE FILTER FOR UP TO 9 CYCLES */
*****/
FILTLIST SPECIAL_DATRC9 INCLUDE('1999000')
/*****
/* DEFINE FILTER FOR TAPE UNITS */
*****/
FILTLIST TAPEUNIT INCLUDE(TAPE*, T3420* T3490*,
                          '3420','3480','3490',
                          T3590*, '3590')
/*****
/* SELECT VALID MANAGEMENT CLASS */
*****/
SELECT
  WHEN (4EXPDT = 4SPECIAL_DATE9 66,
        4UNIT = 4TAPEUNIT) SET 4MGMTCLAS = 'M99000'
  WHEN (4EXPDT = 4SPECIAL_DATE9 66,
        4UNIT = 4TAPEUNIT) SET 4MGMTCLAS = 'M99009'
  OTHERWISE SET 4MGMTCLAS = 'NORMAL'
END
END

```

Figure 4-7. Assign Management Class to Data Set

SS212.0

Notes:

You can use the DFSMS management class automatic class selection (ACS) routine to assign a management class to a data set that is system-managed by DFSMS. In addition to the EXPDT value, DFSMS has many other sources of information available to the routine for decision-making logic. The slide shows a simple example of special expiration dates.

Types of Retention Policies

A volume serial number can be used as a matching criteria in a vital record specification.

Specific volume serial number

RMM ADDVRS VOLUME(NSCH00) LOCATION(SAFE) COUNT(5)

Generic volume serial number

RMM ADDVRS VOLUME(NSC*) LOCATION(SAFE) COUNT(5)

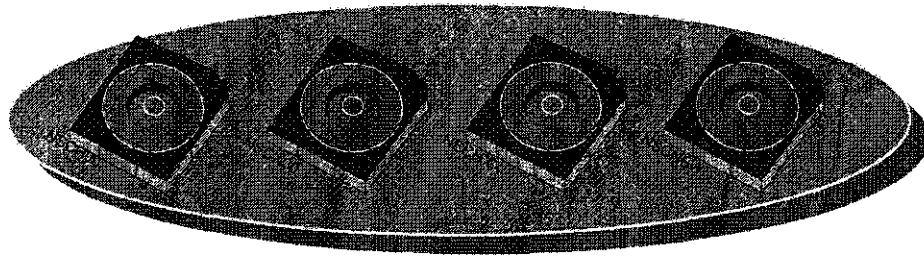


Figure 4-8. Types of Retention Policies

SS212.0

Notes:

Volume based vital record specifications can be used to retain volumes based exclusively on their volume serial numbers and regardless of the data set names that are included.

While being available, this type of control is used in real life not very often.

How DFSMSrmm Matches Data Set Names to Data Set Masks (VRSEL(OLD))

- 1 **A reserved name**
RMM ADDVRS DSNAME('ABEND') COUNT(1)
RMM ADDVRS DSNAME('OPEN') DAYS COUNT(2)
- 2 **A management class name**
RMM ADDVRS DSNAME('M99000') WHILECATALOG
- 3 **A full qualified data set name**
RMM ADDVRS DSNAME('SCHLUM.BACKUP.DATA')
RMM ADDVRS DSNAME(BACKUP.DATA)
- 4 **A data set name mask that includes a % or a ^**
RMM ADDVRS DSNAME('SCHLUM.BACKUP%.DATA')
RMM ADDVRS DSNAME('SCHLUM.BACKUP.DATA.D^AAAAA.T^AAAAA')
- 5 **Any data set name mask that includes single ***
RMM ADDVRS DSNAME('SCHLUM.BACK*.DATA')
- 6 **Any data set name mask that includes an ***
RMM ADDVRS DSNAME('SCHLUM.*.DATA')
- 7 **A data set name mask that includes ****
RMM ADDVRS DSNAME('SCHLUM.**')
RMM ADDVRS DSNAME('**.BACKUP.DATA')
- 8 **A vital record specification management value**
RMM ADDVRS DSNAME('D99000') WHILECATALOG
- 9 **A data set name mask of ****
RMM ADDVRS DSNAME('***')

Figure 4-9. How DFSMSrmm Matches Data Set Names to Data Set Masks (VRSEL(OLD))

SS212.0

Notes:

The special names ABEND and OPEN take precedence over the actual name of the data set. The DFSMS management class is also used if there is a matching vital record specification.

When creating VRS policies, that include masking characters, more than one policy might apply to a data set. DFSMSrmm uses the policy that most closely matches the data set name.

Most-specific matching works like generic data set profiles in RACF: When DFSMSrmm compares the data set name from left to right with the matching data set masks character by character, the most-specific match is with the mask that matches the most characters exactly in consecutive order without a break. If, when you use this method two or more masks match the same number of characters in the data set name, the kind of masking characters determines the closest match as shown in the figure.

How DFSMSrmm Matches Data Set Names to Data Set Masks (VRSEL(NEW))

- 1 A reserved name
RMM ADDVRS DSNAME('ABEND') DAYS COUNT(1)
RMM ADDVRS DSNAME('OPEN') DAYS COUNT(2)
- 2 A full qualified data set name
RMM ADDVRS DSNAME('SCHLUM.BACKUP.DATA')
RMM ADDVRS DSNAME('BACKUP.DATA')
- 3 A data set name mask that includes a % or a ^
RMM ADDVRS DSNAME('SCHLUM.BACKU%.DATA')
RMM ADDVRS DSNAME('SCHLUM.BACKUP.DATA.D^AAAAA.T^AAAAA')
- 4 Any data set name mask that includes single *
RMM ADDVRS DSNAME('SCHLUM.BACK*.DATA')
- 5 Any data set name mask that includes an *
RMM ADDVRS DSNAME('SCHLUM.*.DATA')
- 6 A data set name mask that includes **
RMM ADDVRS DSNAME('SCHLUM.**')
RMM ADDVRS DSNAME('**.BACKUP.DATA')
RMM ADDVRS DSNAME('*.**')
- 7 A management class name
RMM ADDVRS DSNAME('M99000') WHILECATALOG
- 8 A vital record specification management value
RMM ADDVRS DSNAME('D99000') WHILECATALOG
- 9 A data set name mask of **
RMM ADDVRS DSNAME('**')

Figure 4-10. How DFSMSrmm Matches Data Set Names to Data Set Masks (VRSEL(NEW)) SS212.0

Notes:

When OPTION VRS(NEW) is specified in EDGRMMxx, the matching order is slightly different than with OPTION VRSEL(OLD). The management class has moved close to the end of the matching order, just above the management value.

If you still have the choice being at the beginning of your implementation of DFSMSrmm, we strongly recommend that you use VRSEL(NEW). As you will see later on, there are some nice features exclusively available with VRSEL(NEW).

How DFSMSrmm Matches Job Names to Job Name Masks

OPTION VRSJOBNAME(1)

Job name !! Data set name

Job name is the primary value used to match the data set to a vital record specification.

Example:

Data set name mask	Job name mask	Term to be compared
.B.C	SAVE	SAVE!!.B.C
A.*.C	SAVE	SAVE!!A.*.C
A.B.*	SAVE*	SAVE*!!A.B.*

Data set name: A.B.C Job name: SAVE



Figure 4-11. How DFSMSrmm Matches Job Names to Job Name Masks

SS212.0

Notes:

The resolution of job name masks is the same as for data sets names, except that a job name is not made up of multiple qualifiers.

If both a data set name mask and a job name mask are specified in a vital record specification, it depends on the setting of option VRSJOBNAME in member EDGRMMxx how DFSMSrmm matches a data set.

When VRSJOBNAME(1) is specified, DFSMSrmm uses the data set name and job name to match a data set to a vital record specification. Job name is the primary value used to match the data set to a vital record specification. If there is no match, a match by data set name only is acceptable.

How DFSMSrmm Matches Job Names to Job Name Masks

OPTION VRSJOBNAME(2)

Data set name !! Job name

If a data set matches multiple vital record specifications, then DFSMSrmm uses a job name to further qualify the term to be compared.

Example:

Data set name mask	Job name mask	Term to be compared
*.B.C	SAVE	*.B.C!!SAVE
A.*.C	SAVE	A.*.C!!SAVE
A.B.*	SAVE*	A.B.*!!SAVE*

Data set name: A.B.C Job name: SAVE



Figure 4-12. How DFSMSrmm Matches Job Names to Job Name Masks... SS212.0

Notes:

When VRSJOBNAME(2) is specified, DFSMSrmm uses the data set name to match a data set to a vital record specification. If a data set matches multiple vital record specifications with the same data set name, then DFSMSrmm uses a job name to further qualify the data set name. If there is no match, DFSMSrmm does not apply a policy to the data set.

OPTION VRSJOBNAME(...) has an effect on all vital record specifications that use job name masks, so think about your decision for that setting carefully. The examples we have looked at, show clearly that when you change this option later on, a new matching vital record specifications will probably come up with your data sets.

An Asterisk (*) in a Data Set Name Mask

A single asterisk (*) used as a qualifier represents one or more characters.

Data set name mask	Matching	Unmatched
A.B.*	ABA	AB
	ABBB	ABBB
	ABCCCC	ABGCD
A*.C	AAC	BC
	ABBC	ABBB
	ABCCC	ABCCD
*.B.C	ABBC	AB
	BBBC	ABBB
	CCCCBC	ABCCD

Figure 4-13. An Asterisk (*) in a Data Set Name Mask

SS212.0

Notes:

An asterisk (*) can be used to represent a single qualifier in a data set name. It indicates that, while there must be a qualifier, any valid qualifier is acceptable. Each of the examples indicates that the data set name must be made up of exactly three qualifiers.

An Asterisk (*) in a Data Set Name Mask

A single asterisk (*) used within a qualifier represents zero or more characters

Data set name mask	Matching	Unmatched
A.B.C*	A.B.CA A.B.CBB A.B.CCC	AB ABB ABCD
A.B*.C	ABAC ABBC ABCCC	BC BB BCD
A*.B.C	ABAC ABBC ABCCC	AB ABB ABCD

Figure 4-14. An Asterisk (*) in a Data Set Name Mask

SS212.0

Notes:

When an asterisk is used as part of a qualifier, it indicates that zero or more characters can be substituted for the asterisk. It can be used as a suffix or a prefix. In our examples, we use it as a suffix.

Which is the best matching vital record specification for data set A.B.C?

A Double Asterisk (**) in a Data Set Name Mask

A double asterisk represents zero or more qualifiers. At the end of the mask, .** indicates to accept zero or more qualifiers.

Data Set Name Mask	Matching	Unmatched
A.B.**	AB ABBB ABCCC	A AB ABCDE
A.**.C	AC ABBC ACCC	BC AB ACB
**.B.C	BC ABBC AAABBBB.C	A AB A.B.C.D

Figure 4-15. A Double Asterisk (**) in a Data Set Name Mask

SS212.0

Notes:

Whenever a double asterisk (**) is coded as a qualifier in a data set name mask, zero or more qualifiers are acceptable.

Not Sign (^) in a Data Set NameMask

Not sign (^) is a place holder for a single character in a pseudo-GDG data set name mask which must be present.

Data Set Name Mask	Matching	Unmatched
A.B.C^	ABCA ABCB ABCC	AB ABD ABCD
A.B^.C	ABAC ABBC ABDC	ABC ABE ABED
A^.B.C	ABAC ABBC ACBC	AB ABD ABCD

Figure 4-16. Not Sign (^) in a Data Set Name Mask

SS212.0

Notes:

Data set name masks can also contain a masking character that represents a single-position place holder. This means that any valid character is acceptable in that position, and that one must exist. In this example we use the ^ symbol (not sign) to represent a single value that is to be ignored.

Percent Sign (%) in a Data Set Name Mask

Percent sign (%) is a place holder for a single character in data set name mask which must be present.

Data Set Name Mask	Matching	Unmatched
A.B.C%	ABCA ABCB ABCC	AB ABD ABCE
A.B%.C	ABAC ABBC ABCC	ABD ABE ABCD
A%.B.C	ABC ABD ABE	AB ABF ABCD

Figure 4-17. Percent Sign (%) in a Data Set Name Mask

SS212.0

Notes:

Data set name masks can also contain a masking character that represents a single-position place holder. This means that any valid character is acceptable in that position, and that one must exist. In this example we use the percent sign (%) to represent a single value that is to be ignored. While the not and the percent signs both seem to do the same thing, we will see shortly how they differ.

Count

Count can be a number from 1 to 99999.

1 to 99998

means to retain the number of
Cycles
Days
Volumes

99999

means to retain
All CYCLES of a data set
All volumes specified
For ever
No limit

Figure 4-18. Count

SS212.0

Notes:

DFSMSrmm will retain all of the data sets or volumes for a matching vital record specification unless a lesser value than 99999 is specified.

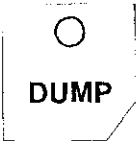
Retaining DFSMShsm Tapes

To avoid inadvertent release, retain DFSMShsm tapes using vital record processing.

```
RMM ADDVRS DSNNAME('mprefix.**') -  
COUNT(99999)
```



```
RMM ADDVRS DSNNAME('bprefix.**') -  
COUNT(99999)
```



```
RMM ADDVRS DSNNAME('authid.**') -  
COUNT(99999)
```

Figure 4-19. Retaining DFSMShsm Tapes

SS212.0

Notes:

You can retain all DFSMShsm tapes that are created during migration, backup, and dump processing by vital record specifications. For those volumes that require no movement, you can use the VRSs as shown in the picture.

CYCLE with DELAY

This example retains the three latest cycles or incidences of the data set.

RMM ADDVRS DSNAME('SCHLUM.BACKUP.DATA') -
COUNT(3) LOCATION(LOCAL) DELAY(1)

SCHLUM.BACKUP.DATA	012234	creation date 2002/002	
<u>SCHLUM.BACKUP.DATA</u>	<u>023456</u>	<u>creation date 2002/004</u>	<u>LOCAL</u>
<u>SCHLUM.BACKUP.DATA</u>	<u>054567</u>	<u>creation date 2002/010</u>	<u>LOCAL</u>
<u>SCHLUM.BACKUP.DATA</u>	<u>054389</u>	<u>creation date 2002/023</u>	<u>HOME</u>

Figure 4-20. CYCLE with DELAY

SS212.0

Notes:

When duplication of data sets occurs, DFSMSrmm retains the latest versions of the data set name. The first example retains three cycles of a data set (cycle control is the default and not coded in our vital record specification). The first VRS specified retains the three most recent of the four matching data sets.

DELAY specifies how many days the movement of a volume to a different location is to be delayed. For example, if today were 2002/023, we would see the above results.

Data Set Name Mask with Ignore Character

This example retains only the latest cycle or incidence of the data sets in a data set group which is formed of all data sets beginning with SCHLUM.BACKUP.DAT.

```
RMM ADDVRS DSNAME('SCHLUM.BACKUP.DAT^') -
COUNT(3) LOCATION(LOCAL) DELAY(1)
```

SCHLUM.BACKUP.DATA	012234	creation date 2002/002	
SCHLUM.BACKUP.DATA	023456	creation date 2002/004	
SCHLUM.BACKUP.DATA	034567	creation date 2002/010	
<u>SCHLUM.BACKUP.DATA</u>	<u>054389</u>	<u>creation date 2002/023</u>	<u>LOCAL</u>
SCHLUM.BACKUP.DATACOPY	045967	creation date 2002/002	
SCHLUM.BACKUP.DATACOPY	059876	creation date 2002/004	
SCHLUM.BACKUP.DATACOPY	029485	creation date 2002/010	
SCHLUM.BACKUP.DATACOPY	076296	creation date 2002/023	
SCHLUM.BACKUP.DATB	045967	creation date 2002/002	
SCHLUM.BACKUP.DATB	059876	creation date 2002/004	
<u>SCHLUM.BACKUP.DATB</u>	<u>029485</u>	<u>creation date 2002/011</u>	<u>LOCAL</u>
<u>SCHLUM.BACKUP.DATB</u>	<u>076296</u>	<u>creation date 2002/023</u>	<u>HOME</u>

Figure 4-21. Data Set Name Mask with Ignore Character

SS212.0

Notes:

In this example, we use a *not* sign (^) mask. This tells DFSMSrmm to retain the three most recently created data sets whose names match the vital record specification. Note that this keeps one that ends in DATA and two that end in DATB. If DFSMSrmm ignored the character instead of the *not* sign in the sense, that it only verifies the existence of the character regardless of the exact content, then the SCHLUM.BACKUP.DATA and SCHLUM.BACKUP.DATB data set names would be regarded as equal. In a later example, we will see how the situation is different when we use a percent sign mask.

Again, in this example, the movement of the volume that contains the most recent data set is delayed by one day.



Generation Data Group (GDG) Base Names

The keyword GDG specifies that the data set name is a generation data group (GDG) base name.
DFSMSrmm checks for the standard version qualifier, GnnnnVnn.

```
RMM ADDVRS DSNAME('SCHLUM.BACKUP.DATA') GDG COUNT(99999) CYCLES

SCHLUM.BACKUP.DATA.G0001V00 012234      creation date 2002/002
SCHLUM.BACKUP.DATA.G0002V00 023456      creation date 2002/004
SCHLUM.BACKUP.DATA.G0003V00 054567      creation date 2002/010
...
SCHLUM.BACKUP.DATA.G9997V00 012234      creation date 2002/356
SCHLUM.BACKUP.DATA.G9998V00 023456      creation date 2002/35/
SCHLUM.BACKUP.DATA.G9999V00 054567      creation date 2002/360
SCHLUM.BACKUP.DATA.G0001V00 054389      creation date 2002/102
SCHLUM.BACKUP.DATA.G0002V00 043986      creation date 2002/204
```

Figure 4-22. Generation Data Group (GDG) Base Names SS212.0

Notes:

If you are using GDG version numbering, DFSMSrmm keeps only the latest version of each generation.
You must not specify the last qualifier of the real generation data set names in the name mask when using the keyword, GDG.

Pseudo - GDG Data Set Names

Another way to match generation data sets, is to use a mask including not signs (^) in place of the digits in the last qualifier of the true data set name.

```
RMM ADDVRS DSNAME('SCHLUM.BACKUP.DATA.G^^^^V^^') -
COUNT(99999) CYCLES
```

```
SCHLUM.BACKUP.DATA.G0001V00 012234    creation date 2002/002
SCHLUM.BACKUP.DATA.G0002V00 023456    creation date 2002/004
SCHLUM.BACKUP.DATA.G0003V00 054567    creation date 2002/010
...
SCHLUM.BACKUP.DATA.G9999V00 012234    creation date 2002/356
SCHLUM.BACKUP.DATA.G9998V00 023456    creation date 2002/357
SCHLUM.BACKUP.DATA.G9999V00 054567    creation date 2002/360
SCHLUM.BACKUP.DATA.G0001V00 054389    creation date 2002/102
SCHLUM.BACKUP.DATA.G0002V00 043986    creation date 2002/204
```

Figure 4-23. Pseudo - GDG Data Set Names

SS212.0

Notes:

The data set name mask in the example matches the data set names of true generation data sets. The only way to achieve this is to use *ignore* characters. For that reason all data set name masks containing *ignore* characters are called pseudo-GDGs.

A pseudo-GDG is similar to a GDG VRS in that COUNT applies to the most recent matching data sets and not to each unique combination. However, a pseudo-GDG does the most retain duplicate names if they are covered by the count, while for generation data sets, on the other hand, DFSMSrmm does not retain duplicate generations.

Pseudo - GDG Data Set Names

A pseudo-GDG is a collection of data sets, that use the same data set name pattern and, that DFSMSrmm manages like a GDG.

RMM ADDVRS DSNAME('SCHLUM.BACKUP.DATA.D^T^') COUNT(99999)-
CYCLES

<i>SCHLUM.BACKUP.DATA.D97002.T1254</i>	<i>012234</i>	<i>creation date 2002/002</i>
<i>SCHLUM.BACKUP.DATA.D97004.T0332</i>	<i>023456</i>	<i>creation date 2002/004</i>
<i>SCHLUM.BACKUP.DATA.D97010.T1848</i>	<i>054567</i>	<i>creation date 2002/010</i>
...		
<i>SCHLUM.BACKUP.DATA.D97356.T2359</i>	<i>012234</i>	<i>creation date 2002/356</i>
<i>SCHLUM.BACKUP.DATA.D97357.T0032</i>	<i>023456</i>	<i>creation date 2002/357</i>
<i>SCHLUM.BACKUP.DATA.D97360.T1617</i>	<i>054567</i>	<i>creation date 2002/360</i>
<i>SCHLUM.BACKUP.DATA.D98102.T0944</i>	<i>054389</i>	<i>creation date 2002/102</i>
<i>SCHLUM.BACKUP.DATA.D98204.T1323</i>	<i>043986</i>	<i>creation date 2002/204</i>

Figure 4-24. Pseudo - GDG Data Set Names

SS212.0

Notes:

You must use the NOGDG operand (which is the default) when defining a pseudo-GDG vital record specification.

If the count is less than 99999, DFSMSrmm retains the most recent name, regardless of any duplication.

CYCLES

Cycle control retains data sets based on cycles or copies of data sets.

```
RMM ADDVRS  DSNNAME('SCHLUM.BACKUP.DATA.G^^^^V^^) COUNT(3) CYCLES

SCHLUM.BACKUP.DATA.G0001V00 012234      creation date 2002/002
SCHLUM.BACKUP.DATA.G0002V00 023456      creation date 2002/003
SCHLUM.BACKUP.DATA.G0003V00 054567      creation date 2002/003
SCHLUM.BACKUP.DATA.G0004V00 012234      creation date 2002/003
SCHLUM.BACKUP.DATA.G0005V00 023456      creation date 2002/004
SCHLUM.BACKUP.DATA.G0006V00 054567      creation date 2002/004
SCHLUM.BACKUP.DATA.G0007V00 054368      creation date 2002/005
SCHLUM.BACKUP.DATA.G0008V00 043986      creation date 2002/005
```

Figure 4-25. CYCLES SS212.0

Notes:

Cycle control retains the most recent data sets according to cycle count. In our example two volumes are created on 2002/004. The oldest data set seems to be the one ending on G0005V00, due to the standard ordering of generation data sets.

CYCLES and BYDAYSCYCLE

BYDAYSCYCLE specifies retaining all instances of a data set created on a single day as a single cycle.

RMM ADDVRS DSNNAME('SCHLUM.BACKUP.DATA.G^^^^V^^) COUNT(3) -
BYDAYSCYCLE

SCHLUM.BACKUP.DATA.G0001V00	012234	creation date 2002/002	
SCHLUM.BACKUP.DATA.G0002V00	023456	creation date 2002/003	1
SCHLUM.BACKUP.DATA.G0003V00	054567	creation date 2002/003	1
SCHLUM.BACKUP.DATA.G0004V00	012234	creation date 2002/003	1
SCHLUM.BACKUP.DATA.G0005V00	023456	creation date 2002/004	2
SCHLUM.BACKUP.DATA.G0006V00	054567	creation date 2002/004	2
SCHLUM.BACKUP.DATA.G0007V00	054389	creation date 2002/005	3
SCHLUM.BACKUP.DATA.G0008V00	043986	creation date 2002/005	3

Figure 4-26. CYCLES and BYDAYSCYCLE SS212.0

Notes:

You can also choose BYDAYSCYCLE in order to have DFSMSrmm regard all instances of a data set created on a single day as one cycle.

Defining Storage Locations in EGDRMMxx

```

LOCDEF
  LOCATION(...)
    DISTANT
    LOCAL
    REMOTE
    Installation_defined
    SHELF
    System_managed_library_name

  MEDIANAME(...)
    3420
    3480
    REEL
    ...

```

Figure 4-27. Defining Storage Locations in EGDRMMxx

SS212.0

Notes:

DFSMSrmm has three built-in storage locations: LOCAL, DISTANT, and REMOTE. All three are managed in a similar way, in that a specific number of slots (bins) are allocated and are all treated alike in that a slot holds one volume, regardless of the shape of the media. In addition to these three, DFSMSrmm recognizes each IBM system-managed tape library as a unique storage location by its DFSMS library name. Tape volumes in the computing facility that are not in an IBM system-managed library are considered to be in the location SHELF. Many installations require more than three storage locations (vaults) for movement of tape volumes.

Medianame can be any 1 to 8 character name you choose to describe a media name, type of media, a shape, or size. The media names you use on the LOCDEF commands should be the same as or a subset of the media names you use for your installation VLPOOL definitions. When you specify MEDIANAME(*) using the parmlib LOCDEF command or the RMM ADDBIN or RMM ADDRACK subcommands, any volume with any media name can be sent to the location and DFSMSrmm does not segregate the volumes by shape.

LOCDEF TYPE and MANAGEMENTTYPE

LOCDEF
 TYPE(...)
 LIBRARY
 STORAGE
 STORAGE,HOME

MANAGEMENTTYPE(...)
 BINS
 NOBINS

Figure 4-28. LOCDEF TYPE and MANAGEMENTTYPE

SS212.0

Notes:

Use the operand TYPE to identify the type of location you are defining. The value can be either STORAGE or LIBRARY. This operand is optional. If MEDIANAME or MANAGEMENTTYPE are specified, only TYPE(STORAGE) is valid. You can identify a storage location as a home location by specifying TYPE(STORAGE,HOME). When you identify a storage location as a home location, you can manage volumes in a storage location like volumes that reside in a LIBRARY location. You can:

- Schedule release actions for volumes when they return to their storage home location
- Return volumes to scratch status while they reside in their storage home location

Locations other than IBM system-managed libraries and SHELF are *storage locations*. If the location of the volume at a storage location is not the responsibility of the sending location, then MANAGEMENTTYPE(NO BINS) can be specified. However, the location must be able to accept the MEDIANAME associated with the volume.

LOCDEF PRIORITY

LOCDEF PRIORITY (1-9999)	
100	REMOTE location
200	DISTANT location
300	LOCAL location
2000	Installation-defined storage type
4800	Automated tape libraries
4900	Manual tape libraries
5000	SHELF location

Figure 4-29. LOCDEF PRIORITY SS212.0

Notes:

DFSMSrmm manages all data sets on a volume. When the movement policies for two or more data sets on a volume conflict, DFSMSrmm uses a priority scheme to determine where the volume should be scheduled for movement. The priority of a location can be changed from its default value in DFSMSrmm. Lower numbers have higher priorities. Priority is used to determine where to move a volume when a volume is assigned multiple destinations based on matching to two or more vital record specification definitions. You can override this value by specifying a priority on the vital record specification. If priority is omitted, the priority is based on the location type as shown in the figure.

An installation can define its own storage location names (one to eight positions in length beginning with an alphabetic character). For example:

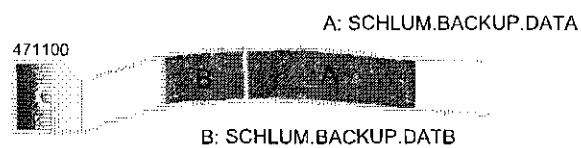
```
/* LOCDEF - Add a location definition */
LOCDEF LOCATION(MIKESLOC) MEDIANAME(3490) TYPE(STORAGE) -
MANAGEMENTTYPE(BINS)
```

Example of Priority Based Vaulting

If two or more data sets reside on the same tape volume and match different vital record specifications, vaulting applies according to priority.

RMM ADDVRS DSNAME('SCHLUM.BACKUP.DATA') -
COUNT(3) LOCATION(LOCAL)

RMM ADDVRS DSNAME('SCHLUM.BACKUP.DATB') -
COUNT(3) LOCATION(REMOTE)



To which location will volume 471100 be moved?

Figure 4-30. Example for Priority Based Vaulting

SS212.0

Notes:

DFSMSrmm processes all data set records. For a multiframe volume, a movement conflict will occur if VRS policies for different data sets result in the need to move the volume to two or more storage locations. DFSMSrmm resolves this conflict by assigning the volume to the highest-priority location, according to the rule: the lower the number, the higher the priority.

Based on the default settings for priority, volume 471100 will go to REMOTE.

VRS with Priority

A priority value which is specified with the ADDVRS command will override the priority defined in the LOCDEF command for a location.

```
RMM ADDVRS DSNAME('SCHLUM.BACKUP.DATB') -  
COUNT(3) LOCATION(REMOTE)
```

```
RMM ADDVRS DSNAME('SCHLUM.BACKUP.DATA') -  
COUNT(3) LOCATION(LOCAL) DELAY(1) PRIORITY(99)
```

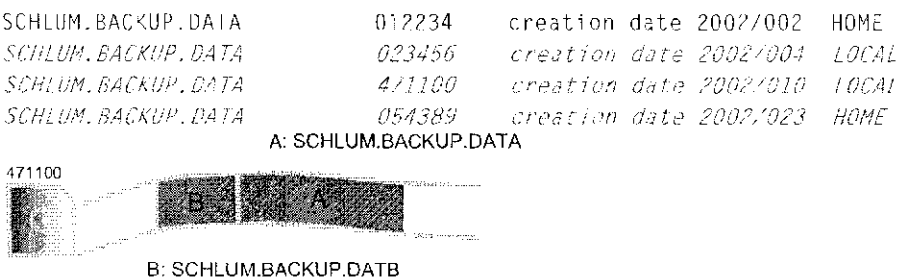


Figure 4-31. Priority

SS212.0

Notes:

The priority value can be changed via the PRIORITY option on the VRS policy. In our example, SCHLUM.BACKUP.DATA has been given a higher priority, 99, which it would normally have had for location LOCAL (300). Volume 471100 will be moved to LOCAL.

Example: A Full Qualified Data Set Name

RMM ADDVRS DSNAME('SCHLUM.BACKUP.DATA') COUNT(2)

SCHLUM.BACKUP	090003	creation date 2002/013
SCHLUM.BACKUP	039966	creation date 2002/020
SCHLUM.BACKUP	043288	creation date 2002/032
SCHLUM.BACKUP.DATA	023456	creation date 2002/014
SCHLUM.BACKUP.DATA	034467	creation date 2002/021
SCHLUM.BACKUP.DATA	054382	creation date 2002/033
SCHLUM.BACKUP.DATB	058765	creation date 2002/015
SCHLUM.BACKUP.DATR	098531	creation date 2002/022
SCHLUM.BACKUP.DATB	031998	creation date 2002/034
SCHLUM.BACKUP.DATACOPY	059876	creation date 2002/016
SCHLUM.BACKUP.DATACOPY	029485	creation date 2002/023
SCHLUM.BACKUP.DATACOPY	076296	creation date 2002/035
SCHLUM.BACKUP.DATBCOPY	059876	creation date 2002/017
SCHLUM.BACKUP.DATBCOPY	029485	creation date 2002/024
SCHLUM.BACKUP.DATBCOPY	076296	creation date 2002/036
SCHLUM.BACKUP.COPYDATA	022947	creation date 2002/018
SCHLUM.BACKUP.COPYDATA	073397	creation date 2002/025
SCHLUM.BACKUP.COPYDATA	052134	creation date 2002/037

Figure 4-32. Example: A Full Qualified Data Set Name SS212.0

Notes:

When fully qualified data set names are coded, DFSMSrmm retains the most recent ones.

Example: Data Set Mask That Includes a *

RMM ADDVRS DSNAME('SCHLUM.BACKUP.DAT*') COUNT(2)		
SCHLUM.BACKUP	090003	creation date 2002/013
SCHLUM.BACKUP	039966	creation date 2002/020
SCHLUM.BACKUP	043288	creation date 2002/032
SCHLUM.BACKUP.DAT	023456	creation date 2002/014
SCHLUM.BACKUP.DAT	034567	creation date 2002/021
SCHLUM.BACKUP.DAT	054389	creation date 2002/033
SCHLUM.BACKUP.DATB	058765	creation date 2002/015
SCHLUM.BACKUP.DATB	028531	creation date 2002/022
SCHLUM.BACKUP.DATB	031998	creation date 2002/034
SCHLUM.BACKUP.DATACOPY	059876	creation date 2002/016
SCHLUM.BACKUP.DATACOPY	028485	creation date 2002/023
SCHLUM.BACKUP.DATACOPY	016290	creation date 2002/035
SCHLUM.BACKUP.DATBCOPY	059876	creation date 2002/017
SCHLUM.BACKUP.DATBCOPY	029485	creation date 2002/024
SCHLUM.BACKUP.DATBCOPY	016296	creation date 2002/036
SCHLUM.BACKUP.COPYDATA	022947	creation date 2002/018
SCHLUM.BACKUP.COPYDATA	073397	creation date 2002/025
SCHLUM.BACKUP.COPYDATA	052134	creation date 2002/037

Figure 4-33. Example: Data Set Mask That Includes a * SS212.0

Notes:

When an asterisk or a percent sign is used in a mask, the count represents the number of each unique combination of data sets to be retained. In our example, 8 of the 18 data sets are retained. Cycle control applies to each of the four groups of matching data set names. The first group of data sets does not match, because three qualifiers are required. The last group of data sets does not match, because the last qualifier does not meet the form as requested by the data set name mask.

Example: Data Set Mask that Includes a %

RMM ADDVRS DSNAME('SCHLUM.BACKUP.DAT%') COUNT(2)		
SCHLUM.BACKUP	090903	creation date 2002/013
SCHLUM.BACKUP	039966	creation date 2002/020
SCHLUM.BACKUP	043288	creation date 2002/032
SCHLUM.BACKUP.DAT	023456	creation date 2002/014
SCHLUM.BACKUP.DAT	023467	creation date 2002/021
SCHLUM.BACKUP.DAT	054289	creation date 2002/033
SCHLUM.BACKUP.DATB	058765	creation date 2002/015
SCHLUM.BACKUP.DATB	048531	creation date 2002/022
SCHLUM.BACKUP.DATB	031928	creation date 2002/034
SCHLUM.BACKUP.DATACOPY	059876	creation date 2002/016
SCHLUM.BACKUP.DATACOPY	079485	creation date 2002/023
SCHLUM.BACKUP.DATACOPY	076296	creation date 2002/035
SCHLUM.BACKUP.DATBCOPY	059876	creation date 2002/017
SCHLUM.BACKUP.DATBCOPY	029485	creation date 2002/024
SCHLUM.BACKUP.DATBCOPY	076296	creation date 2002/036
SCHLUM.BACKUP.COPYDATA	022947	creation date 2002/018
SCHLUM.BACKUP.COPYDATA	073397	creation date 2002/025
SCHLUM.BACKUP.COPYDATA	052134	creation date 2002/037

Figure 4-34. Example: Data Set Mask that Includes a % SS212.0

Notes:

In this example, a percent sign is used. Since it is a single character place holder, we retain four of the 18 data sets shown, because the first and last three do not match the mask.

Through cycle control, DFSMSrmm considers only data sets with names match exactly. If several data set names match the data set name mask, DFSMSrmm sorts them by data set name before applying cycle control to each group separately. This is why we see in our example that a total of four data sets are retained by the specification on the part of the vital record specification, of a count of two.

Which data sets would be retained if the *not* sign (^) had been used?

Example: Data Set Mask that Includes a ^

RMM ADDVRS DSNAME('SCHLUM.BACKUP.DAT^') COUNT(2)

SCHLUM.BACKUP	090003	creation date 2002/013
SCHLUM.BACKUP	039966	creation date 2002/020
SCHLUM.BACKUP	043288	creation date 2002/032
SCHLUM.BACKUP.DATA	023456	creation date 2002/014
SCHLUM.BACKUP.DATB	034567	creation date 2002/021
SCHLUM.BACKUP.DATB	054389	creation date 2002/033
SCHLUM.BACKUP.DATB	058765	creation date 2002/015
SCHLUM.BACKUP.DATB	098531	creation date 2002/022
SCHLUM.BACKUP.DATB	031999	creation date 2002/034
SCHLUM.BACKUP.DATBCOPY	059876	creation date 2002/017
SCHLUM.BACKUP.DATBCOPY	029485	creation date 2002/024
SCHLUM.BACKUP.DATBCOPY	076296	creation date 2002/036
SCHLUM.BACKUP.COPYDATA	022947	creation date 2002/018
SCHLUM.BACKUP.COPYDATA	073397	creation date 2002/025
SCHLUM.BACKUP.COPYDATA	052134	creation date 2002/037

Figure 4-35. Example: Data Set Mask that Includes a SS212.0

Notes:

By changing the last position to a *not* sign (^), from a percent sign, only two data sets are retained rather than four.

Because of the *not* sign (^) in the data set name mask, DFSMSrmm considers the data set names of SCHLUM.BACKUP.DATA and SCHLUM.BACKUP.DATB as equal! So the policy is applied to *one* group of *four* data sets, leading to two data sets which are retained.

Example: Data Set Mask that Includes %%%%%

RMM ADDVRS DSNAME('SCHLUM.BACKUP.DAT%%%%%') COUNT(2)		
SCHLUM.BACKUP	090003	creation date 2002/013
SCHLUM.BACKUP	039966	creation date 2002/020
SCHLUM.BACKUP	043288	creation date 2002/032
SCHLUM.BACKUP.DAT	023456	creation date 2002/014
SCHLUM.BACKUP.DAT	034567	creation date 2002/021
SCHLUM.BACKUP.DAT	054389	creation date 2002/033
SCHLUM.BACKUP.DATB	058765	creation date 2002/015
SCHLUM.BACKUP.DATB	098531	creation date 2002/022
SCHLUM.BACKUP.DATB	031998	creation date 2002/034
SCHLUM.BACKUP.DATACOPY	059876	creation date 2002/016
SCHLUM.BACKUP.DATACOPY	029485	creation date 2002/023
SCHLUM.BACKUP.DATACOPY	076296	creation date 2002/035
SCHLUM.BACKUP.DAT9COPY	059876	creation date 2002/017
SCHLUM.BACKUP.DAT9COPY	029485	creation date 2002/024
SCHLUM.BACKUP.DAT9COPY	076296	creation date 2002/036
SCHLUM.BACKUP.COPYDATA	022947	creation date 2002/018
SCHLUM.BACKUP.COPYDATA	073397	creation date 2002/025
SCHLUM.BACKUP.COPYDATA	052134	creation date 2002/037

Figure 4-36. Example: Data Set Mask that Includes %%%%%

SS212.0

Notes:

In this example we use multiple percent signs. There are a total of six matching data sets, but the count is to retain the most recent two of each unique name. Therefore, we retain four data sets.

Examples: Data Set Mask that Includes a ^

RMM ADDVRS DSNAME('SCHLUM.BACKUP.DAT^^^^')COUNT(2)

SCHLUM.BACKUP	090003	creation date 2002/013
SCHLUM.BACKUP	039966	creation date 2002/020
SCHLUM.BACKUP	043288	creation date 2002/032
SCHLUM.BACKUP.DATA	023456	creation date 2002/014
SCHLUM.BACKUP.DATA	034567	creation date 2002/021
SCHLUM.BACKUP.DATA	054389	creation date 2002/033
SCHLUM.BACKUP.DATB	058765	creation date 2002/015
SCHLUM.BACKUP.DATB	098531	creation date 2002/022
SCHLUM.BACKUP.DATB	031998	creation date 2002/034
SCHLUM.BACKUP.DATACOPY	059876	creation date 2002/016
SCHLUM.BACKUP.DATACOPY	029485	creation date 2002/023
SCHLUM.BACKUP.DATACOPY	076296	creation date 2002/035
SCHLUM.BACKUP.DATBCOPY	059876	creation date 2002/017
SCHLUM.BACKUP.DATBCOPY	029485	creation date 2002/024
SCHLUM.BACKUP.DATBCOPY	076296	creation date 2002/036
SCHLUM.BACKUP.COPYDATA	022947	creation date 2002/018
SCHLUM.BACKUP.COPYDATA	073397	creation date 2002/025
SCHLUM.BACKUP.COPYDATA	052134	creation date 2002/037

Figure 4-37. Examples: Data Set Mask that Includes a ^^^^^ SS212.0

Notes:

By changing the percent signs to *not* signs (^) and the count to three, we see that only two data sets are retained. This is the manner in which generation data sets in a generation data group are retained. When a *not* sign (^) is used in a mask, we call it a *pseudo GDG* vital record specification.

Example: * for a Single Qualifier

RMM ADDVRS DSNAME('SCHLUM.BACKUP.*') COUNT(2)		
SCHLUM.BACKUP	090003	creation date 2002/013
SCHLUM.BACKUP	039966	creation date 2002/020
SCHLUM.BACKUP	043288	creation date 2002/032
SCHLUM.BACKUP.DATA	023456	creation date 2002/014
SCHLUM.BACKUP.DATA	034567	creation date 2002/021
SCHLUM.BACKUP.DATA	054382	creation date 2002/023
SCHLUM.BACKUP.DATB	058165	creation date 2002/015
SCHLUM.BACKUP.DATB	098531	creation date 2002/022
SCHLUM.BACKUP.DATB	031998	creation date 2002/024
SCHLUM.BACKUP.DATACOPY	059876	creation date 2002/016
SCHLUM.BACKUP.DATACOPY	029485	creation date 2002/025
SCHLUM.BACKUP.DATACOPY	076296	creation date 2002/035
SCHLUM.BACKUP.DATBCOPY	059876	creation date 2002/017
SCHLUM.BACKUP.DATBCOPY	029485	creation date 2002/024
SCHLUM.BACKUP.DATBCOPY	076296	creation date 2002/036
SCHLUM.BACKUP.COPYDATA	022947	creation date 2002/018
SCHLUM.BACKUP.COPYDATA	073397	creation date 2002/026
SCHLUM.BACKUP.COPYDATA	052134	creation date 2002/037

Figure 4-38. Examples: Data Set Mask that Includes a ^^^^^ SS212.0

Notes:

A single asterisk represents a single qualifier. Only the first set of data sets does not match the vital record specification. Only the most recent two of the matching names in each group of data set names are retained, thereby resulting in the retention of 10 data sets altogether.

Example: ** for Zero or More Qualifiers

RMM ADDVRS DSNAME('SCHLUM.BACKUP.**') COUNT(2)

SCHLUM.BACKUP	090003	creation date 2002/013
<i>SCHLUM.BACKUP</i>	<i>039966</i>	<i>creation date 2002/020</i>
<i>SCHLUM.BACKUP</i>	<i>043208</i>	<i>creation date 2002/032</i>
SCHLUM.BACKUP.DATA	023456	creation date 2002/014
<i>SCHLUM.BACKUP.DATA</i>	<i>024567</i>	<i>creation date 2002/021</i>
<i>SCHLUM.BACKUP.DATA</i>	<i>054389</i>	<i>creation date 2002/033</i>
SCHLUM.BACKUP.DATB	058765	creation date 2002/015
<i>SCHLUM.BACKUP.DATB</i>	<i>098631</i>	<i>creation date 2002/022</i>
<i>SCHLUM.BACKUP.DATB</i>	<i>031998</i>	<i>creation date 2002/034</i>
SCHLUM.BACKUP.DATACOPY	059876	creation date 2002/016
<i>SCHLUM.BACKUP.DATACOPY</i>	<i>074489</i>	<i>creation date 2002/023</i>
<i>SCHLUM.BACKUP.DATACOPY</i>	<i>076296</i>	<i>creation date 2002/035</i>
SCHLUM.BACKUP.DATBCOPY	059876	creation date 2002/017
<i>SCHLUM.BACKUP.DATBCOPY</i>	<i>072485</i>	<i>creation date 2002/024</i>
<i>SCHLUM.BACKUP.DATBCOPY</i>	<i>076296</i>	<i>creation date 2002/036</i>
SCHLUM.BACKUP.COPYDATA	022947	creation date 2002/018
<i>SCHLUM.BACKUP.COPYDATA</i>	<i>073397</i>	<i>creation date 2002/025</i>
<i>SCHLUM.BACKUP.COPYDATA</i>	<i>052123</i>	<i>creation date 2002/037</i>

Figure 4-39. Example: ** for Zero or More Qualifiers SS212.0

Notes:

By changing the asterisk (*) in the previous example to a double asterisk (**), we pick up the first two data sets of each group in our list and thereby retain 12 data sets.

4.2 Examples

Examples: Expiration Date and Cycles

An expiration date is applied by JCL.

```
//          LABEL=RETPD=7
```

A retention policy is applied by a vital record specification.

```
RMM ADDVRS DSN('SCHLUM.SAVE') COUNT(5) GDG
```

A volume is released if both retention mechanisms retire.

Monday	<u>G0001V00</u>	<u>EXPD1=2002/100</u>
Tuesday	<u>G0002V00</u>	<u>EXPD1=2002/101</u>
Wednesday	<u>G0003V00</u>	<u>EXPD1=2002/102</u>
Thursday	<u>G0004V00</u>	<u>EXPD1=2002/103</u>
Friday	<u>G0005V00</u>	<u>EXPD1=2002/104</u>
Saturday		
Sunday		
Monday	<u>G0006V00</u>	<u>EXPD1=2002/105</u>

Figure 4-40. Examples: Expiration Date and Cycles SS212.0

Notes:

Many tape data sets are actually generation data sets. This means that the last qualifier of the name is in the form, GnnnnV00. If more than one data set of the same generation are created (for example due to a re-run), normally only the most recent is to be retained. DFSMSrmm has the GDG parameter for DSNAME VRS policies, which tells DFSMSrmm that a last qualifier in the form, GnnnnVnn, is to be added to the data set name specified. In this example, that means retaining the five most recent unique generations.

Examples: Expiration Date and Cycles

If the same rules apply, and two generation data sets are created each day, the result is as follows:

Monday	G0001V00	EXPDT=2002/100
	G0002V00	EXPDT=2002/100
Tuesday	G0003V00	EXPDT=2002/101
	G0004V00	EXPDT=2002/101
Wednesday	G0005V00	EXPDT=2002/102
	G0006V00	EXPDT=2002/102
Thursday	G0007V00	EXPDT=2002/103
	G0008V00	EXPDT=2002/103
Friday	G0009V00	EXPDT=2002/104
	G0010V00	EXPDT=2002/104
Saturday		
Sunday		
Monday	G0011V00	EXPDT=2002/107
	G0012V00	EXPDT=2002/107

Figure 4-41. Examples: Expiration Date and Cycles

SS212.0

Notes:

DFSMSrmm releases volumes if both of the following criteria are met:

There is no longer a matching VRS

The expiration date of the volume is reached or exceeded.

In our example, the volumes containing G0003V00 through G0007V00 are retained only because the expiration dates of the volumes are not reached or exceeded.

Examples: Expiration Date and Cycles

If the same rules apply, and a public holiday occurs while one generation data set is created each day, the result is as follows:

Monday	00001000	EXPDT=2002/100
Tuesday	00002000	EXPDT=2002/100
Wednesday	public holiday	
Thursday	00003000	EXPDT=2002/100
Friday	00004000	EXPDT=2002/100
Saturday		
Sunday		
Monday	00005000	EXPDT=2002/100

Figure 4-42. Examples: Expiration Date and Cycles

SS212.0

Notes:

VRS is a retention mechanism. If the policy matches a data set, it is retained. Actually, DFSMSrmm retains the volume where the data set was written on. The matching VRS even retains the volume if its expiration date is exceeded.

Example: Movement

```
RMM ADDVRS DSNAME('SCHLUM.*.DATA') GDG COUNT(99999) -
  LOCATION(HOME) STORENUMBER(1) NEXTVRS(SAVE3)-
  WHILECATALOG
```

```
RMM ADDVRS NAME(SAVE3) STORENUMBER(3)-
  LOCATION(SAVE)
```

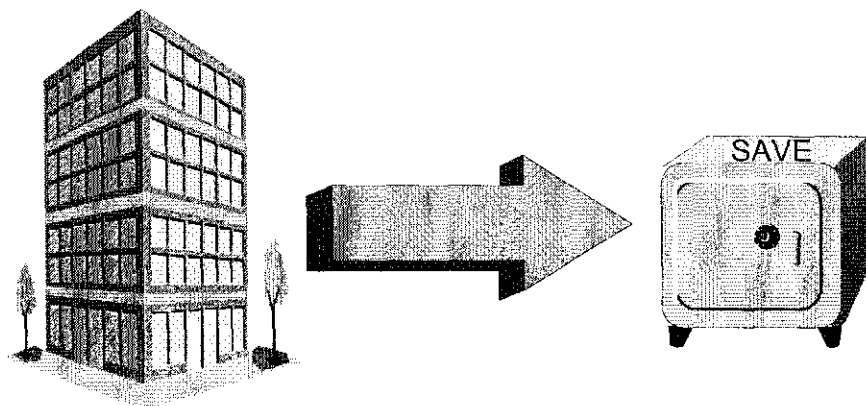


Figure 4-43. Example: Movement

SS212.0

Notes:

If the tape data represents multiple backup copies and you want to retain the most recent one in the data center, the storage location, HOME, can be used. Recall that HOME represents the home location of a volume: either SHELF, or an IBM system-managed tape library, or a storage location for which you specified TYPE(STORAGE,HOME) in the LOCDEF statement.

The first vital record specification retains all data sets but stores the most recent at the home location. The name type VRS is chained by NEXTVRS and moves the next three data sets to a locally defined location called SAVE. If there are more than four data sets retained by this rule, all remaining data sets are kept at their home location.

Example: Movement to Several Locations

```
RMM ADDVRS DSNAME('SCHLUM*.DATA') GDG COUNT(99999) -  
LOCATION(HOME) STORENUMBER(1) WHILECATALOG NEXTVRS(SAVE3)  
RMM ADDVRS NAME(SAVE3) STORENUMBER(3) LOCATION(SAVE) -  
NEXTVRS(RMT3)  
RMM ADDVRS NAME(RMT3) STORENUMBER(3) LOCATION(REMOTE)
```

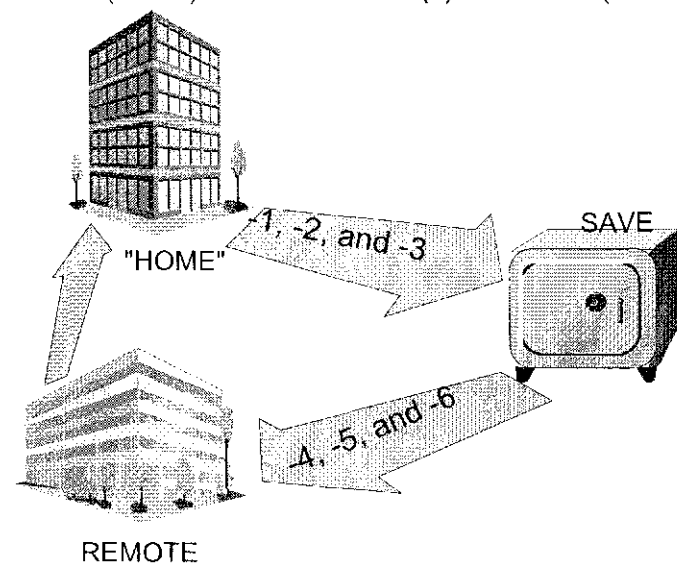


Figure 4-44. Example: Movement to Several Locations

SS212.0

Notes:

Let's expand on our last example. As tapes at SAVE are displaced by more-recent volumes, they can be sent to another location, REMOTE in our example. Tapes displaced from REMOTE return to their home location.

Running DFSMSrmm with DFSMSHsm

The retention and movement of TAPECOPY tapes or DUPLEX tapes is assisted by the following vital record specifications.

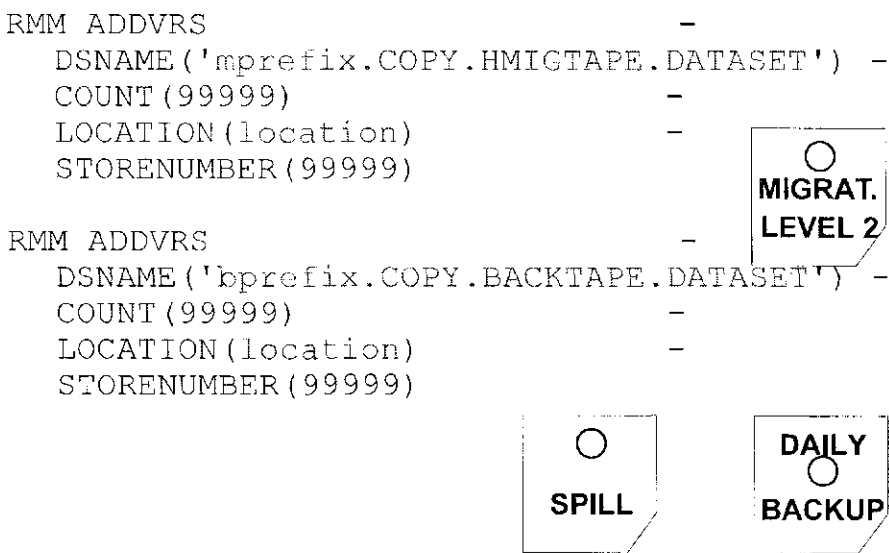


Figure 4-45. Running DFSMSrmm with DFSMSHsm SS212.0

Notes:

The DFSMSHsm TAPECOPY command and the DUPLEX tape feature create tapes called alternate tapes. Both the TAPECOPY command and the DUPLEX tape feature use the same naming convention for the alternate tapes.

To retain alternate tapes and to move them to another location, you can define two vital record specifications, one for backup copies and one for migration copies as shown in the picture.

Combining Retention in a Vital Record Specification

	DSNAME	VOLSER	NAME	CYCLES	DAYS	BYDAYSCYCLE	EXTRADAYS	LASTREF	WHILECATAL.	UNTILEXPIRED	OPEN	ABEND	JOBNAME	RELEASE	DELETEDATE	NEXTVRS	ANDVRS
DSNAME	X	X	X				X				X	X					
VOLSER	X	X					X				X	X					
NAME	X	X					X				X	X					
CYCLES				X		X	X	X					X	X			
DAYS				X	X	X	X	X					X	X			
BYDAYSCYCLE					X	X	X	X					X	X			
EXTRADAYS	X	X		X	X	X	X	X	X								X
LASTREFERENCEDAYS				X	X	X	X										
WHILECATALOGED							X										
UNTILEXPIRED							X										
OPEN	X	X	X														
ABEND	X	X	X														
JOBNAME			X														
RELEASE		X	X														
DELETEDATE																	
NEXTVRS																	X
ANDVRS							X										

Figure 4-46. Combining Retention in a Vital Record Specifications SS212.0

Notes:

You can combine some VRS retention options with others. This chart is designed to assist you in determining which combinations are valid (check) or invalid (X).

4.3 Updates of DFSMSrmm CDS during VRSEL

What Happens during VRSEL?

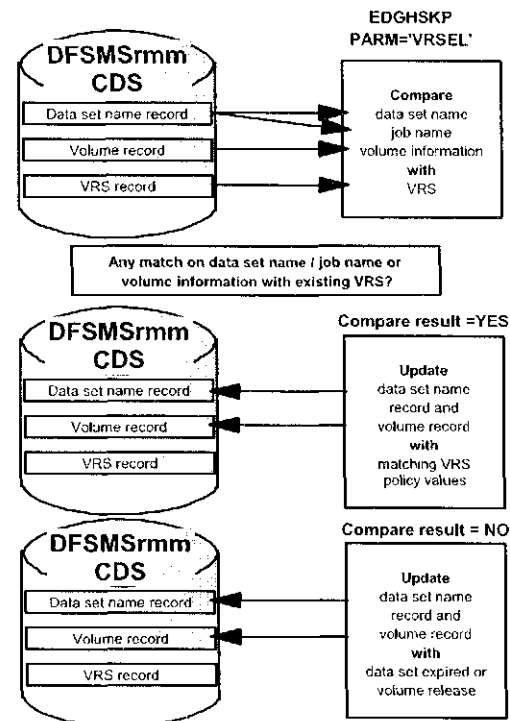


Figure 4-47. What Happens during VRSEL?

SS212.0

Notes:

Only during VRSEL are all vital record specifications cross-checked with all data set name records and all volume records, in order to find the most specific matching VRS. This match is based on the name of the data set or volume (if any). If there is a match, the name of the matching VRS in the data set record is recorded, the data set and the volume are marked as VRS protected, and the retention date for the data set and the volume is calculated. In this way, the data-set or the volume record, or both, are updated to reflect the new situation.

If there is no match for a volume or any data set it contains, the volume is only protected by its expiration date.

If a data set or a volume is no longer retained by a vital record specification, data-set or volume record, or both, are updated in order to indicate that they are not retained by a VRS. During the next run of EXPROC, these volumes are eligible for expiration processing.

Data Set Record with Matching VRS

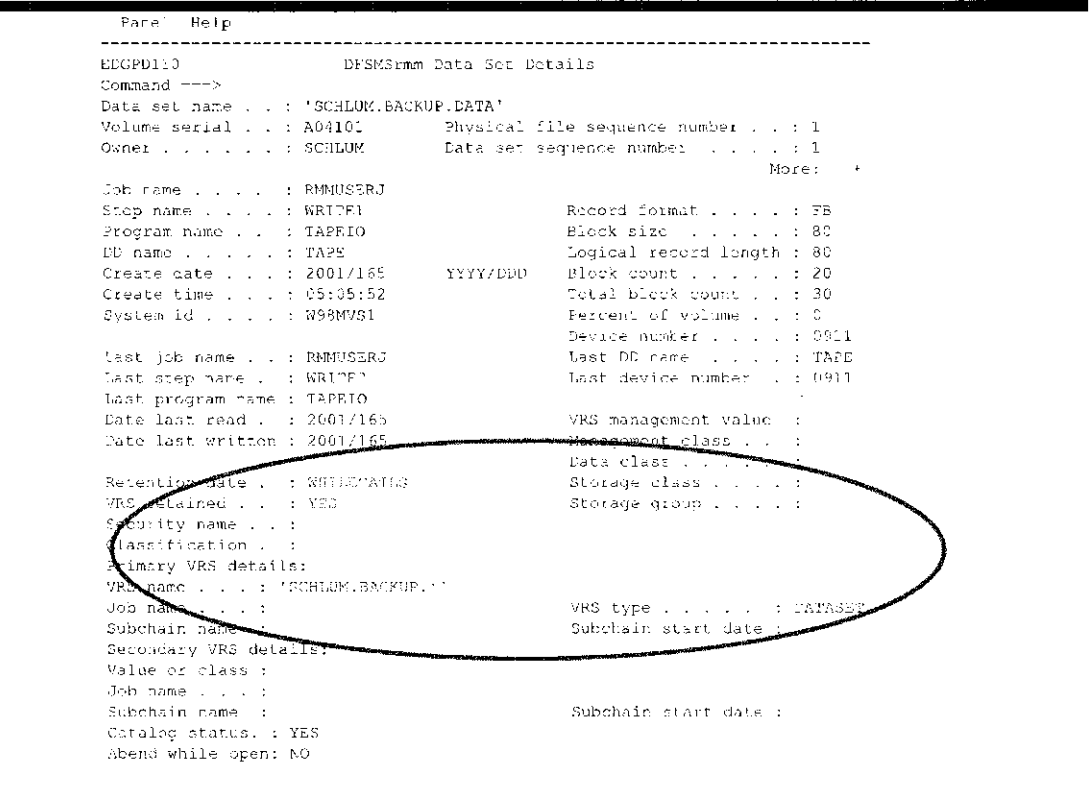


Figure 4-48. Data Set Record with Matching VRS SS212.0

Notes:

Display of the data set record will show more details about possible updates during VRSEL. The retention date indicates when retention is expected to stop. If an exact date cannot be calculated, a phrase such as WHILECATLG (to indicate that catalog control is active) is displayed.

The following details also appear:

- Whether the volume is VRS retained (yes or no)
- The VRS name
- The VRS type

Volume Record Retained by VRS

```
Panel: Help
-----
DFSMSrmm Volume Details - A04101
Command ===>
Volume ..... : A04101          Rack number ..... :
Media name ..... : 3480        Status ..... : MASTER
Volume type ..... : PHYSICAL    Stacked count ..... : 0
Retention date ..... : 001103100 Retention date ..... : 0001/100
Set retained ..... : 100        Original expiration date :
Description ..... :
Data set name ..... : 'SCHMM.BACKUP.DATA'
Media type ..... : CBT          Release actions:
Label ..... : SL               Return to SCRATCH pool : YES
Current version :               Replace volume ..... : NO
Required version :             Return to owner ..... : NO
Density ..... : 10CB          Initialize volume ..... : NO
Recording format : 16TRACK      Erase volume ..... : NO
Compression ..... : YES        Notify owner ..... : NO
Attributes ..... : NONE        Expiry date ignore ..... : NO
Availability ..... : OFFLINE    Scratch immediate ..... : NO
Owner ..... : SCHMM            Owner access ..... : AFTER
Assigned date ..... : 200/105   Assigned time ..... : 05:05:00
File 1 system ID :
Security name ..... :
Classification ..... :
Account number ..... : T.B.IOM,,,SYSPROG
Jobname ..... : RMJ00000       KVS use ..... : YES
Job location ..... :           VM use ..... : NO
Previous volume :             Last changed by ..... : NONE
Volume access list :          Next volume ..... :
User ..... :                 Access ..... : NONE
User ..... :                 User ..... :
User ..... :                 User ..... :
User ..... :                 User ..... :
User ..... :                 User ..... :
User ..... :                 User ..... :
User ..... :                 User ..... :
Volume use count ..... : 0      Volume used (KB) ..... : 0
Capacity (MB) ..... : 0        Percent full ..... : 0
Create date ..... : 1998/294    Create time ..... : 05:00:00
System ID ..... : M00
```

Figure 4-49. Volume Record Retained by VRS

SS212.0

Notes:

If the volume has been retained by VRS because of at least one VRS-retained data set, and if you look at the volume record details for such a volume, you will find the latest retention date of all VRS-retained data sets on the volume.

Selecting Matching VRS for Data Sets (New)

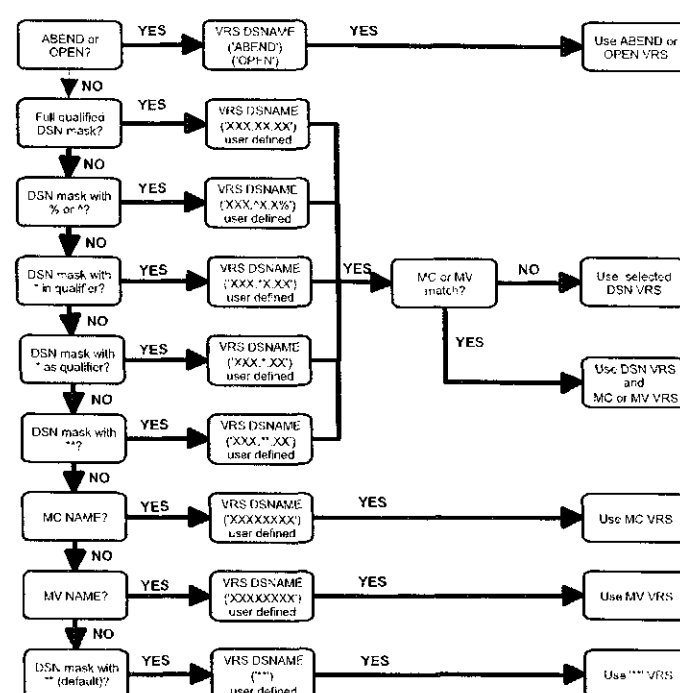


Figure 4-50. Selecting Matching VRS for Data Sets (NEW)

SS212.0

Notes:

You must remember the matching order for vital record specifications shown at the beginning of this unit. When a management class or a management value is assigned to a data set, the matching VRS for such an issue will be searched during VRSEL, too. Only in the case of a match are the data set record or the volume record, or both, updated to indicate that the data set or the volume, or both, are VRS protected. If the matching vital record specification applies a policy only to the data set, it is as if a match based on the real data set name applied. In some cases there may be volumes indicating a primary VRS due to a matching data set name, and a secondary VRS due to a matching management class or value.

The retention criteria for both vital record specifications are processed in parallel. Only if neither of these vital record specifications will retain the data set any longer is the volume eligible for expiration processing.

Processing Vital Records during VRSEL

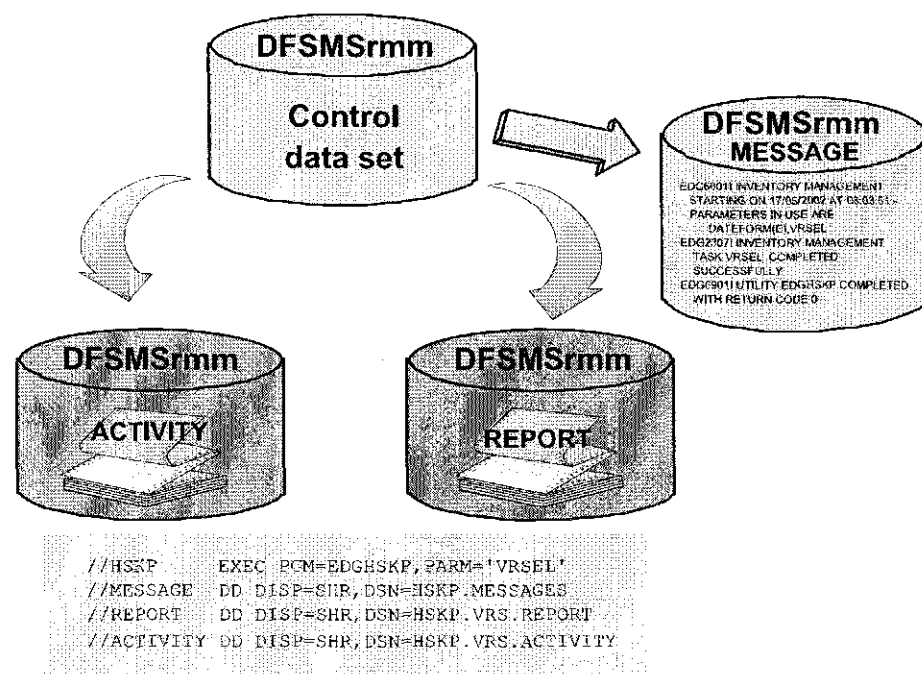


Figure 4-51. Processing Vital Records during VRSEL

SS212.0

Notes:

DFSMSrmm has a verification process for VRS changes. The VRSEL process will check to see whether any VRS changes have been made and not yet verified. Changes must be verified before VRSEL will actually process data sets and volumes for VRS retention and movement. This option can be disabled by coding **VRSCCHANGE(INFO)** into your option statement in EDGRMMxx.

The verify run produces an activity file. While mapping macros are provided for the records created, EDGJACTP in SYS1.SAMPLIB is designed to produce formatted reports by using ICETOOL reporting functions.

EDGRMMxx OPTIONS in Connection with VRSEL

VRCHANGE(...)	
INFO	No additional processing or action is required.
VERIFY	Any changes made to vital record specifications must be verified by running EDGHSKP.
VRSMIN(count, action)	
0	Minimum number of VRS records expected in
to	DFSMSrmm CDS, if there are less VRS records
2147483647	encountered, issues message EDG2229I with different return codes depending on action.
FAIL	RC=8
INFO	RC=0
WARN	RC=4
VRSEL(...)	
OLD	Ignore new functions when reclaimed in VRS.
NEW	Honor new functions in VRS released with DFSMS/MVS 1.4 DFSMSrmm.
CATRETPD(...)	
0 - 9999	Minimum retention time (hours) for data sets not cataloged but protected by a VRS indicating WHILECATALOG

Figure 4-52. EDGRMMxx Options in Connection with VRSEL

SS212.0

Notes:

Use VRSMIN to specify a minimum number of vital record specifications required by inventory management vital record processing, and the action to be taken by DFSMSrmm when the minimum number of vital record specifications are not defined. DFSMSrmm counts the vital record specifications used by vital record processing. DFSMSrmm does not count vital record specifications that are deleted during vital record processing. *Count* can be 0 to 2,147,483,647. Specify a count value of 0 to disable this checking. The default count value is 1. Specify *action* to control the action DFSMSrmm takes when the *count* is not reached. *Action* can be FAIL, INFO, or WARN. If the minimum number of VRS is not encountered during inventory management, DFSMSrmm issues message EDG2229I to the MESSAGE file.

EDGRMMxx OPTION CATRETPD

The following examples demonstrate, how CATRETPD can help you to prevent the inadvertent release of volume.

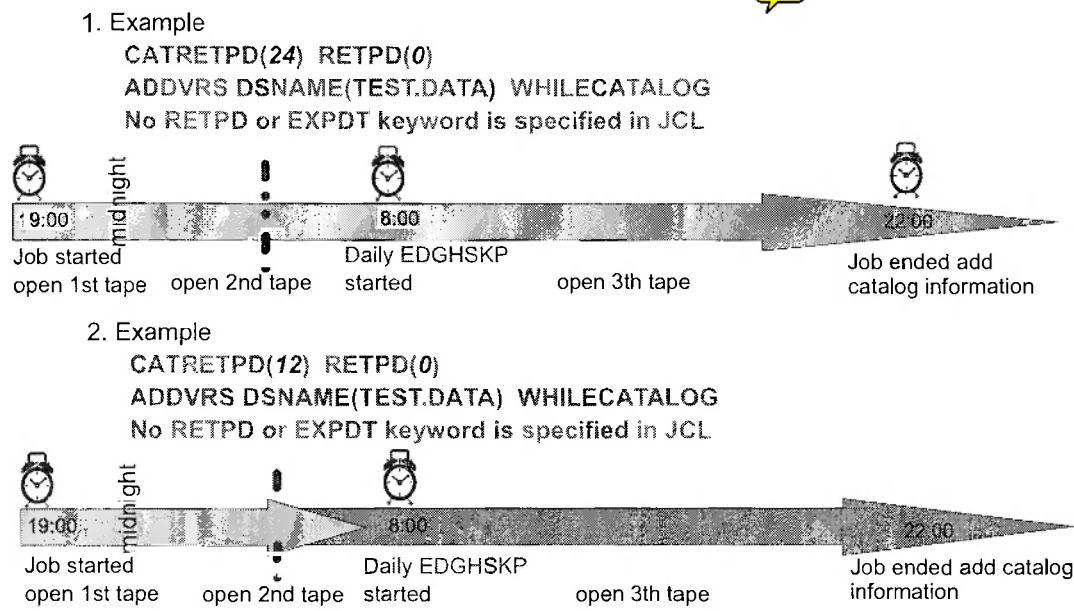


Figure 4-53. EDGRMMxx OPTION CATRETPD SS212.0

Notes:

Tape data sets are not cataloged until step termination time (even though they can be closed by the program). You can specify, in hours, how long a data set should be retained for after it has been created, if it has not been cataloged and if it matches a vital record specification that includes the WHILECATALOG operand.

The DFSMSrmm default setting is 12 hours.

EDGHSKP REPORT File

Contents

- Formatted report
- All VRS retained data sets and volumes

Browse it or print it

Entries are presented

- By matching VRS
- By data set group

Details include:

- Primary and secondary VRS and current subchain
- Number of copies, versions, or cycles

Figure 4-54. EDGHSKP REPORT File

SS212.0

Notes:

The report file contains a detailed report of DFSMSrmm vital record specification processing.

EDGHSKP REPORT File Example

REMOVABLE MEDIA MANAGER										VITAL RECORDS RETENTION REPORT		PAGE 0120		
(C) IBM CORPORATION 1993,1994												TIME 06:17:23 DATE 03/22/2001		
JOB NAME	DATA SET OR VOLUME NAME (CONTINUED)	OWNER	TYPE	RTN	C/X	DELETE	PLY	GROUP	STRTM	LOCATION	MODE			
	.GGG..ICANT.*	X002	DSN	000001	1	000001/1999	0	000001	000001	LOCAL				
JOB NAME	DATA SET NAME (CONTINUED)	200VBS	20	NAME	2500	2500	VOLUME	MSG	OWNER	CURRENT	REQUIRED	PLY	RETDATE	RENAME
X015121	DT01.GGG.DS000001.DS000001.DS000001		1	1	010100		1	0001	LOCAL	LOCAL		300	WHILECATG *	
NUMBER OF DATA SETS RETAINED (GROUP STORE) =												1		
X015122	DT01.GGG.DS000001.DS000001.DS000001		1	1	010100		1	0001	LOCAL	LOCAL		300	WHILECATG *	
NUMBER OF DATA SETS RETAINED (GROUP STORE) =												1		
X015123	DT01.GGG.DS000001.DS000001.DS000001		1	1	010100		1	0001	LOCAL	LOCAL		300	WHILECATG *	
NUMBER OF DATA SETS RETAINED (GROUP STORE) =												1		
JOB NAME	DATA SET OR VOLUME NAME (CONTINUED)	OWNER	TYPE	RTN	C/X	DELETE	PLY	GROUP	STRTM	LOCATION	MODE			
	.GGG..IMKCPY2.*	X002	DSN	000001	1	000001/1999	0	000001	000001	LOCAL				
JOB NAME	DATA SET NAME (CONTINUED)	200VBS	20	NAME	2500	2500	VOLUME	MSG	OWNER	CURRENT	REQUIRED	PLY	RETDATE	RENAME
Q000000	DS01.GGG.DS000001.IMKCPY2.DS000001		1	1	000000		1	000000	LIB1	LOCAL		300	WHILECATG *	
Q000000	DS01.GGG.DS000001.IMKCPY2.DS000001		1	1	000000		1	000000	LIB1	LOCAL		300	WHILECATG *	
Q000000	DS01.GGG.DS000001.IMKCPY2.DS000001		1	1	000000		1	000000	LIB1	LOCAL		300	WHILECATG *	
Q000000	DS01.GGG.DS000001.IMKCPY2.DS000001		1	1	000000		1	000000	LIB1	LOCAL		300	WHILECATG *	
NUMBER OF DATA SETS RETAINED (GROUP STORE) =												4		

Figure 4-55. EDGHSKP REPORT File Example

SS212.0

Notes:

Use the report to perform these tasks:

- Check the vital record specifications that match to data sets and volumes.
- Identify the versions of the data sets that are being retained.
- Check the required location for each data set and volume.

EDGHSKP ACTIVITY File

Contents

- Header record - Run time values
- Data set record - Details of changes

Browse it or report on it

- EDGJACTP sample reports on it
 - Summaries: Vital status, retention date, matching VRS, VRS subchain
 - Detailed reports by changes

Changes listed:
 Vital status
 Retention date
 Matching VRS
 VRS subchain
 Catalog status

Figure 4-56. EDGHSKP ACTIVITY File

SS212.0

Notes:

The ACTIVITY file is a pre-allocated direct access storage device (DASD) data set, like the REPORT file. The ACTIVITY file is not intended to be a report. The ACTIVITY file contains detailed information about changes made to data sets during vital record processing. The DFSMSrmm-supplied sample EDGJHKPA shows the JCL to allocate the ACTIVITY file, as well as other DFSMSrmm inventory management data sets. The DFSMSrmm-supplied sample EDGJACTP shows the JCL to report on the contents of the ACTIVITY file.

The ACTIVITY file is a variable-blocked file with the record length set to the largest record created by DFSMSrmm. The system determines the block size of the ACTIVITY file. See *z/OS DFSMSrmm Reporting* for a mapping of the ACTIVITY file.

DFSMSrmm writes an activity record for data set changes only when a change is identified in the ACTD_CHANGE section of the record. During vital record processing, if an ACTIVITY file is allocated, DFSMSrmm writes information about changes made to the matching vital record specification, the vital record status, and the retention date to the ACTIVITY file.

EDGHSKP ACTIVITY File Report

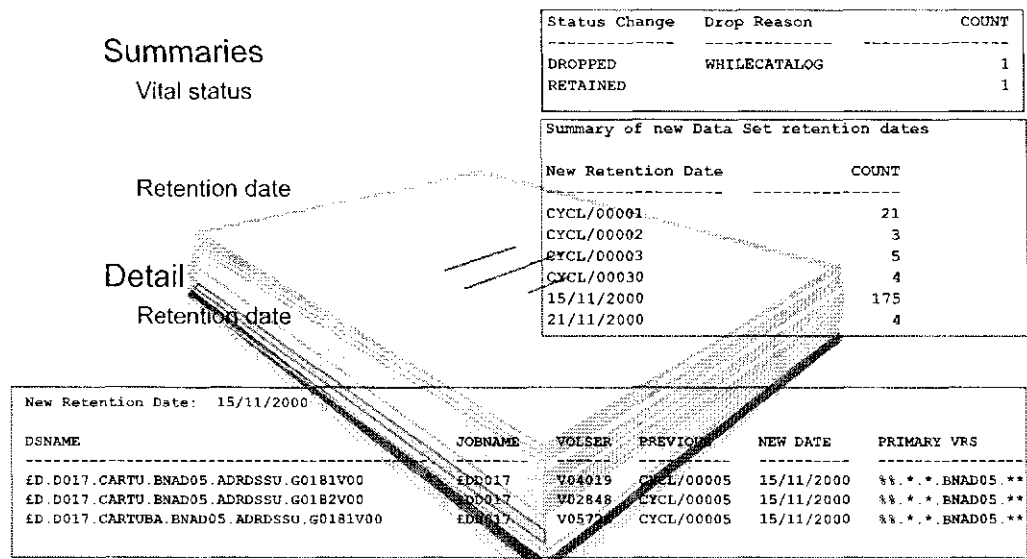


Figure 4-57. EDGHSKP ACTIVITY File Report

SS212.0

Notes:

You can view the ACTIVITY file online. To print the ACTIVITY file, use a product such as DFSORT or DFSORT's ICETOOL to selectively format and print fields. DFSMSrmm provides a sample job EDGJACTP in SAMPLIB that shows how to selectively format and print fields. The sample EDGJACTP produces reports in pairs: a report containing detailed information and a summary report that is broken down by category and a count within each category. The reports focus on the different types of changes that DFSMSrmm makes to data set records during inventory management. For example, DFSMSrmm can change the vital record specification or vital record specification subchain that retains the data sets. You can use these reports to help you understand the updates that DFSMSrmm is making to data sets that are based on matching vital record specifications.

Checkpoint

Indicate whether each of the following statements is true or false.

1. Vital record specification records are required in DFSMSrmm in order to retain data sets.
2. Data sets may be retained by job name, but a data set name is also required.
3. When matching data set records to VRS policies, DFSMSrmm searches for matches in what order?
 - a. By full qualified data set name
 - b. Reserved name of ABEND and OPEN
 - c. By management class name
 - d. By data set name mask containing a single percent or not sign (^)
 - e. By management value
 - f. By data set name mask containing a single asterisk
 - g. By the data set name mask, **
 - h. By a data set name mask that includes ** (but not ** by itself)
4. If multiple VRS policies apply to a multi-file volume for movement, DFSMSrmm determines the volume location based solely on the priority of the location as specified by the LOCDEF statements or by the defaults implemented in the code of DFSMSrmm if not specified in a LOCDEF.
5. A VRS with the GDG parameter will match only those data sets whose last qualifier meets the standard MVS generation data set name format.
6. A VRS is required in order to have DFSMSrmm schedule movement of a volume.

Figure 4-58. Checkpoint

SS212.0

Notes:

Write down your answers here:

- 1.
- 2.
- 3.

- 4.
- 5.
- 6.

Unit Summary

Now that you have completed this unit, you should be able to:

- Define VRS policies for retention and movement
- Identify VRS retention options for retaining data sets
- Use DFSMSrmm VRS masking facility for retaining data sets
- Identify updates of the control data set during VRSEL

Notes:

Unit 5. Advanced DFSMSrmm VRS Processing

What This Unit Is About

This unit describes several enhancements to DFSMSrmm VRS processing.

What You Should Be Able to Do

After completing this unit, you should be able to:

- Use chains of vital record specifications
- Use VRS definitions that include special volume release options
- Use VRS definitions for volume movement
- Understand what happens during the DSTORE processing of the DFSMSrmm inventory management
- Enable the use of management class for all tape data sets by calling SMS automatic class selection routines from DFSMSrmm

How You Will Check Your Progress

Accountability: Checkpoint questions

References

- | | |
|-----------|--|
| SC26-7404 | <i>DFSMSrmm Guide and Reference</i> |
| SC26-7405 | <i>DFSMSrmm Implementation and Customization Guide</i> |
| SC26-7405 | <i>DFSMSrmm Reporting</i> |

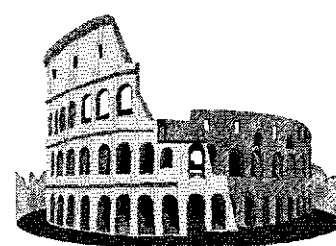
Objectives

After completing this unit, you should be able to:

- Choose advanced DFSMSrmm VRS processing
- Identify the set of EDGRMMxx options to use advanced VRS processing
- Use VRS definitions for volume movement
- Understand DSTORE processing during inventory management
- Enable use of management class for all tape data sets by calling SMS automatic class selection routines from DFSMSrmm

Notes:

VRS Enhancements



Use option VRSEL(OLD/NEW) to control how DFSMSrmm inventory management vital record selection processing (VRSEL) uses policy information in vital record specifications.

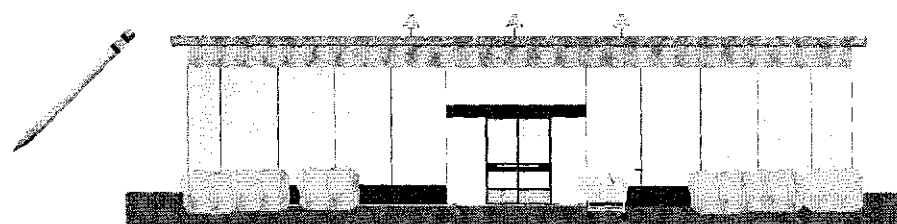


Figure 5-2. VRS Enhancements

SS212.0

Notes:

VRS processing of DFSMSrmm provides some nice features that are available only when the VRSEL(NEW) option is selected.

Selecting VRSEL(NEW) Processing

Specify **OPTION VRSEL(NEW)** in parmlib member **EDGRMMxx** to process

- Retention information in NAME type vital record specifications
- Release options defined in vital record specifications (*EXPIRYDATEIGNORE* and *SCRATCHIMMEDIATE*)
- Vital record specifications chained using the RMM **ADDVRS ANDVRS** operand

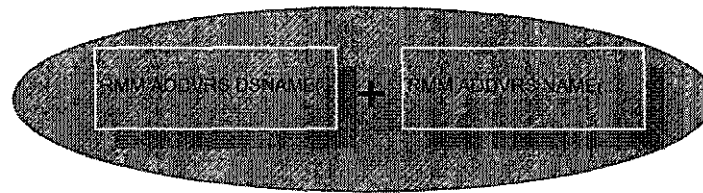


Figure 5-3. Selecting VRSEL(NEW) Processing

SS212.0

Notes:

Specify the **VRSEL(NEW)** option in **EDGRMMxx** if you want **DFSMSrmm** to process retention information in NAME vital record specifications, release options defined in vital record specifications, and vital record specifications chained using the RMM **ADDVRS ANDVRS** operand.

There may even be different retention rules in each part of a VRS chain.

Selecting VRSEL(NEW) Processing (Cont.)

Specify OPTION VRSEL(NEW) in parmlib member EDGRMMxx to process

- The location CURRENT in a vital record specification that is used to avoid movement of a volume
- The specification EXTRADAYS in a name type vital record specification
- The specification BYDAYSCYCLE in a vital record specification

Figure 5-4. Selecting VRSEL(NEW) Processing

SS212.0

Notes:

Specify the location CURRENT to keep a volume where it resides, thus avoiding volume movement in a certain part of a VRS chain.

Use EXTRADAYS to specify the number of days since a name vital record specification started to retain the data set. The number of extra days is specified with the COUNT operand. EXTRADAYS is the number of days since the NAME vital record specification started to retain the data set. The number of days depends on when the previous vital record specification stopped retaining the data set and the time when vital record processing is run. EXTRADAYS can only be used in a NAME vital record specification.

BYDAYSCYCLE specifies retaining all instances of a data set created on a single day as a single cycle. BYDAYSCYCLE is mutually exclusive with CYCLES, DAYS, LASTREFERENCEDAYS, and EXTRADAYS. BYDAYSCYCLE can be used on data set vital record specifications and name vital record specifications.

Example: Retention Policy

A data set should be retained if a catalog entry exists but no longer than 10 days since its creation.

```
DSN('SCHLUM.BACKUP.**')  
WHILECATALOG  
DAYS  
COUNT(10)
```

Figure 5-5. Example: Retention Policy

SS212.0

Notes:

If you combine more than one piece of retention information in a single VRS, all conditions must be true in order for the data set to be retained by that VRS. In our example the data set will be retained as long as a catalog entry exists, but for no longer than 10 days after its creation. If the catalog entry is dropped on the second day, the data set will no longer be retained.

You can consider the two conditions to be connected by a boolean AND: Only if both conditions are true does the VRS retain the data set.

Another question also arises: How could you combine conditions, for example, to keep a data set for 10 days after its creation but only if it has been referenced within the last two days? ADDVRS does not accept both the DAYS and LASTREFERENCEDAYS keywords, because both need to be completed with a count specification. In conclusion: VRSEL(OLD) allows little control when it comes to combining two or more retention criteria.

Example: More Complex Retention Policy

A data set should be retained for *at least* 10 days since its creation and as long as a catalog entry exists.

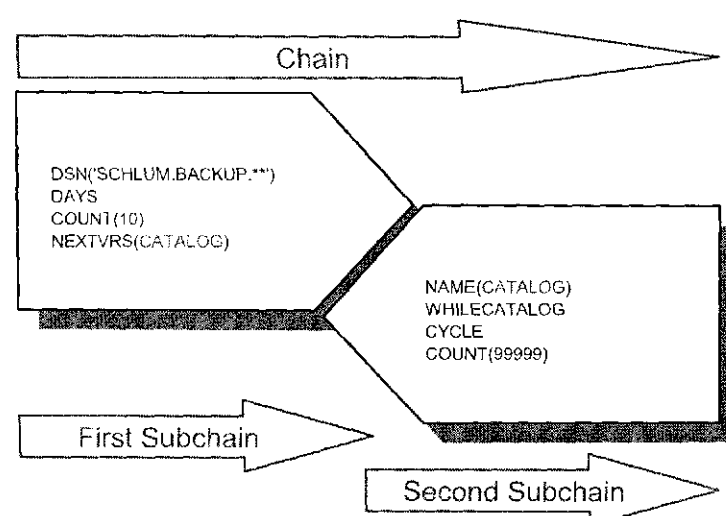


Figure 5-6. Example: More Complex Retention Policy

SS212.0

Notes:

By using VRSEL(NEW), you can link two or more retention criteria, which will be applied in sequence to the data set.

In our example the data set is retained by the first subchain for the first 10 days after its creation. On the 11th day, control moves to the next VRS in the chain, which keeps the data set as long as a catalog entry exists and regardless of how many days have passed since its creation.

All VRSs linked together with NEXTVRS are called a chain. Each time a VRS linked with NEXTVRS contains different retention criteria, we speak of the beginning of a new subchain.

Our example seems to prove that using NEXTVRS to link two or more VRSs together works like the boolean OR. If we exchange the two conditions, the resulting retention policy does not change: The data set is kept while being cataloged, regardless of how many days have passed since its creation. After the catalog entry has been dropped, the data set would continue to be retained if it is not at least ten days old. But in general, this is not true.

Example: Add More Complexity

A data set should be retained for *at least 10 days* and as long as a catalog entry exists and if being referenced within the last two days (regardless of the catalog status and of the number of days since creation).

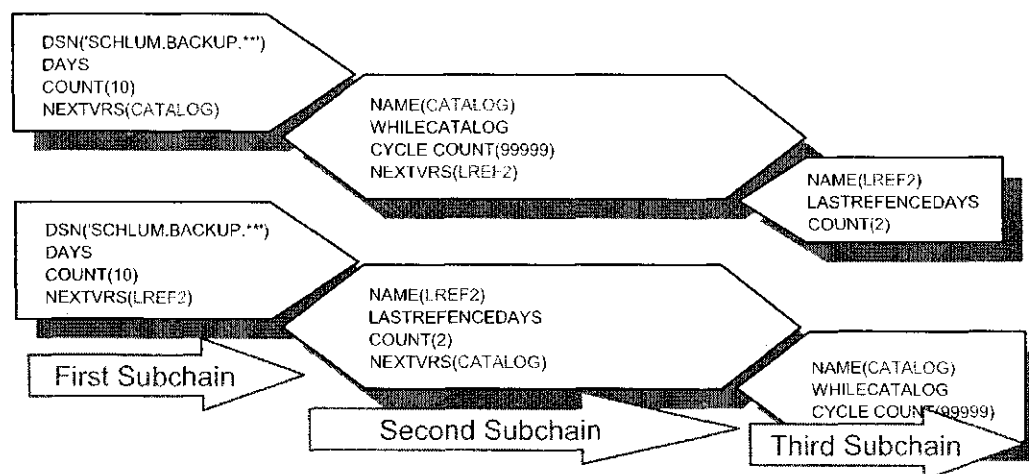


Figure 5-7. Example: Add More Complexity

SS212.0

Notes:

To add more complexity, we add the condition to retain the data set if being referenced during the last two days. The first chain does exactly what we want: The data set is kept for at least 10 days after its creation. On the 11th day, control moves to the second subchain, which retains the data set as long as a catalog entry exists. If the catalog entry is dropped, control moves to the third subchain, which retains the data set if has been referenced within the last two days.

For a variation on the theme, let's look at the second chain. Does it represent the same retention policy? No: If the data set is not referenced after its creation, control moves on day 11 to the third subchain in example 2. If the data set is referenced four weeks after creation while it is still cataloged, and if the catalog entry is dropped a few minutes later, the data set will be no longer retained by the chain in example 2. In example 1, control moves to the third subchain four weeks after creation, and the data set will be retained for at least two more days, or even longer if it is referenced again within the next two days.

It follows from these examples that you must be careful when discussing service-level agreements!

Data Set Record with Matching VRS

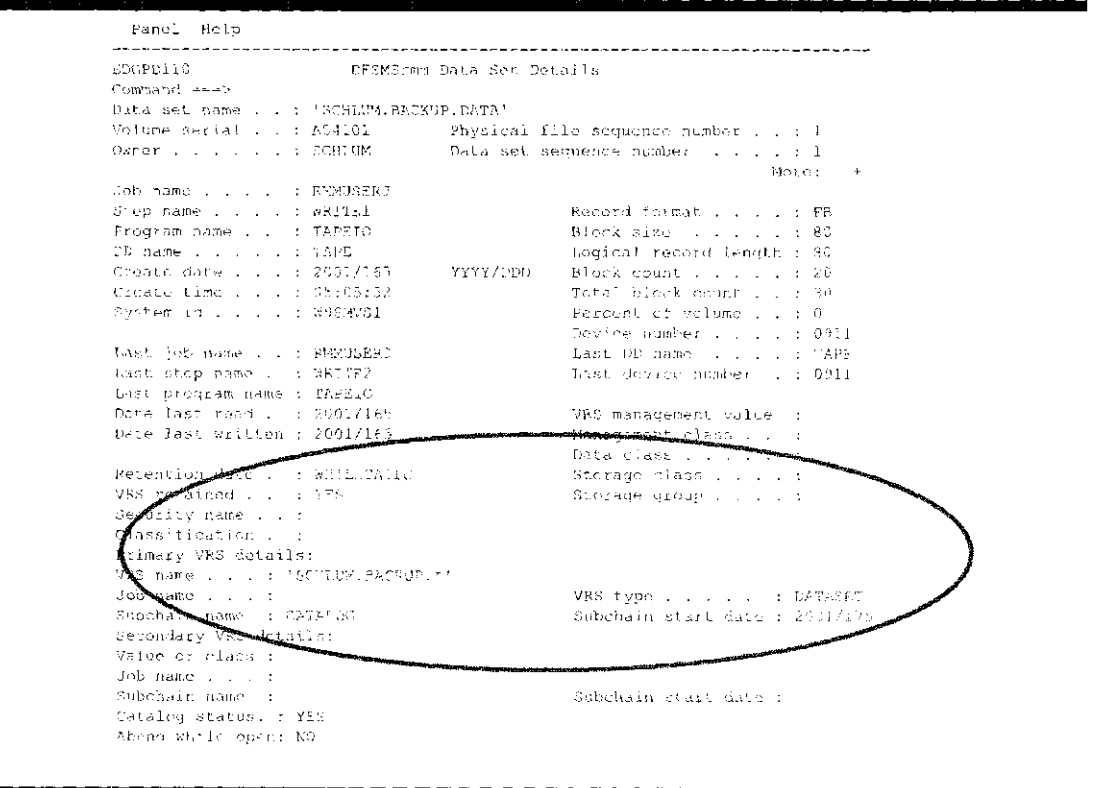


Figure 5-8. Data Set Record with Matching VRS SS212.0

Notes:

If you look at the data set details by using the ISPF application of DFSMSrmm, you might see a situation such as that shown above.

The data set is retained by a VRS chain starting with VRS 'SCHLUM.BACKUP.*', which is of DSNAME type. Currently the subchain that starts with VRS CATALOG retains the data set leading to the retention date, WHILECATLG. Control moved to the CATALOG subchain on the date reported in the **subchain start date**.

Example: Add More Complexity (Cont.)

A data set should be retained for *at least 10 days* and if also being referenced during the last two days (regardless of the status of the catalog entry).

If at least one of these conditions doesn't apply any longer, the data set should be retained for as long as a catalog entry exists.

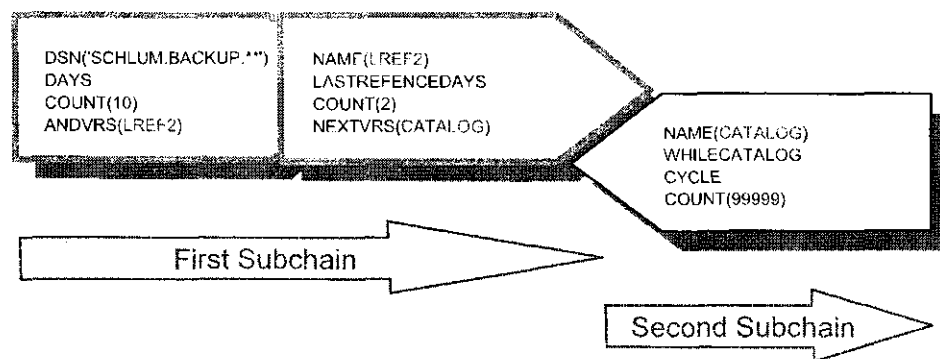


Figure 5-9. Example: Add More Complexity (Cont.)

SS212.0

Notes:

Using ANDVRS in the first VRS of the chain links the second VRS in a different manner: Both retention criteria in the first two VRSs must be met in order for the data set to be retained. This is why the first subchain now consists of two vital record specifications.

All VRSs linked by ANDVRS are part of the same subchain.

If one of these two conditions represented by subchain 1 is no longer met, control moves to the second subchain, which retains the data set as long as it is being cataloged.

The sequence of vital record specifications linked by ANDVRS does not matter. This subchain represents a true boolean AND.

Like with NEXTVRS you can link only to a name type VRS. A chain can contain more than one VRS including the keyword ANDVRS.

An Example of a Complex Chain

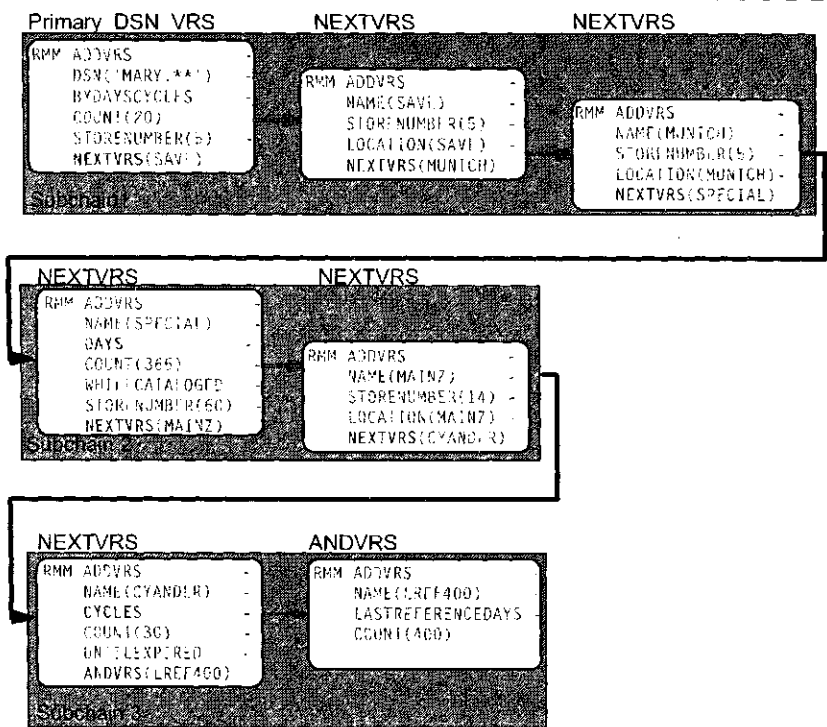


Figure 5-10. An Example of a Complex Chain

SS212.0

Notes:

While you might compose chains that are even more complex, we do not recommend this unless it is absolutely necessary.

The first subchain consists of three VRSs because they represent a single retention rule (BYDAYSCYCLE). The two VRSs named SAVE and MUNICH do nothing more than control volume movement. Note that the first and last five cycles are kept at the HOME location.

For cycles exceeding the count of 20, control moves to the next subchain, which retains data sets if it is no more than 1 year old and has been cataloged. Some of these data sets (and the volumes they are residing in) will be moved to the location, MAINZ.

When the catalog entry is dropped, or when the data sets are more than 1 year old, control moves to subchain 3, which retains another 30 cycles if the volume expiration date has not been reached and if the data sets have been referenced within the last 400 days.

Who can explain if a volume would be moved or retained unexpectedly?

Example: Grace Period for DFSMSHsm Tapes

In this example, a chain of VRSs is used to provide a grace period before the release of DFSMSHsm tapes. This VRS chain should be combined with OPTION TVEXTPURGE(EXPIRE).

```
RMM ADDVRS DSN('mprefix.**') -  
COUNT(99999) UNTILEXPIRED NEXTVRS(EXTRA7)  
RMM ADDVRS NAME(EXTRA7) EXTRADAYS COUNT(7)
```

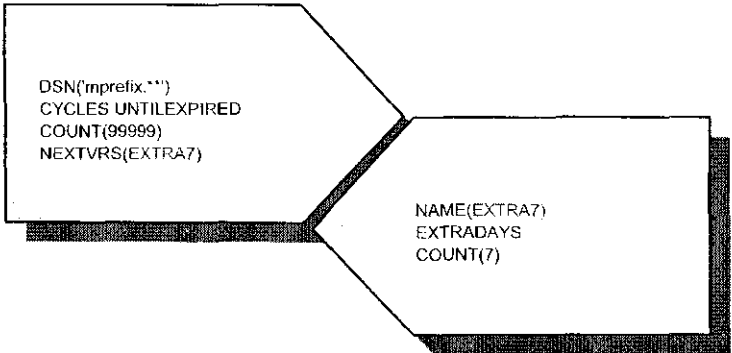


Figure 5-11. Example: Grace Period for DFSMSHsm Tapes SS212.0

Notes:

While most customers use the TVEXTPURGE(RELEASE) option to release HSM tapes instantly when requested by DFSMSHsm, you might specify a grace period to keep them for a few extra days before the return to scratch.

VRS Enhancements

These ADDVRS operands are honored during inventory management when you specify the DFSMSrmm parmlib option VRSEL(NEW) operand.

LOCATION(CURRENT)

Use CURRENT to avoid moving of a volume.

RELEASE(EXPIRYDATEIGNORE)

The associated volume expiration date should be ignored.

RELEASE(SCRATCHIMMEDIATE)

The volume release actions should be treated in a special way (cannot be combined with the release actions REPLACE and ERASE).

Figure 5-12. VRS Enhancements

SS212.0

Notes:

There might be some rare situations when you want to keep a volume at its current location while the VRS controls a data set.

The options used when a volume is released are very interesting. A volume must be VRS selected to receive a release option of EXPIRYDATEIGNORE or SCRATCHIMMEDIATE. When the volume is no longer VRS retained, these options are used by inventory management EXPROC while the volume is being processed for release. EXPIRYDATEIGNORE will allow the release of volumes even if the volume expiration date has not been reached. SCRATCHIMMEDIATE will release the volume without a second inventory management EXPROC run as long as the only pending release action is return to scratch.

Note that these indicators will not be reset before the return to scratch even if the matching VRS changes.

Cycle of a Volume

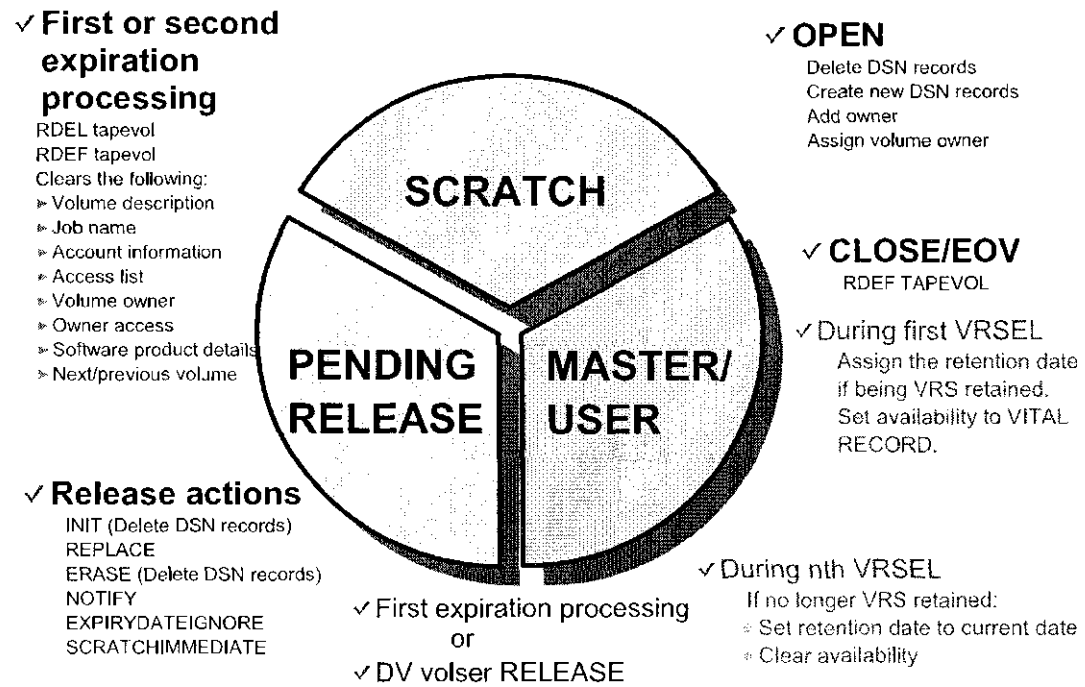


Figure 5-13. Cycle of a Volume

SS212.0

Notes:

A volume record is updated

- During open, close, end-of-volume processing
- When the volume is VRS retained
- When VRSEL detects that the volume is no longer VRS retained
- During EXPROC processing
- By DFSMSrmm subcommand processing

At the moment we are most interested in updates during VRSEL.

Processing of the Release Action SCRATCHIMMEDIATE

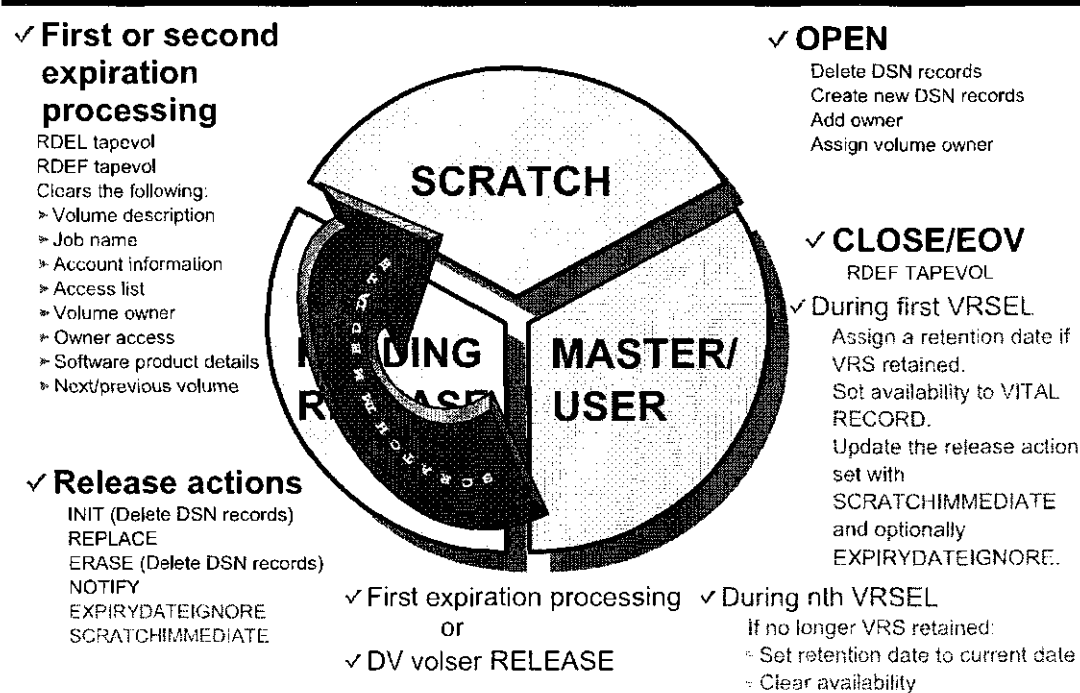


Figure 5-14. Processing of the Release Action SCRATCHIMMEDIATE

SS212.0

Notes:

During first VRSEL, if the matching VRS includes RELEASE(SCRATCHIMMEDIATE) or RELEASE(EXPIRYDATEIGNORE), or both, the release action set of a master or a user volume is updated with SCRATCHIMMEDIATE or EXPIRYDATEIGNORE, or both. These indications will remain unchanged until the volume releases to scratch.

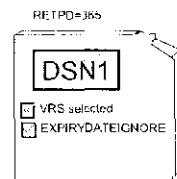
After some time, VRSEL processing will detect that the volume is no longer VRS protected. VRSEL will reset the retention date to the current date and clear availability. Normally, during the next EXPROC the availability of a volume is updated to indicate PENDING RELEASE if the volume expiration date has been reached. But If the release action set for the volume indicates SCRATCHIMMEDIATE and if the expiration date of the volume has been reached, the volume will be released immediately unless another pending release action is requested (for example, ERASE).

If the release action set of the volume indicates EXPIRYDATEIGNORE, release processing for the volume will take place regardless of its expiration date.

You can also combine both release options.

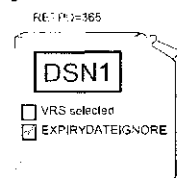
Example: VRS Including EXPIRYDATEIGNORE

**ADDVRS DSN('OPEN') DAYS COUNT(2) -
RELEASE(EXPIRYDATEIGNORE)**



The inventory-management VRSEL assigns VRS 'OPEN' because DSN1 is still being written.

2 days later:



The inventory-management VRSEL clears availability and EXPROC sets availability to PENDING RELEASE.

The volume will be released during the next inventory management EXPROC in spite of the volume expiration date not being reached.

Figure 5-15. Example: VRS including EXPIRYDATEIGNORE

SS212.0

Notes:

We do not recommend the use of RELEASE(EXPIRYDATEIGNORE) in a VRS that is dedicated to open data sets, because that might lead to the unexpected release processing of volumes containing data sets being accidentally open during VRSEL. Remember that the EXPIRYDATEIGNORE characteristic in the release action set of the volume will not be reset before the release to scratch.

Retaining Volumes

How does DFSMSrmm retain volumes if they are retained by a vital record specification?

Expiration date reached	UNTILEXPIRED specified in the matching VRS	Volume availability after VRSEL	Release action set of volume indicates EXPIRYDATEIGNORE SCRATCHIMMEDIATE	Volume status after VRSEL+EXPROC
<u>NO</u>	NO	-	NONE	MASTER
<u>NO</u>	YES	-	NONE	MASTER
<u>NO</u>	YES	VITAL RECORD	NONE	MASTER
<u>NO</u>	YES	VITAL RECORD	EXPIRYDATEIGNORE	MASTER
<u>NO</u>	YES	-	BOTH	SCRATCH
<u>YES</u>	NO	VITAL RECORD	NONE	MASTER
<u>YES</u>	NO	VITAL RECORD	SCRATCHIMMEDIATE	MASTER
<u>YES</u>	NO	-	SCRATCHIMMEDIATE	SCRATCH
<u>YES</u>	<u>YES</u>	-	EXPIRYDATEIGNORE	MASTER
<u>YES</u>	<u>YES</u>	-	BOTH	SCRATCH
<u>YES</u>	NO	-	BOTH	SCRATCH
<u>YES</u>	NO	-	SCRATCHIMMEDIATE	SCRATCH

Figure 5-16. Retaining Volumes SS212.0

Notes:

The table summarizes some situations and shows the resulting volume status. All volumes have been VRS selected, and some are still protected by a VRS. In each row the reason for the resulting volume status is marked by an underscore. In some situations the volume was VRS selected before the start of inventory management, but the row titled Volume availability after VRSEL does not read VITAL RECORD because VRSEL clears availability.

One possible reason for the cessation of VRS selection in rows 9 and 10 is that the volume expiration date was reached and the matching VRS included UNTILEXPIRED.

Row 5 shows a case similar to the one on the last visual.

Combining Keywords in a Vital Record Specification

	DSNAME	VOLSER	NAME	CYCLES	DAYS	BYDAYSCYCLE	EXTRADAYS	LASTREF.	WHILECATAL.	UNTILEXPIRED	OPEN	ABEND	JOBNAME	RELEASE	DELETEDATE	NEXTVRS	ANDVRS
DSNAME		X	X				X				X	X					
VOLSER	X		X				X				X	X	X	X			
NAME	X	X					X				X	X	X	X			
CYCLES						X	X	X									
DAYS				X		X	X	X									
BYDAYSCYCLE				X	X		X	X									
EXTRADAYS	X	X		X	X	X		X	X								X
LASTREFERENCEDAYS				X	X	X	X										
WHILECATALOGED							X										
UNTILEXPIRED							X										
OPEN	X	X	X														
ABEND	X	X	X														
JOBNAME		X	X														
RELEASE		X	X														
DELETEDATE																	
NEXTVRS																	X
ANDVRS							X									X	

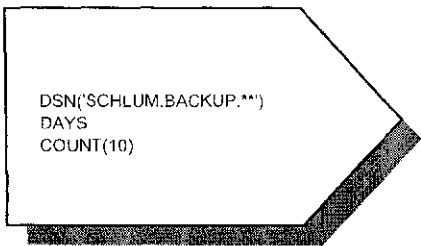
Figure 5-17. Combining Keywords in a Vital Record Specification SS212.0

Notes:

The table indicates possible combinations in a single ADDVRS subcommand.

Using a Primary and a Secondary VRS

A data set should be retained for *at least* 10 days and as long as a catalog entry exists.



```
PROC 1 $MGMTCLAS
/*****
/* DEFINE FILTER FOR KEEP WHILE CATALOGED (99000) */
/*****
FILTLIST SPECIAL_DATE9 INCLUDE('1999000')
/*****
/* DEFINE FILTER FOR UP TO 9 CYCLES */
/*****
FILTLIST SPECIAL_DATEC9 INCLUDE('199900*')
/*****
/* DEFINE FILTER FOR TAPE UNITS */
/*****
FILTLIST TAPEUNIT INCLUDE(TAPE*, T3420* T3490*,
                          '3420','3480','3490',
                          T3590*, '3590')
/*****
/* SELECT VALID MANAGEMENT CLASS */
/*****
SELECT
  WHEN ($EXPDT = $SPECIAL_DATE9 &&
        $UNIT = $TAPEUNIT) SET $MGMTCLAS = 'M99000'
  WHEN ($EXPDT = $SPECIAL_DATEC9 &&
        $UNIT = $TAPEUNIT) SET $MGMTCLAS = 'M90009'
  OTHERWISE SET $MGMTCLAS = 'NORMAL'
END
END
```

Figure 5-18. Using a Primary and a Secondary VRS

SS212.0

Notes:

DFSMSrmm provides another solution that allows you to set up the retention policy we covered earlier:

You can assign a VRS of the DSNAME type as a primary VRS during VRSEL. If a management class is also assigned (by means either of the SMS automatic class selection routine or of the CHANGEDATASET subcommand), the matching VRS of the DSNAME type will be used during VRSEL as a secondary VRS.

Selecting Matching VRS for Data Sets

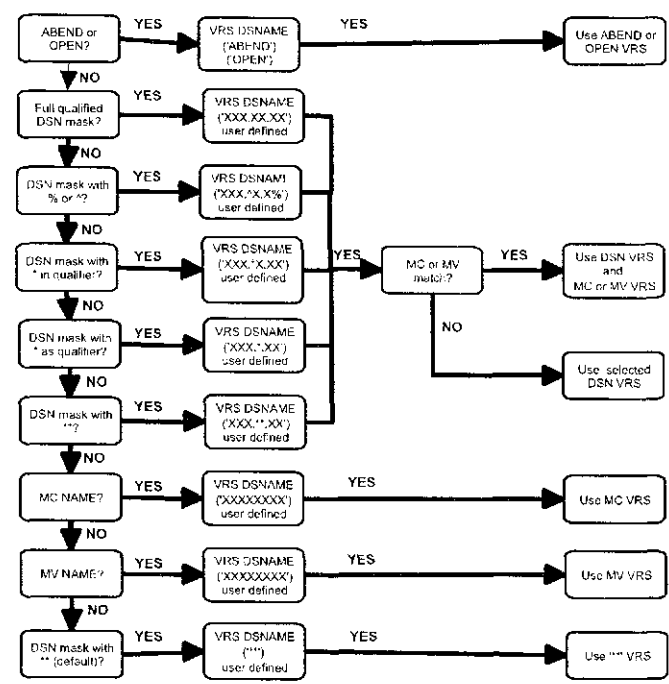


Figure 5-19. Selecting Matching VRS for Data Sets

SS212.0

Notes:

If a primary VRS and either a management class or a management value match a data set, both VRSs are processed in parallel to retain the data set if DFSMSrmm was initialized with the VRSEL(NEW) option.

Data Set Details Including a Secondary VRS

```
Panel: 0000
-----
DFSMSrmm Data Set Details
Command ==>
Data set name . . . : 'SCHLUM.BACKUP.DAT'
Volume serial . . . : A00101      Physical file sequence number . . . : 1
Owner . . . . . : SCHLUM        Data set sequence number . . . : 1
                                   More: 1

Job name . . . . . : RMVUSSTRT
Step name . . . . . : WR1101
Program name . . . : TAPE10
PD name . . . . . : TAPE
Create date . . . . : 2001/105      YYYY/CDD
Create time . . . . : 00:00:00
System ID . . . . . : W08MVS01

Record format . . . . : FB
Block size . . . . . : 80
Logical record length : 80
Block count . . . . . : 20
Total block count . . : 30
Percent of volume . . : 0
Device number . . . . : 0911
Last DB name . . . . : T000
Last device number . . : 0911

Test job name . . . : RMVUSSTRT
Last step name . . . : WR1101
Test program name : TAPE10
Date last read . . . : 2001/145
Date last written : 2001/145

VRS management value :
Management class . . : M99000
Data class . . . . . :
Storage class . . . . :
Storage group . . . . :

Retained data . . . : SE LATCHED
VRS retained . . . : YES
Security name . . . :
Classification . . . :
Secondary VRS details:
VRS name . . . . . : 'SCHLUM.BACKUP.*'
Job name . . . . . :
Subchain name . . . :
Secondary VRS details:
Value of class : 'M99000'
Job name . . . . . :
Subchain name . . . :
Catalog status . . : YES
Record while open : NO

VRS type . . . . . : DSNAME
Subchain start date :
Subchain start date :
```

Figure 5-20. Data Set Details Including a Secondary VRS SS212.0

Notes:

If you look at the data set details by using the DFSMSrmm ISPF application, you might see a situation like that shown above.

The data set is retained by a VRS 'SCHLUM.BACKUP.*' which is of type DSNAME. This VRS is reported under heading **Primary VRS details**. The data set is also retained by a secondary VRS, M99000, which is of the DSNAME type. This VRS is reported under heading **Secondary VRS details**. The name of the management class was assigned by the DFSMS automatic class selection routine during allocation or open processing. During VRSEL processing, DFSMSrmm verifies the matching VRS, M99000, and enables its use by updating the secondary VRS filed **value or class**.

You can also use VRS chains as primary and secondary VRSs. The current active subchains are reported in fields **Subchain name** and **Subchain start date**.

For secondary VRSs only, a VRS of the DSNAME type can be used. If while defining VRS M99000 you inadvertently chose a type of NAME, the management class will not match this VRS.

Excerpts from a VRSEL Report

DFSMSRMME: VRSEL REPORT										1
IBM CORP. CORPORATION 1997, 1998										
VRSEL: DATA SET OR VOLUME NAME										

1	2	3	4	5	6	7	8	9	10	11
DATA NAME	DATA SET NAME	2NDVRS	2NDNAME	ESCL	ORCL	VSCL	VSCL	VSCL	VSCL	VSCL

1	2	3	4	5	6	7	8	9	10	11
1	2	3	4	5	6	7	8	9	10	11
1	2	3	4	5	6	7	8	9	10	11
NUMBER OF DATA SETS RETAINED (CURRENT SOURCE)										

Figure 5-21. Excerpts from a VRSEL Report

SS212.0

Notes:

Vital record processing provides two columns in a VRSEL report to show a matching secondary VRS. The headings of these columns are **2ndVRS** and **2ndNAME**.

The **2ndVRS** field lists the name of the management value vital record specification or management class vital record specification when a data set matches to both a primary and a secondary vital record specification. If the management class vital record specification or the management value vital record specification is currently retaining the data set, DFSMSRmm puts the name of the vital record specification that is retaining the data set in the **2ndName** field.

The **2ndName** field is always filled when a data set is retained by a secondary vital record specification. When vital record specifications are chained using ANDVRS, DFSMSRmm puts the name of the first vital record specification in this field. This field is blank if the data set is retained by a primary vital record specification.

Using Multifile and Multivolume Data Sets

Volume 1



Volume 2



Volume 3



Figure 5-22. Using Multifile and Multivolume Data Sets

SS212.0

Notes:

For a volume set such as the one shown above, DFSMSrmm allows some customization for retention and movement control.

Note that stacking data sets that have a large range of retention periods such as the data sets in our example is not recommended. We have chosen this example only for demonstration purposes.

Unless it has been disabled by the EDGUX100 exit, DFSMSrmm records information for all data sets on a volume. It keeps track of each data set in a multifile/multivolume set. Each volume is managed individually by the data it contains.

Reporting of Multifile and Multivolume Data Sets

	VOLUME 1	VOLUME 2	VOLUME 3
V e n d o r	DSN 1 DSN 2 DSN 3 DSN 4 DSN 5	DSN 1	DSN 1
D F S M S r m m	DSN 1 DSN 2	DSN 2 DSN 3 DSN 4	DSN 4 DSN 5

Figure 5-23. Reporting of Multifile and Multivolume Data Sets SS212.0

Notes:

DFSMSrmm reports each volume and its contents rather than showing all data sets as being in the first volume of a multifile/multivolume set.

Using Multifile and Multivolume Data Sets

Result after six days

Volume 1 RETPD = 5480 days

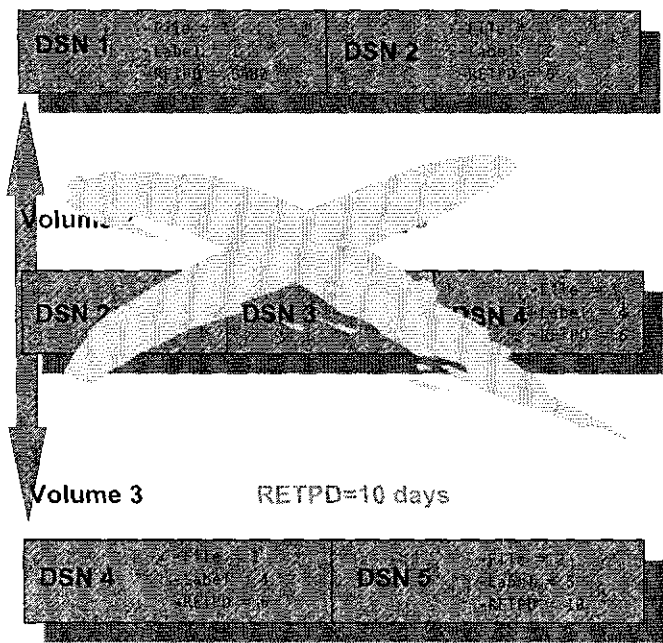


Figure 5-24. Using Multifile and Multivolume Data Sets SS212.0

Notes:

Since DFSMSrmm manages each volume by its own contents, it sets the volume expiration dates based on the data sets for that volume. In our example volume 1 is set to expire at 5480 days (FILE1), while volume 2 is at 6 (FILE4) and volume 3 at 10 (FILE5). After one week, DFSMSrmm will expire volume 2 and retain volumes 1and 3.

Reporting of Multifile and Multivolume Data Sets

Result after six days

	VOLUME 1	VOLUME 2	VOLUME 3
V e n d o r	DSN 1 DSN 2 DSN 3 DSN 4 DSN 5	DSN 1	DSN 1
D F S M S r m m	DSN 1 DSN 2	DSN 4 DSN 5	

Figure 5-25. Reporting of Multifile and Multivolume Data Sets SS212.0

Notes:

Expiration processing carried out 6 days after these tape files were created, and DFSMSrmm then has volumes 1 and 3 in inventory, while volume 2 is awaiting release. If we managed the volume set based on the first file, all files would be retained for 5480 days.

Using Multifile and Multivolume Data Sets

Result after ten days

Volume 1 RETPD = 5480 days



Volume 2



Volume 3 RETPD = 10 days



Figure 5-26. Using Multifile and Multivolume Data Sets SS212.0

Notes:

After another four days, DFSMSrmm would cause volume 3 to expire.

Reporting of Multifile and Multivolume Data Sets

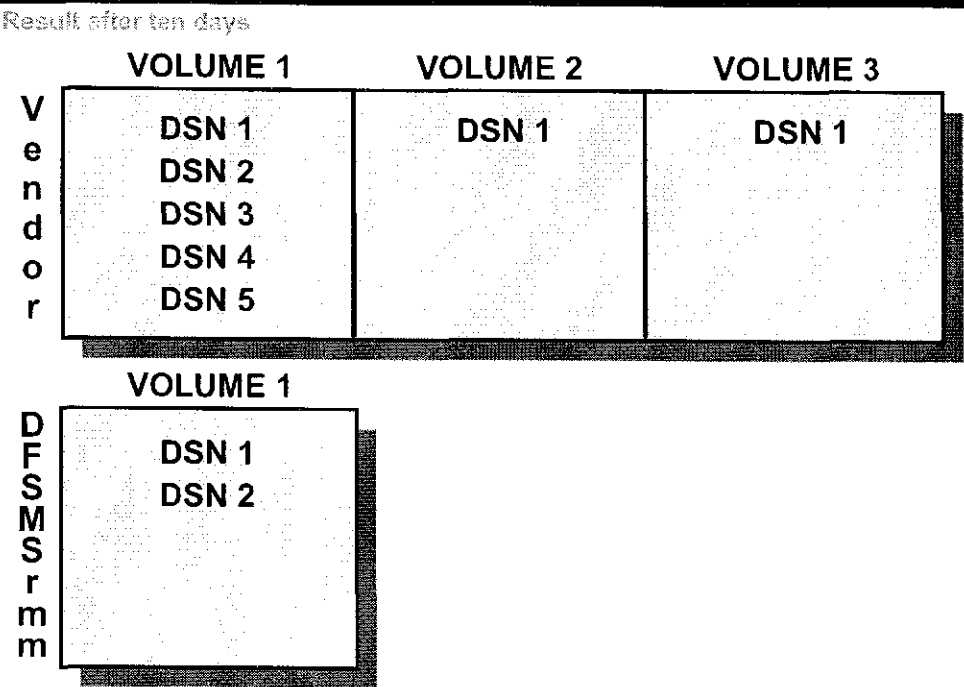


Figure 5-27. Reporting of Multifile and Multivolume Data Sets SS212.0

Notes:

After 10 days, DFSMSrmm would show only volume 1 as unexpired. It would also indicate that there is a second physical file in the volume.

RETAINBY(SET) - EXPDT

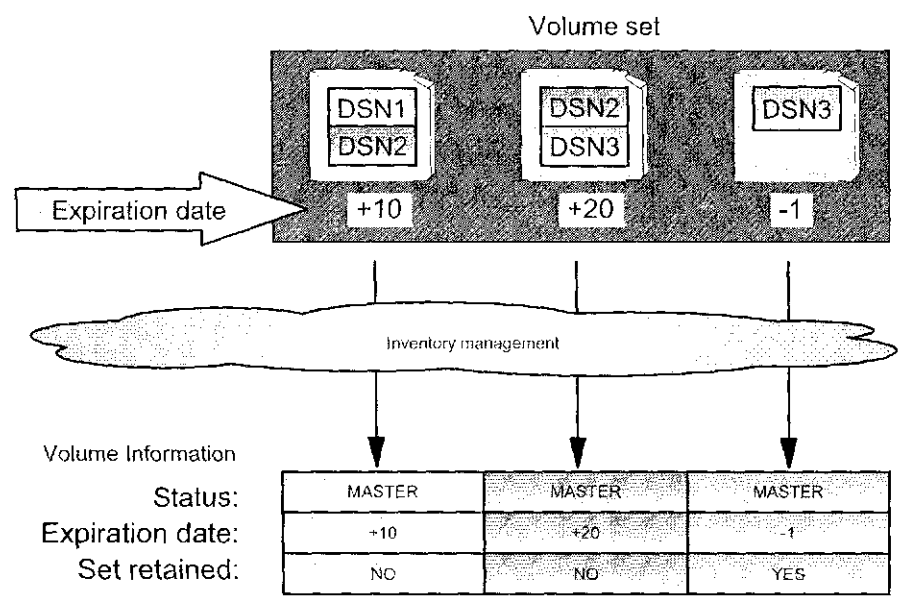


Figure 5-28. RETAINBY(SET) - EXPDT SS212.0

Notes:

To have DFSMSrmm retain tapes like other tape management systems do, use the RETAINBY(SET) option. As you can see in the example shown above, DFSMSrmm will retain the whole volume set until the last volume expires.

Note that the default setting is the RETAINBY(VOLUME) option.

The RETAINBY option has been available since OS/390 DFSMS 2.10.

RETAINBY(SET) - VRS

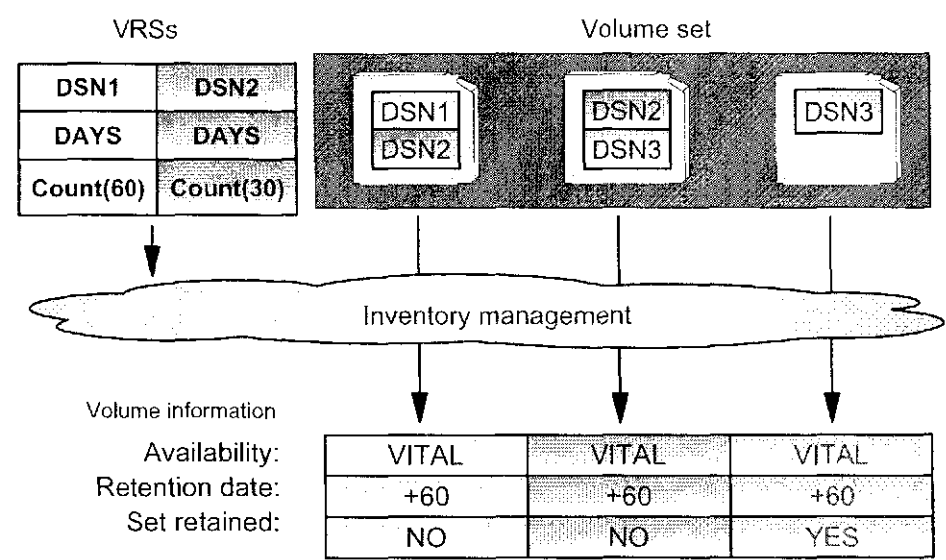


Figure 5-29. RETAINBY(SET) - VRS

SS212.0

Notes:

The RETAINBY(SET) option works in a similar way for volume sets retained by a vital record specification.

RETAINBY(SET) MOVEBY(SET)

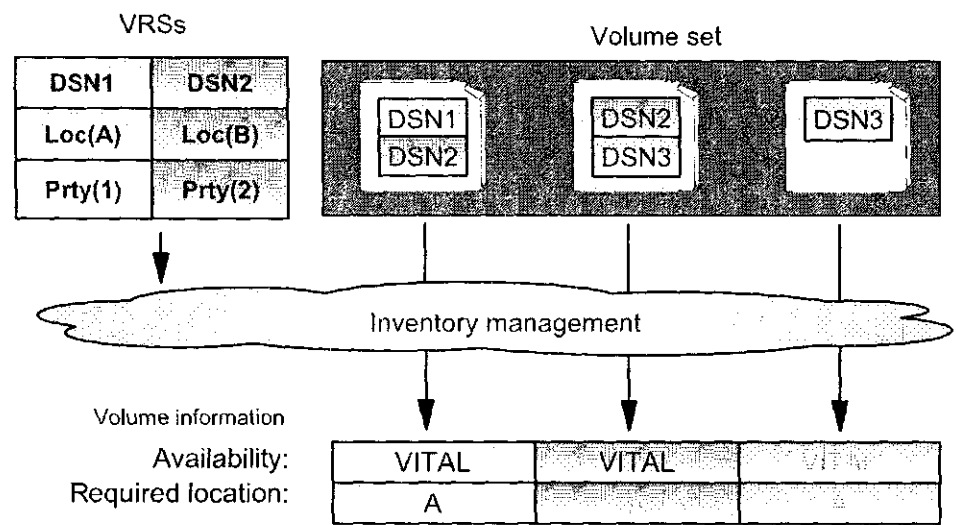


Figure 5-30. RETAINBY(SET) MOVEBY(SET) SS212.0

Notes:

To customize DFSMSrmm volume movement processing you can specify the MOVEBY option in EDGRMMxx. MOVEBY(SET) will request the movement of all volumes of the set to storage location A.

MOVEBY(VOLUME) requests volume 1 to location A and volume 2 to location B, while volume 3 would remain at the home location.

Movement of Volumes

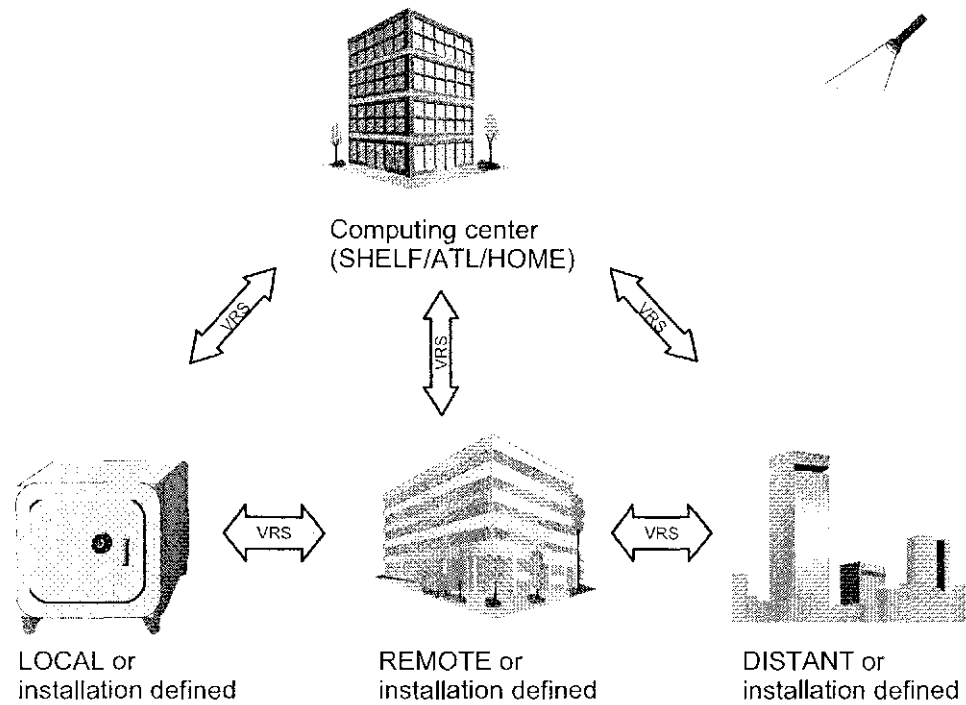


Figure 5-31. Movement of Volumes

SS212.0

Notes:

When a volume is defined to DFSMSrmm, the location to which it is assigned, SHELF or an IBM system-managed tape library, is its home location. Volumes are scheduled for movement based upon specifications made in VRS policies. When a volume is at a storage location and is no longer retained there by a VRS, DFSMSrmm inventory management schedules it to return to its home location. DFSMSrmm offers the ability to move a volume from one storage location to another by having one VRS policy connect to another (NEXTVRS).

Example: Movement to Several Locations

```

RMM ADDVRS DSNAME('SCHLUM.*.DATA') GDG COUNT(99999) -
LOCATION(HOME) STORENUMBER(1) NEXTVRS(SAVE3)-
WHILECATALOG
RMM ADDVRS NAME(SAVE3) STORENUMBER(3) LOCATION(SAVE)-
NEXTVRS(RMT3)
RMM ADDVRS NAME(RMT3) STORENUMBER(3) LOCATION(REMOTE)

```

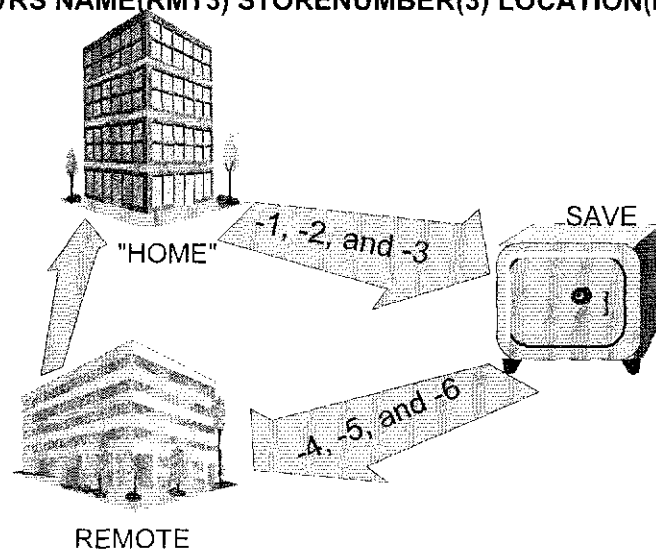


Figure 5-32. Example: Movement to Several Locations

SS212.0

Notes:

The first VRS retains all data sets but stores the most recent at home and the next three at a locally defined location called SAVE. As tapes at SAVE are displaced by more recent volumes, they can be sent to another location, REMOTE in our example. Tapes displaced from REMOTE return HOME.

CDS Changes during VRSEL

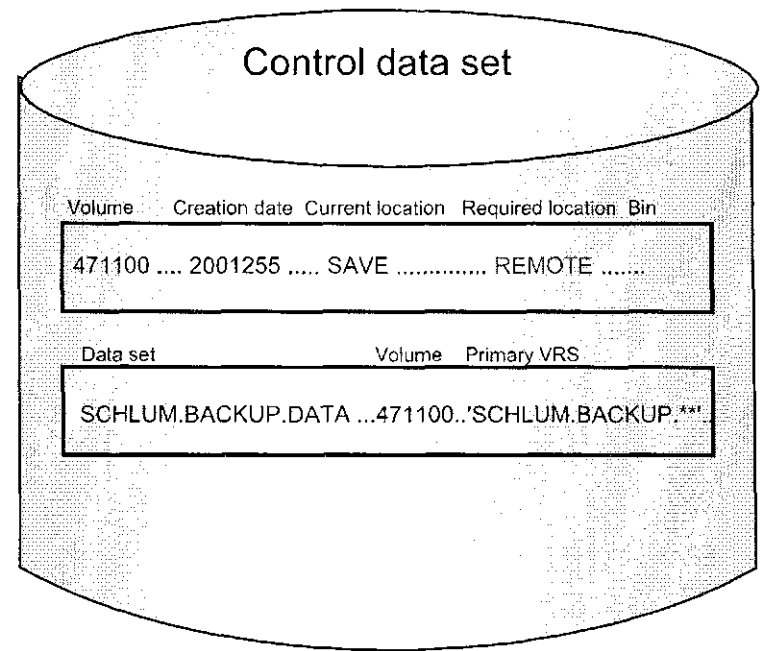


Figure 5-33. CDS Changes during VRSEL

SS212.0

Notes:

During VRSEL of inventory management the need for a volume to move is detected by DFSMSrmm. The required location field of the volume record is updated with the name of the target storage location.

Volume Record Indicating Required Location

```
Panel Help
-----
DFSMSrmm Volume Details - A04101
Command * *
Volume . . . . . : A04101      Back number . . . . . :
Media name . . . . : 3480      Status . . . . . : MASTER
                                   More:
Volume type . . . . : PHYSICAL   Stacked count . . . . . : 0
Retention date . . . : 00/00/00  Expiration date . . . . . : 2500/170
Get retained . . . . : NO        Original expiration date . .
Description . . . . :
Data set name . . . . : 'COCHLUM.BACKUP.DATA.00289V00'
Media type . . . . . : CSE       Archive actions:
Label . . . . . : SI            Return to SCRATCH pool . . : YES
Current version . . . :          Replace volume . . . . . : NO
Required retention . . . :        Return to label . . . . . : NO
Density . . . . . : LPTF        Initialize volume . . . . . : NO
Recording format . . . : ISTRACK  Erase volume . . . . . : NO
Corruption . . . . . : YES       Notify owner . . . . . : NO
Attribution . . . . . : NONE     Expiry date ignore . . . . . : NO
Availability . . . . . : YES/NO  Serial inmediate . . . . . : NO
Owner . . . . . : EXHILUM        Owner address . . . . . : ALPFR
Assigned date . . . . : 200 /160  Assigned time . . . . . : 05:05:52
.
.
.
.
Move mode . . . . . : AUTO       Movement tracking date . . . : 00/1/198
                                   In transit . . . . . : NO
Location:  Current:  Destination:  Old:  Required:  New:
-----
Name . . . . : SAME        STORP  REMOTE  SAME
Type . . . . : SCORE       STORE
Bin number . . : V10250
Media name . . :
Product details:
Product number . . :
Level . . . . . :
Feature code . . . :
Enter DSCROLL commands for more volume information, or END command to CANCEL
```

Figure 5-34. Volume Record Indicating Required Location SS212.0

Notes:

These volume details can be seen only directly after VRSEL processing has finished and before DSTORE processing has completed volume-update process for an outstanding movement. New information is contained in the required location field indicating in our example the need to move to the REMOTE storage location.

Processing Vital Records during VRSEL

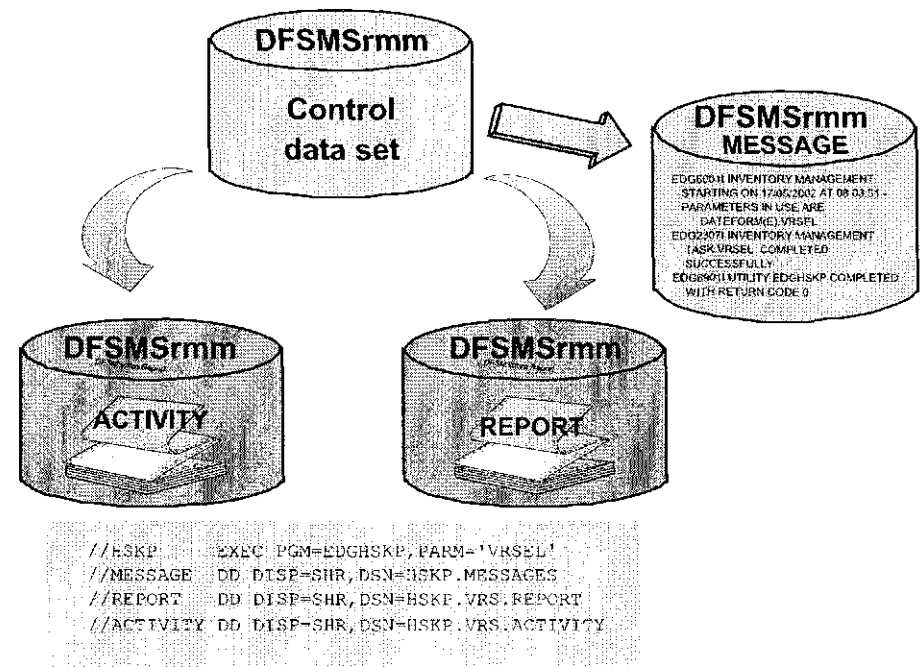


Figure 5-35. Processing Vital Records during VRSEL SS212.0

Notes:

You may recall VRSEL processing during inventory management as introduced in Unit 2. VRSEL provides a report data set that will show the required location of a volume, if that information is available.

For testing purposes, you can run VRSEL with PARM='VRSEL,VERIFY' to produce the output data sets, but not to update records in the control data set.

Figure 5-36. Excerpts from a VRSEL Report SS212.0

The **Required location** field reports the destination of the volume based on the matching vital record specification. When there are multiple data sets in a volume, the destination of the latter is calculated by referring to the priority of each of the required locations identified for it. The destination of the volume is calculated based on the location priorities as described in Unit 4. If there are multiple data sets in a volume and if the volume is retained through the use of multiple vital record specifications, **the required location listed in the report might be different from the destination to which DFSMSrmm attempts to move the volume.**

CDS Changes during DSTORE

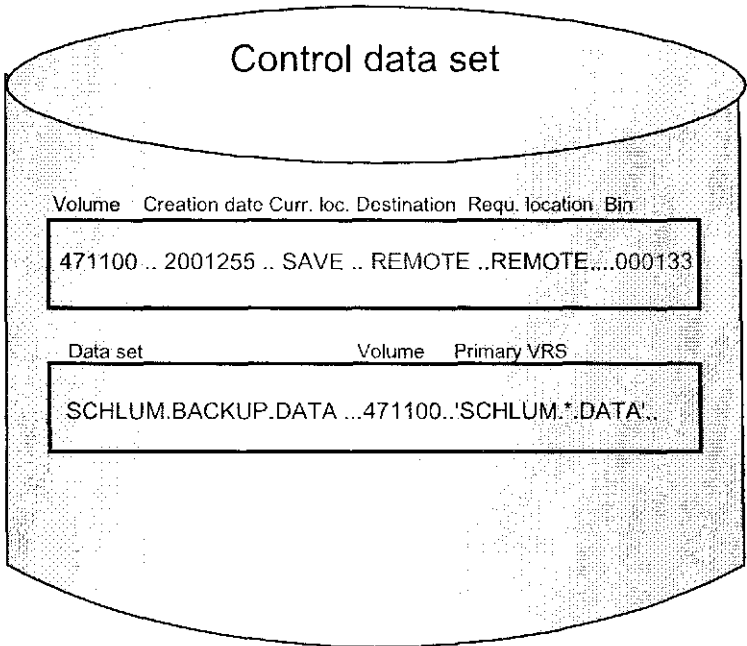


Figure 5-37. CDS Changes during DSTORE?

SS212.0

Notes:

The destination field in the volume record will be updated with the name of the required storage location during DSTORE. A bin number will be assigned for storage locations that are defined to DFSMSrmm with shelf management.

Outstanding Move of a Volume

```
Panel Help
-----
DFSMSrmm Volume Details - A04101
Command ===>
Volume ..... : A04101      Back number ..... :
Media name ..... : 3450      Status ..... : MAGNETIC
                               Bytes ..... :
Volume type ..... : 9910 CAL      Stacked mount ..... : 0
Retention date ..... : 01/05/03    Expiration date ..... : 0001/179
Ret. retained ..... : NO          Original expiration date ..... :
Description ..... :
Data set name ..... : 'SCHLICK.BACKUP.DAT.0009500'
Media type ..... : CST          Release actions:
Label ..... : SL              Return to SCRATCH pool ..... : YES
                                Replace volume ..... : NO
                                Return to owner ..... : NO
                                Initialize volume ..... : NO
                                Erase volume ..... : NO
                                Notify owner ..... : NO
                                Expire date ignore ..... : NO
                                Scratch immediate ..... : NO
                                Owner access ..... : ALLDEV
                                Assigned time ..... : 00:00:00
                                ..... :
                                ..... :
                                ..... :
Move code ..... : AUTO          Movement tracking date ..... : 0001/000
                                In transit ..... : YES
Location:      Current:      Destination:      Old:      Required:      None:
-----
Name ..... : 0400      000000      000000      000000
Type ..... : 000000      000000      000000      000000
File number ..... : 000000      000000      000000      000000
Serial name ..... :
Product serials:
Product number ..... :
Serial ..... :
Product code ..... :
Enter SCROLL command for more volume information, or END command to QALCY.
```

Figure 5-38. Outstanding Move of a Volume SS212.0

Notes:

Note that the destination must be confirmed after a volume move has completed. Use the CHANGEVOLUME subcommand to confirm volume movement:

CHANGEVOLUME * CONFIRMMOVE

This subcommand performs global confirmation of all outstanding moves to all storage locations. Processing at the volume level will be carried out during the next inventory management. When confirmation is performed, destination location information is copied to the **current location** field and the **destination location** and **required location** fields are cleared. This is a prerequisite to any further move of the volume to a different location.

Performing Storage Location Management

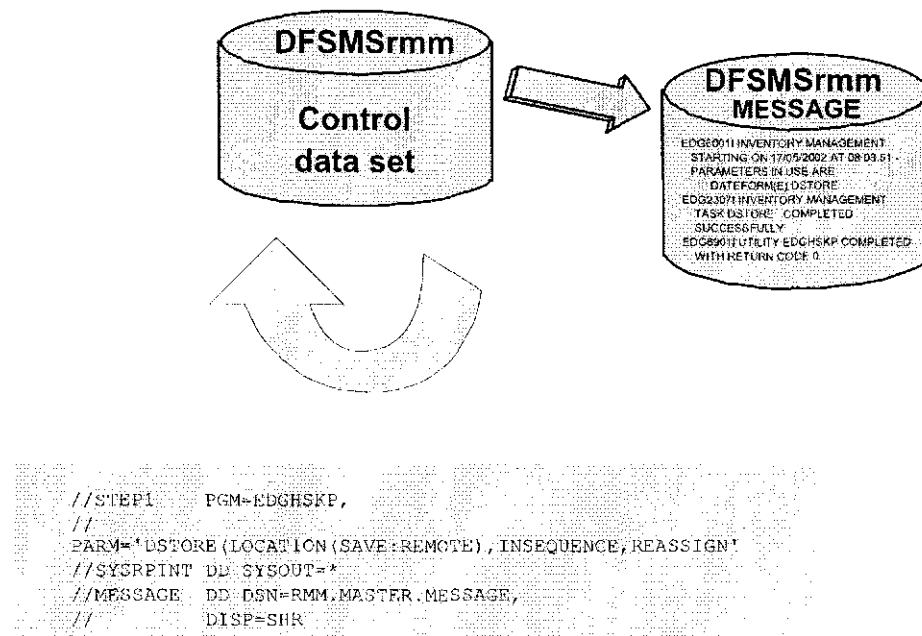


Figure 5-39. Performing Storage Location Management

SS212.0

Notes:

DSTORE processing is performed for all locations if you do not requested otherwise. Support for DSTORE by location is available if the PTFs reported in OW47993 are installed.

INSEQUENCE applies only to those volumes that are required to move from a non-bin-managed to a bin-managed location. Storage location management processing assigns volumes to available bins in volume and bin sequence, starting with the lowest volume serial number and the lowest bin number. All bins that become available in a single run of storage location management processing can be re-used for other volumes. Bins can become available during storage location management processing under these conditions:

Global confirm move processing.

Volumes starting a move out of the bin with parmlib option REUSEBIN(STARTMOVE) specified.

The REUSEBIN(CONFIRMMOVE) Option

Sequence of movements during DFSMSrmm storage location management (DSTORE) while EDGRMMxx option REUSEBIN(CONFIRMMOVE) is in use.

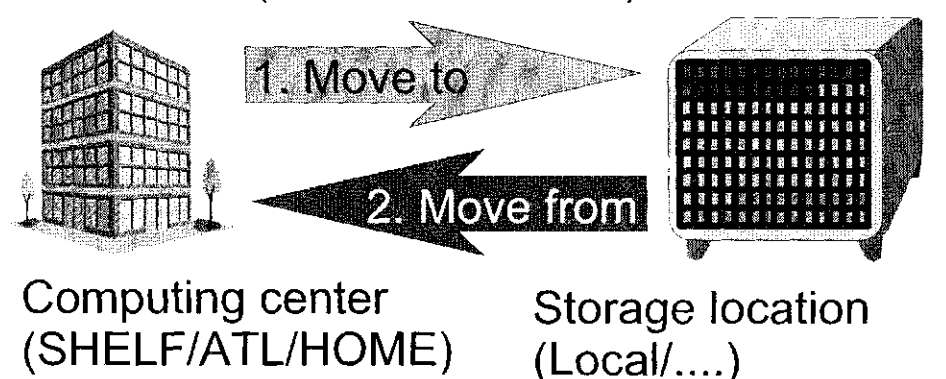


Figure 5-40. The REUSEBIN(CONFIRMMOVE) Option

SS212.0

Notes:

When DFSMSrmm schedules a move for a volume, it is marked IN TRANSIT until the move has been confirmed (for volumes in an automated IBM system-managed tape library, this is when they actually get ejected from the library). For shelf-managed locations, this means that the slot (bin) occupied by a volume being moved elsewhere is not available for re-use until the move is confirmed. Option REUSEBIN(CONFIRMMOVE) is the default setting. Volumes are not in their home location until they are confirmed to be at home or inserted into an IBM system-managed tape library.

When inventory management analyzes VRS policies for movement, any outstanding volume changes for a volume are processed first. For example, if you vault daily, the tapes confirmed for return from a prior run will have their slots, made available at a storage location for re-use.

The REUSEBIN(STARTMOVE) Option

Sequence of movements during DFSMSrmm storage location management (DSTORE) while EDGRMMxx option REUSEBIN(STARTMOVE) is in use.

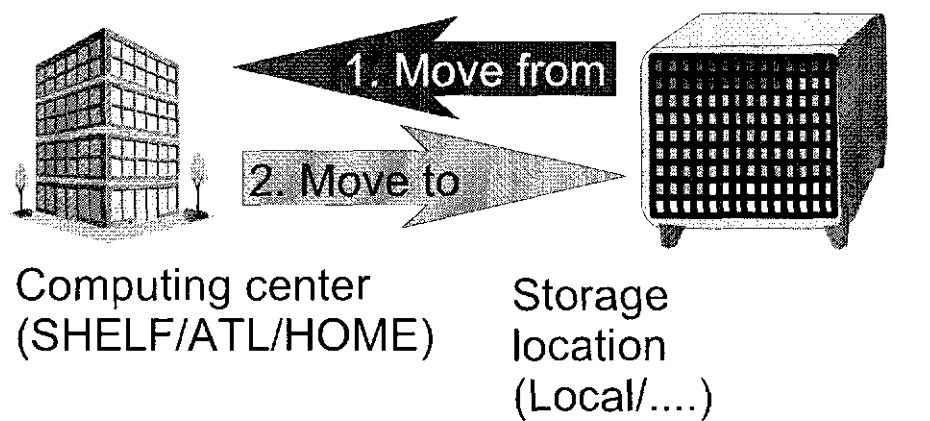


Figure 5-41. The REUSEBIN(STARTMOVE) Option

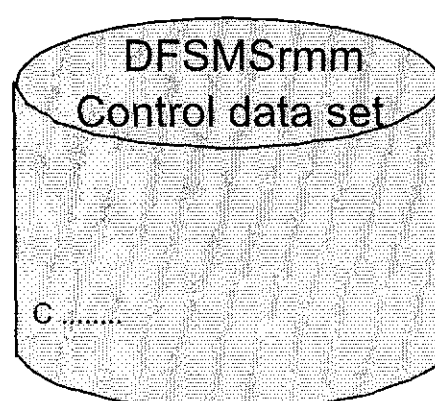
SS212.0

Notes:

Bins are put to the best use if INSEQUENCE and REASSIGN are specified during DSTORE processing and if the REUSEBIN(STARTMOVE) parmlib option is also specified. Fewer empty bins need to be defined because bins are available for re-use at the start of a volume move.

If REASSIGN is also specified, the volumes that restart their move are merged in sequence with those volumes that have just started their move. Freed bins are merged with empty bins.

Enabling Extended Bin Support



```
//STEP1  PGM=EDGUTIL,
//        PARM='UPDATE'
//SYSPINT DD SYSOUT=*
//MASTER DD DISP=SHR,
//        DSN=RMM.MASTER.DSET
//SYSIN   DD *
CONTROL EXTENDED BIN(YES)
/*
```

Figure 5-12. Enabling Extended Bin Support

SS212.0

Notes:

The REUSEBIN(STARTMOVE) option is honored only if extended bin support is enabled. When extended bin support is enabled, DFSMSrmm records additional volume and bin information to keep track of the location of a volume when the latter is moving either from or to a bin-managed storage location.

DFSMSrmm keeps track of the following information for a volume:

- Destination bin number
- Destination bin media name for a volume
- Current bin number
- Current bin media name
- Old bin number
- Old bin media name

When a volume starts moving to a bin-managed storage location, DFSMSrmm updates the destination bin fields with the bin number and the bin media name. When the move has been confirmed, DFSMSrmm updates the current bin fields with the destination bin fields. If a volume moves from a bin-managed storage location, DFSMSrmm does not change the

Student Notebook

current bin fields until the move has been confirmed. DFSMSrmm changes the old bin number to the current bin number and clears the current bin number when a move is confirmed. When extended bin support is not enabled and a volume is moving, DFSMSrmm shows the source bin in the old bin fields and the target bin in the current bin fields.

Volume Movement and Inventory Reports

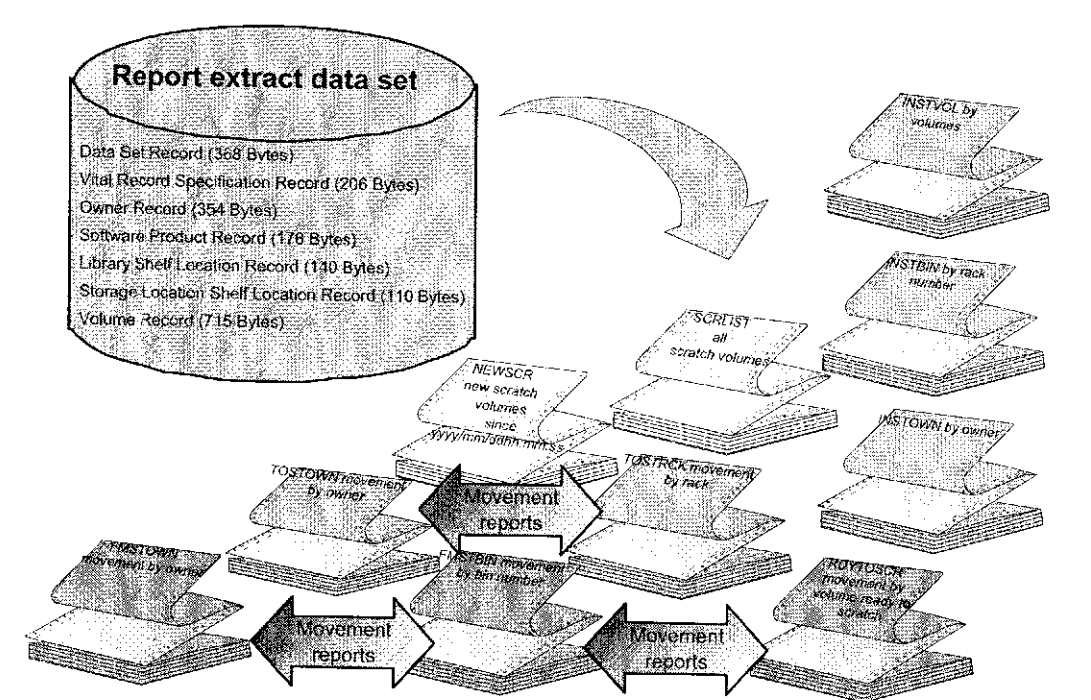


Figure 5-43. Volume Movement and Inventory Reports SS212.0

Notes:

To create volume movement reports, you can use the reporting facilities introduced in Unit 2.

Ejecting Volumes from Automatic Libraries

```
// EXEC PGM=IKJEFT01
//SYSTSPRT DD SYSOUT=*
//SYSTSIN DD *
RMM SEARCHVOLUME VOLUME(*) OWNER(*) LIMIT(*) -
LOCATION(atlname) DESTINATION(storename) -
INTRANSIT(N) CLIST('RMM CHANGEVOLUME ',' EJECT(BULK)')
EXEC EXEC.RMM.CLIST
/*
```

Figure 5-44. Ejecting Volumes from Automatic Libraries

SS212.0

Notes:

DFSMSrmm search subcommands provide a powerful option that allows you to create a command for each object that matches the search criteria. The keyword CLIST allows to specify a prefix and a suffix string for the matching volumes. The generated command strings are written to a dynamically allocated output data set, EXEC.RMM.CLIST. The name might be prefixed by a TSO user ID, depending on the TSO profile. To execute the generated commands, you can run the data set as a CLIST.

While our example illustrates the case of a system-managed tape library where you can request ejects by the DFSMSrmm CHANGEVOLUME subcommand, any other tape library can also benefit from the proposed procedure. Most other manufacturers provide a batch utility to initiate ejections of volumes that can use the generated data set as input. Of course you will need different prefix and suffix strings on the CLIST keyword.

The Policy Management Problem

SMS ACS assigns policies for DASD

Storage group (SG) used to place new data sets
Management class (MC) used to set policy attributes

SMS ACS assigns policies for SMS tape

SG directs data sets to a library
MC recorded by RMM
Limited attributes for tape data sets

RMM assigns policies for all tape

VRS management value (VRS MV) set by EDGUX100
Pooling via VLPOOL or EDUX100 places data sets
Policy attributes are in RMM VRS

R10 provides solution options

Centralize policy assignment in SMS ACS
Feed RMM policy decisions into ACS

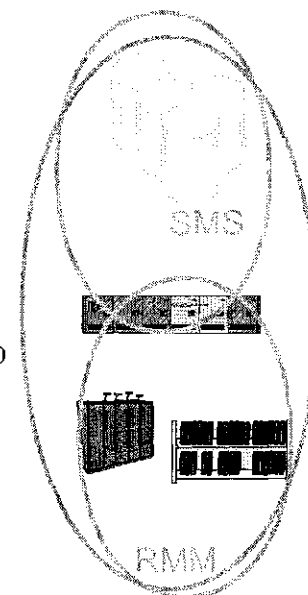


Figure 5-45. The Policy Management Problem

SS212.0

Notes:

We have seen that there are several options that allow you to bind one or more VRSs to a data set. It's up to you to choose what method you want to use. The enhancement described above is not limited to the management of a VRS, but let's focus on that issue now.

Most customers use management values to support special expiration dates. This method has been chosen in many conversion projects because there was no other way to continue supporting of special expiration dates. There is nothing wrong with this approach but what do you think of a replacement? You can now centralize policy assignment in SMS automatic class selection routines.

DFSMSrmm R10 Included

SMS ACS for tape

- RMM parmlib controls for SMS support
 - Selectively enable pre-ACS and SMS ACS
- Extended support for decisions within ACS processing
 - RMM scratch pool and policy input during allocation
 - RMM calls SMS MC and SG routines for non-system managed tape

Consolidate tape decisions in ACS

- Base decisions on RMM input
 - RMM and EDGUX100 values input to ACS
- Direct RMM using ACS decisions
 - No input from RMM or EDGUX100

Decision at data set level

- Enables phased implementation

Notes:

To receive a management class during allocation, a tape data set must be written on a device that is part of a system-managed tape library. For devices outside system managed tape libraries, when ACS routines are called during allocation, ACS processing ends after storage class ACS routine because no storage class is assigned in this case. You can now tell DFSMSrmm to call the management class ACS routine directly during open processing for data sets being written on devices outside a system managed tape library to receive a management class. If a management class is successfully assigned, EDGUX100 will not be called on to resolve the possible need for a management value. This is an excellent opportunity to move the decision on policy from EDGUX100 to ACS routines without the need to change EDGUX100 too frequently.

SMS ACS Support

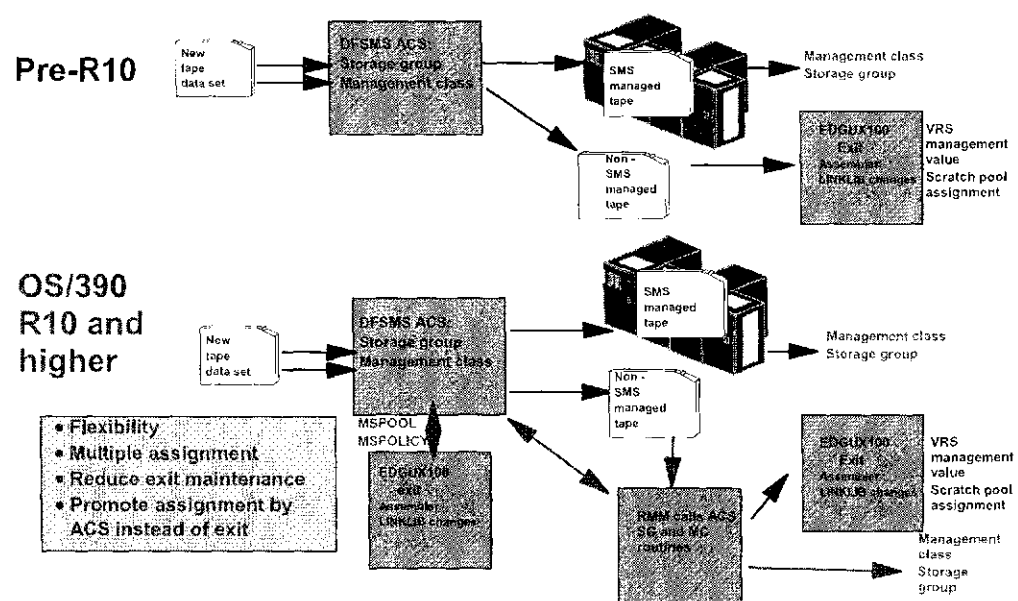


Figure 5-47. SMS ACS Support

SS212.0

Notes:

With this enhancement, EDGUX100 is called only to resolve the need for a management value if no management class was assigned while DFSMSrmm was calling the management class ACS routine.

EDGUX100 Capabilities

Request ignore of a volume
Determine the scratch pool
Control cartridge loader
Assign VRS policy by name
Update/provide EXPDT
Control data set recording
Sticky label customization or production
Set ACS input variables during pre-ACS

Notes:

We discussed EDGUX100 in Unit 3. The assignment of a management value is only one of many options that can be used with this exit, which is called during SMS pre-ACS processing and data set open.

EDGUX100 Environment

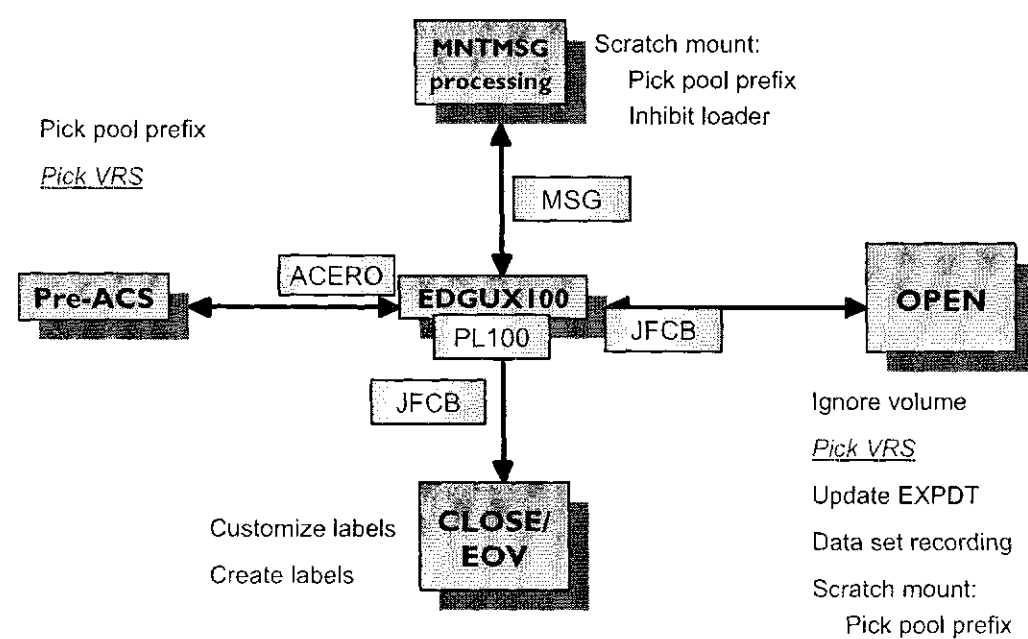


Figure 5-49. EDGUX100 Environment

SS212.0

Notes:

In many cases, you might use several functions that EDGUX100 provides. Coding of this exit can be a very difficult task. It is a good idea to reduce the use of this exit to a minimum of functions.

Recommendations

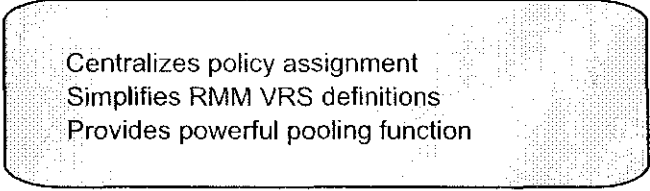
Do not use RMM pre-ACS support.

This extends the life of EDGUX100 customization.

Use SMS management class to replace VRS MV.

Replace RMM VRS DSNAMES with ACS FILTLISTs.

Use SMS storage group to replace RMM pool prefix.



Centralizes policy assignment
Simplifies RMM VRS definitions
Provides powerful pooling function

Figure 5-50. Recommendations

SS212.0

Notes:

To centralize policy assignment, we recommend the use of a management class to replace a VRS management value.

You can replace VRS data set name masks with filter lists in automatic class selection routines.

We will discuss pooling in Unit 7.

Using SMS Management Class

Simplify VRS policies

- Move data set name masks to ACS filter lists
- One VRS CHAIN implements one service level

Code MC ACS routine to process

- ⌘ RMMVRS environment call for non-SMS tape
- ⌘ Regular allocation for SMS tape
- Enable via parmlib option SMSACS(YES)

Assign policies by name

- Move decisions out of EDGUX100 exit
- Cleanup VRS definitions

Figure 5-51. Using SMS Management Class

SS212.0

Notes:

Using management classes to implement service level agreements not only moves decisions out of EDGUX100 but also allows to reduce the number of VRSs.

SMS MC Considerations

Data sets retained by different VRSs

- Matching on data set name
- Matching on SMS MC
- Matching on VRS management value

After implementation all new data sets use SMS MC

- Mixed matching is ok *except* when using *CYCLES*
- Use the same VRS to avoid problems
 - Use CD subcommand to set MC
 - Delete the Data Set Name VRS

MC(CYCLES)	VRS(A,B,C)
MC(CYCLES)	VRS(A,B,C)
MC(CYCLES)	VRS(A,B,C)
MV(CYCLES)	

Figure 5-52. SMS MC Considerations

SS212.0

Notes:

Combining two vital record specifications for a data set by using the concept of primary and secondary VRSs works fine in general as we have seen earlier in this Unit. But what if both VRSs use cycle control? The results might be confusing, therefore you shouldn't use that.

SMS MC Considerations

Data sets retained by different VRSs

- Matching on data set name
- Matching on SMS MC
- Matching on VRS management value

After implementation all new data sets use SMS MC

- Mixed matching is ok *except* when using *CYCLES*
- Use the same VRS to avoid problems
 - Use CD subcommand to set MC
 - Delete the Data Set Name VRS

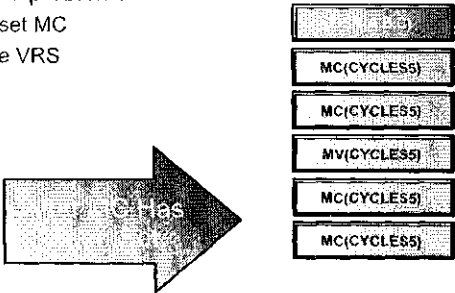


Figure 5-53. SMS MC Considerations SS212 0

Notes:

Processing is more clear if you only use one VRS at a time. You still have the choice to link several VRSs by using NEXTVRS.

Implementing SMS MC

Simplify VRS policies
Move data set name masks to ACS filter lists
One VRS chain implements one service level

```
RMM AS DSN('A.*') DAYS COUNT(365) NEXTVRS(E2)  
RMM AS NAME(E2) EXTRADAYS COUNT(2)  
  
RMM AS DSN('B.*') DAYS COUNT(365) NEXTVRS(E2)  
  
RMM AS DSN('C.*') DAYS COUNT(365) NEXTVRS(E2)  
  
RMM AS DSN('MCXYZ') DAYS COUNT(365) NEXTVRS(E2)
```

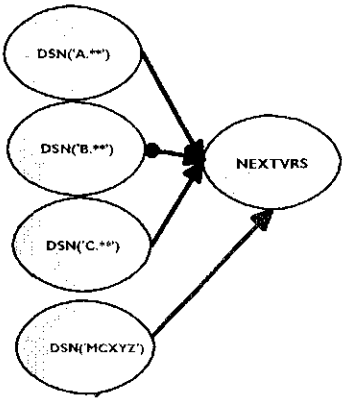


Figure 5-54. Implementing SMS MC

SS212.0

Notes:

The idea is to replace several vital record specifications that provide the same service for different data sets by a single management class that in turn represents a service-level agreement.

Coding the Management Class ACS Routine

Code MC ACS routine to process

- ⌘ RMMVRS environment call for non-SMS tape
- ⌘ Regular allocation for SMS tape

Enable via parmlib option SMSACS(YES)

```
FILELIST XYZ INCLUDE(A,"B","C")
....
IF &ACSENVIR = 'RMMVRS' THEN
DO
  SELECT (&DSN)
  WHEN (&XYZ) SET &MGMTCLAS = 'MCXYZ'
  END
  IF &EXPDT = '1999000' THEN SET &MGMTCLAS = 'CATALOG'
...
END
```

Parmlib:
OPTION SMSACS(YES)

Cmd:
F DFRMM,M=xx

Figure 5-55. Coding the Management Class ACS Routine

SS212.0

Notes:

To enable a call for a management class ACS routine, specify the SMSACS(YES) option in EDGRMMxx. The default setting for SMSACS is NO.

Cleanup Activities

Assign policies by name

- Move decisions out of EDGUX100 exit
 - EDGUX100 is not called when MC is assigned
 - EDGUX100 VRS MV assignments can be removed at convenient time
(Next release, or, next system update)

Cleanup VRS definitions

- Delete VRSs with DSNAMES masks
 - These have been replaced by MC names
 - Only when no longer in use
- Consider VRSMIN(count) parmlib option

Notes:

Checkpoint

Indicate whether each of the following statements is true or false.

1. Chaining of vital record specifications is only available when you specify the VRSEL(NEW) option in EDGRMMxx.
2. Both data set vital record specifications and name vital record specifications can contain retention information when you specify the VRSEL(NEW) option in EDGRMMxx.
3. A subchain contains always only one vital record specification that specifies a retention information.
4. DFSMSrmm moves by default all volumes in the same multi-volume set that are retained by vital record specifications, and all volumes are retained in the same location selected by the vital record specification priority.
5. When a volume moves out of a bin, DFSMSrmm does not reuse this bin until the move has been confirmed unless you specify the REUSEBIN(STARTMOVE) option in EDGRMMxx.
6. A management class can be used to match a vital record specification for data sets in both system-managed volumes and non-system-managed volumes.

Figure 5-57. Checkpoint

SS212.0

Notes:

Write down your answers here:

- 1.
- 2.
- 3.
- 4.
- 5.
- 6.

Unit Summary

Now that you have completed this unit, you should be able to:

- Choose advanced DFSMSrmm VRS processing
- Identify the set of EDGRMMxx options to use advanced VRS processing
- Use VRS definitions for volume movement
- Understand DSTORE processing during inventory management
- Enable use of management class for all tape data sets by calling SMS ACS from DFSMSrmm

Notes:

Unit 6. Miscellaneous DFSMSrmm Topics

What This Unit Is About

This unit covers in detail a number of topics that preceding units have referred to, such as how DFSMSrmm handles the catalog status information of tape data sets, which is an important issue for how long housekeeping takes.

What You Should Be Able to Do

After completing this unit, you should be able to:

- Identify the mechanism whereby catalog status is tracked
- Modify the EDGRMMxx parmlib member so that it can use the catalog status information during inventory management
- Describe the interface between DFSMSrmm and CATALOG address space
- Describe related DFSMSrmm activities that occur during inventory management
- Maintain the DFSMSrmm control data set
- Use EDGINERS for initialization and erasure of tape volumes
- Explain the role DFSMSrmm plays in providing tape-data-set security

How You Will Check Your Progress

Accountability: Checkpoint questions

References

- | | |
|-----------|--|
| SC26-7404 | <i>DFSMSrmm Guide and Reference</i> |
| SC26-7405 | <i>DFSMSrmm Implementation and Customization Guide</i> |

Objectives

After completing this unit, you should be able to:

- Identify the mechanism whereby catalog status is tracked
- Modify the EDGRMMxx parmlib member to exploit catalog synchronization
- Describe the interaction between CATALOG and DFSMSrmm address space during inventory management
- Maintain the DFSMSrmm control data set
- Use EDGINERS for initialization and erasure of tape volumes
- Explain the role DFSMSrmm plays in providing tape-data-set security

Notes:

6.1 Catalog Status Tracking

Non-Shared Catalogs

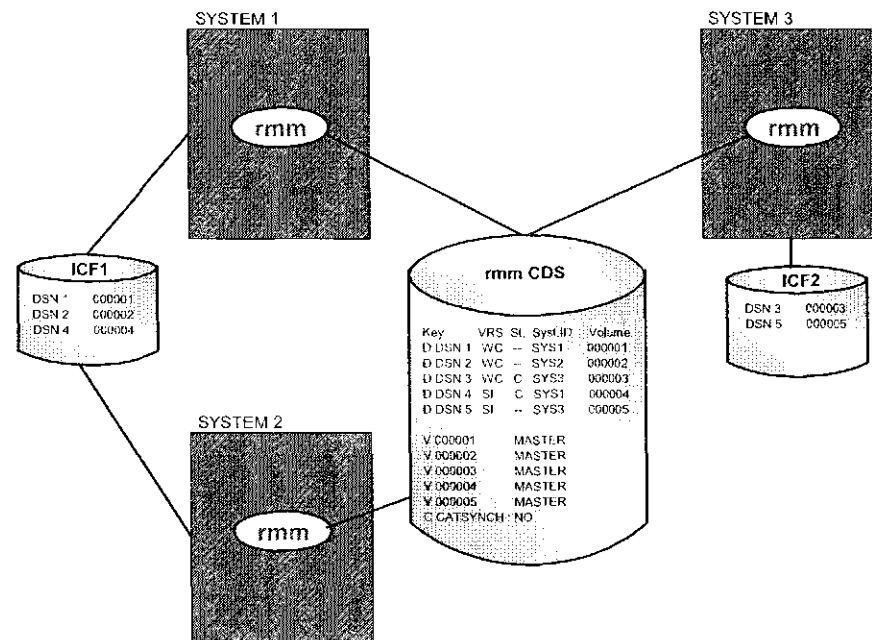


Figure 6-2. Non-Shared Catalogs

SS212.0

Notes:

When customers want to share tape pools across the enterprise, the DFSMSrmm CDS must be shared, too. It might seem at first as though, this is not a big issue. But today more and more tape data sets are managed based on their catalog status. Because DFSMSrmm has to find out while it is carrying out VRSEL whether the data set in question is still cataloged, the question arises where to carry out inventory management.

Performing catalog locate processing during VRSEL to resolve the catalog status of data sets, DFSMSrmm cannot find a number of catalog entries because the catalogs are not connected to each system, no matter where you run inventory management. Under the circumstances, an inadvertent release of volumes can happen.

If you are not using catalog synchronization, which is the default, DFSMSrmm looks for the catalog entry (which is called a catalog locate) to determine the catalog status of each data set that is managed by a VRS including WHILECATALOG during housekeeping VRSEL. Active catalog synchronization relies on the catalog status tracked in data set name records in DFSMSrmm CDS, thus obviating the need for catalog locates.

EDGRMMxx Settings

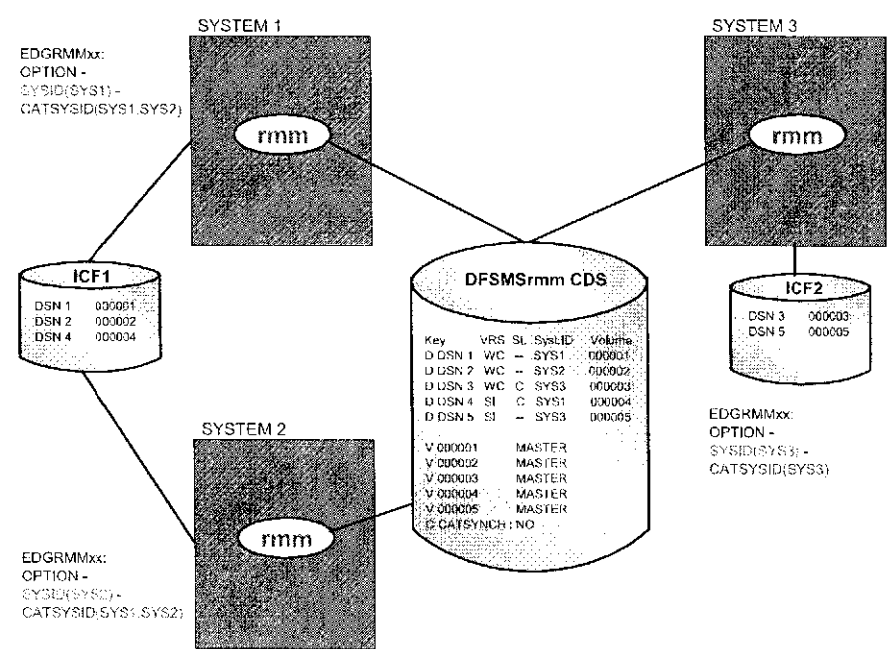


Figure 6-3. EDGRMMxx Settings

SS212.0

Notes:

To enable catalog synchronization during housekeeping, use the *CATSYSID* option. There is no default setting for this option. Specify the system IDs of the systems sharing the same catalogs in the EDGRMMxx members of the related systems.

In our example systems, SYS1 and SYS2 share catalogs ICF1 and ICF2. Tell DFSMSrmm this fact by specifying *OPTION CATSYSID(SYS1,SYS2)* in the EDGRMMxx members of systems SYS1 and SYS2. The sysid option defaults to the SMF ID and can be omitted. SYS3 is not connected to ICF1 and ICF2. This system uses only ICF3, which is not shared with SYS1 and SYS2. EDGRMMxx contains the *OPTION CATSYSID(SYS3)*.

In case of a full shared catalog environment, simply specify *OPTION CATSYSID(*)* in all members EDGRMMxx.

Data sets DSN 1, DSN 2 and DSN 5 were created before these systems supported catalog status tracking. Data sets DSN 3 and DSN 4 show the catalog status set to C (cataloged) in their corresponding data set name records in DFSMSrmm CDS, thereby indicating that they were created in an environment with catalog status tracking support. The system ID reported in the data set name record is the ID of the *creating system*.

Initial Catalog Synchronization (SYS1/SYS2)

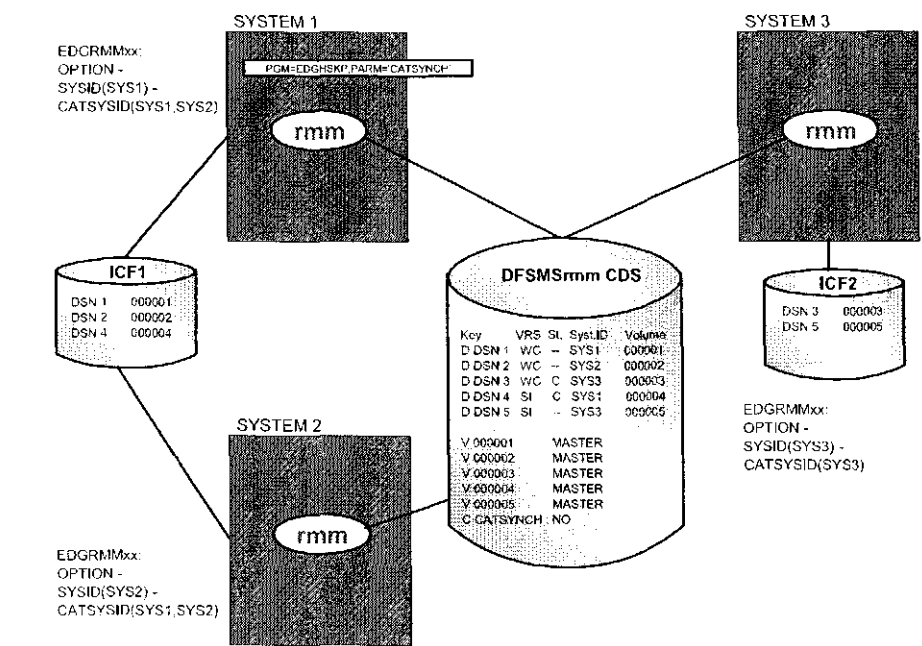


Figure 6-4. Initial Catalog Synchronization

SS212.0

Notes:

This feature was introduced with DFSMS/MVS 1.5.0. All participating systems must use DFSMS/MVS 1.5 or higher. Catalog status is tracked for all data sets created after taking DFSMS/MVS 1.5 (or higher) in production. Because data sets may already have been created, you have to request catalog synchronization explicitly.

You have to restart DFSMSrmm to activate the new OPTION settings. A run of EDGHSKP with PARM='CATSYNCH' will resolve catalog status of all data sets with a creating system ID equal to one of the names specified in OPTION CATSYSID using catalog search interface.

You can also start with a run of EDGHSKP with PARM='CATSYNCH,VERIFY', which will not change the catalog status but will report each difference found between a data set name record in the DFSMSrmm CDS and the occurrence of this data set in an ICF catalog.

Recommendation: It is a good idea to check message-file space before running EDGHSKP with CATSYNCH, because a message is generated for each data set when its catalog status is being updated.

Update Catalog Status in DFSMSrmm CDS

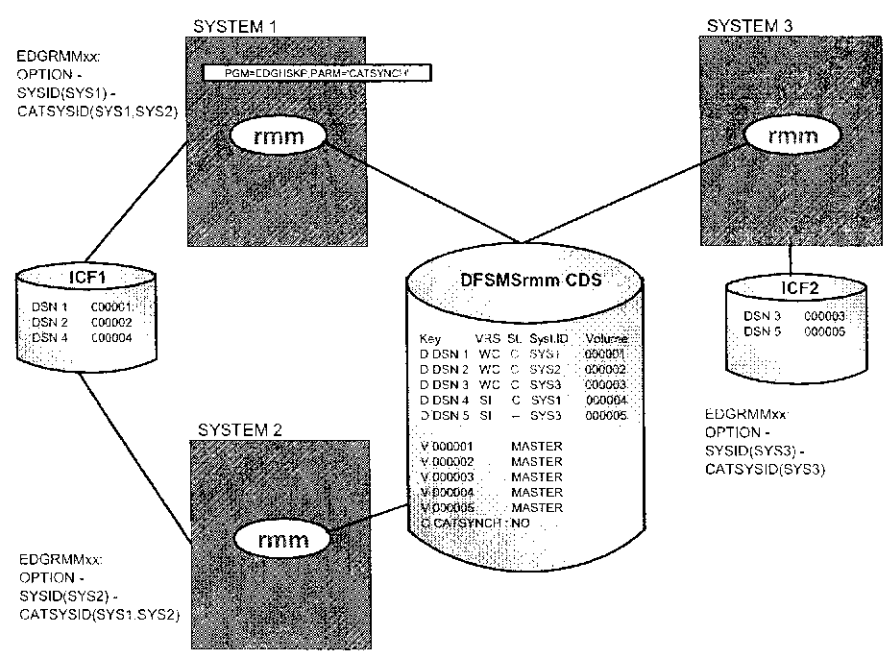


Figure 6-5. Update Catalog Status in DFSMSrmm CDS

SS212.0

Notes:

As a result of the catalog synchronization during the run of EDGHSKP, the catalog status of DSN 1 and DSN 2 has changed to C. DSN 5 was not resolved because the creating system ID did not match any of the strings specified in OPTION CATSYSID in EDGRMMxx on the system where EDGHSKP was run.

Initial Catalog Synchronization (SYS3)

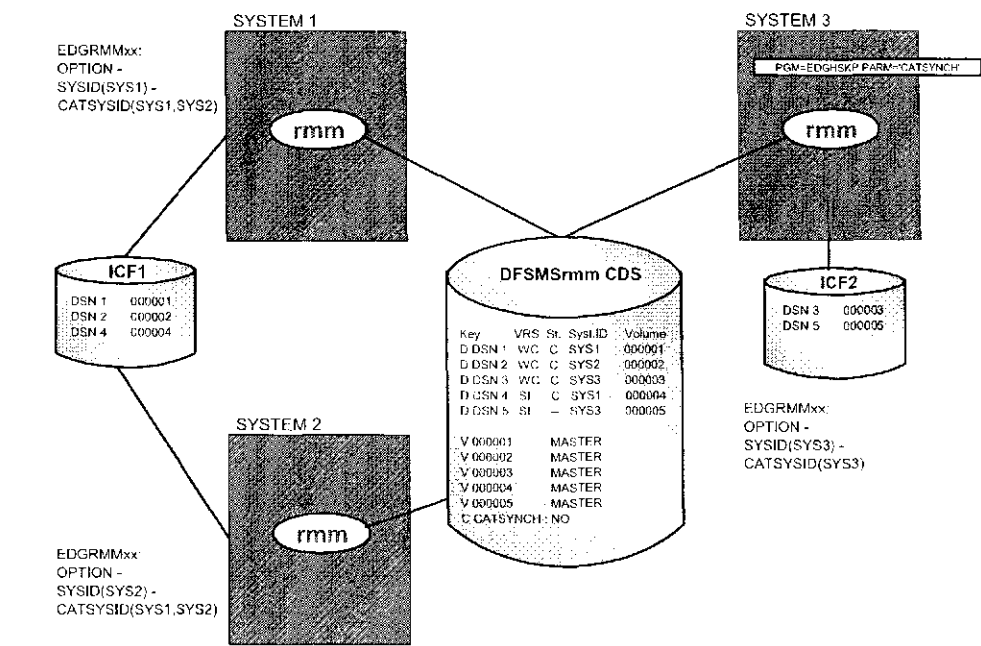


Figure 6-6. Initial Catalog Synchronization (SYS3)

SS212.0

Notes:

We have to repeat the catalog synchronization by using EDGHSKP with PARM='CATSYNCH' on system 3, which will resolve catalog status of all data sets with a creating system id equal to SYS3, which is the name specified in OPTION CATSYSID.

You can also start with a run of EDGHSKP with PARM='CATSYNCH,VERIFY', which will not change the catalog status but will report each difference found between a data set name record in the DFSMSrmm CDS and the occurrence of this data set in an ICF catalog.

Update Catalog Status in DFSMSrmm CDS

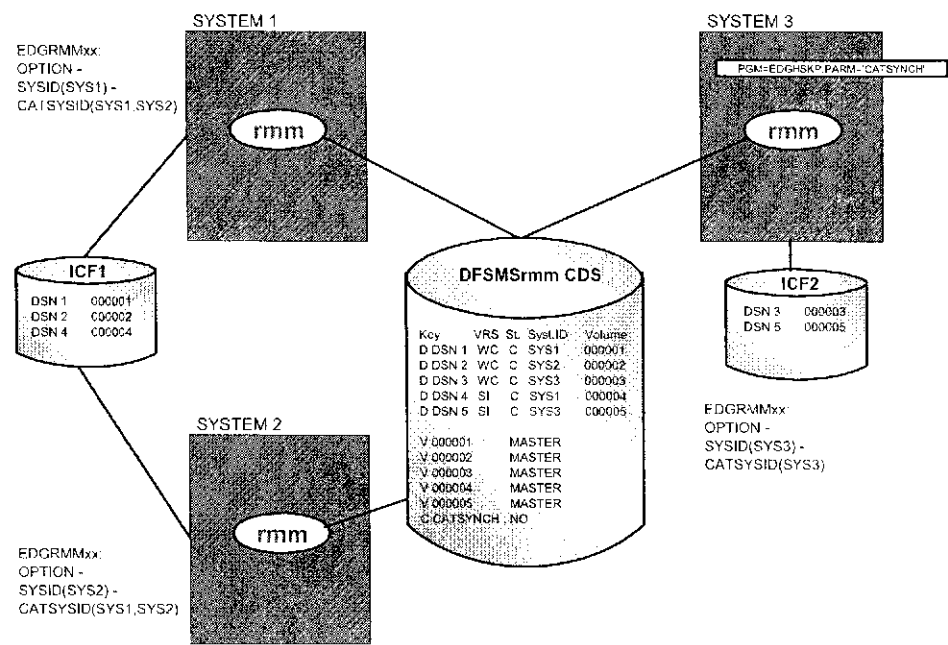


Figure 6-7. Update Catalog Status in DFSMSrmm CDS

SS212.0

As a result of the catalog synchronization during the run of EDGHSKP, the catalog status of DSN 5 has changed to C.

Final Catalog Synchronization

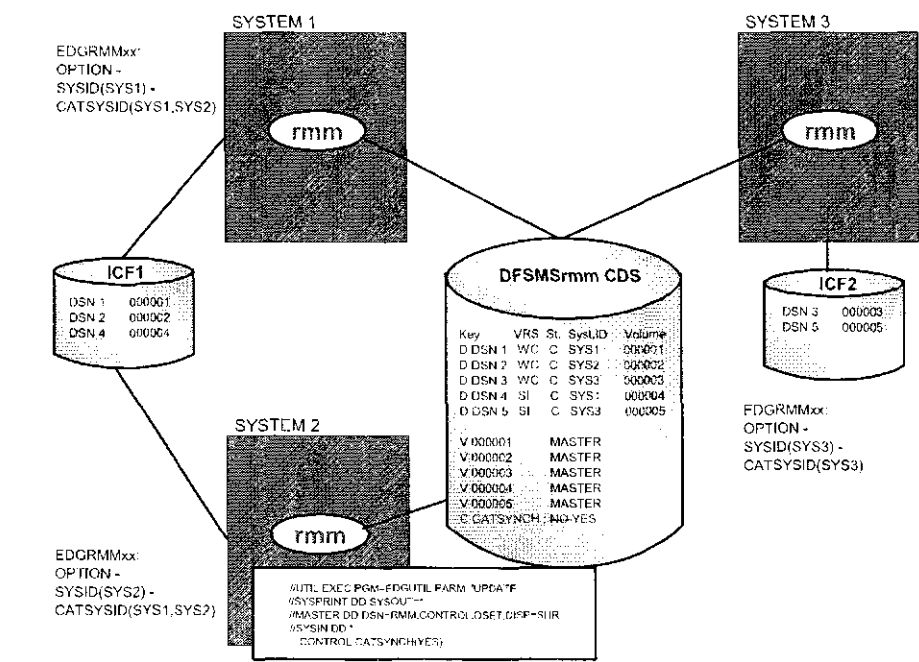


Figure 6-8. Final Catalog Synchronization

SS212.0

Notes:

To enable catalog synchronization while DFSMSrmm is carrying out inventory management, we need a change to the control record in the DFSMSrmm CDS by using EDGUTIL with PARM='UPDATE'. DFSMSrmm sets the date and time of the last successful catalog synchronization in the control record to confirm that all catalogs are synchronized. In the picture, we have noted instead **CATSYNCH: YES**.

DFSMSrmm using Catalog Synchronization

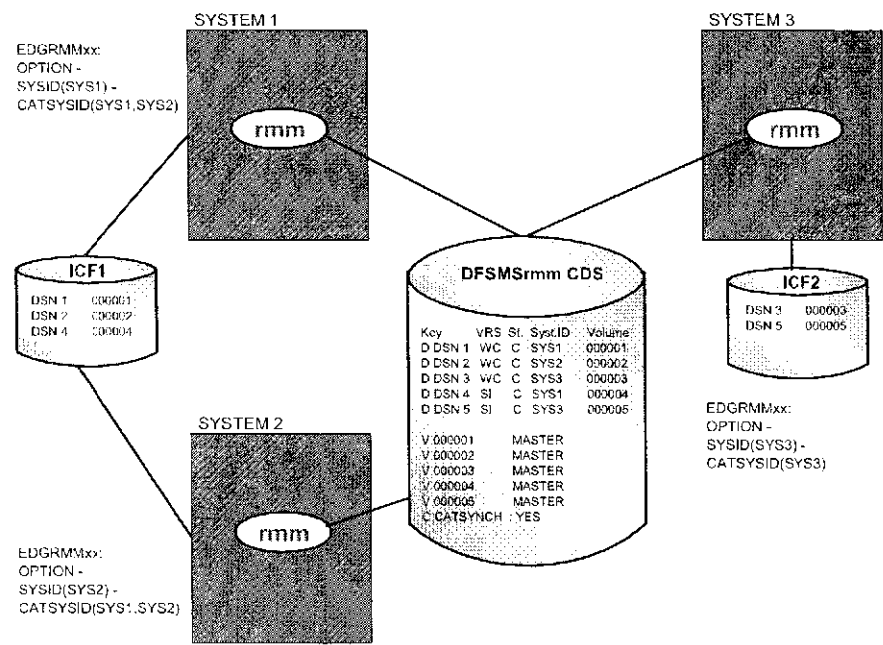


Figure 6-9. DFSMSrmm using Catalog Synchronization

SS212.0

Notes:

Catalog synchronization is now fully functional and will help you save time during future runs of housekeeping.

The set-up of catalog synchronization is generally considered a one-time action.

Warning: After setting OPTION CATSYSID(...) in EDGRMMxx you can run housekeeping VRSEL only if the DFSMSrmm CDS control record indicates CATSYNCH(YES)! If you have not finished all the steps before the next normal housekeeping, you should revert to EDGRMMxx members that do not contain OPTION CATSYSID(...).

Uncatalog of a Tape Data Set by Application

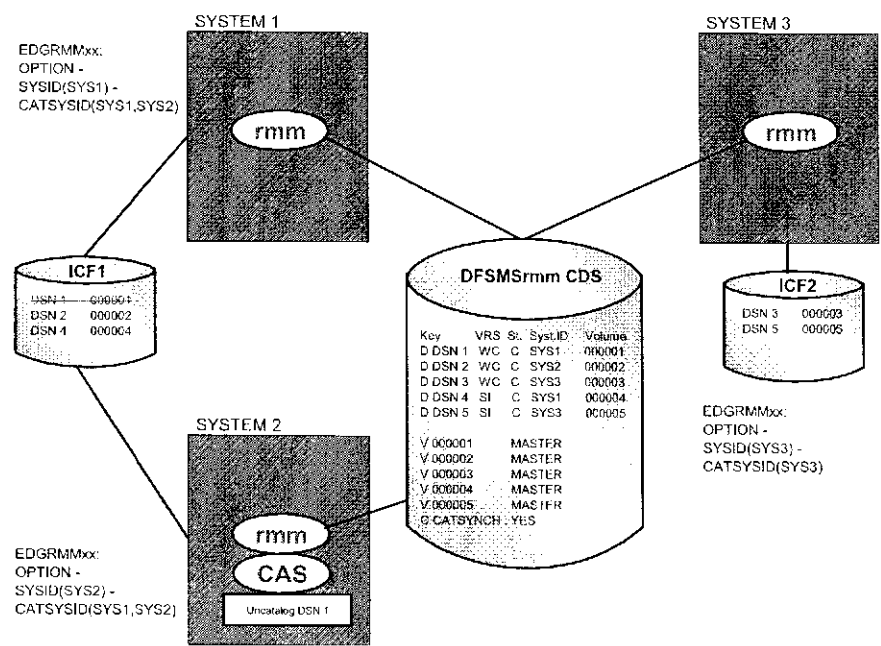


Figure 6-10. Uncatalog of a Tape Data Set by Application

SS212.0

Notes:

When a tape data set is uncataloged by an application, the Catalog Address Space (CAS) will remove the related record from the ICF catalog.

Change Catalog Status in DFSMSrmm CDS

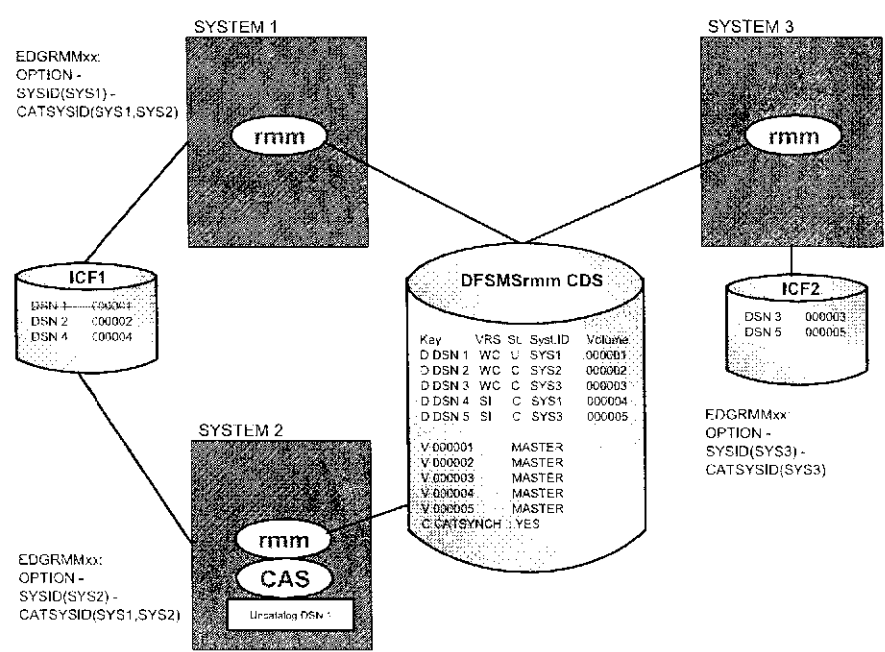


Figure 6-11. Change Catalog Status in DFSMSrmm CDS

SS212.0

Notes:

CAS will **always** inform DFSMSrmm about changes to the catalog status of tape data sets. This process works independently of the setting of both OPTION CATSYSID(...) and the CATSYNCH status in the control record in DFSMSrmm CDS.

Thus, if between two normal housekeeping runs you cannot finish all the steps required to set up catalog synchronization and thus have to begin the set-up a second time, you do not have to repeat any steps you completed the first time.

Uncatalog of a Tape Data Set by Application

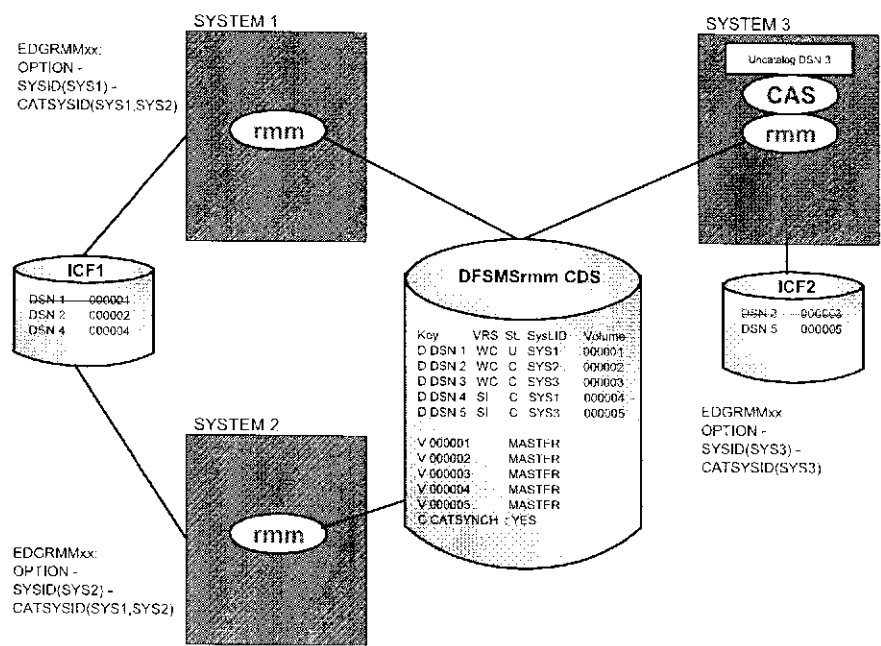


Figure 6-12. Uncatalog of a Tape Data Set by Application SS212.0

Notes:

The picture shows the same procedure for a data set on System 3.

Change Catalog Status in DFSMSrmm CDS

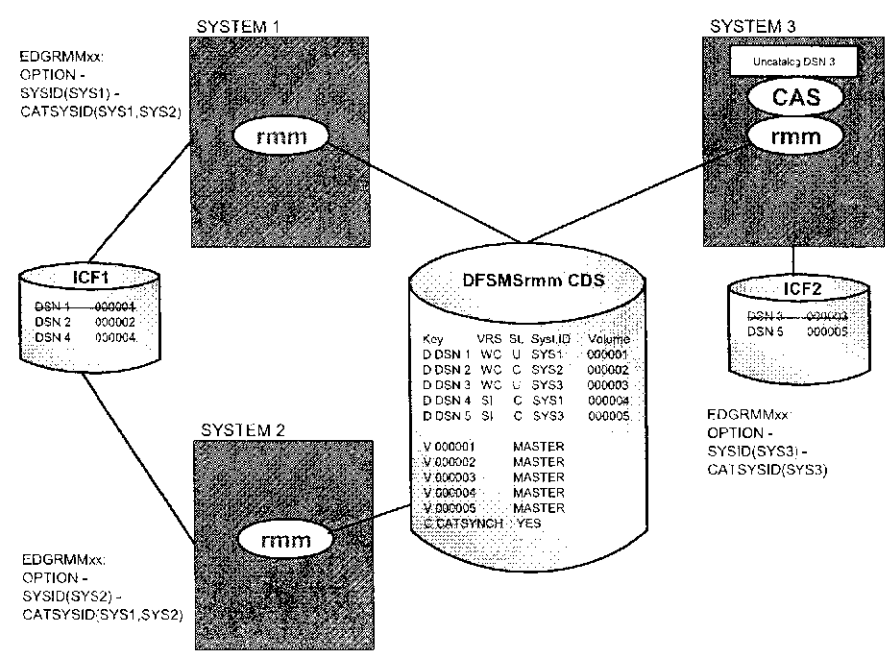


Figure 6-13. Change Catalog Status in DFSMSrmm CDS

SS212.0

Notes:

Catalog status tracking is always carried out on any system that runs DFSMSrmm at the minimum level, DFSMS/MVS V1.5.

DFSMSrmm Housekeeping on SYS2 (1 of 2)

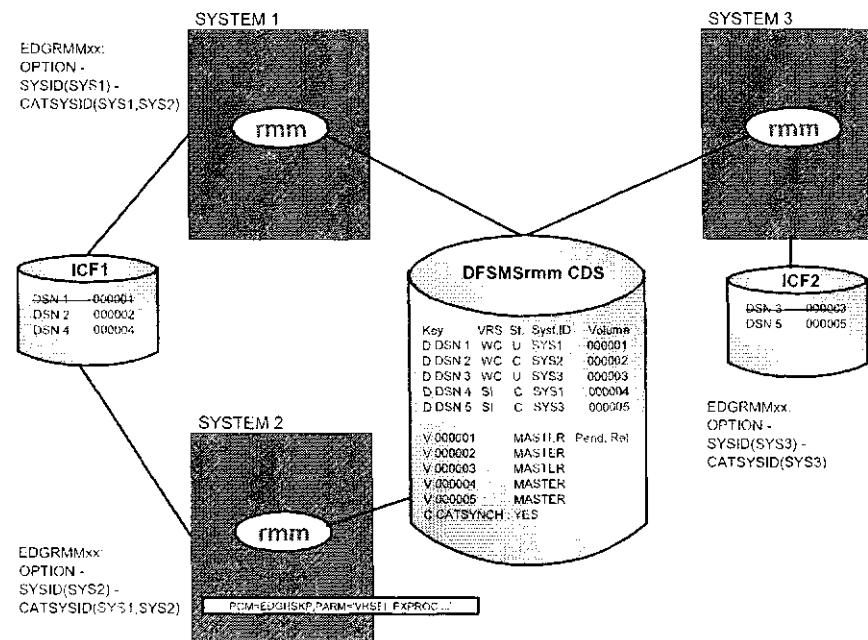


Figure 6-14. DFSMSrmm Housekeeping on SYS2 (1 of 2)

SS212.0

Notes:

During a normal housekeeping run that includes all the steps usually needed, let's say on system 2, no catalog locates will be made during VRSEL, both because the CATSYSID(...) option is specified and catalog synchronization is active (CATSYNCH=YES) in the control record).

During EXPROC we will see the availability of volume V00001 being set to PENDING RELEASE. DFSMSrmm first checks the matching *create system ID* in the data set record of DSN 1, which is the first data set on volume V00001. During EXPROC, a match with one of the system IDs contained in OPTION CATSYSID(...) is a prerequisite to managing the data set and the volume in which it resides. Being retained up to that point by a matching VRS with WHILECATALOG specified (shortcut WC means WHILECATALOG), DSN 1 is being dropped and DFSMSrmm starts release processing for volume V00001.

Housekeeping ignores DSN 3 on volume V00003 although the volume is no longer retained by VRS because the creating system ID does not match one of the IDs in the CATSYSID option.

DFSMSrmm Housekeeping on SYS2 (2 of 2)

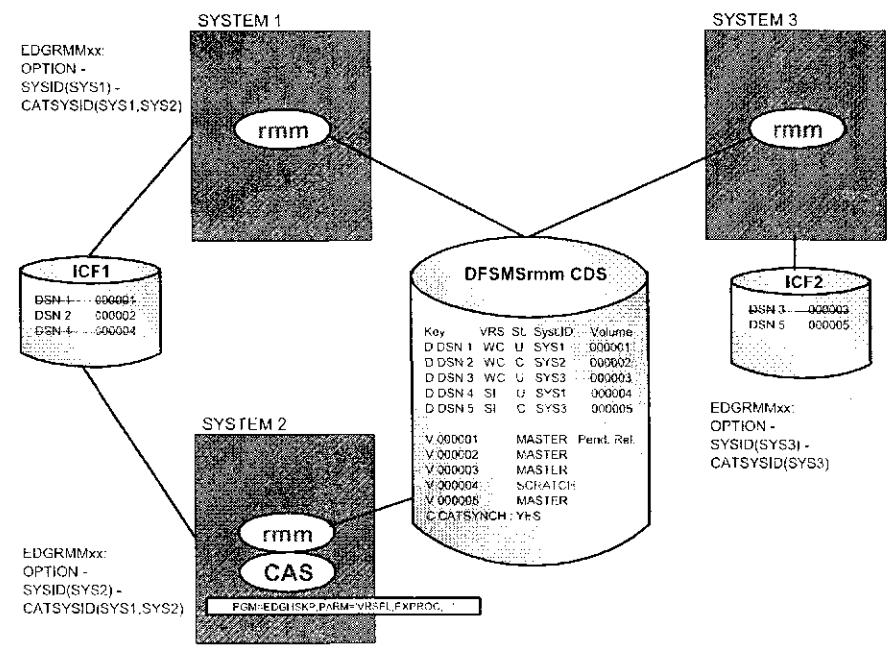


Figure 6-15. DFSMSrmm Housekeeping on SYS2 (2 of 2)

SS212.0

Notes:

Volume V00004 is released because its expiration date has been reached and the vital record specification no longer protects the data set. DFSMSrmm sends a request to uncatalog the data set to CAS and changes the catalog status in the data set record of DSN 4 to U (uncataloged), because of the UNCATALOG option being set to Y, which is the default in protect mode. The shortcut *SI* in the column with headline *VRS* the data set record means *SCRATCHIMMEDIATE*.

Volume V00005 remains unchanged although its expiration date has also been reached or exceeded. EXPROC doesn't affect this volume, because the creating system ID in the first data set on volume V00005 (DSN 5) doesn't match any of the IDs specified with *OPTION CATSYSID(...)*.

DFSMSrmm Housekeeping on SYS3

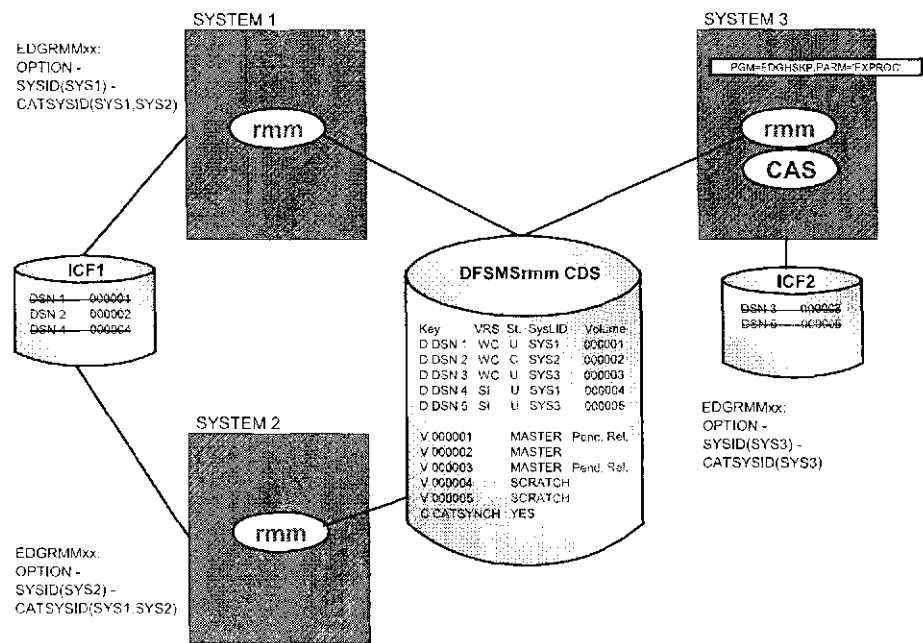


Figure 6-16. DFSMSrmm Housekeeping on SYS3

SS212.0

Notes:

In order to release volumes containing data sets with a creating system ID other than SYS1 or SYS2, you also have to run EXPROC on system SYS3.

The availability of volume V00003 is set to PENDING RELEASE, since it is no longer retained by VRS.

While volume V00005 is no longer retained by a VRS and the release option set indicates scratch immediate, the volume status changes to SCRATCH.

Tape DS Deletion while RMM Being Stopped

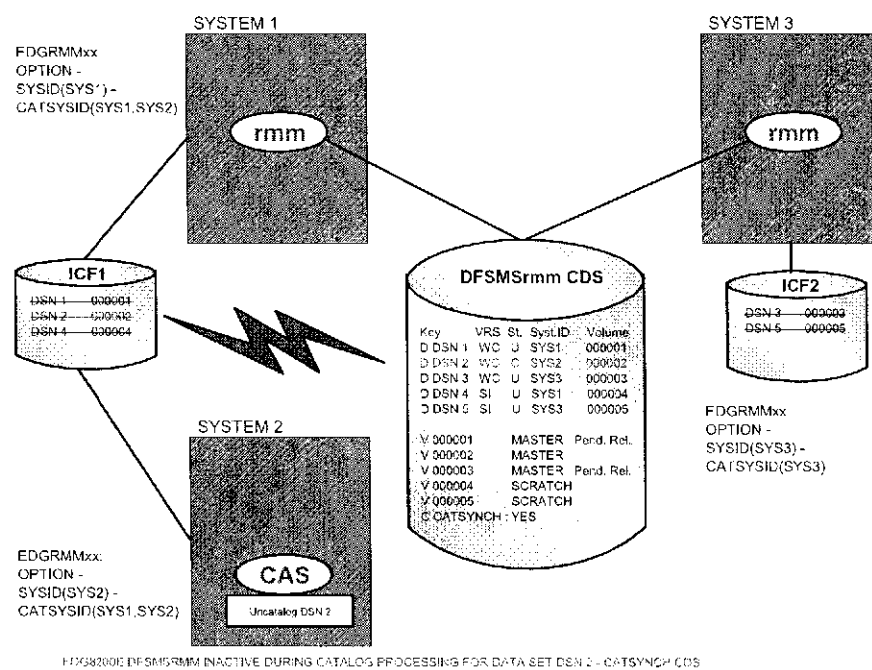


Figure 6-17. Tape DS Deletion while RMM Being Stopped

SS212.0

Notes:

In the unlikely event that DFSMSrmm cannot track catalog updates, messages EDG8200E and EDG8201E are issued. Here are the steps to take in response:

1. Use EDGUTIL with PARM=UPDATE and specify CATSYNCH(NO) through the SYSIN DD card. This prevents DFSMSrmm from relying on the catalog information in the control data set.
2. Resynchronize the DFSMSrmm CDS and catalogs.

Remember that housekeeping will be impossible as long as CATSYNCH(NO) is active, unless you remove OPTION CATSYSID(...) from EDGRMMxx members.

6.2 Maintaining the Control Data Set

Reorganizing the DFSMSrmm CDS

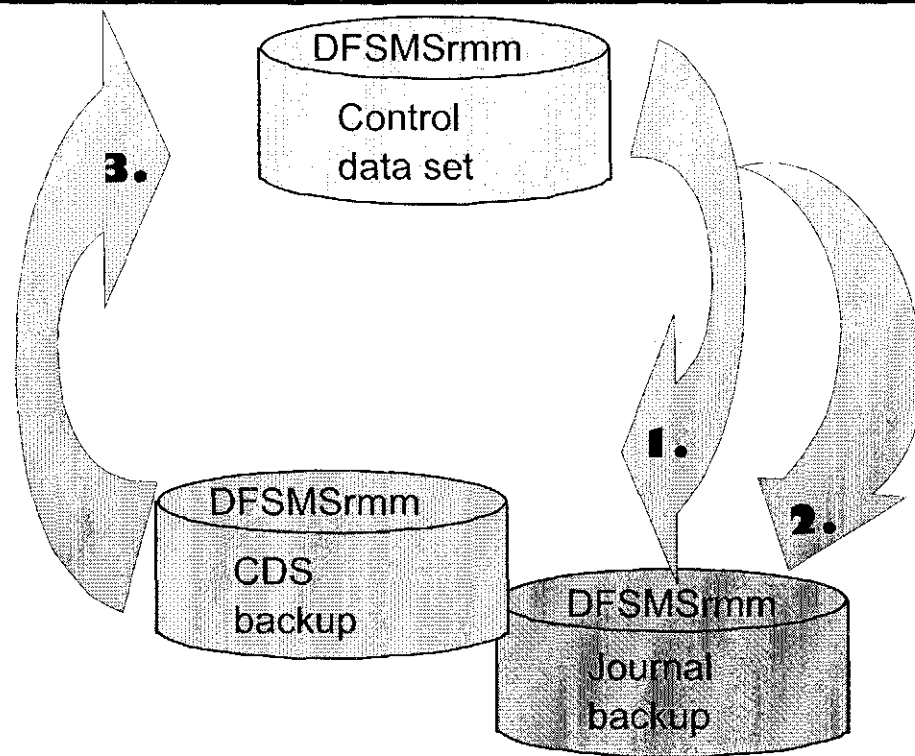


Figure 6-18. Reorganizing the DFSMSrmm CDS

SS212.0

Notes:

Among the uses to which the DFSMSrmm utility EDGBKUP is put is the reorganization of the control data set. The following is a JCL model:

```
//RSTR      EXEC PGM=EDGBKUP,PARM='BACKUP(REORG) '
//SYSPRINT DD  utility message data set
//BACKUP    DD  backup data set
//MASTER    DD  control data set
//JOURNAL    DD  journal data set
//JRNLBKUP  DD  journal backup data set
```

Security and Audit Reports

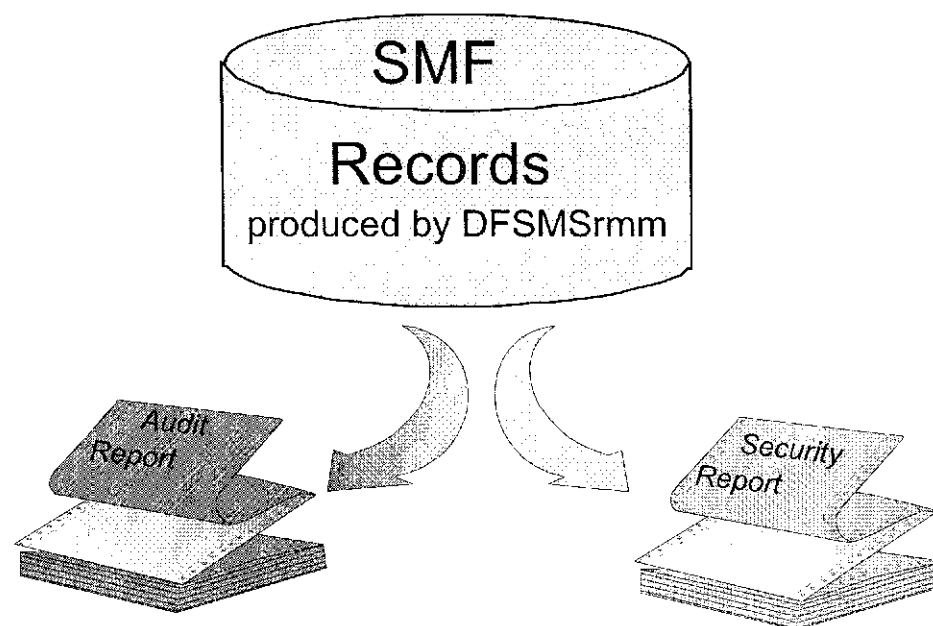


Figure 6-19. Security and Audit Reports

SS212.0

Notes:

If you have specified user SMF record numbers for control data set updates and/or classified data access, you can use the EDGAUD utility to produce audit trail reports:

```
//AUDREPT EXEC PGM=EDGAUD,
//      PARM='SMFAUD(nnn),SMFSEC(nnn),SEC(''security classification'')'
//SYSPRINT DD program messages
//SMFIN DD input data set of SMF records
//AUDREPT DD audit report
//SECREPT DD security report
//SYSOUT DD DFSORT messages
//SYSIN DD select statements for audit report
```

Refer to *Creating Security and Audit Reports: EDGAUD* in the *DFSMSrmm Implementation and Customization Guide*, SC26-7405, for further details.

Restoring the Control Data Set

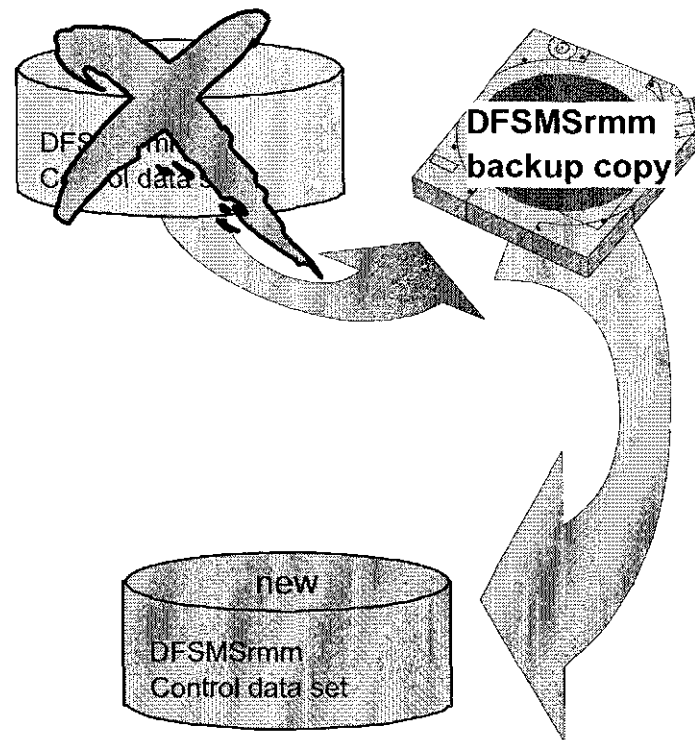


Figure 6-20. Restoring the Control Data Set

SS212.0

Notes:

The following JCL example restores a control data set from the most recent backup copy without forward recovery:

```
//EDGBKUP EXEC PGM=EDGBKUP,PARM='RESTORE'
//SYSPRINT DD SYSOUT=*
//MASTER DD DISP=SHR,DSN=RMM.CDS
//*
//BACKUP DD DISP=SHR,DSN=BACKUP.CDS(0)
/*
```

Forward Recovering the Control Data Set

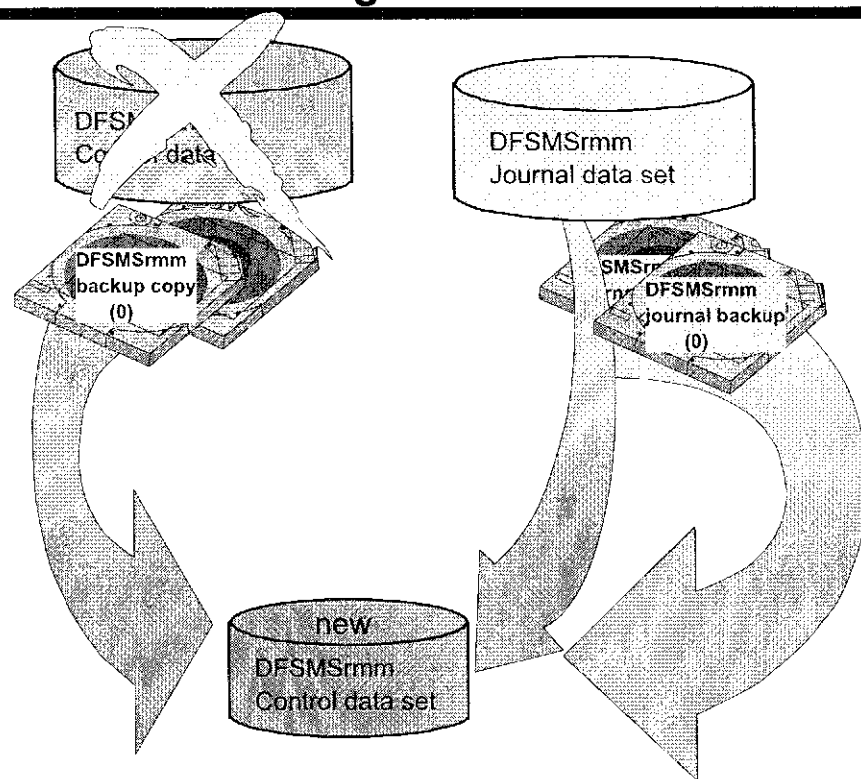


Figure 6-21. Forward Recovering the Control Data Set

SS212.0

Notes:

To recover a control data set and perform forward recovery, a journal data set is needed. In the following code example, DFSMSdss was used to back up the control data set and journal and the CONCURRENT option was used. In this case, two journals are required, the latest backup and the current journal.

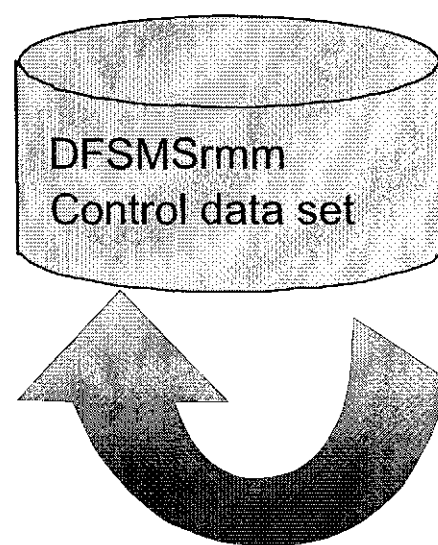
```
//EDGBKUP EXEC PGM=EDGBKUP, PARM='RESTORE'
//SYSPRINT DD SYSOUT=*
//MASTER DD DISP=SHR, DSN=RMM.CDS
//*
//BACKUP DD DISP=SHR, DSN=BACKUP.CDS(0)
//*
//JOURNAL DD DISP=SHR, DSN=BACKUP.JRNL(0)
// DD DISP=SHR, DSN=RMM.JOURNAL
//*
```

You can use the JOURNAL DD statement to concatenate multiple journal data sets. You can use a restored or nonrestored control data set, if automatic forward recovery failed, or

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if you have already restored the control data set but have not used journal data for forward recovery. The DFSMSrmm subsystem must be stopped or quiesced during forward recovery of the active control data set.

Verifying the Control Data Set



```
//STEP1  PGM=EDGUTIL,
//        PARM='VERIFY (ALL) '
//SYSPRINT DD SYSOUT=*
//MASTER DD DISP=SHR,
//        DSN=RMM.MASTER.DSET
```

Figure 6-22. Verifying the Control Data Set

SS212.0

Notes:

The DFSMSrmm utility EDGUTIL can be used to check the contents of the control data set. For VERIFY processing, EDGUTIL reads sequentially through the different record types in the control data set. The record types are identified by the VERIFY options you specify. For each record DFSMSrmm validates key fields and cross-checks information with related records in the control data set. If any discrepancies are identified an informational message is issued to the SYSPRINT file. The **ALL** option in our example includes DSN, OWNER, PP, RACK, STORE, VOLCAT, VOLUME, and VRS. These can be used individually if desired.

Mending the Control Data Set

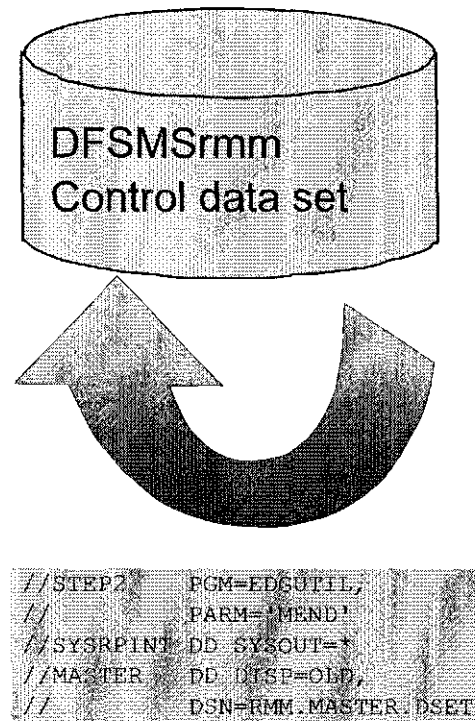


Figure 6-23. Mending the Control Data Set

SS212.0

Notes:

The **MEND** function should be used with caution. The following is taken from the *DFSMSrmm Implementation and Customization Guide*, SC26-7405:

Recommendations:

- Always run the MEND function on a VSAM copy of the control data set first. Taking a back up of the control data set is essential because the control data set is unusable if MEND processing fails for any reason.
- Do not run the MEND function when DFSMSrmm is running on the same system and is using a control data set with the same name as the control data set that the MEND function is expected to correct.
- Do not enable stacked volume support until all systems using the control data set are on a supporting release level.

You can use EDGUTIL VERIFY as described in "Verifying the Contents of the Control Data Set" on page 325 to find most control data set errors. Obtain guidance from IBM service if you are using EDGUTIL MEND to fix the errors found in the DFSMSrmm control data set.

When you use the EDGUTIL MEND function to fix errors for system-managed tape volumes, MEND updates the DFSMSrmm control data set from information from the TCDB and the Library Manager database.

EDGUTIL VERIFY DSN

DFSMSrmm validates data set information.

Files must be in sequence without gaps.

The associated volume record must be available.

Volume records of the previous and the next volume must be available.

The owner record must be available.

Notes:

When DSN is verified, the checks listed are made.

EDGUTIL VERIFY Owner

DFSMSrmm validates owner information.

All volumes stored in the owner records must be available.

Figure 6-25. EDGUTIL VERIFY Owner

SS212.0

Notes:

When an OWNER record is processed by VERIFY, it makes sure all the volume records associated with the owner are present.

EDGUTIL VERIFY PP

DFSMSrmm validates product information.

All volumes associated to a product must be available.

The owner record must be available.

Notes:

This is the VERIFY processing for program product records.

EDGUTIL VERIFY Rack

DFSMSrmm validates rack number information.

The number of racks must be the same as the number of racks that is stored in the DFSMSrmm control record.

If the status of the rack is *empty*, no volume can be assigned to this rack.

If the status of the rack is *in use*, the volume record of the associated volume must be available.

Figure 6-27. EDGUTIL VERIFY Rack

SS212.0

Notes:

The verification of a rack record is as listed.

EDGUTIL VERIFY Store

DFSMSrmm validates the bin number information.

The total number of bins must be the same as the number of bins that is stored in the DFSMSrmm control record (only for built-in storage locations).

If the status of the bin is *empty*, no volume can be assigned to this bin number.

If the status of the bin is *in use*, the bin number stored in the volume record must match the volume serial number that is associated with the bin.

Figure 6-28. EDGUTIL VERIFY Store

SS212.0

Notes:

For a storage location with bin management, VERIFY checks bin numbers against volume records.

6.3 Initializing and Erasing Volumes

Initializing and Erasing Volumes

EDGINERS can run in automatic processing mode.

```
//INITAUTO PGM=EDGINERS,  
//          PARM='MEDIANAME(3480),VERIFY'  
//SYSPRINT DD SYSOUT=*  
//TAPE      DD UNIT=(TAPE,,DEFER)
```

Exec parameters:
BATCH(number_of_batches)
COUNT(count)
INITIALIZE/ERASE
MEDIANAME/POOL/LOCATION(value)
VERIFY/NOVERIFY
WRONGLABEL(FAIL/IGNORE/PROMPT/RMMPROMPT)

Figure 6-29. Initializing and Erasing Volumes

SS212.0

Notes:

The DFSMSrmm utility, EDGINERS, should be used to initialize and erase volumes defined to DFSMSrmm. EDGINERS will run in automatic or manual mode. In automatic mode, DFSMSrmm uses the PENDING actions indicated in the volume records. Volumes can be selected by media name, pool, or location (such as a system-managed tape library). DFSMSrmm will initialize up to 99 volumes in one batch. Multiple batches can be specified.

EDGINERS can also be used to ERASE and reinitialize tapes. DFSMSrmm erases volumes by using the hardware security erase feature, where supported. When erasing volumes, DFSMSrmm also reinitializes them so that the correct volume labels are written and the volumes are ready for re-use. If the hardware security erase feature is not available, DFSMSrmm overwrites volumes with a bit pattern, hex FF.

Initializing and Erasing Volumes

EDGINERS can run in manual processing mode.

```
//INITMAN  PGM=EDGINERS
//SYSPRINT DD  SYSOUT=*
//TAPE      DD  UNIT=(TAPE,,DEFER)
//SYSIN     DD  *
```

Sysin commands:

```
INIT/ERASE
VOLUME(volser,new_volser)
LABEL(SL,AL,NL)
MEDIANAME(media_name)
RACK/POOL(value)
OWNERTEXT
ACCESS(ccode)
```

Figure 6-30. Initializing and Erasing Volumes

SS212.0

Notes:

EDGINERS can also be used in manual mode. In this case, a SYSIN statement is used to supply control statements. For example, to re-label a tape you would code the following:

```
//INIT      EXEC PGM=EDGINERS
//SYSPRINT  DD  SYSOUT=*
//TAPE      DD  UNIT=(tape,,DEFER)
//SYSIN     DD  *

INIT VOLUME(CURR01,NEW001) LABEL(SL)
```


6.4 Authorizing DFSMSrmm Users and Ensuring Security

OWNER Information

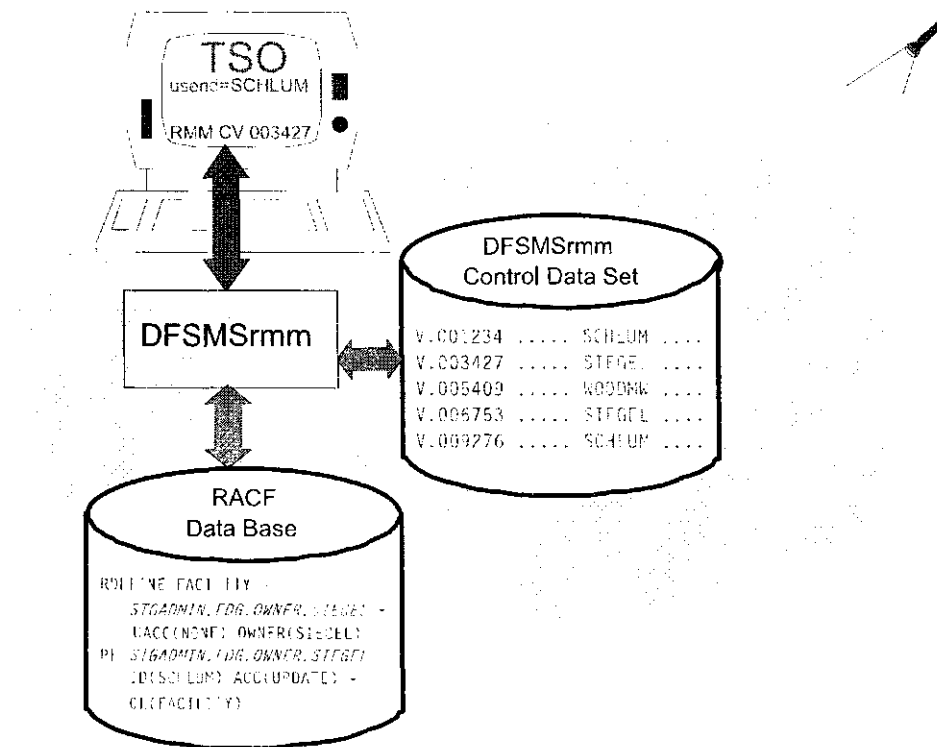


Figure 6-31. OWNER Information

SS212.0

Notes:

DFSMSrmm relies on the security interface to determine what actions a user can perform. As mentioned earlier, the primary control is through the STGADMIN.EDG.MASTER facility class profile. Update access to this profile allows owners to change many fields in their owned volumes.

Remember, that anyone who has CONTROL access to STGADMIN.EDG.MASTER can perform actions on any volume managed by DFSMSrmm.

DFSMSrmm records owner information for a non-scratch volume. When a volume is defined to DFSMSrmm as other than a scratch volume, the command used has an owner option. If an owner is not provided, the volume is owned by the user adding it into the CDS. DFSMSrmm automatically records the owner for volumes going from scratch to master status. The user ID that RACF returns becomes the owner unless RACF returns an asterisk, in which case the owner is the job name. If necessary, DFSMSrmm creates an owner record.

General users can use DFSMSrmm to:

- Manually request a scratch volume
- Change information about an owned volume or data set
- Update information about their owner ID
- Manually release an owned volume
- Create lists of resources and display information about most resources defined to DFSMSrmm

Access to another user's volumes can be gained by defining a RACF owner profile STGADMIN.EDG.OWNER.ownerid. Those granted update access to the profile may then perform general user functions on the volumes that are owned by user *ownerid*.

Security of Data on Tape

Security for access to data sets on tape is basically the business of DFSMSdfp.

Related RACF settings

SETROPTS TAPEDSN

Enables a RACHECK at open of any tape data set by checking of data set profiles for tape data sets like data sets on DASD.

Class TAPEVOL active

This provides protection at the tape volume level.

Discrete or generic profiles set up the rules for accessing data on the volume.

You have to define profiles in class TAPEVOL.

Both kinds of protection mechanism may be used individually or together.

The best protection is achieved by using both controls together.

DFSMSrmm offers an interface to RACF. Use the TPRACF option in EDGRMMxx with the following settings:

- A Automatic definition and deletion of RACF profiles when volume status changes
- P Automatic definition and deletion of RACF profiles when volume status changes
- N Interface is disabled

Notes:

To avoid manual administration of profiles in RACF class TAPEVOL, you can tell DFSMSrmm to automatically manage definition and deletion of such profiles at the right time.

The TPRACF option P tells DFSMSrmm to cover a tape volume even by a profile while the volume is in scratch status.

DFSMSrmm and Security of Data on Tape

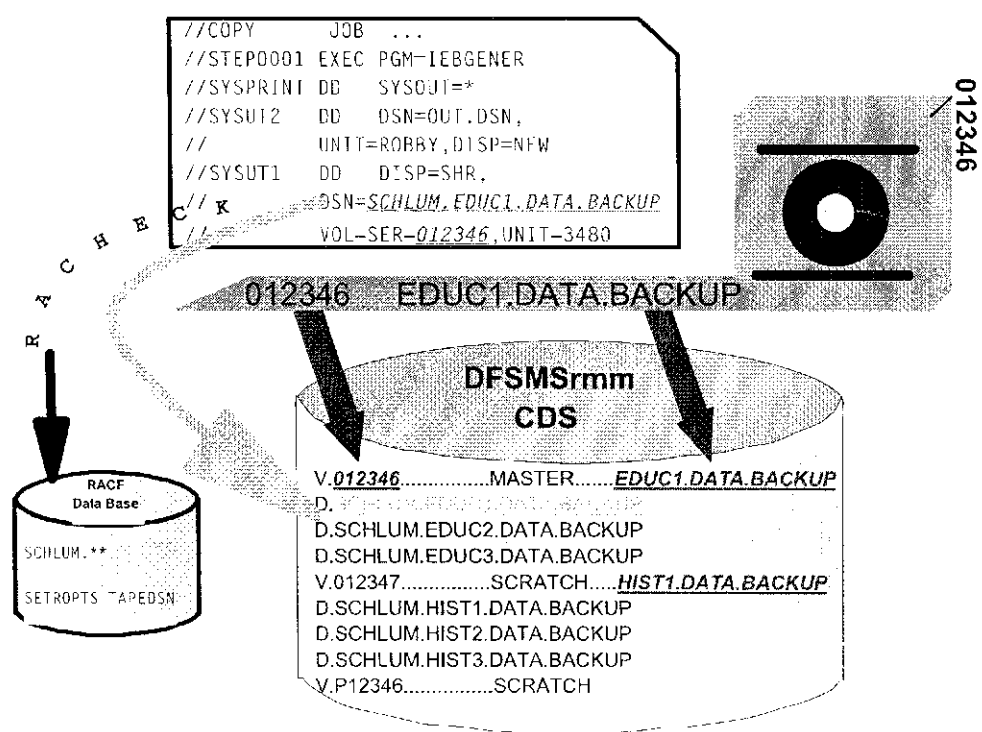


Figure 6-33. DFSMSrmm and Security of Data on Tape SS212.0

Notes:

Running in protect mode, DFSMSrmm validates the full 44 character data set name during open of an existing tape data set, thus making sure that accurate data set names must be used for RACHECK processing.

Could We Use a Different Name During Open?

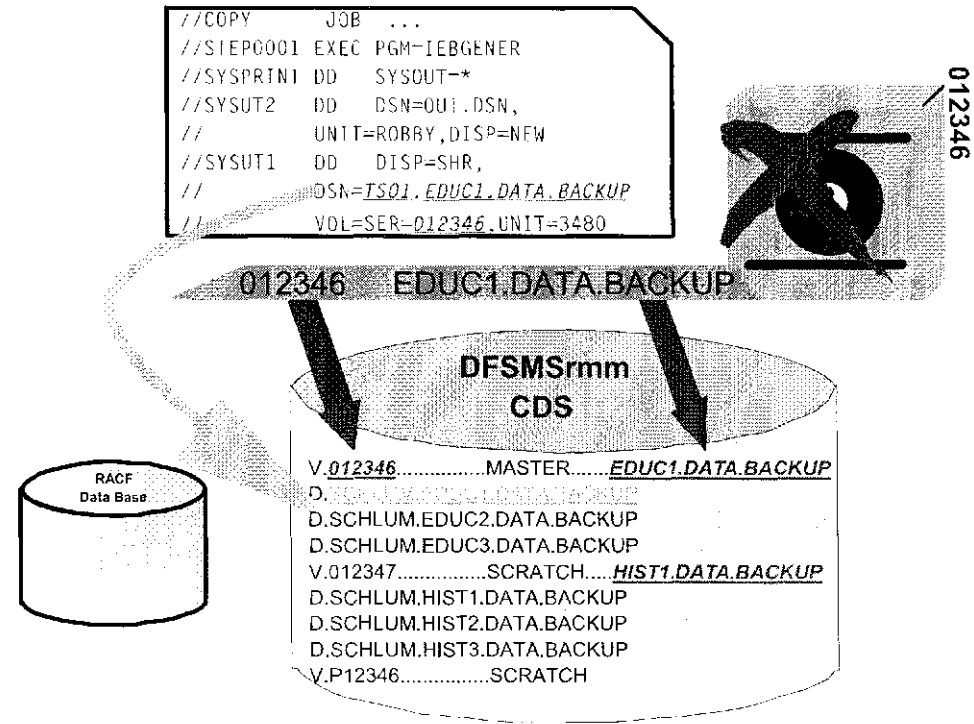


Figure 6-34. Could We Use a Different Name During Open?

SS212.0

Notes:

If a user tries to use a different name during opening of an existing data set than the name that was recorded by DFSMSrmm when the data set was created, DFSMSrmm will reject the volume if it runs in protect mode.

Possible Security Exposure?

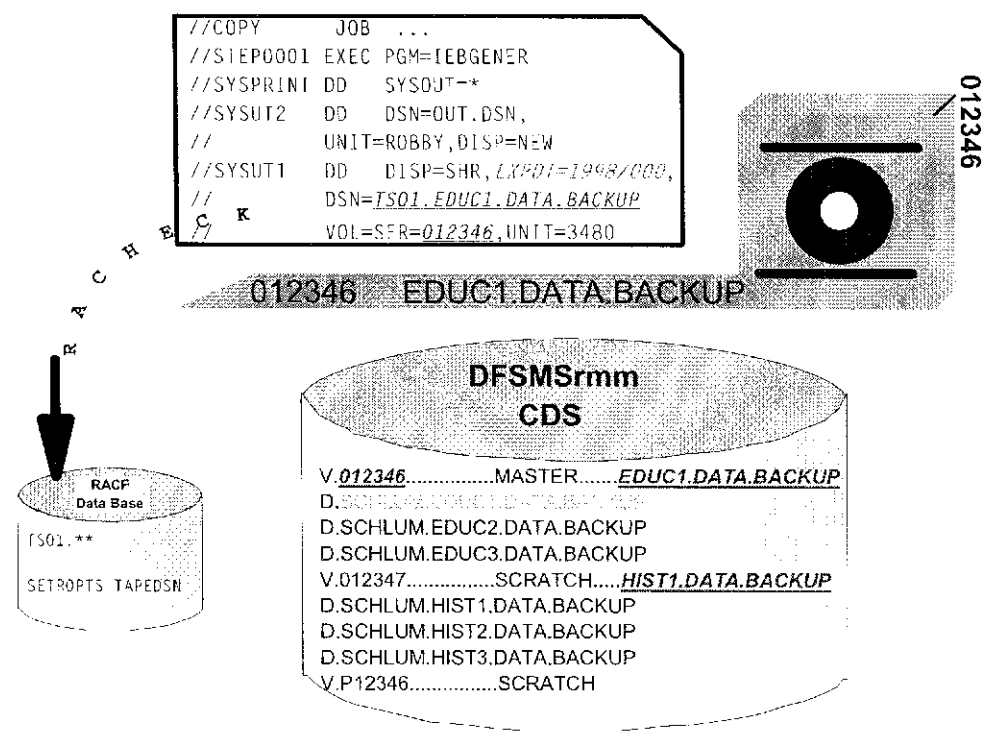


Figure 6-35. Possible Security Exposure? SS212.0

Notes:

While volumes are not covered by profiles in the RACF TAPEVOL class, a user might prefix the data set name with his user ID to gain access to the data set. If DFSMSrmm runs not in protect mode or if it is bypassed, the only verification that DFSMSdfp can perform is the match with the last 17 characters of the data set name as specified in the header on the tape.

Recommendation for RACF Settings

Regarding the security of data on tape, DFSMSrmm supports any RACF settings.

For the best protection of tape data (and to avoid a data security exposure like the one we showed on the last visual) we recommend:

Class TAPEVOL active

and

SETROPTS TAPEDSN

Figure 6-36. Recommendation for RACF Settings

SS212.0

Notes:

To provide maximum security for data on tape, you should follow the recommendations that are shown in the visual.

Better Protection with TVTOC in TAPEVOL

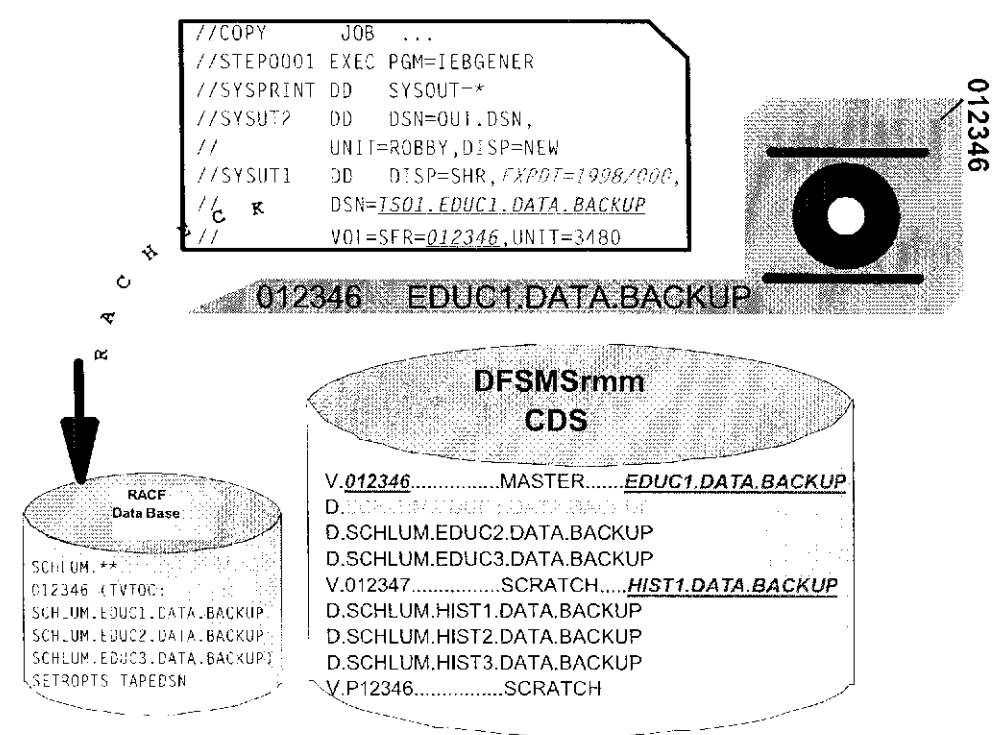


Figure 6-37. Better Protection with TVTOC in TAPEVOL SS212.0

Notes:

Volumes covered by profiles in the RACF TAPEVOL class are better protected if the profiles include a tape volume table of contents (TVTOC). When DFSMSrmm verification is bypassed during open processing, DFSMSdftp can verify the data set name through the use of the TVTOC information.

Life Cycle of a Tape Volume

The DFSMSrmm RACF interface is active at:

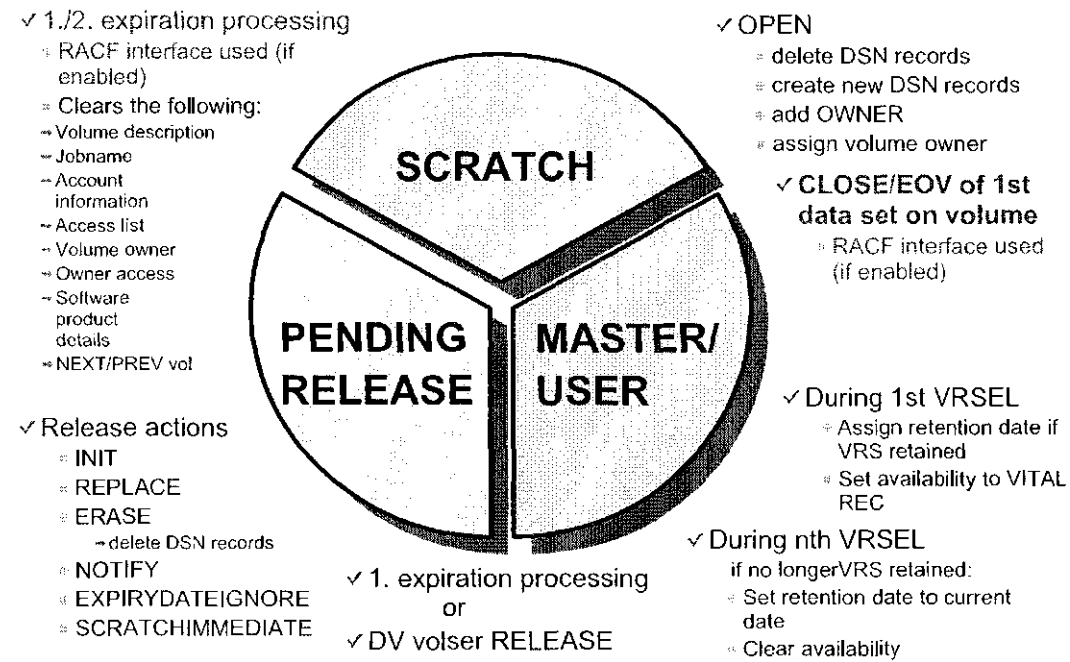


Figure 6-38. Life Cycle of a Tape Volume

SS212.0

Notes:

DFSMSrmm interfaces with RACF during CLOSE or EOVS processing of volumes and during expiration processing.

EDGRMMxx OPTION TPRACF

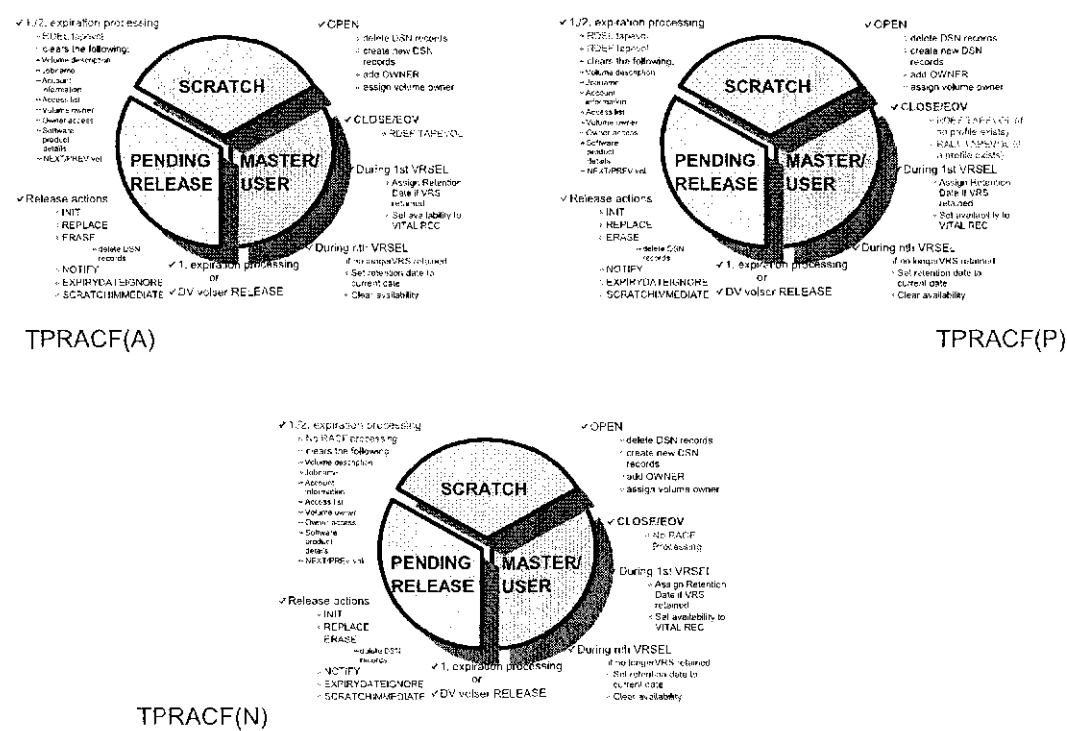


Figure 6-39. EDGRMMxx OPTION TPRACF

SS212.0

Notes:

While running with the TPRACF(P) option, DFSMSrmm covers tape volumes with profiles in RACF class TAPEVOL even while they are in scratch status. Such profiles do not protect the tape volumes against being written, thus anybody can use a scratch volume to write data. This is the only difference between the TPRACF(P) and TPRACF(A) options.

TPRACF and RACF Settings

Protection is at the level of	TAPEDSN only	TAPEVOL active only	TAPEVOL active and TAPEDSN	NOTAPEDSN and TAPEVOL not active
TPRACF(A)	data sets	volumes	data sets and volumes	no protection
TPRACF(P)	data sets	volumes	data sets and volumes	no protection
TPRACF(N)	data sets	volumes	data sets and volumes	no protection

DFSMSrmm creates and deletes profiles	TAPEDSN only	TAPEVOL active only	TAPEVOL active and TAPEDSN	NOTAPEDSN and TAPEVOL not active
TPRACF(A)	deletes for data sets	creates and deletes for volumes	creates and deletes for volumes	
TPRACF(P)	deletes for data sets	creates and deletes for volumes	creates and deletes for volumes	
TPRACF(N)				

Figure 6-40. TPRACF and RACF Settings

SS212.0

Notes:

The tables summarize the different levels of protection that are gained by the various combinations of RACF and DFSMSrmm settings as well as the actions DFSMSrmm provides in a particular environment.

DFSMSrmm Commands and TPRACF Settings

If class TAPEVOL is active, DFSMSrmm ADDVOLUME command will do the following:

- TPRACF(A/P)
 - Create TAPEVOL profile
 - Build access list using the OWNER, USER and ACCESS information from volume record

- TPRACF(N)
 - No action

If TAPEDSN is set, DFSMSrmm DELETEDATASET will do the following:

- TPRACF(A/P)
 - Delete tape data set profile

- TPRACF(N)
 - No action

Figure 6-41. DFSMSrmm Commands and TPRACF Settings

SS212.0

Notes:

For a complete list of actions see *z/OS DFSMSrmm Implementation and Customization Guide*.

DFSMSHsm Tape Security

You can combine DFSMSHsm SETSYS TAPESECURITY(RACF) with both EDGRMMxx options TPRACF(P) or OPTION TPRACF(A).

No conflicts because DFSMSHsm always acts as though it were closer to RACF than DFSMSrmm.

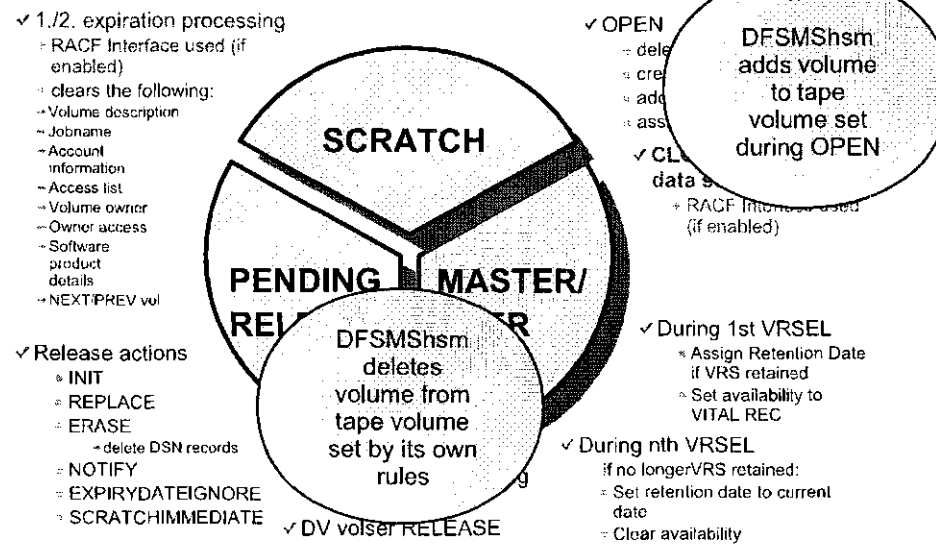


Figure 6-42. DFSMSHsm Tape Security

SS212.0

Notes:

For details of using SETSYS TAPESECURITY(RACF) see *z/OS DFSMSHsm Implementation and Customization Guide, SC35-0418*.

Checkpoint

Indicate whether each of the following statements is true or false.

1. When you have correct level of DFSMSrmm code installed, DFSMSrmm dynamically records all catalog updates for tape data sets in the control data set.
2. When you run DFSMSrmm with user catalogs and the DFSMSrmm control data set synchronized, DFSMSrmm does not need to issue catalog locates during inventory management to find out if a tape data set is cataloged. This is the default setting in EDGRMMxx if all catalogs are shared.
3. Before you can recover the DFSMSrmm control data set, DFSMSrmm must be stopped and the subsystem interface reset.
4. The DFSMSrmm utility EDGINERS always must be used in automatic mode to initialize a tape that is managed by DFSMSrmm.
5. DFSMSrmm checks a user's authority to process a data set based on the full data set name it recorded when the data set was created.

Figure 6-43. Checkpoint

SS212.0

Notes:

Write down your answers here:

- 1.
- 2.
- 3.
- 4.
- 5.

Unit Summary

Now that you have completed this unit, you should be able to:

- Identify the mechanism whereby catalog status is tracked
- Modify the EDGRMMxx parmlib member to exploit catalog synchronization
- Describe the interaction between CATALOG and DFSMSrmm address space during inventory management
- Maintain the DFSMSrmm control data set
- Use EDGINERS for initialization and erasure of tape volumes
- Explain the role DFSMSrmm plays in providing tape-data-set security

Notes:

Unit 7. Tape Volume Pools and Tape Libraries

What This Unit Is About

This unit describes DFSMSrmm support for tape volume pools and tape libraries.

What You Should Be Able to Do

After completing this unit, you should be able to:

- Define and use DFSMSrmm tape volume pools
- Understand the customization needs in DFSMSrmm if you use tape libraries
- Use support for stacked volumes in a VTS
- Use EDGUTIL to verify and synchronize the contents of the DFSMSrmm CDS and the TCDB and the Library Manager data base

How You Will Check Your Progress

Accountability: Checkpoint questions

References

SC26-7404	<i>DFSMSrmm Guide and Reference</i>
SC26-7405	<i>DFSMSrmm Implementation and Customization Guide</i>

Objectives

After completing this unit, you should be able to:

- Use pools of tape volumes
- Understand DFSMSrmm processing in an environment with tape libraries
- Identify the customization need for library partitioning
- Use support for stacked volumes
- Use EDGUTIL to verify and synchronize the contents of the DFSMSrmm CDS, the TCDB, and the Library Manager data base

Notes:

7.1 DFSMSrmm Support for Pools of Tape Volumes

Libraries and Locations

DFSMSrmm can manage

- **System-managed tape libraries as part of the removable media library**
 - IBM TotalStorage Enterprise Automated Tape Library
 - IBM TotalStorage Virtual Tape Server
- **Non-system-managed tape libraries**
 - Traditional tape libraries
 - Other robots

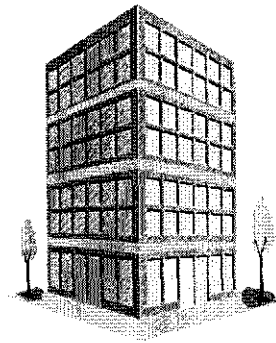


Figure 7-2. Libraries and Locations

SS212.0

Notes:

DFSMSrmm automatically manages tape volumes and data sets. A volume can reside in an IBM system-managed library or in traditional (manual) tape libraries. As mentioned earlier, each IBM system-managed library is known to DFSMSrmm as a storage location. Volumes in the computing center, but not those in an IBM system-managed library, are considered SHELF volumes. To DFSMSrmm, robotic libraries that are not IBM system-managed appear as shelf managed volumes. This includes IBM 3494 TotalStorage Enterprise Automated Tape Libraries using Basic Tape Library Support (BTLS) as well as robot systems of other manufacturers.

Libraries and Locations (Cont)

Storage locations can be

- LOCAL
- DISTANT
- REMOTE
- Installation_defined_location_name

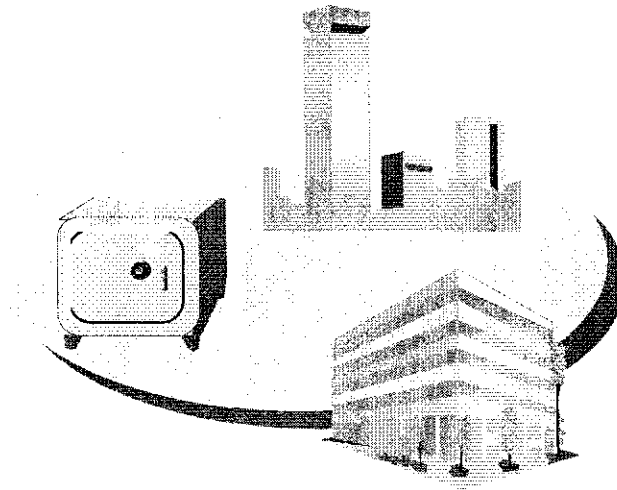


Figure 7-3. Libraries and Locations

SS212.0

Notes:

Tape can also be stored elsewhere than in the computing center. DFSMSrmm has three predefined locations: LOCAL, DISTANT, and REMOTE. Recall that you can define your own location names and management criteria by using the LOCDEF statement in your EDGRMMxx parmlib member. LOCDEF also supports to use a storage location as home location, which offers to use an installation defined location as home location for volumes in a library of other manufacturers.

Organizing the Removable Media Library

Organizing the library by pools

Systems or subset of systems

MVS or VM usage

Using rack pools to hold foreign tapes

Media names your installation defines,
like 3480, TAPE, MINIREEL, CASSETTE

Defining pools in parmlib member EDGRMMxx by the
VLPOOL command

Default VLPOOL command

VLPOOL PREFIX(*) RACF(N) TYPE(S) -
EXPDTCHECK(O) MEDIANAME(3480) -
DESCRIPTION('DEFAULT POOL')

Figure 7-4. Organizing the Removable Media Library

SS212.0

Notes:

For shelf volumes, you can organize your library by a variety of means. By using the VLPOOL statements, you can set up both rack and scratch pools. In addition, you can use the REJECT statement to restrict the use of volumes on selected systems. If there are no VLPOOL statements in your parmlib member, all volumes added to DFSMSrmm will belong to the default pool as shown.

EDGRMMxx VLPOOL

DESCRIPTION	Describes the pool (between 1 and 32 characters)	
EXPDTCHECK	N	DFSMSrmm does not check or validate the expiration date of data sets on a tape
	Y	DFSMSrmm ensures that all unexpired tape data sets are not overwritten
	O	DFSMSrmm takes no action
MEDIANAME	Name or type of media for all volumes in this pool	
NAME	Pool name that DFSMSrmm uses to update operator messages and tape drive displays for non-specific tape mount requests (can be a 1 to 8 character name, but the first character must be an alphanumeric or national character)	

Figure 7-5. EDGRMMxx VLPOOL

SS212.0

Notes:

Each VLPOOL command defines a pool. Each generic shelf location defined by PREFIX must be unique. If you do not specify any VLPOOL commands, or do not specify a VLPOOL command for PREFIX(*), DFSMSrmm automatically creates a default pool definition that includes all the shelf locations in them. The defaults for this pool are:

```
VLPOOL RACF(N) TYPE(S) EXPDTCHECK(O) MEDIANAME(3480) -
DESCRIPTION('DEFAULT POOL') PREFIX(*)
```

IBM recommends that you use EXPDTCHECK(N) when running a DFSMSrmm-managed scratch tape pool. DFSMSrmm lets you re-use tapes only when it is certain the information on them is no longer required. DFSMSrmm can easily automate the re-use of tapes if they are still expiration protected—for example, when they are re-used as scratch tapes.

The MEDIANAME need not be an esoteric name. It is meant to describe shelf locations for a unique media type, not a hardware tape drive.

NAME must meet the storage group name if pooling control is done by storage group assignment. Name is also used on tape drive displays for scratch mount.

EDGRMMxx VLPOOL (Cont.)

PREFIX		Specifies a generic shelf location that is used in operator messages, TSO subcommands, and the DFSMSrmm ISPF dialog.
RACF	Y	You want DFSMSrmm to create RACF tape profiles
	N	You do not want DFSMSrmm to create tape profiles
SYSID	ID	Associates the scratch pool you are defining with a particular system.
TYPE	R	A rack pool
	S	A scratch pool

Figure 7-6. EDGRMMxx VLPOOL (Cont.) SS212.0

Notes:

The prefix can be from one to five positions in length. DFSMSrmm supports two types of pools, scratch and rack. Non-specific requests cannot be satisfied by a scratch tape that is part of a rack pool. A scratch pool is used to satisfy these requests. The volume remains in the same DFSMSrmm scratch pool regardless of status (scratch, master, and so on), because it occupies always the same shelf space.

Using SYSID restricts a scratch pool to a particular system (the SYSID must match that specified in the OPTION statement). DFSMSrmm enforces a match of the SYSID when validating non-specific tape mounts. Only scratch volumes from a pool associated with a specific system can satisfy non-specific mount requests for that system. DFSMSrmm rejects volumes from other pools on that system.

If TPRACF is enabled (Y or P specified), the RACF(N) can be used to prevent DFSMSrmm from creating a profile in the TAPEVOL class.

EDGRMMxx MNTMSG

Defining mount and fetch messages

ID

1 - 128 start position of the message identifier

MSGID

message text 1 to 12 characters long

RACK

1 - 128 / 999 position to insert the RACK number

VOLUME

1 - 128 position of the volume serial number

Examples:

```

IEF501N M 4B4 SC0516 SL DFSM11 DFSM BEG BACKTAP2 DATASET - RACK=SC0516
IEF233A M 500 DP3RM1 NLTAP ML HUC001 - RACK=PP0021
IEF233A M 4B4 PRIVAT DFSM11 DFSM - POOL=NSM**
IEF233A M 4C3 PRIVAT DFSM11 DFSM - POOL=SCRATCH1

IEF502E K 4B5 SC1041 SL REASAVE ESABASE DSMV3 DATSTCH REBASE
RACK=SC1041
IEF234E K 4B4 SC0516 PVI DFSM11 DFSM - RACK=SC0516
  
```

Figure 7-7. EDGRMMxx MNTMSG

SS212.0

Notes:

MNTMSG causes DFSMSrmm to modify the message, add the rack number of a volume to the message if there is a specific mount request, and add the pool name or the pool prefix or the storage group name to the message if there is a scratch mount request.

DFSMSrmm needs to know the starting positions of the message ID and the volume serial number, the rack number and which message ID to intercept:

MNTMSG MSGID(IEF233A) ID(1) VOLUME(15) RACK(999)

Samples are provided in a sample parmlib member that forms part of the installation verification procedures (see member EDGIVPPM in SYS1.SAMPLIB).

RACK/VLPOOL/VOLSER

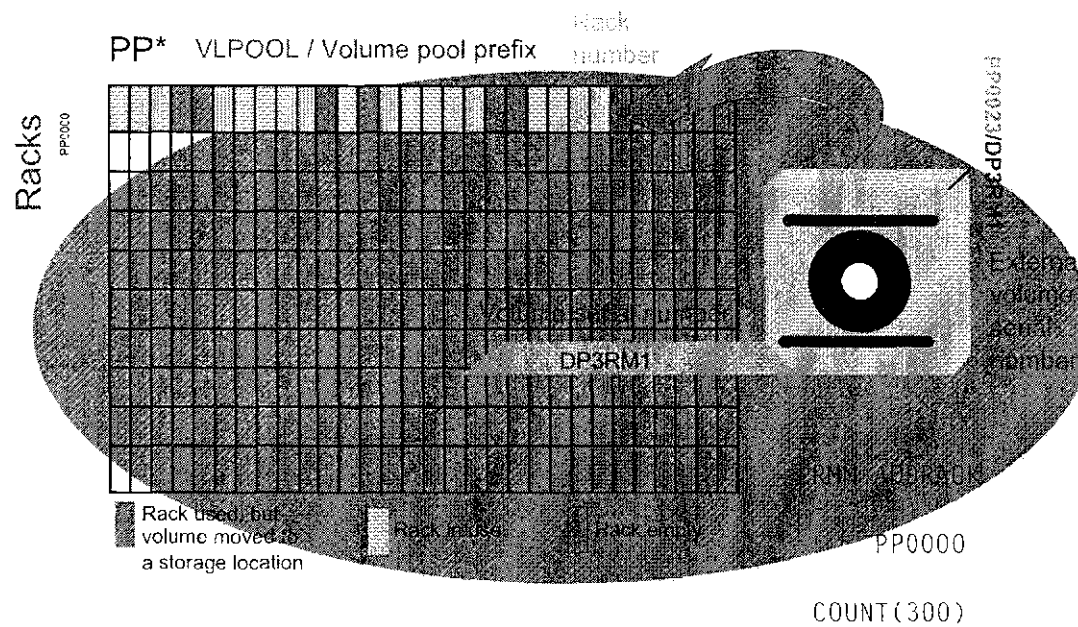


Figure 7-8. RACK/VLPOOL/VOLSER

SS212.0

Notes:

Tapes received for use on the system already have volume labels. These are normally used for reading, and they remain in master status. For example, a new release of a program product can be received on several tape volumes. To have DFSMSrmm manage these and not allow them to be mounted for scratch use (non-specific request for a new tape), create a rack pool via the VLPOOL definition for a series of shelf locations reserved for this type of situation, PP* in our example.

A series of actual slots (racks) need to be defined to DFSMSrmm to hold the volumes. When a tape is received, it can then be added into one of the shelf locations. Perform the following steps to define a rack pool and to add a new volume into rack location PP0023:

1. Add this statement in the EDGRMMxx member:

```
VLPOOL TYPE(R) MEDIANAME(CART) DESCRIPTION('PRIVATE POOL') -
PREFIX(PP*)
```

Note: Do not forget to choose your preferred RACF and EXPDTCHECK values. The media name CART is meant to signify any square, cartridge type tape volume.

2. Define 25 shelf locations that are in the *private pool*.

```
RMM ADDRACK PP0001 COUNT(25)
```

3. Define a new volume as part of the *private pool*.

```
RMM ADDVOLUME DP3RM1 RACK(PP0023) STATUS(MASTER) ...
```

There are many other parameters that should be specified such as OWNER and EXPDT.

To define a scratch pool where the rack ID match the volsers, you might have the following:

1. Add this statement in the EDGRMMxx member:

```
VLPOOL TYPE(S) MEDIANAME(ECCST) DESCRIPTION('ECCST SCRATCH POOL') -  
PREFIX(S*)
```

Note: Do not forget to choose your preferred RACF and EXPDTCHECK values. The media name ECCST is meant to signify the cartridges for 3490E, 36 Track recording.

2. This makes available 100 shelf locations, S00001 through S00100.

```
RMM ADDRACK S00001 COUNT(100)
```

Note: This definition is optional because DFSMSrmm does not depend on predefined racks for volumes in scratch pools.

3. Define 100 new scratch volumes.

```
RMM ADDVOLUME S00001 STATUS(SCRATCH) INIT(Y) COUNT(100) -  
LOCATION(SHELF) MEDIATYPE(ECCST)
```

Note: In this example, the tapes are not pre-initialized. We have indicated this with the INIT(Y) parameter. The DFSMSrmm utility EDGINERS is used to initialize these tapes. Virtual volumes in a VTS need no initialization by utility.

MEDIANAME

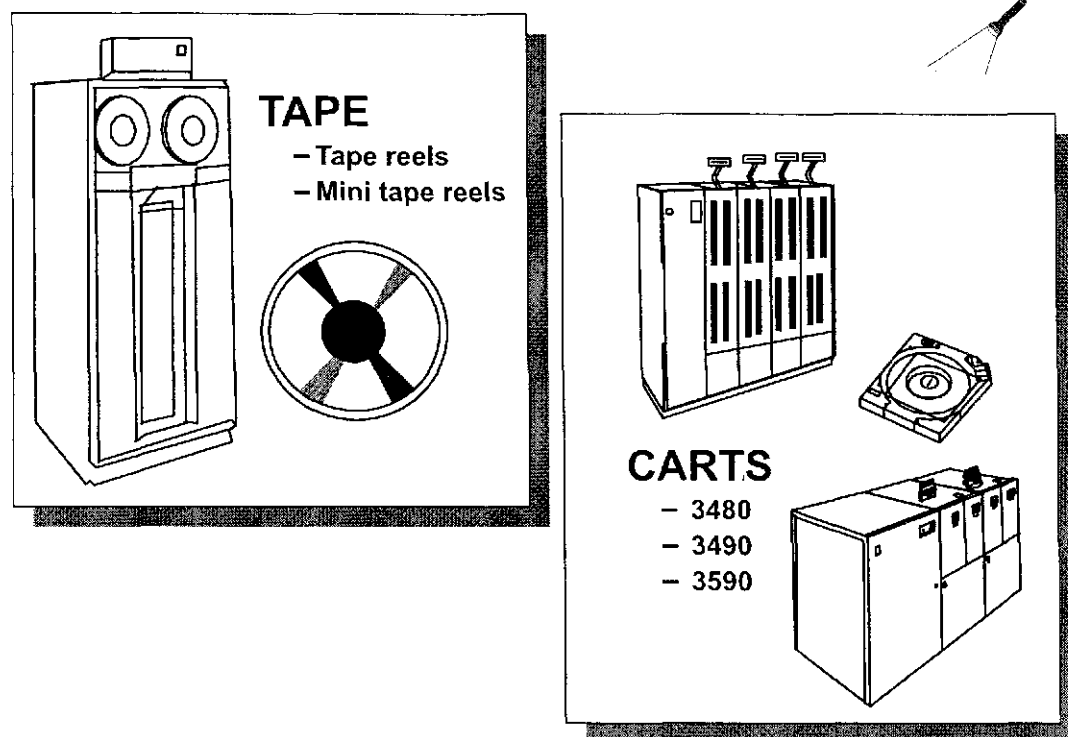


Figure 7-9. MEDIANAME

SS212.0

Notes:

DFSMSrmm media names are not related to MVS tape hardware names. You can specify any name because DFSMSrmm does not check if it is a device type that has been defined to MVS. To allow more flexibility regarding which media can reside in a pool, select media names that reflect shelving requirements such as size and shape. For example, you can specify CART, ROUND, SQUARE, 3490, 3590, TAPE, OPTICAL, CASSETTE, and so on.

MEDIANAME can be used as a filter criterion for EDGINERS and for vaulting. Normally, there is no need for specific controls by MEDIANAME. If this is the case, we recommend using the default, 3480.

To distinguish one media type from another at the volume level, the MEDIATYPE field in the volume record is automatically recorded to indicate the following:

CST	Cartridge system tape (MEDIA1)
ECCST	Enhanced capacity cartridge system tape (MEDIA2)
HPCT	High performance cartridge tape (MEDIA3)
EHPCT	Extended high performance cartridge tape (MEDIA4)

MEDIANAME (Cont)

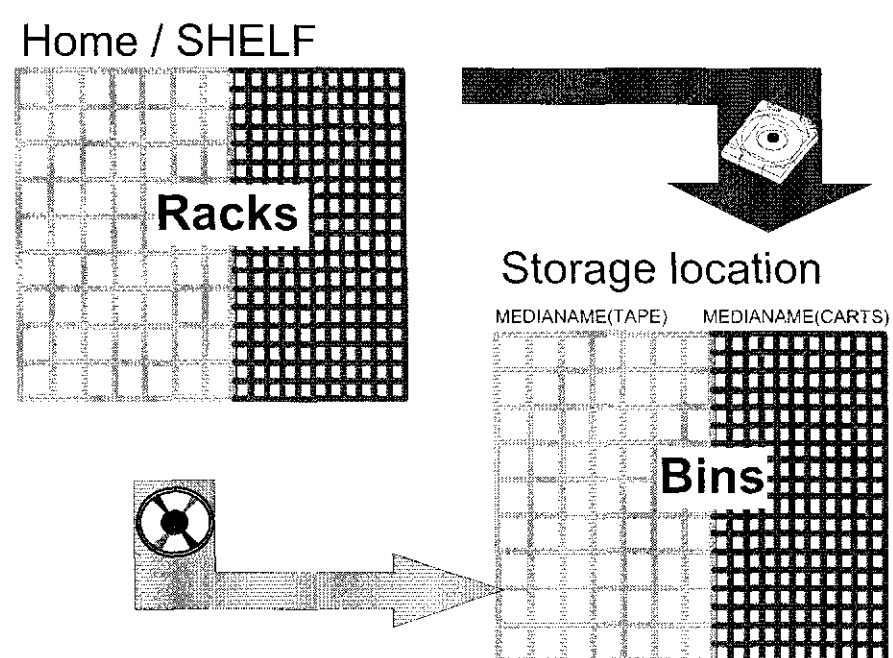


Figure 7-10. MEDIANAME (Cont.)

SS212.0

Notes:

The storage location that receives volumes from the computing center needs to be defined (LOCDEF) before it can accept the media name associated with the volume being moved. CARTS and TAPE (from our prior example) have different shelving requirements. A number of slots (bins) would likely be set aside for each media category at a storage location.

During inventory management DFSMSrmm checks the media name for the location and prevents volume movement when the media name does not match. DFSMSrmm issues message EDG2412E for each volume that cannot be moved because its media name is not supported at the location.

MEDIANAME (Cont)

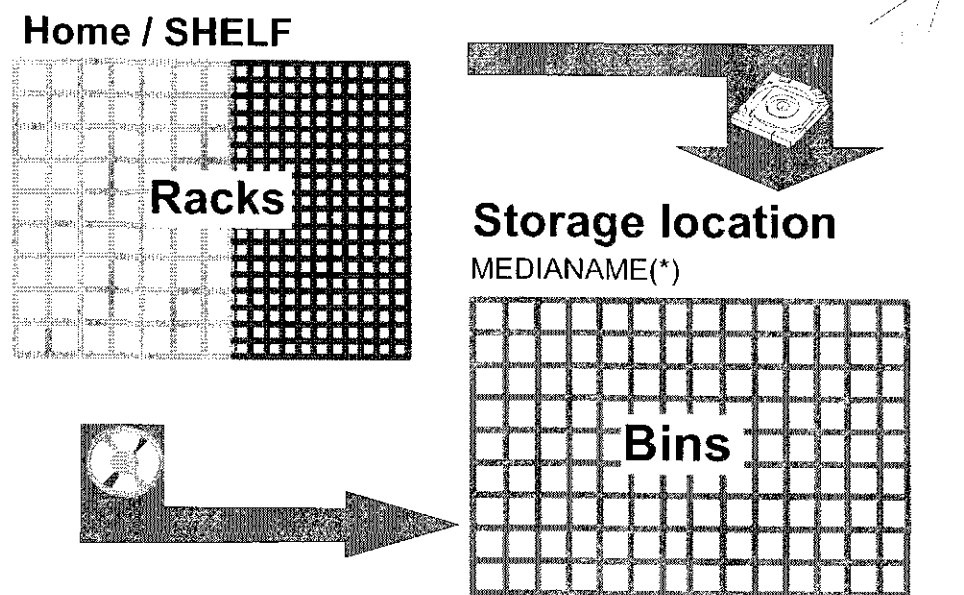


Figure 7-11. MEDIANAME (cont.)

SS212.0

Notes:

If the physical volume characteristics do not matter, a storage location can be defined so that it accepts all media names by using an asterisk (*).

Implementing System-Managed Tape

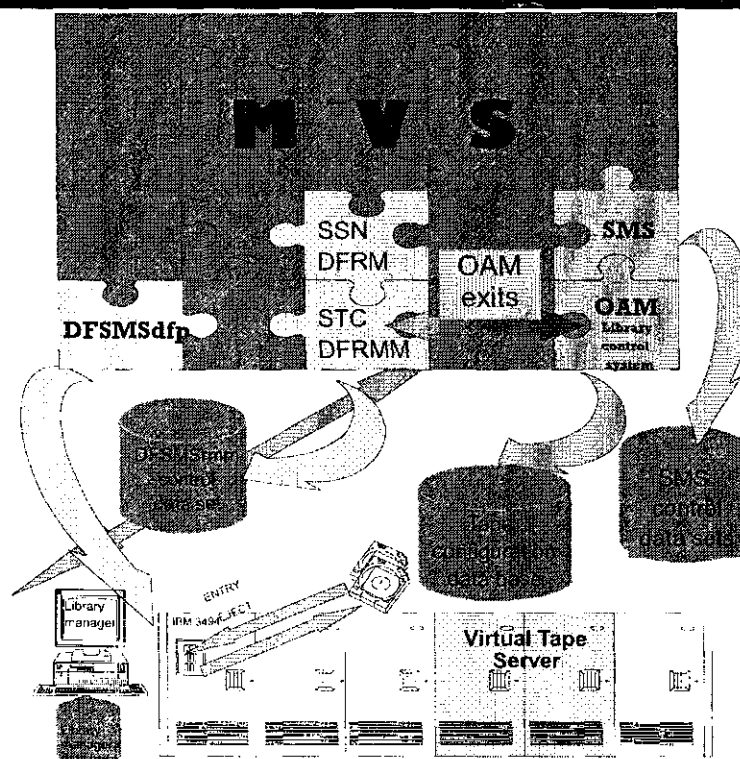


Figure 7-12. Implementing System-Managed Tape

SS212.0

Notes:

IBM system-managed tape libraries have two inventories of their volumes:

1. The tape configuration data base (TCDB) contains a record for each library and one for each volume it contains. The volume record indicates which library contains the volume, and its status (scratch or private). The TCDB is on the MVS host.
2. The IBM TotalStorage Enterprise Automated Tape Library (3494) has an inventory of its contents, which indicates where the cartridge is stored, and its status.

DFSMSdfp supports the 3494 and VTSs as library devices. It uses the object access method (OAM) address space in conjunction with the storage management subsystem (SMS). The OAM address space contains, in part, the library control system (LCS).

The tape library must be defined in DFSMS and new allocation assignments must be made via the SMS automatic class selection routines.

In addition to publications, IBM Education and Training offers course number ES590, *Analyzing and Managing DFSMS Tape Data*, which covers how to implement system-managed tape.

Using DFSMSrmm Programming Interfaces

Exit	DFSMSrmm interface	DFSMSrmm uses the
EDGDFHSM	EDGTVEXT	DFSMSshm tape volume exit to manage the tapes that DFSMSshm uses
EMODVOL1	EDGEMVOL	DFSMSdfp end-of-volume label editor installation exit to control tape volume labeling
CBRUXCUA	EDGLCSUX	OAM change use attribute exit to manage the volumes in system-managed tape libraries
CBRUXEJC	EDGLCSUX	OAM eject exit to manage the volumes in system-managed tape libraries
CBRUXENT	EDGLCSUX	OAM entry exit to manage the volumes in system-managed tape libraries
CBRUXVNL	EDGLCSUX	OAM volume-not-in-library installation exit to process tape volumes that are not resident in system-managed tape libraries but that are needed in order for processing to continue
IGXMSGEX	EDGMSGEX	DFSMSdfp MSCDISP exit to update tape drive displays and control the use of cartridge loaders
OMODVOL1	EDGOMVOL	DFSMSdfp open volume label editor installation exit to control tape volume labeling
IATUX71	EDG3X71	JES3 exit to determine what processing is needed for JES3 messages and tape drive display processing

Figure 7-13. Using DFSMSrmm Programming Interfaces

SS212.0

Notes:

Starting with DFSMS/MVS Version 1, Release 4, DFSMSshm calls DFSMSrmm directly via a new interface, EDGTVEXT. ARCTVEXT should NOT be used in DFSMSshm to interface with DFSMSrmm. For DFSMS/MVS 1.3, DFSMSrmm supplies an ARCTVEXT that should be enabled in DFSMSshm. This exit, as supplied by DFSMSrmm, calls EDGDFHSM. Refer to the chapter titled *Using DFSMSrmm Programming Interfaces*, in your *DFSMS/MVS DFSMSrmm Implementation and Customization Guide*, SC26-4932, for more details.

What DFSMSrmm Provides

Detailed information for volumes

Location name and type

Media type, format, compaction, and read compatibility

Manual cartridge entry for manual tape libraries (MTL)

A ready-to-run interface to OAM exits via EDGLCSUX

Notification of DFSMSrmm at cartridge-entry time

Confirmation of entry, and the optional automatic definition of volumes in the DFSMSrmm CDS

Support for physical and logical volume types (determined automatically by library)

Optional rejection of any volume if not predefined in CDS

Tracking and controlling of volume location

Updating of volume records at entry and eject

Automatic confirmation of moves (INTRANSIT, entry)

Return-to-scratch processing updates TCDB

Selection of PARMLIB control through the use of OPTION

SMSTAPE

Figure 7-14. What DFSMSrmm provides

SS212.0

Notes:

DFSMSrmm commands can copy the volume information that exists in the TCDB to create or update the volume record in its CDS. On the other hand, when you change the volume information in the CDS by using a DFSMSrmm command (CHANGEVOLUME), the corresponding volume entry in the TCDB is updated accordingly.

Parmlib Controls for SMS Tape

OPTION SMSTAPE

UPDATE

EXITS	DFSMSrmm overrides volume status during entry processing.
SCRATCH	DFSMSrmm updates TCDB during EXPROC for volumes returning to scratch.
COMMAND	DFSMSrmm TSO subcommands update TCDB for change of status, TDSI and owner information, eject processing and manual cartridge entry processing.
PURGE	Control for update processing of TCDB during eject processing
YES	Force delete of volume records
NO	Keep the volume records
ASIS	Volume record disposition is determined by eject request or library defaults

Figure 7-15. Parmlib Controls for SMS Tape

SS212.0

Notes:

Remember that, with special OPMODE settings, SMSTAPE specifications might be ignored.

If DFSMSrmm is running in protect mode, any setting of OPTION SMSTAPE(UPDATE(...)) in EDGRMMxx is ignored. In this case, DFSMSrmm always works as if you specified EXITS, SCRATCH, and COMMAND.

The setting of OPTION SMSTAPE(PURGE(...)) in EDGRMMxx is ignored if OPMODE(M) is specified.

All other combinations will be honored and might help you to prevent DFSMSrmm from changing the content of TCDB while testing DFSMSrmm during implementation.

Manual Tape Library

Associates library with specific drives

Drives must not be in automated tape libraries.

Human operator mounts volumes.

Volumes are system-managed.

DFSMSrmm adds volumes into the TCDB

Simulates entry processing.

Storage group associated with MTL can be used for scratch pools.

Figure 7-16. Manual Tape Library

SS212.0

Notes:

Manual tape libraries provide the opportunity to control the allocation of tape drives outside system-managed tape libraries through the use of the same technique known for devices inside system-managed tape libraries.

Library Partitioning - REJECT

Enables library sharing with other systems
 Other OS/390 and z/OS systems or other platforms
 Correct scratch categories are important

Parmlib control using REJECT ANYUSE(prefix)

At *entry* time

OAM leaves in INSERT category if rejected or
 Automatic definition in CDS

At *open* time

Use is rejected

Can be ignored by using EDGUX100

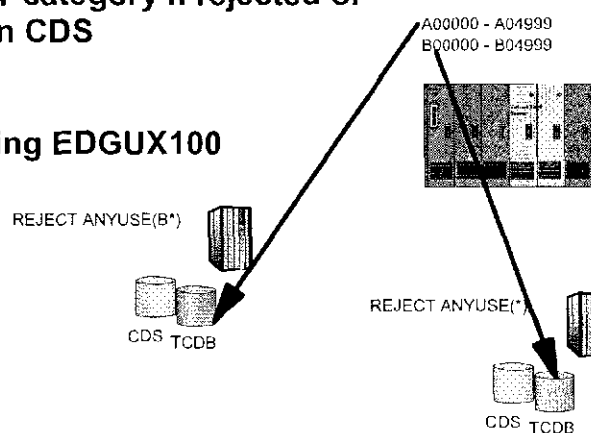


Figure 7-17. Library Partitioning - REJECT

SS212.0

Notes:

You can use REJECT to prevent a volume not defined to DFSMSrmm from being defined in a system-managed tape library. The REJECT ANYUSE(*prefix*) operand prevents a volume from being defined in the system-managed tape library on the current system. The REJECT OUTPUT(*prefix*) operand allows you to define the volume to the system-managed tape library but only use the volume for input processing.

When you enter a volume into a system-managed tape library, if the volume is defined to DFSMSrmm and you have specified the USE operand without MVS, or the volume matches a specified REJECT ANYUSE(*prefix*), EDGLCSUX sets a return code of 12 to pass to OAM. OAM leaves the volume in the system-managed tape library in the INSERT category; it does not create a volume entry in the TCDB. The volume is then available for another sharing system to process the insert request. The sharing system could be another VM or MVS system.

Note that REJECT ANYUSE is used at entry time and at open time.

EDGRMMxx REJECT

REJECT

- ANYUSE not available for read or write processing
- * Reject all volumes not defined to DFSMSrmm
- A* or DFSMSrmm can reject a range
12* of tapes only if you first define them to
 DFSMSrmm.
- OUTPUT not available for write processing
- * Prohibit writing to all volumes not defined to
 DFSMSrmm
- B* or A range of volumes not
C14 available for write processing

Examples:

```
REJECT ANYUSE(*)  
REJECT ANYUSE (VM1*), OUTPUT (VM*)
```

Figure 7-18. EDGRMMxx REJECT SS212.0

Notes:

Most installations will probably use REJECT ANYUSE(*) to suppress the use of foreign tapes by using EXPDT=98000 unless users are authorized by STGADMIN.EDG.IGNORE.tape profile in the FACILITY class. In this case, EDGUX100 is required in order to allow bypassing of DFSMSrmm during processing of foreign tapes.

The Variety of Library Partitioning Options

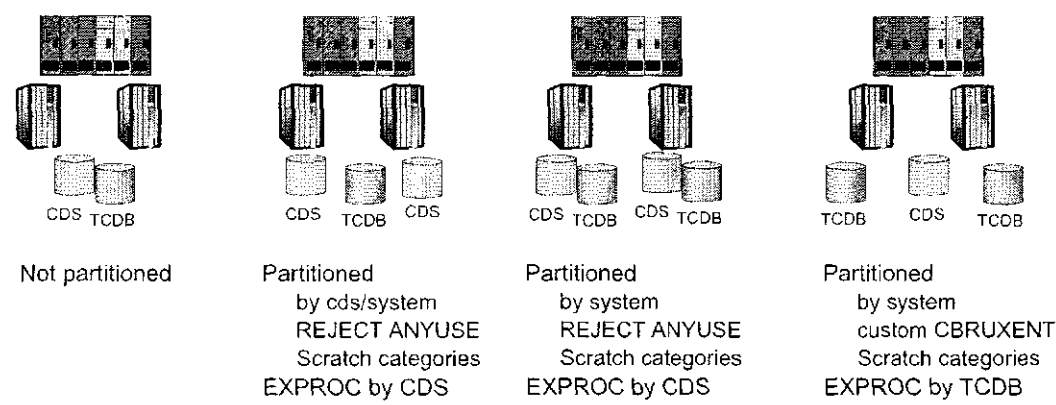


Figure 7-19. The Variety of Library Partitioning Options SS212.0

Notes:

Remember that DFSMSrmm updates the TCDB both during inventory management and while it is processing commands. In most situations you will need to run inventory management EXPROC on each system to allow the management of all volumes.

For entry processing you will need appropriate REJECT ANYUSE definitions or even have to customize the CBRUXENT exit to avoid the intermixing of volumes.

Each system has to use its own scratch categories.

Library Partitioning

There are two ways to control entry processing:

REJECT ANYUSE(*)

All volumes undefined in DFSMSrmm CDS will be left in
INSERT category.
You must pre-define volumes to be used.

No REJECTS or selective REJECTS

All volumes are added automatically to DFSMSrmm CDS.
Volume status meets library default entry status.
In case you need a different status, you have to change it
manually change after entry of the volume.
Pre-define volume with desired status.

Any alternative requires CBRUXENT exit coding.

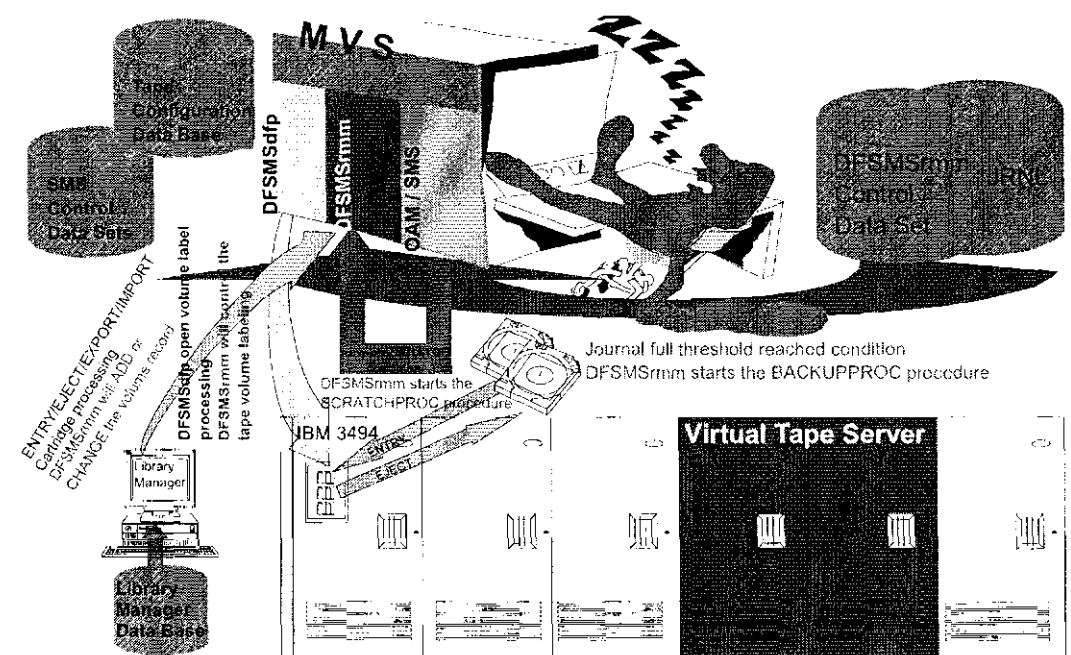
Figure 7-20. Library Partitioning

SS212.0

Notes:

REJECT ANYUSE(*) allows the simple but reliable control through the pre-definition of volumes in the proper CDS in each situation.

If you need special control, you must use your own code in CBRUXENT.



SS212.0

The visual gives an overview of the interface options available with DFSMSrmm and system-managed tape libraries.

The second option is the automatic replenishment of the scratch pool when a low-on-scratch condition in a system managed tape library occurs. Note that the low-on-scratch condition might not be cleared automatically by DFSMSrmm.

Low-on-Scratch in a System-Managed Tape Library

EDGRMMxx OPTION SCRATCHPROC(procname)
 Identify the procedure to be automatically started when CBR3660A is issued

EDGXPROC
 Sample name of the procedure DFSMSrmm starts to replenish scratch volumes in an automated tape library dataserwer.
 Sample JCL

```
//EDGXPROC PROC OPT=EXPROC
//EDGHSKP PGM=EDGHSKP,PARM='&OPT.',
//          TIME=1440,REGION=4096K
//SYSPRINT DD SYSOUT=*
//MESSAGE DD DISP=SHR,DSN=RMM.MESSAGE.DSET
```

Figure 7-22. Low-on-Scratch in a System-Managed Tape Library

SS212.0

Notes:

When defining the library in the DFSMS configuration, you must specify the low threshold of the scratch volumes for each media type. When during normal system processing one of the thresholds is reached, a message, CBR3660A, is issued by OAM. Specifying the name of a SCRATCHPROC in EDGRMMxx enables DFSMSrmm to respond automatically in this situation.

The idea is to release, during this special run of EXPROC, some volumes that were already marked as *pending release* in previous runs of inventory management. Note that many customers have organized their housekeeping in such a way that no pending release volumes remain in the control data set after normal inventory management has finished. The simplest way to do this is to run a second EXPROC in the housekeeping job. Another option is to use a VRS, indicating SCRATCHIMMEDIATE.

Pre-ACS Interface and SMS ACS Support

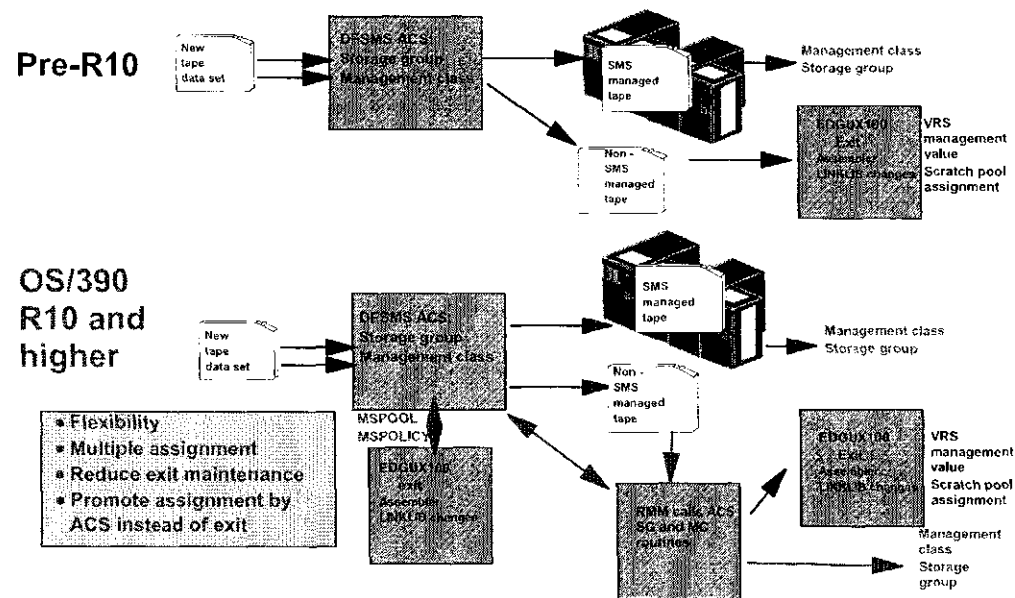


Figure 7-23. Pre-ACS Interface and SMS ACS Support

SS212.0

Notes:

As we have seen in Unit 5, you can specify `OPTION SMSACS(YES)` to make DFSMSrmm call SMS automatic class-selection routines for the assignment of a management class. This also applies to the assignment of a storage group that can be used to determine an acceptable scratch pool. With pre-releases of OS/390 DFSMS, this was available only with EDGUX100.

When you run DFSMSrmm with system-managed tape libraries, you need only one scratch pool in DFSMSrmm, because the IBM TotalStorage Enterprise Automated Tape Library supports only one scratch pool per system.

Using a tape robot that supports more than one scratch pool to be selected by applications, you need modified mount messages to identify the requested scratch pool. DFSMSrmm can take the decision for a pool based on the job name, or the data set name, or both.

Assignment of Pool and VRS Policy

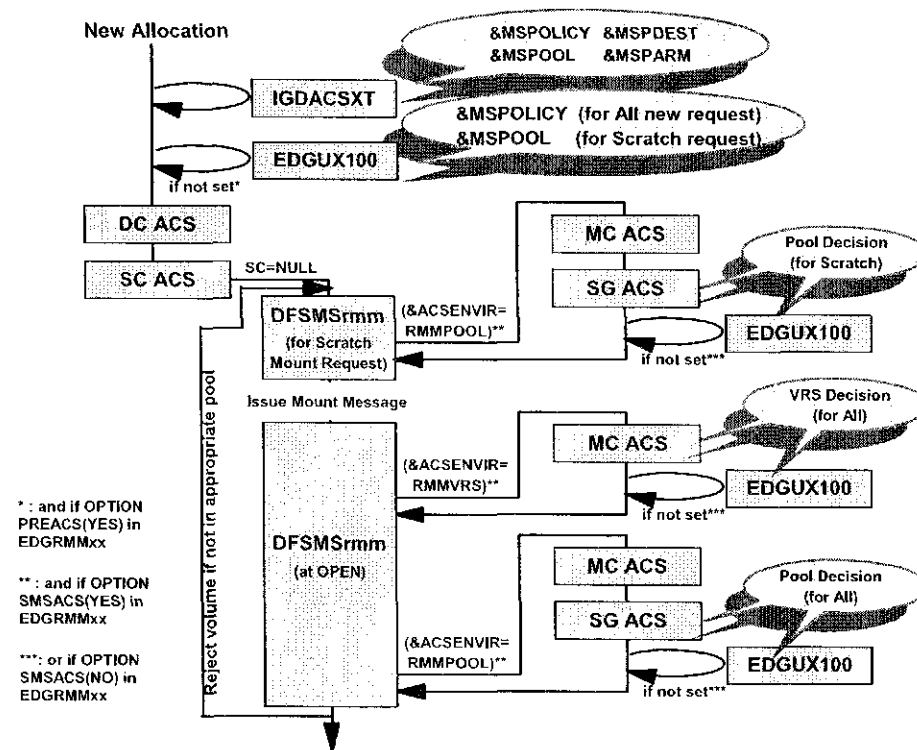


Figure 7-24. Assignment of Pool and VRS Policy

SS212.0

Notes:

During the allocation of a new tape data set, first IGDACSXT (pre-ACS routine exit) is called, if it is available. EDGUX100 will then be called if OPTION PREACS(YES) in EDGRMMxx is specified and &MSPOLICY (and &MSPOL for scratch requests) is not set.

The allocation calls ACS services that start with data-class and storage-class routines. If a storage class is assigned, management-class and storage-group ACS routines are called, thus leading to a data set on a system-managed volume (not shown in the visual).

If a storage class is not assigned for a scratch request, DFSMSrmm calls management-class and storage-group SMS ACS routines during mount processing to determine the pool if OPTION SMSACS(YES) is specified in EDGRMMxx. During execution of ACS routines, the read-only ACS variable &ACSENVIR contains 'RMMPOOL' to allow special code for pool determination. Note that the management-class assignment is only for temporary use, to help you make a storage-pool decision during a call for the storage-group ACS routine. This is because, during a call for the storage-group routine, not all read-only variables are available (for example, job name), whereas all of these variables can be used during a call for the management-class ACS routine. The idea is to determine

the pool during the call of the management-class ACS routine and to notify the storage-group ACS routine via a dummy management class that is not used for any other purpose.

If the storage group is not set, or if SMSACS(NO) is specified in EDGRMMxx, EDGUX100 is called to determine the pool based on the job name or the data set name, or both.

A mount message is then issued, including information on the requested pool if MNTMSG is specified in EDGRMMxx for the mount message ID.

During data set open for scratch requests as well as for specific requests, if SMSACS(YES) is specified in EDGRMMxx, DFSMSrmm calls the management-class ACS routine directly to determine the management class for the VRS policy. During this call, the read-only variable &ACSENVIR contains 'RMMVRS' to allow separate code for a VRS policy decision. If a management class is not set or if a management-class ACS routine is not called because of OPTION SMSACS(NO) in EDGRMMxx, EDGUX100 may be called to assign a management value.

Also during data set open, the mounted volume for a scratch request must be verified if it is to be contained in the requested pool. For a description of what happens, see the details on mount processing above. If the pool does not match, the volume is rejected and mount processing starts again.

In conclusion, decisions taken in SMS ACS routines have precedence over EDGUX100, which is not called in these cases.

EDGUX100 Capabilities

You can use EDGUX100 to

- Request ignore of a volume
- *Determine the scratch pool*
- Control a cartridge loader
- Assign a VRS policy by name
- Update or provide an expiration date
- Record only the first data set name in a tape volume
- Produce and customize sticky labels
- Set ACS input variables during pre-ACS

Figure 7-25. EDGUX100 Capabilities

SS212.0

Notes:

While EDGUX100 provides powerful capabilities, management of exits is much more complicated than coding SMS ACS routines.

EDGUX100 Environment

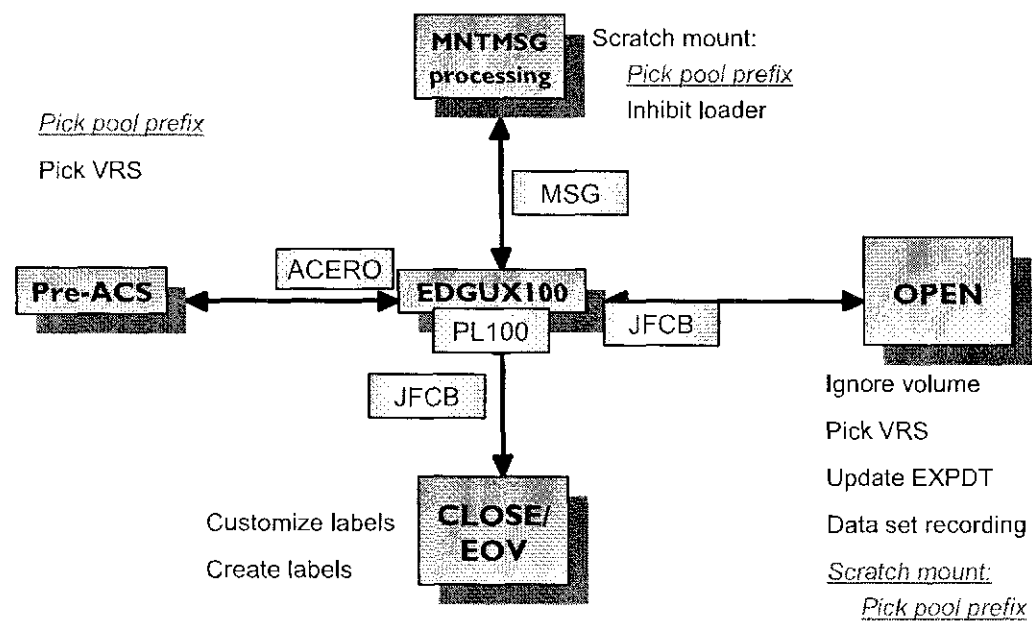


Figure 7-26. EDGUX100 Environment

SS212.0

Notes:

EDGUX100 is called in a range of situations. Increased exploitation of the possibilities EDGUX100 offers entails more-frequent code changes, and some of these can result in a malfunction. It's a good idea to remove some of the decisions from EDGUX100 if possible.

Recommendations

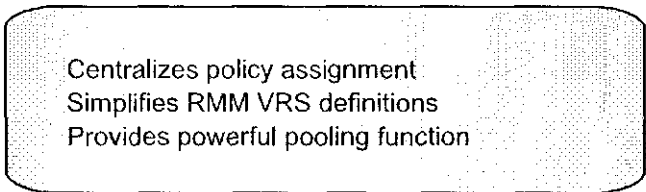
Do not use RMM pre-ACS support.

This extends the life of EDGUX100 customization.

Use the SMS management class to replace VRS MV.

Replace the RMM VRS DSNAME masks with ACS FILTLISTs.

Use the SMS storage group to replace RMM pool prefix.



Centralizes policy assignment
Simplifies RMM VRS definitions
Provides powerful pooling function

Figure 7-27. Recommendations

SS212.0

Notes:

These are the recommendations that DFSMSrmm developers are making.

Using SMS Storage Group

Use VLPOOL to aid pool management.

VLPOOL PREFIX(prefix) NAME(sgname)

ADDVOLUME POOL(prefix)

Volume is assigned to sgname.

Use the same SG Name on multiple VLPOOLS

or

Override default processing.

ADDVOLUME SGNAME(sgname)

Code MC & SG ACS routines to process

- RMMPOOL environment call for non-SMS tape
 - Regular allocation for SMS tape
- Enable using parmlib options SMSACS(YES).

Move pooling decisions out of EDGUX100 exit.

Data set and job name masks added to ACS filter lists.

SG = Pool name

Notes:

The steps required to use a storage group for pool management are as follows:

1. Connect volumes to a storage group.
2. Define a storage group of type tape in your SMS configuration.
3. Assign a storage group by the storage-group ACS routine.

Remember that if a storage group is assigned by the ACS routine, EDGUX100 is no longer called, thus, in fact, you have removed a decision out of the exit.

Understanding RMM Pools

VLPOOL PREFIX(A*) SYSID(A) NAME(A) TYPE(S) VLPOOL PREFIX(9*) SYSID(A) NAME(A) TYPE(S) VLPOOL PREFIX(P*) SYSID(B) NAME(B) TYPE(S) VLPOOL PREFIX(M*) NAME(C) TYPE(S)	System Based Multiple/system Any system pool Type=Scratch
VLPOOL PREFIX(1*) NAME(RED) TYPE(R) VLPOOL PREFIX(2*) SYSID(NONE) TYPE(S) VLPOOL PREFIX(3*) NAME(BLU) TYPE(R) VLPOOL PREFIX(4*) TYPE(R)	UX100 Assigned Only one pool Any type/system
VLPOOL PREFIX(X*) SYSID(X) NAME(SG1) TYPE(S) VLPOOL PREFIX(Y*) SYSID(Y) NAME(SG1) TYPE(S) VLPOOL PREFIX(Z*) NAME(SG2) TYPE(R) VLPOOL PREFIX(W*) NAME(SG3) TYPE(S)	Storage Groups Multiple/SG Any type/system

Figure 7-29. Understanding RMM Pools

SS212.0

Notes:

System based pools allow multiple pools for each system but you can use system assignments only for scratch pools.

EDGUX100 assignment allows for only one pool, though of any type, per decision. SYSID is ignored in this case.

You can use storage groups to compose pools from more than one VLPOOL definition, and you can use any type of pool. SYSID is ignored in this case.

Scratch Mount Validation

Validation of scratch mounts is carried out in the following priority order:

1. System-managed ATL
Leave validation to system
 2. SG assigned, or, system-managed MTL
RMM scratch volume
SG name matches or VLPOOL SG name matches
 3. EDGUX100 assigned pool prefix
RMM scratch volume
VLPOOL prefix matches
Volume must not belong to a storage group
 4. RMM system based pooling
RMM scratch volume
Volume must not belong to a storage group
- Volume is rejected if it has a SG name.**

Figure 7-30. Scratch Mount Validation

SS212.0

Notes:

For system-managed tape libraries, scratch mount validation is carried out by the system. The library manager receives a mount from category and selects only a scratch volume from the requested category number.

For validation outside system-managed tape libraries, any assigned storage group takes priority.

If no storage group has been assigned, EDGUX100 can select a scratch pool. Any scratch volume of the selected pool is acceptable, but the volume must not belong to a storage group.

If EDGUX100 does not manage pool assignment, system-based pooling applies. Any scratch volume of the selected pool is acceptable, but the volume must not belong to a storage group.

Implementing SMS SG

Use VLPOOL to aid pool management.

VLPOOL PREFIX(prefix) NAME(sgname)

```
VLPOOL PREFIX(X*) SYSID(X) NAME(SG1) TYPE(S)
VLPOOL PREFIX(Y*) SYSID(Y) NAME(SG1) TYPE(S)
VLPOOL PREFIX(Z*) NAME(SG2) TYPE(R)
VLPOOL PREFIX(W*) NAME(SG3) TYPE(S)
```

ADDVOLUME X00001

Volume is assigned to SG1.

ADDVOLUME X00099 POOL(X*)

Volume is assigned to SG1.

You can use the same SG name on multiple VLPOOLS.

ADDVOLUME A00001 STORGRP(SG1)

Override processing and assign specific SG.

A volume with no STORGRP inherits SG from VLPOOL NAME.

Figure 7-31. Implementing SMS SG

SS212.0

Notes:

Use the name keyword in VLPOOL to connect it with a storage group. If the pool name is a valid storage group name, DFSMSrmm uses the VLPOOL NAME value as a default for volumes added to the pool.

You can even override the storage group at the volume level during ADDVOLUME processing.

Implementing SMS SG ...

Existing volumes have no SG

- No need to set a SG name for each volume.
- Update VLPOOL to include a pool NAME.
- Define pool NAMEs as SMS tape storage groups.
- Refresh RMM using new parmlib member.

Defining tape storage groups

- A system-managed library is required.
- You can define a dummy library using ISMF library define.
 - Select a dummy library name.
 - Make up a fake library ID.
- Use this library when adding tape storage groups.

Notes:

For existing volumes without connection to a storage group, you can update the VLPOOL statement in EDGRMMxx with a name that equals a storage group of type TAPE in your active SMS configuration to make them available for SG-based pooling.

Because of consistency checks during the validation of an SMS configuration, you must define all storage groups in that configuration that have been assigned by an ACS routine.

For the definition of a storage group of type TAPE, the storage group needs to be connected with a library. You can define a dummy library that does not represent real hardware. All storage groups used by DFSMSrmm for non-system-managed tape volume pooling can be connected to a single dummy library.

Implementing SMS SG ...

Code MC and SG ACS routines to process

RMMPOOL environment call for non-SMS tape

Regular allocation for SMS tape

Data set name and job name masks added to ACS filter lists

SG = Pool NAME

```

PROC 1 MGMTCLAS
  FILE LIST POOLA INCLUDE(USERA.*.USERY.KEEP.*) -
    EXCLUDE(USERA.POOLB.*)
  IF &ACSENVIR = 'RMMPOOL' THEN
    DO
      SELECT (&DSN)
        WHEN (&POOLA) SET &MGMTCLAS = 'POOLA'
        OTHERWISE SET &MGMTCLAS = 'DEFAULT'
      END
      IF &DSTYPE = TEMP THEN SET &MGMTCLAS = TEMP
    ...
  END
  ...
END

PROC 1 STORGRP
  IF &ACSENVIR = 'RMMPOOL' THEN
    DO
      SELECT (&MGMTCLAS)
        WHEN ('POOLA') SET &STORGRP = 'POOLA'
        WHEN ('DEFAULT') SET &STORGRP = 'DEFAULT'
        WHEN ('TEMP') SET &STORGRP = 'TEMP'
      END
    ...
  END
  ...
END

```

Figure 7-33. Implementing SMS SG ...

SS212.0

Notes:

Note the check for the contents of the &ACSENVIR read-only variable to isolate that part of the code that is dedicated to DFSMSrmm.

If your ACS processing does not set the storage group name, DFSMSrmm processing continues without using storage group names.

The processing that the STORGRP ACS routine performs is limited to the variables it can reference. You might want to make your decisions in the MGMTCLAS ACS routine and use the STORGRP routine to set the STORGRP variable to the selected value as shown in the visual.

Any management class value that you set during ACS processing for the RMMPOOL ACS environment call is ignored by DFSMSrmm processing.

Implementing SMS SG ...

Move pooling decisions out of EDGUX100 exit

Enable using parmlib options SMSACS(YES)

```
Parmlib:
OPTION SMSACS(YES)

Cmd:
F DFRMM,M=xx
```

Cleanup EDGUX100

EDGUX100 is not called when SG is assigned

EDGUX100 POOLTAB entries can be removed at convenient time

Next release, or, next system update

Figure 7-34. Implementing SMS SG ...

SS212.0

Notes:

You can take pooling decisions away from EDGUX100 step by step (pool by pool) once you have enabled DFSMSrmm calls for the SMS ACS routines by OPTION SMSACS(YES) in EDGRMMxx. There is no need to remove the code from EDGUX100 instantly. This flexibility provides a secure and comfortable environment during conversion.

The same flexibility applies to VRS policy decisions.

Migration and Coexistence

Use APAR OW48865 for SMSACS/PREACS options

Simpler than SMS ACS updates to prevent construct assignment when
&ACSENVIR = RMMVRS or &ACSENVIR = RMMPOOL

Update SMS ACS before enabling new function

Selectively process tape data sets
Start with selected users or data sets
Grow when results are proven

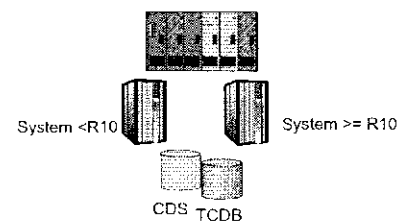


Figure 7-35. Migration and Coexistence

SS212.0

Notes:

If you want to use SMS ACS support neither for pooling nor for VRS policy assignment, simply specify OPTION SMSACS(NO) in EDGRMMxx. Make sure that the service is installed as described in APAR OW48865.

Do you still need EDGUX100?

Let's have a look at the other EDGUX100 functions.

Request ignore of volume

Usually EXPDT=98000 is used. The function will be required less in the future.

Control cartridge loader

There is no current equivalent, but more robots and virtual tape is being used.

Update/provide EXPDT

Used to remove special keyword dates.

Keyword dates are already expired - no longer need to be updated.

Use VRSs to retain all data sets.

Control data set recording

Still required for special tape applications, but there is minimal maintenance once the sample table is updated.

Sticky label customization or production

Optional for RMM disposition control. Use to

- Alter CONFIRMMOVE handling
- Select LOAN location or storage location
- Customize from standard sticky labels

Figure 7-36. Do you still need EDGUX100?

SS212.0

Notes:

You can expect that most installations will still need EDGUX100 due to its other capabilities.

These functions are described in detail in *z/OS DFSMSrmm Implementation and Customization Guide*, SC26-7405.

Stacked Volume Support

This support is included in OS/390 R10.

Migration is recommended because this support simplifies the management of logical volumes.

Enable stacked volume support only once all systems have reached the R10 level.

Otherwise, DFSMSrmm works like the support that's currently available.

Enabling this support builds stacked volume information automatically from

- Logical volumes
- Container names

Figure 7-37. Stacked Volume Support

SS212.0

Notes:

While stacked volumes were supported by DFSMSrmm right from the beginning when the EXPORT feature was included with VTS, it was rather difficult to manage by DFSMSrmm the movement of exported stacked volumes.

VTS Support

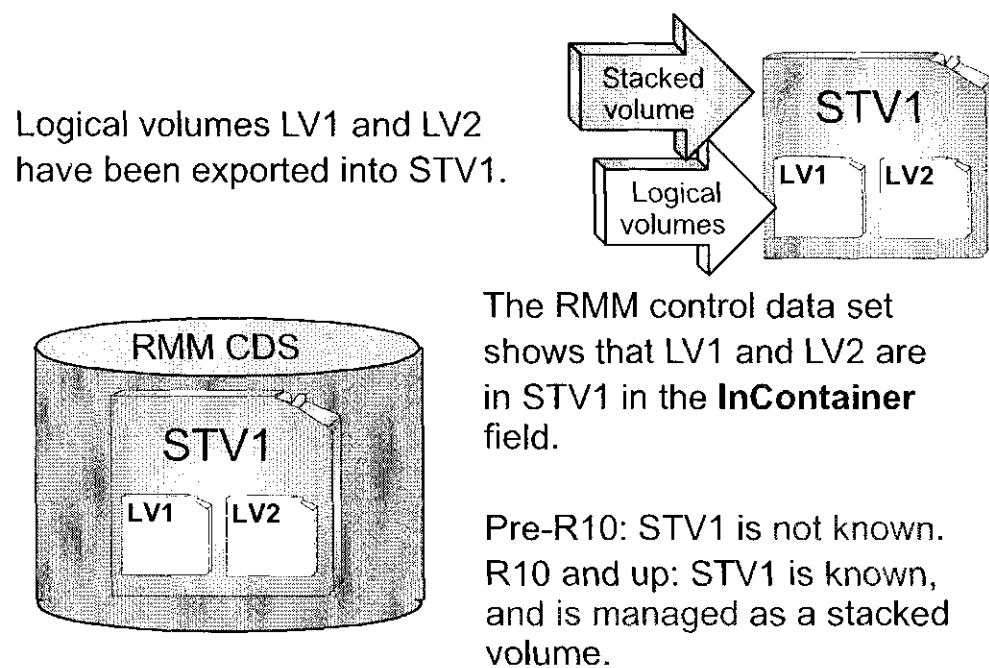


Figure 7-38. VTS Support

SS212.0

Notes:

When logical volumes are exported from VTS, they are no longer part of it, because EXPORT moves the logical volumes to an exported stacked volume that is intended to be ejected from the IBM TotalStorage Virtual Tape Server.

If stacked volume support is not enabled, the name of the exported stacked volume is only reported in **InContainer** field of the logical volumes that have been exported.

If stacked volume support is enabled, the exported stacked volume is automatically defined in DFSMSrmm CDS and managed as a stacked volume. Stacked volumes are managed by DFSMSrmm differently from logical or physical volumes. Stacked volumes normally start and end their movement in a VTS automated tape library. The host system is not notified when stacked volumes enter or leave an automated tape library so stacked volumes cannot be managed the same way that physical volumes in an automated tape library are managed.

VT S Support for 'Stacked' Volumes

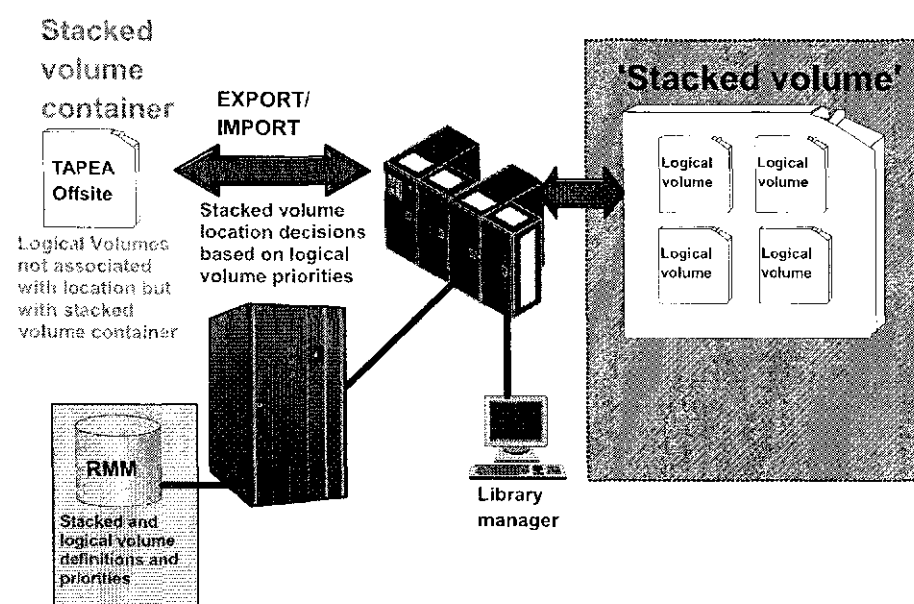


Figure 7-39. VTS Support for Stacked Volumes

SS212.0

Notes:

Based on the requested movement of exported logical volumes, the movement of stacked volumes is managed by DFSMSrmm instead. Logical volumes are not associated with storage locations if stacked volume support is enabled. After export, logical volumes are associated with a stacked volume by the content of the **InContainer** field.

If during an export process multiple storage locations are addressed in the export list file, the logical volumes that are to be exported are sorted by storage location. For each target storage location, a separate stacked volume is used.

The Open or Close Status of Stacked Volumes

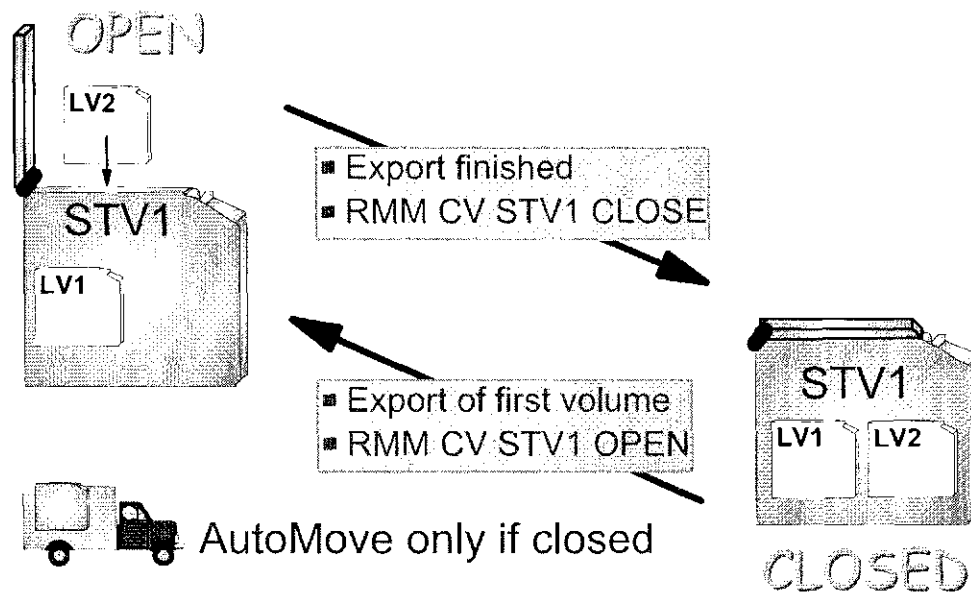


Figure 7-40. The Open or Close Status of Stacked Volumes

SS212.0

Notes:

During export of the first logical volume, DFSMSrmm changes the status of the stacked volume to OPEN. Once the export is complete, the status is changed to CLOSE. Only those stacked volumes whose status is CLOSE are moved automatically.

If necessary, you might type in the subcommands shown in the visual.

Stacked Volume Information

Field	Export value	Import value
Status	Not empty	Empty
Assign date	Set at start	Set at end
Stacked count	Increment	Decrement
Use count	Increment at start	Increment at start
Usage	Sum of contained volumes (KB)	Sum of contained volumes (KB)
Security level	Highest	(No change)
Last read date	(Empty)	Set
Last write date	Set	(No change)

Figure 7-41. Stacked Volume Information

SS212.0

Notes:

All information is automatically recorded by DFSMSrmm.

Box Support from the Library

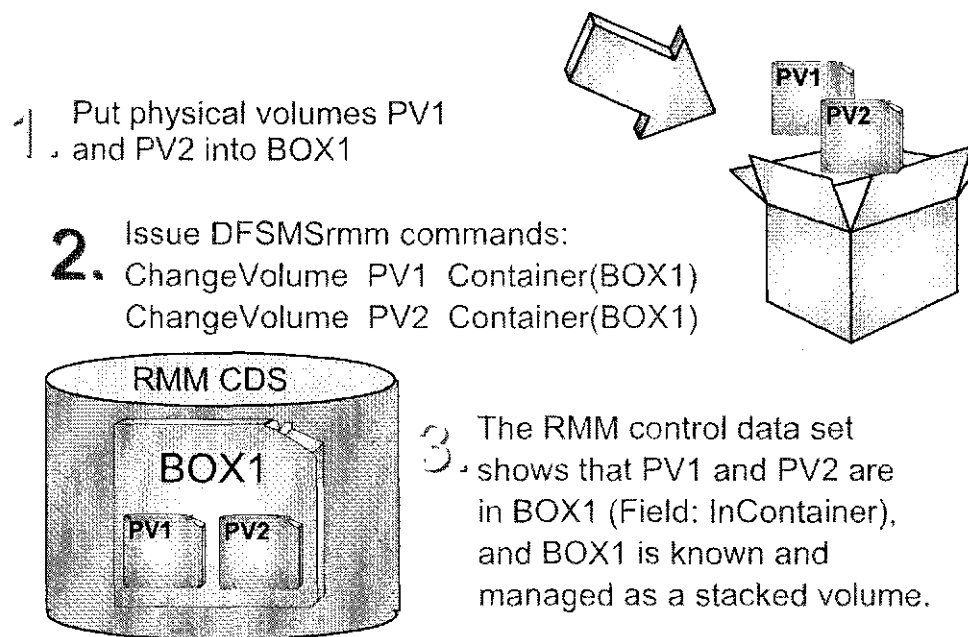


Figure 7-42. Box Support from the Library

SS212.0

Notes:

Box support means that you might really put physical volumes in a box for transport. By using the commands shown in the visual, you can move the box that includes the volumes.

You can use stacked volume support for virtual tape solutions other than VTS. The difference is that you must provide DFSMSrmm with the information that OAM provides to VTS during the export/import process. In this particular case, you must define the stacked and physical volumes to DFSMSrmm as follows:

```
RMM ADDVOLUME stacked_vol TYPE(STACKED) LOCATION(SHELF)
```

```
RMM ADDVOLUME logical_vol TYPE(PHYSICAL/LOGICAL) -  
STATUS(MASTER) CONTAINER(stacked_vol)
```

DFSMSrmm is notified by OAM about import and export processing, and can obtain movement completion information from Library Manager. You must use commands to perform the system all of these steps. If you have already defined physical and stacked volumes, you can add a virtual volume to a container by using CV CONTAINER(name).

Box Support Back at the Library

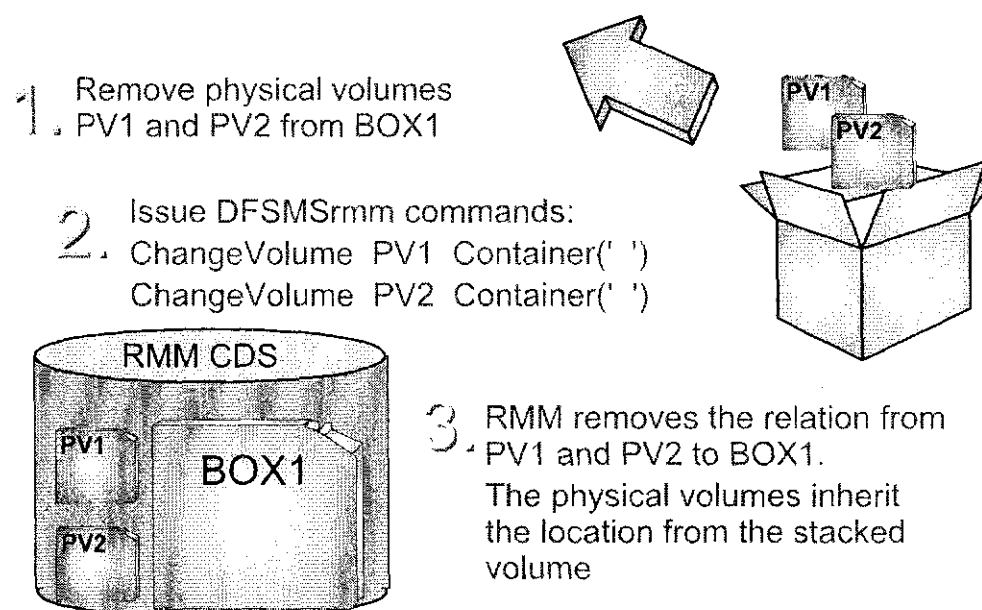


Figure 7-43. Box Support Back at the Library

SS212.0

Notes:

Volumes in a container cannot be moved other than in the container. DFSMSrmm uses the required location and priority of the contained volume to determine the required location for the container or the stacked volume. During VRSEL processing, DFSMSrmm determines where to move the stacked volume, based on the VRS specified for the physical volume.

Once the box is back at the library, the `CHANGEVOLUME` subcommand with the keyword `CONTAINER(' ')` will remove the volumes from the box. The physical volumes inherit the location from the stacked volume.

Three-Way Audit for DFSMSrmm CDS

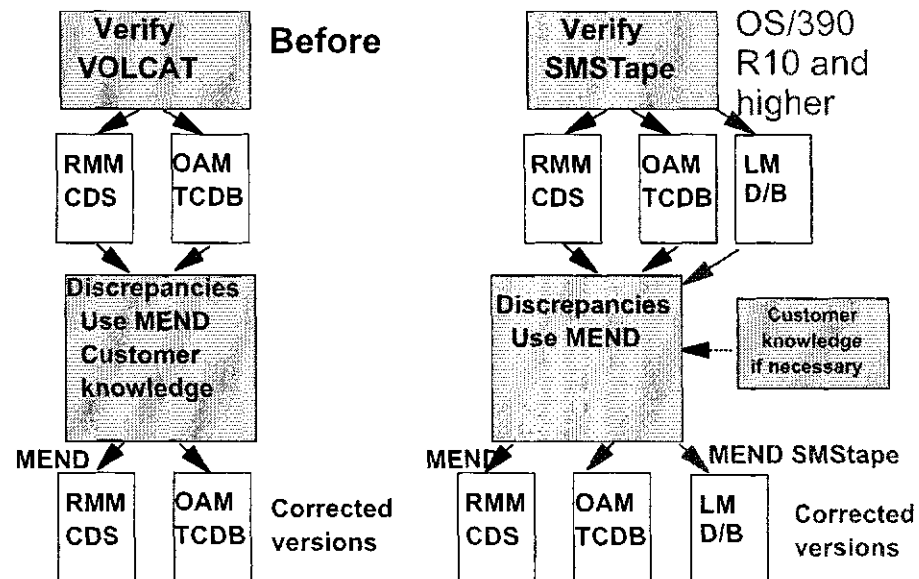


Figure 7-44. Three-Way Audit for DFSMSrmm CDS

SS212.0

Notes:

You can use EDGUTIL VERIFY(VOLCAT) to check the consistency of information in the control data set and the tape configuration database (TCDB). EDGUTIL checks volume status and the library where the volume resides. You can use EDGUTIL VERIFY(SMSTAPE) to perform more detailed checking including the Library Manager database. You can use MEND(SMSTAPE) to update the TCDB and Library Manager database with information from the DFSMSrmm control data set.

Three-Way Audit

EDGUTIL VERIFY(SMSTAPE) is a new option.

The TCDB and LM information is input as well as the DFSMSrmm CDS.

The TCDB is scanned sequentially to identify the volumes missing from the DFSMSrmm CDS.

VERIFY(SMSTAPE) enables an audit of stacked, logical, and physical volumes.

Correction of errors without stopping DFSMSrmm
by using MEND(SMSTAPE).

TCDB and LM are synchronized from the DFSMSrmm CDS.

Corrections are done automatically if the error is clear.

Otherwise, manual correction has to be done.

Correction of DFSMSrmm CDS with DFSMSrmm stopped
by using MEND(SMSTAPE).

TCDB and Library Manager inventory input is used to correct DFSMSrmm CDS errors.

Figure 7-45. Three-Way Audit

SS212 0

Notes:

Use MEND(SMSTAPE) to update the TCDB and the Library Manager database using information from the DFSMSrmm control data set. Use MEND to update the DFSMSrmm control data set based on information from the TCDB and the Library Manager database.

Verifying the Control Data Set

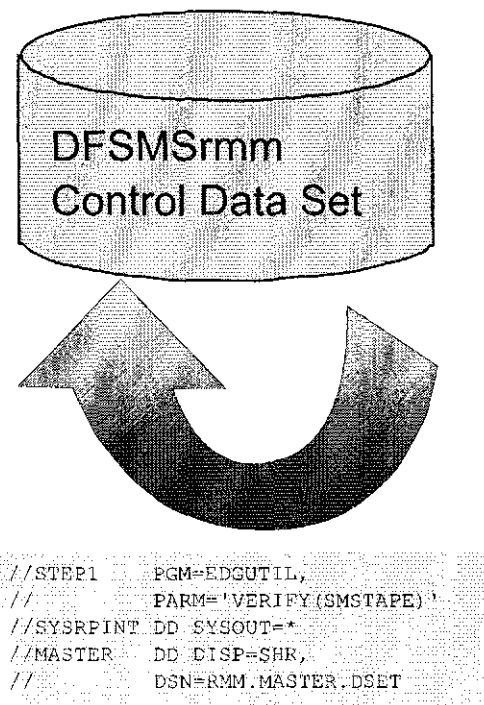


Figure 7-46. Verifying the Control Data Set

SS212.0

Notes:

Before running MEND(SMSTAPE), you should first use the VERIFY(SMSTAPE) option to find information that is not the same in the DFSMSrmm control data set, TCDB, and the Library Manager database. After running VERIFY(SMSTAPE) and before running MEND(SMSTAPE), you can update the control data set using DFSMSrmm TSO subcommands to correct DFSMSrmm information.

Mending the Control Data Set or the TCDB and the Library Manager Data Base

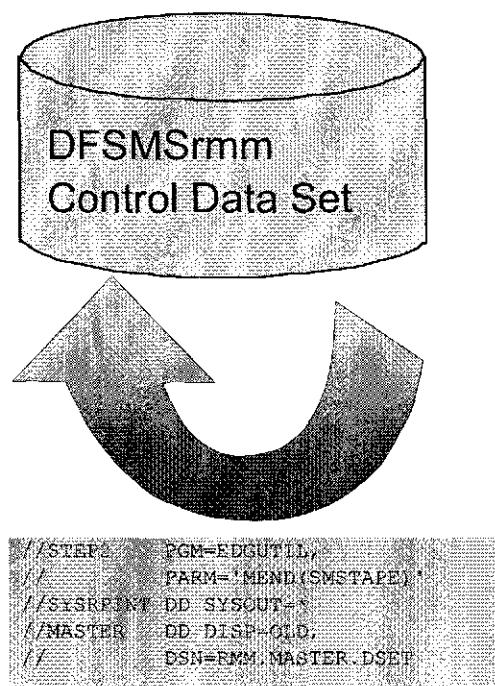


Figure 7-47. Mending the Control Data Set or the TCDB and the Library Manager Data Base

SS212.0

Notes:

Use MEND(SMSTAPE) to update the TCDB and the Library Manager database using information from the DFSMSrmm control data set. Use MEND to update the DFSMSrmm control data set based on information from the TCDB and the Library Manager database.

Storage Location as Home Location

This enhanced support for other robot than system-managed tape libraries came out with z/OS DFSMS 1.3.

Normally, the location is SHELF for volumes not in system-managed tape libraries unless they are moved to vault.

OW47891 provides the support for return to scratch if a volume is not at HOME LOCATION.

Volumes in other robots than system-managed tape libraries can have the name of a storage location as HOME LOCATION.

The LOCDEF option in EDGRMMxx requires TYPE(STORAGE,HOME).

MANAGEMENTTYPE(BINS) and MANAGEMENTTYPE(NO BINS) are supported.

You can use CV commands to set LOCATION and HOME LOACTION as desired for existing volumes.

Figure 7-48. Storage Location as Home Location

SS212.0

Notes:

You can identify a storage location as a home location by specifying TYPE(STORAGE,HOME). When you identify a storage location as a home location, you can manage volumes in a storage location like volumes that reside in a system-managed tape library. You can schedule release actions for volumes when they return to their storage home location and you can return volumes to scratch status while they reside in their storage home location.

When you use storage locations as the home location for volumes, you can decide how volumes are shelf managed. Shelf management is required if you want volumes to be stored in a public slot such as a rack number or a bin number. A shelf location is not required if the volume is stored in a robotic tape library.

To avoid using shelf locations, define the storage location by specifying MANAGEMENTTYPE(NO BINS) and do not define rack numbers that match to the volume serial numbers.

Checkpoint

Indicate whether each of the following statements is true or false.

1. DFSMSrmm has two kinds of volume pools, scratch and rack.
2. An EDGRMMxx member is rejected by DFSMSrmm during startup if there is no VLPOOL statement present.
3. The medianame in DFSMSrmm must match those of a valid MVS generic or esoteric tape device.
4. Currently, if you use mediatype for a volume, it must be CST, ECCST, HPCT, EHPCT or asterisk (*).
5. When you enter a volume into a system-managed tape library, if the volume is not defined to DFSMSrmm and you have specified REJECT ANYUSE(*) in member EDGRMMxx, OAM leaves the volume in the system-managed tape library in the INSERT category; it does not create a volume entry in the TCDB. The volume is then available for another sharing system to process the insert request.
6. Pools can either be selected by EDGUX100, or by a storage group, but you cannot use both methods in parallel.

Figure 7-49. Checkpoint

SS212.0

Notes:

Write down your answers here:

- 1.
- 2.
- 3.
- 4.
- 5.
- 6.

Unit Summary

Now that you have completed this unit, you should be able to:

- Use pools of tape volumes
- Understand DFSMSrmm processing in an environment with tape libraries
- Identify the customization need for library partitioning
- Use support for stacked volumes
- Use EDGUTIL to verify and synchronize the contents of the DFSMSrmm CDS, the TCDB, and the Library Manager data base

Notes:

Appendix A. DFSMSrmm Authorization Control

What This Unit Is About

This unit describes the facility resource classes used by DFSMSrmm for authorization checking.

What You Should Be Able to Do

After completing this unit, you should be able to identify the facility classes used to:

- Protect DFSMSrmm Resources
- Authorize DFSMSrmm Functions

How You Will Check Your Progress

Accountability:

- Checkpoint questions

References

SC26-7405	<i>DFSMSrmm Implementation and Customization Guide</i>
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Objectives

- Identify the facility classes used to:
 - Protect DFSMSrmm Resources
 - Authorize DFSMSrmm Functions

Figure 8-1. Unit Objectives

SS212.0

Notes:

A.1. DFSMSrmm Authorization Control

DFSMSrmm Authorization and Control



Security

Figure 8-2. DFSMSrmm Authorization and Control

SS212.0

Notes:

DFSMSrmm does not perform security checking directly. It uses the Security Authorization Facility (SAF) interface to verify authority to perform functions. Resource Access Control Facility (RACF) or a similar product is required to define user authorization. DFSMSrmm uses Resource Facility Class definitions to allow an installation the ability to control the use of DFSMSrmm functions.

Facility Classes

•STGADMIN.EDG.	FORCE
•STGADMIN.EDG.	HOUSEKEEP
•STGADMIN.EDG.	IGNORE.TAPE.volser
•STGADMIN.EDG.	LABEL.volser
•STGADMIN.EDG.	LISTCONTROL
•STGADMIN.EDG.	MASTER
•STGADMIN.EDG.	NOLABEL.volser
•STGADMIN.EDG.	OPERATOR
•STGADMIN.EDG.	OWNER.userid
•STGADMIN.EDG.	RELEASE
•STGADMIN.EDG.	RESET.SSI
•STGADMIN.EDG.	VRS
•STGADMIN.EDG.	INERS.WRONGLABEL

Figure 8-3. Facility Classes

SS212.0

Notes:

All DFSMSrmm facility class profiles begin with the DFSMS/MVS standard high level qualifier of **STGADMIN**. The second qualifier, **EDG** for DFSMSrmm, identifies the component or program within DFSMS/MVS that checks the profile. We will look at each of these facility class profiles to see what functions they control.

STGADMIN.EDG.MASTER

ACCESS	Description
Not defined	Same as CONTROL access
READ	The following functions can be performed: *List all control data set information except vital record specifications and control information *Search for all control data set information except vital record information *Update your own owner ID details *Request a SCRATCH volume for yourself *Release an owned volume when STGADMIN.EDG.RELEASE resource is not protected
UPDATE	Same as READ access, plus: some non-restricted fields can be updated for owned volumes and data sets as described in <i>DFSMS/MVS DFSMSrmm Guide and Reference</i> , RMM CHANGEVOLUME subcommand information
CONTROL	Same as UPDATE access, plus: You can *DEFINE, CHANGE and DELETE any control data set entries except vital record specifications *List control information when STGADMIN.EDG.LISTCONTROL resource is not protected

Figure 8-4. STGADMIN.EDG.MASTER

SS212.0

Notes:

This is the primary control point for DFSMSrmm functions. The level of access, **READ**, **UPDATE**, or **CONTROL**, determines much of what a user can do in DFSMSrmm. If this profile is not defined or controlled via another (generic) profile, then **anyone** can update DFSMSrmm!

STGADMIN.EDG.OWNER.userid

ACCESS	Description
Not defined	No authority is granted to update volume information except based on STGADMIN.EDG.MASTER access
NONE	Based on STGADMIN.EDG.MASTER access
UPDATE	Some non-restricted fields for volumes and data sets owned by <i>userid</i> can be updated. Also a volume owned by <i>userid</i> be released. See <i>DFSMS/MVS DFSMSrmm Guide and Reference</i> , RMM CHANGEVOLUME subcommand information

Figure 8-5. STGADMIN.EDG.OWNER.userid

SS212.0

Notes:

The resource may be established so that those with update access may manage volumes owned by another **userid**.

STGADMIN.EDG.FORCE

ACCESS	Description
Not defined	Information previously recorded by DFSMSrmm cannot be changed
UPDATE	Information previously recorded by DFSMSrmm can be changed based on access to STGADMIN.EDG.MASTER

Figure 8-6. STGADMIN.EDG.FORCE

SS212.0

Notes:

Access to this class allows previously recorded information to be changed. It is designed to handle special situations such as a tape being written to on another system and returned for processing. If the newly written data has a new data set name, DFSMSrmm rejects the volume since it does not match the name DFSMSrmm has previously recorded.

STGADMIN.EDG.LISTCONTROL

ACCESS	Description
Not defined	Functions are based on STGADMIN.EDG.MASTER access
CONTROL	You are allowed to use the RMM LISTCONTROL subcommand

Figure 8-7. STGADMIN.EDG.LISTCONTROL SS212.0

Notes:

This allows control of the DFSMSrmm LISTCONTROL command which obtains your installation's options and rules. If it is not defined, READ access to MASTER is required to use LISTCONTROL.

STGADMIN.EDG.RELEASE

ACCESS	Description
Not defined	Based on STGADMIN.EDG.MASTER access
READ	You are allowed to use the RMM DELETEVOLUME RELEASE subcommand to release an owned volume

Figure 8-8. STGADMIN.EDG.RELEASE SS212.0

Notes:

Once again, some installations need to be able to restrict certain actions such as scratching a tape. This may be used to prevent an owner from releasing (scratching) an owned tape. If this is not defined, READ access to master allows users to release their own tapes.

STGADMIN.EDG.IGNORE.TAPE.volser

ACCESS	Description
Not defined	Volumes cannot be ignored using the DFSMSrmm installation exit
READ	Volumes that are to be ignored by DFSMSrmm for input requests are allowed to be opened
UPDATE	Volumes that are to be ignored by DFSMSrmm for output requests are allowed to be opened

Figure 8-9. STGADMIN.EDG.IGNORE.TAPE.volser

SS212.0

Notes:

This profile is used in conjunction with the DFSMSrmm exit, EDGUX100. The sample provided uses the special date, 98000, to request that a tape be ignored by DFSMSrmm. When this is requested, DFSMSrmm checks the user's authority to ignore the tape. READ access to the profile allows the tape volser to be read while UPDATE allows the volume to be opened for output.

STGADMIN.EDG.LABEL.volser

ACCESS	Description
Not defined	A volume must be in user status to switch from NL or to change to label types at OPEN time
UPDATE	IBM standard labels can be created on a non- scratch volume for an AL or SL output request
ALTER	IBM standard labels can be created on a scratch volume during a nonspecific volume request for AL or SL

Figure 8-10. STGADMIN.EDG.LABEL.volser SS212.0

Notes:

Occasionally, there may be a need to write a data set using labels that do not match the existing label format. This profile may be used to authorize changing of the label type for a volume. UPDATE versus ALTER controls whether only SL to AL and AL to SL may be done or if an NL tape may be made into an SL tape.

STGADMIN.EDG.NOLABEL.volser

ACCESS	Description
Not defined	A volume must be in user status to switch to NL at OPEN time
UPDATE	You are allowed to destroy labels on a non-scratch volume for a no label output request
ALTER	You are allowed to destroy labels on a scratch volume during nonspecific volume requests for no labels

Figure 8-11. STGADMIN.EDG.NOLABEL.volser

SS212.0

Notes:

Access to this profile allows standard labelled scratch tapes to be written over as unlabeled (NL) tapes. Applications creating transmittal tapes often have this requirement. Only those applications needing this capability should be granted access to this profile.

STGADMIN.EDG.OPERATOR

ACCESS	Description
Not defined	Same as UPDATE access
NONE	No authority is granted to use EDGINERS to initialize and erase volumes
READ	No authority is granted to use EDGINERS to initialize and erase volumes
UPDATE	EDGINERS can be used to initialize and erase volumes

Figure 8-12. STGADMIN.EDG.OPERATOR SS212.0

Notes:

This profile controls the use of the DFSMSrmm utility program EDGINERS which is used to initialize or erase and initialize tapes.

STGADMIN.EDG.INERS.WRONGLABEL

ACCESS	Description
Not defined	Use of the EDGINERS EXEC statement PARM IGNORE and RMMPROMPT parameters is denied
CONTROL	You are allowed to use the EDGINERS EXEC statement PARM IGNORE parameter
UPDATE	You are allowed to use the EDGINERS EXEC statement PARM RMMPROMPT parameter

Figure 8-13. STGADMIN.EDG.INERS.WRONGLABEL SS212.0

Notes:

This profile is also used with EDGINERS. Upon occasion, it may be necessary to reinitialize a tape when the existing volser is not known. EDGINERS has a parameter, **IGNORE**, that allows a tape to be initialized as long as the volser is not known or is in pending release status of INIT. Update access requires an operator reply while control access does not. In either case, messages are logged with the original volser that was overwritten with a new label.

STGADMIN.EDG.RESET.SSI

ACCESS	Description
Not defined	You cannot use the EDGRESET utility to remove DFSMSrmm from the system If you have no security product installed, you can use EDGRESET to remove DFSMSrmm from the system
ALTER	You are allowed to withdraw DFSMSrmm from the system during error recovery or problem during implementation

Figure 8-14. STGADMIN.EDG.RESET.SSI SS212.0

Notes:

In any mode other than manual, DFSMSrmm is maintaining information in the control data set (CDS). If DFSMSrmm is withdrawn from the system and subsequent tape activity occurs, its CDS becomes out of date. This profile controls the ability to stop DFSMSrmm and withdraw it from the system.

STGADMIN.EDG.VRS

ACCESS	Description
Not defined	Same as CONTROL access
READ	You are allowed to list and search for all vital record specifications
CONTROL	Same as READ access, plus: You can define and delete vital record specifications

Figure 8-15. STGADMIN.EDG.VRS SS212.0

Notes:

This controls who is allowed to create and remove life cycle management records in the control data set.

STGADMIN.EDG.HOUSEKEEP



ACCESS	Description
Not defined	Same as READ access
READ	Any of the inventory management facilities can be invoked

Figure 8-16. STGADMIN.EDG.HOUSEKEEP SS212.0

Notes:

Read access to the profile allows the DFSMSrmm housekeeping program and all of its functions to be used.

Suggested Access 1 of 2

Resource STGADMIN.EDG...	General User	Storage Administrator	System Programmer	Librarian	Inventory Management Functions	Operator
FCRCL	-	Update	-	Update	-	-
HOUSEKEEP	-	-	-	-	Read	-
HOUSEKEEP.SP+EXT	-	Read	Read	Read	Read	Read
IGNORE.TAPE..ulser	-	-	-	-	-	-
LABEL.volser	Update	Alter	Alter	Alter	Alter	Alter
LISTCONTROL	Control	Control	Control	Control	-	Control
MASTER	Read	Control	Control	Control	-	Control

Figure 8-17. Suggested Access 1 of 2

SS212.0

Notes:

These are the suggested profile access levels for the different DFSMSrmm user categories.

Suggested Access 2 of 2

Resource STGADMIN.EDG...	General User	Storage Administrator	System Programmer	Librarian	Inventory Management Functions	Operator
MOLABEL.volser	Update	Alter	Alter	Alter	Alter	Alter
OPERATOR	-	-	Update	Update	-	Update
OWNER.usesid	-	Update	-	-	-	-
RELEASE	Read	-	-	-	-	-
RESET.SSI	-	-	-	-	-	Alter
VRS	Read	Control	Control	Control	-	-
INERS.WRONGI.ABFI	-	-	-	Update	-	-

Figure 8-18. Suggested Access 2 of 2 SS212.0

Notes:
These are the suggested profile access levels for the different DFSMSrmm user categories.

Security (1 of 4)

- **Creating Profiles**
 - **Creating a RACF Profile**
 - **RDEFINE FACILITY -**
STGADMIN.EDG.MASTER UACC(READ)
- **Creating a RACF Profile with Audit Options**
 - **RDEFINE FACILITY -**
STGADMIN.EDG.VRS UACC(NONE)
 - **RALT FACILITY -**
STGADMIN.EDG.VRS UACC(NONE) -
GLOBALAUDIT(ALL(UPDATE))
- **Permit**
 - **User SCHLUM needs CONTROL access authority to the facility STGADMIN.EDG.MASTER entity**
 - **PERMIT 'STGADMIN.EDG.MASTER' ID(SCHLUM)**
 - **ACC(CONTROL) CLASS(FACILITY)**

Figure 8-19. Security (1 of 4)

SS212.0

Notes:

A simple profile may be created, such as the one shown for **MASTER**. If RACF audit trails are desired, that can be done as shown for the **VRS** profile. The last example shows how to grant access that would otherwise not be available to the defined profile.

Security (2 of 4)

- **RACF User ID**
 - **Assigning DFSMSrmm a RACF user ID**

```
•RDEFINE STARTED DFRMM
      STDATA(USER(=MEMBER)
              GROUP(STC)
              PRIVILEGED(NO)
              TRACE(NO)
              TRUSTED(NO) )

•or ICHRINØ3
```

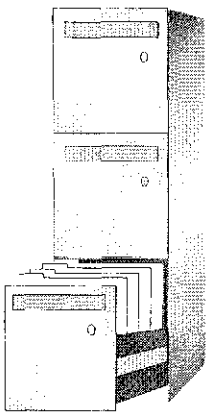


Figure 8-20. Security (2 of 4)

SS212.0

Notes:

Started tasks, such as DFSMSrmm, do not have an explicit job name. They should be defined to RACF based on the procedure name used to start them. In this example, we have used the IBM suggested example, DFRMM, as the procedure name and identified it to RACF.

Security (3 of 4)

- **Define data set profiles**

- **for the DFSMSrmm environment**

- ADDGROUP RMM SUBGROUP(SYS1) OWNER(SYS1)
 - ADDSD 'RMM.*' UACC(NONE)
 - PERMIT 'RMM.*' ID(.....) ACCESS(ALTER)

- **Final activities**

- SETROPTS GENERIC(FACILITY) REFRESH
 - SETROPTS RACLIST(FACILITY) REFRESH
 - SETROPTS GENERIC(DATASET) REFRESH
 - SETROPTS GENERIC(STARTED) REFRESH

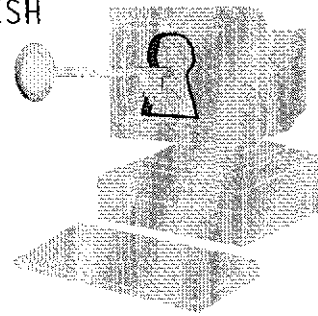


Figure 8-21. Security (3 of 4)

SS212.0

Notes:

The example illustrates defining a generic data set profile for the DFSMSrmm data sets such as the CDS and journal. Only the IDs in the PERMIT list will be allowed to work with the data sets outside of DFSMSrmm. For example, system support will occasionally need to reorganize the DFSMSrmm control data set.

The **REFRESH** examples tell RACF to update its information for the areas we have changed.

Security (4 of 4)

- Create the RACF started task table, if **STARTED** class not active

- Example

```
ICHRIN03 CSECT
ENTRIES DC AL2(ENT+X'800') NUMBER OF ENTRIES FOLLOWING
        SPACE 3
*      DC C < PROC >< USER >< GROUP> <FLAG> < RESERVED
>
        SPACE 3
FIRST DC C'ANOPX11 STCNETV STCGRP
',X'00',X'0000000000000000'
...
LENGTH DS 0C
        DC C'DFRMM STCDFRM STCGRP
',x'00',X'0000000000000000'
...
LAST DS 0C
LEN EQU LENGTH-FIRST
LAS EQU LAST-FIRST
ENT EQU LAS/LEN
END ICHRIN03
```

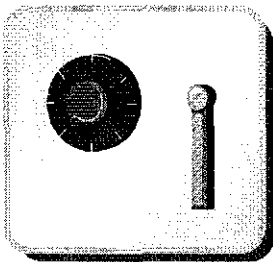


Figure 8-22. Security (4 of 4)

SS212.0

Notes:

This is an example of using ICHRIN03. This is needed only where the **STARTED** class for RACF is not available.

Checkpoint

1. A user having CONTROL to STGADMIN.EDG.MASTER may perform all DFSMSrmm functions.
2. If no STGADMIN.EDG profiles are defined and access is not granted via a generic profile, all DFSMSrmm functions may be performed by everyone.
3. The general recommendation is that all users should have READ access to STGADMIN.EDG.MASTER and STGADMIN.EDG.VRS.

Figure 8-23. Checkpoint

SS212.0

Notes:

Write down your answers here:

- 1.
- 2.
- 3.

Unit Summary

- Facility classes are used to:
 - Protect DFSMSrmm Resources
 - Authorize DFSMSrmm Functions

Figure 8-24. Unit Summary

SS212.0

Notes:

Appendix B. Checkpoint Solutions

Unit 1

1. False
2. True
3. False
4. True

Unit 2

1. False
2. True
3. True
4. False
5. False

Unit 3

1. False
2. True
3. True
4. False
5. d
6. False
7. True

Unit 4

1. False
2. True
3. 3, 1, 2, 4, 7, 5, 8, 6
4. False
5. True
6. True

Unit 5

1. False
2. True
3. False
4. False
5. True
6. True

Unit 6

1. True
2. False
3. False
4. False
5. False

Unit 7

1. True
2. False
3. False
4. True
5. True
6. False

Appendix A

1. False
2. False
3. True

Bibliography

Manuals:

SC26-7404	<i>DFSMSrmm Guide and Reference</i>
SC26-7405	<i>DFSMSrmm Implementation and Customization Guide</i>
SC26-7406	<i>DFSMSrmm Reporting</i>
SY27-9615	<i>DFSMSrmm Diagnosis Guide</i>

Technical Bulletins:

SG24-4998	<i>Converting to DFSMSrmm - A Practical Guide</i>
SG24-6241	<i>Converting to DFSMSrmm from CA-1</i>
SG24-6242	<i>Converting to DFSMSrmm from TLMS</i>
SG24-6243	<i>Converting to DFSMSrmm from CONTROL-T</i>
SG24-6244	<i>Converting to DFSMSrmm from ICF-usercatalog</i>
SG24-6275	<i>Converting to DFSMSrmm from Automevia/ZARA</i>
SG24-5983	<i>DFSMSrmm Primer</i>





DFSMSrmm Implementation

(Course Code SS21)

Student Exercises

ERC 2.0

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Student Exercises

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Student Exercises

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Student Exercises

Exercises Description

Exercise instructions - This section contains what you should accomplish. There are no definitive details on how to perform the tasks. You are given the opportunity to work through the exercise given what you learned in the unit presentation.

Exercise 1. DFSMSrmm ISPF Application

What This Exercise is About

This lab introduces you to the DFSMSrmm ISPF application you can use to help manage and analyze both contents of the CDS and the DFSMSrmm system wide options. The experience you gain in this lab should help make subsequent labs easier, allowing you to concentrate on what is happening rather than the mechanics involved.

What You Should Be Able to Do

At the end of the lab, you should be able to:

- Navigate through the ISPF application
- Perform simple search processing for volumes and data sets
- Display volume and data set details
- Switch from foreground to background mode

Introduction

The exercise offers a guided walk through DFSMSrmm's ISPF application to become familiar with the tool.

Required Materials

SS21 Student Guide

SC26-7405 *DFSMSrmm Guide and Reference*

Exercise Instructions

Preface

- All exercise steps of this chapter depend on the availability of a terminal supporting 3270 emulation for each team of students and connectivity to the lab system. Your instructor will provide you with the information how to connect with the lab system.

Step 1

- ___ 1. Logon as TSO. Your instructor will provide you with your group number n and the logon password.
- ___ 2. Start the DFSMSrmm ISPF application using TSO RMMISPF from any command line. Note that the primary option menu appears as shown in chapter 1 of the student notes of the course material. You now have the choice of selecting a path like USER, LIBRARIAN, STORAGE ADMINISTRATOR, and so on.
- ___ 3. Finish the application. Now, please, start ISMF. In which mode does ISMF currently operate (*end user* or *storage administrator*)? Select DFSMSrmm from the primary option menu of ISMF. What is the look of the primary option menu of DFSMSrmm now?
- ___ 4. Leave DFSMSrmm as well as ISMF again. Now start the DFSMSrmm application for a third time using command TSO RMMISPF LIBRARIAN. What is the look of DFSMSrmm now? Note that the primary option menu appears depending on the option used with the call of RMMISPF. For details see SC26-7404 *z/OS DFSMSrmm Guide and Reference*.
- ___ 5. To perform a data set search, use the fast path command DATASET SEARCH from the command line of the current menu. Note that the data set search option is not offered in the librarian menu. We will perform a data set search later in this exercise. For details on using fast path commands see Appendix in SC26-7404 *z/OS DFSMSrmm Guide and Reference*.
- ___ 6. Now use another fast path command entering VOLUME SEARCH in the command line.

Step 2

- ___ 1. Perform a volume search for volumes that show your TSO userid as owner. Don't forget to use a fitting LIMIT search argument because otherwise you will not see all of the 14 pre-defined volumes in the resulting volume list.
- ___ 2. Fill in the missing values in Table 1 using line commands in column 1 of the volume list. Note that some values are not recorded in the volume record but in the data set record. Line commands V, I, and L will help you to find the requested informations. Use PF1 for help if needed. The table just should help you to walk through the panels of volume details and data set details. You need not fill out all cells of the

table to finish this part of the exercise successfully. The table should help you to take a guided walk through the panels that provide informations about volumes and data sets.

Table 1:

Volume	Expiration Date	Data Set Name		
0n0000			Volume Status	Volume Availability
0n0001			Creating Job	Volume Location
0n0002			DS Creation Date	DS Last Reference Date
0n0003			Volume Status	Media Type
0n0004			Label Type	Assign Date
0n0005			DS Sequence #	Previous Volume
0n0006			Media Name	Density
0n0007			Block Count	Creating Job
0n0008			Recording Format	Special Attributes
0n0009			Compaction	Abend while open
0n0010			Number of Data Sets	Next Volume
0n0011			Volume Type	File 1 system ID
0n0012			DS Block Size	Release Action
0n0013			Vol. Availability	Volume Location

Student Exercises

- ___ 3. Leave the volume list now and make a new selection to perform a data set search. Create a data set list containing all data sets residing on volumes that show your TSO userid as owner. Don't forget to use a fitting LIMIT search argument because otherwise you will not see all the 15 data sets in the resulting data set list. Spend some minutes to try out the line commands available in the list.
- ___ 4. Why does the data set list contain 15 entries? Remember that the volume search you did before presented only 14 volumes that contain your user ID as the volume owner.

Step 3

- ___ 5. Take a look at the OPTIONS of the DFSMSrmm ISPF application. You may choose individual settings for processing, viewing, and sorting. See SC26-7404 *z/OS DFSMSrmm Guide and Reference* for details. Now, please, change processing options to B (background).
- ___ 6. Repeat the actions described under 1.) of step 2. A volume search will not be processed at once (in foreground), but the TSO RMM subcommand that is needed in order to perform the volume search is generated and stored in a data set for later processing (in background).
- ___ 7. Now leave the DFSMSrmm ISPF application. There will be presented an additional panel to allow a selection of processing options for the commands that were generated. Choose option **B** to browse the generated command. End browse and choose **S** to submit a job for processing of the volume search in batch TSO environment. Check the job output for results using SDSF. You will find the output of the job using the H option in SDSF. Select the job by using the line command S.

END OF LAB

Exercise 2. Housekeeping and Reporting

What This Exercise is About

This lab covers basic housekeeping issues and reporting facilities of DFSMSrmm.

What You Should Be Able to Do

At the end of the lab, you should be able to:

- Evaluate the results of a housekeeping run including EXPROC, RPTEXT, and BACKUP processing
- Perform standard DFSMSrmm reporting using EDGRPTD and report generator.

Introduction

Using the same volumes and data sets as in exercise 1, we take a look at what happens with those volumes during housekeeping. After performing inventory management, you will produce reports using standard batch utilities of DFSMSrmm and report generator which is part of the DFSMSrmm ISPF application.

Required Materials

SS21 Student Notebook

SC26-7406 z/OS DFSMSrmm Reporting

Exercise Instructions

Preface

- All exercises of this chapter depend on the availability of a terminal supporting 3270 emulation for each team of students and connectivity to the lab system. Your instructor will provide you with information how to connect with the lab system.

Step 1

- ___ 1. Logon as TSO_n. Your instructor will provide you with your group number *n* and the logon password (see Exercise 1).
- ___ 2. Start the DFSMS_{rmm} ISPF application using TSO RMMISPF from any command line. Perform a volume search for volumes that have your TSO userid as owner. Don't forget to use a fitting LIMIT search argument because otherwise you will not see all of the 14 pre-defined volumes in the resulting volume list.
- ___ 3. Select appropriate values where offered in Table 2. Line command V will help you to find the requested informations. Use PF1 for help if needed. Predict the status and the availability of each volume after the end of the inventory management run and fill in your predictions in Table 2.

Table 2:

Volume	Expiration date	Status after end of housekeeping	Availability after end of housekeeping
0n0000	Reached / not reached	Master / User / Scratch	- / Pending release
0n0001	Reached / not reached	Master / User / Scratch	- / Pending release
0n0002	Reached / not reached	Master / User / Scratch	- / Pending release
0n0003	Reached / not reached	Master / User / Scratch	- / Pending release
0n0004	Reached / not reached	Master / User / Scratch	- / Pending release
0n0005	Reached / not reached	Master / User / Scratch	- / Pending release
0n0006	Reached / not reached	Master / User / Scratch	- / Pending release
0n0007	Reached / not reached	Master / User / Scratch	- / Pending release
0n0008	Reached / not reached	Master / User / Scratch	- / Pending release
0n0009	Reached / not reached	Master / User / Scratch	- / Pending release

Table 2:

Volume	Expiration date	Status after end of housekeeping	Availability after end of housekeeping
0n0010	Reached / not reached	Master / User / Scratch	- / Pending release
0n0011	Reached / not reached	Master / User / Scratch	- / Pending release
0n0012	Reached / not reached	Master / User / Scratch	- / Pending release
0n0013	Reached / not reached	Master / User / Scratch	- / Pending release

- ___ 4. Wait until your instructor has run the inventory management. Now verify your predictions by displaying volume details of your volumes. To do this, repeat the volume search as described in 2.).

Step 2

- ___ 5. RPTEXT processing created an extended extract file in data set SS21.XREPORT.EXTRACT.FILE.L1 during inventory management. This is the base for the reporting utilities you will now use. Submit the jobs that are contained in the data set TSOOn.EX1.JCL.
- ___ 6. Have a look at the reports that were created by the jobs. Some of the reports are empty because in our current environment we have not covered all situations that are possible subjects of reporting. During the following exercises you will have the opportunity to repeat reporting to find an example of some of the reports that are currently empty.
- ___ 7. Now use report generator to produce some reports. You will find in report generator a number of pre-defined reports. These are the same reports as offered by extended reporting. Generate the JCL for some reports of your choice and submit them. Also, try to tailor a pre-defined report in report generator by adding or deleting a column. Follow the steps as described in Unit 2 of your student notes.
- ___ 8. Next, perform the volume search again as described in 2.). By using the line command C, you can change the characteristics of a volume. Switch the status of volume 0n0000 back to MASTER. This can be an important action to rescue a volume from being overwritten. If users want to read the volume again, they cannot request the volume when it is in status SCRATCH, because DFSMSrmm will reject specific requests for scratch volumes.
- ___ 9. Verify your change by re-displaying the volume details. What is the expiration date of the volume now?
- ___ 10. Repeat execution of EDGRPTD by submitting the job in TSOOn.EX1.JCL(EDGRPTD) again. Why does the scratch volume report still contain volume 0n0000?

END OF LAB

Student Exercises

Exercise 3. Implementation Tasks

What This Exercise is About

This lab covers some implementation tasks that are assisted by the DFSMSrmm customization wizard.

What You Should Be Able to Do

At the end of the lab, you should be able to use the DFSMSrmm customization wizard to generate jobs that will perform some of the implementation tasks which are needed when you start to implement DFSMSrmm.

Introduction

When you prepare DFSMSrmm for the first start of the started task, you must perform actions like creating a procedure for the started task, creating an EDGRMMxx member, or defining some profiles in RACF. The DFSMSrmm customization wizard assists you by creating the necessary statements or jobs to perform these initial steps.

Required Materials

SS21 Student Notebook

Exercise Instructions

Preface

- All exercises of this chapter depend on the availability of internet access and a browser like NETSCAPE or MICROSOFT internet explorer in your classroom.

Visit the web site that provides the DFSMSrmm customization wizard

- ___ 1. Choose one of the web sites noted on page 3-6 in your student notes.
- ___ 2. Choose customized setup.
- ___ 3. Select all offered links to specify information which will be used to customize the generated customization tasks.
- ___ 4. After having entered all necessary information, press Build Configuration Tasks button.
- ___ 5. Next, select 'All Definition Files combined together.'
- ___ 6. Have a look at the generated stuff and go 'Back to Interviews' if needed or store the generated output on your local machine.

END OF LAB

Exercise 4. Retention Management using Vital Record Specifications

What This Exercise is About

This lab covers how to use vital record specifications for the retention management of tape volumes.

What You Should Be Able to Do

At the end of the lab, you should be able to:

- Define a vital record specification that is matching a data set by name
- Use ADDVRS statements for appropriate retention control of data sets and volumes
- Identify the matching data set(s) for a given vital record specification data set name mask
- Predict the resulting retention management for data sets and volumes based on a matching vital record specification
- Interpret the contents of a VRSEL report
- Understand the contents and use of an activity file

Introduction

DFSMSrmm uses the vital record specification function to define and enforce the retention and movement of tape data. A vital record specification defines how long a data set or volume should be kept and where it should be kept. When a data set is protected by a VRS, the volume it resides on is not subject to expiration processing. When all data sets on a volume are no longer protected by vital record specifications, DFSMSrmm retains the volume until the volume expiration date unless EXPIRYDATEIGNORE being set in the volume release action set. We have pre-defined a set of volumes and data sets for each lab team. The expiration dates of all volumes have been exceeded, so all volumes would result in availability of pending release without of VRS control. The following exercise instructions contain specifications how the VRS protection should work and it is up to you to define the appropriate vital record specifications. Once these VRSS are defined, you will be asked to predict which volumes are to be retained and which are to be released. Inventory management will be

Student Exercises

executed to perform expiration processing and you will then use DFSMSrmm ISPF interface or commands to verify the accuracy of the predictions.

Required Materials

SS21 Student Notebook

Exercise Instructions

Preface

- All exercises of this chapter depend on the availability of a terminal supporting 3270 emulation for each team of students.

Defining Vital Record Definitions

We have pre-defined a set of volumes and data sets for each lab team. The expiration dates of all volumes have been exceeded, so all volumes would result in an availability of pending release without of VRS protection.

Volume	Volume EXPDT	Data set	Data set creation date	Data set last reference date
0n0000	today-1	TSON.DAYS.EXP	today -10	today - 10
0n0001	today-1	TSON.DAYS.NOEXP	today - 2	today - 2
0n0002	today-1	TSON.LREF.EXP	today - 10	today - 10
0n0003	today-1	TSON.LREF.NOEXP	today - 10	today - 10
0n0004	today-1	TSON.GDG.G0001.V00	today - 7	today - 7
0n0005	today-1	TSON.GDG.G0002.V00	today - 6	today - 6
0n0006	today-1	TSON.C01V0001	today - 9	today - 9
0n0007	today-1	TSON.C01V0002	today - 9	today - 9
0n0008	today-1	TSON.CYC.C01V0001	today - 9	today - 9
0n0009	today-1	TSON.CYC.C01V0002	today - 9	today - 9
0n0010	today-1	TSON.SPEC.EXP.EXAMPLE	today - 9	today - 9
0n0011	today-1	TSON.SPEC.NOEXP.EXAMPLE	today - 9	today - 9
0n0012	today-1	TSON.CATALOG.NOT.EXAMPLE	today - 9	today - 9
0n0013	today-1	TSON.CATALOG.EXAMPLE	today - 9	today - 9

- ___ 1. Logon to TSO as described in Exercise 1. Use DFSMSrmm ISPF dialog or TSO subcommands to define the following vital record specifications:
- ___ 2. Define a data set VRS that fulfills the following requirements:
 - Match data sets whose first two qualifiers are 'TSON.DAYS'
 - Keep the data sets five days since creation
- ___ 3. Define a data set VRS that fulfills the following requirements:
 - Match data sets whose first two qualifiers are 'TSON.LREF'
 - Keep a data set as long as it has been referenced within the last five days
- ___ 4. Define a data set VRS that fulfills the following requirements:
 - Match data sets whose first two qualifiers are 'TSON.CYC'
 - Keep one version of the data set

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- ___ 5. Define a data set VRS that fulfills the following requirements:
 - Match GDG data sets whose GDG base name is 'TSOn.GDG'
 - Keep one member of the GDG
- ___ 6. Define a data set VRS that fulfills the following requirements:
 - Match data sets whose first qualifier is 'TSOn' and the second qualifier is eight characters long with the first character being a 'C' and the fourth character being a 'V'. For other characters in the second qualifier use the ignore character as a masking character which, in fact, makes up a pseudo-GDG definition.
 - Keep one version of the pseudo-GDG
- ___ 7. Define a data set VRS that fulfills the following requirements:
 - Match data sets whose first three qualifiers are 'TSOn.SPEC.NOEXP'
 - Expire the vital record specification tomorrow
- ___ 8. Define a data set VRS that fulfills the following requirements:
 - Match data sets whose first two qualifiers are 'TSOn.CATALOG'
 - Keep a data set as long as it is cataloged
- ___ 9. After having completed your definitions, you should see the following vital record specifications (Perform a VRS search for vital record specifications that show your userid as the owner):

Table 3: VRS Definitions

Data set name	Type	Retention type	Count	Delete date
ABEND	DSN	Days	8	1999/365
TSOn.DAYS.**	DSN	Days	5	1999/365
TSOn.LREF.**	DSN	Lastreferencedays	5	1999/365
TSOn.CYC.C%%V%%%	DSN	Cycles	1	1999/365
TSOn.GDG	GDG	Cycles	1	1999/365
TSOn.C^V^	PGDG	Cycles	1	1999/365
TSOn.SPEC.EXP	DSN	Specific date	99999	today - 1
TSOn.SPEC.NOEXP	DSN	Specific date	99999	today + 1
TSOn.CATALOG.**	DSN	Cycles / Whilecatalog	99999	1999/365

- ___ 10. Note that there is an additional VRS which was pre-defined for each team 'TSOn.SPEC.EXP' to show that if its delete date is exceeded, a VRS will be deleted during inventory management.
- ___ 11. Also, a VRS 'ABEND' was predefined. You are not the owner of this VRS, so you will not see this VRS in the resulting list of your VRS SEARCH action, but it might affect your predictions of the inventory management run. Keep in mind when you predict the result of the following inventory management run that, if any of your data sets

was closed by abend, this VRS will supersede the VRS that matches the true data set name.

Prediction of the housekeeping results

1. Before doing a real housekeeping run, you should predict which volumes will be retained by a VRS. Use the information in the table below together with the matching VRSs. The volumes that are not protected by a VRS will be updated by inventory management to show an availability of pending release because their expiration dates are being reached. Please fill in the cells in the most right column of the table by writing *RETAIN* or *RELEASE*.

Table 4: Retention of Volumes by VRSs

Volume	Volume EXPDT	Data set	Data set creation date	Data set last reference date	Volume is still retained or will be released
0n0000	today-1	TSOn.DAYS.EXP	today-10	today - 10	
0n0001	today-1	TSOn.DAYS.NOEXP	today- 2	today - 2	
0n0002	today-1	TSOn.LREF.EXP	today- 10	today - 10	
0n0003	today-1	TSOn.LREF.NOEXP	today- 10	today - 10	
0n0004	today-1	TSOn.GDG.G0001.V00	today- 7	today - 7	
0n0005	today-1	TSOn.GDG.G0002.V00	today- 6	today - 6	
0n0006	today-1	TSOn.C01V0001	today- 9	today - 9	
0n0007	today-1	TSOn.C01V0002	today- 9	today - 9	
0n0008	today-1	TSOn.CYC.C01V0001	today- 9	today - 9	
0n0009	today-1	TSOn.CYC.C01V0002	today- 9	today - 9	
0n0010	today-1	TSOn.SPEC.EXP.EXAMPLE	today- 9	today - 9	
0n0011	today-1	TSOn.SPEC.NOEXP.EXAMPLE	today- 9	today - 9	
0n0012	today-1	TSOn.CATALOG.NOT.EXAMPLE	today- 9	today - 9	
0n0013	today-1	TSOn.CATALOG.EXAMPLE	today- 9	today - 9	

2. Wait until your instructor has run the inventory management. Now verify your predictions by displaying volume details of your volumes. To do this, perform a volume search. Alternatively, you can try the CONFIRM option on the librarian menu and select the line that shows a status of PENDING.
3. Check the content of the report file SS21.REPORT.FILE.L4 that was created during the inventory management run. Search for your VRSs and the matching data sets. If you don't find the expected results, try to find out what was wrong with your VRSs.
4. Check for the VRS TSOn.SPEC.EXP.EXAMPLE, which should have been deleted during the inventory management.
5. Why is volume 0n0010 still retained?
6. Create a report of the contents of the activity file by using the job in your TSOn.EX4.JCL(EDJACTP). Check for the appearance of your data sets and

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volumes in all the reports that are contained in the output of the job. The activity file was created during the trial run of inventory management.

END OF LAB

Exercise 5. Movement Management using Vital Record Specifications

What This Exercise is About

This lab covers how to use vital record specifications to move tape volumes.

What You Should Be Able to Do

At the end of the lab, you should be able to:

- Use ADDVRS statements to set up policies to move volumes
- Chain vital record specifications by using the name type VRSs to define a sequence of moves
- Predict the resulting destinations of volumes based on a matching vital record specification
- Interpret the contents of a VRSEL report
- Understand the contents and use of an activity file

Introduction

A vital record specification can be used to manage the movement of tape volumes. When a VRS is defined for a data set or a volume, the storage location where it is to be retained is specified. The movement and retention of volumes in and out of storage locations is done during DFSMSrmm inventory management. During vital record processing, DFSMSrmm sets the required location for each volume using information from vital record specifications. The volume does not move to the required location during vital record processing. During storage location management processing, DFSMSrmm sets the destination for a volume using the required location information if that volume can be moved. If a move is required, and the move is to a shelf-managed storage location, DFSMSrmm assigns an empty bin in the target location to the volume based on the media name and LOCDEF information.

Each volume record contains the home location information, which is by default the starting location of the volume (the location derived from the ADDVOLUME subcommand). A volume is kept at a storage location as long as a VRS that specifies the location information retains the volume. When a volume is no longer protected by a VRS

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and the current location is different than its home location, a move to its home location is scheduled.

In our lab, we use the built-in storage locations LOCAL, DISTANT and REMOTE. Built-in storage locations are shelf-managed storage locations, thus you have to provide bins in each storage location in order to move volumes to these storage locations successfully.

Required Materials

SS21 Student Notebook

Exercise Instructions

Preface

- All exercises of this chapter depend on the availability of a terminal supporting 3270 emulation for each team of students.

Defining Vital Record Specifications and Bins

We have pre-defined a set of volumes and data sets for each lab team as shown in the table below. The current location of all volumes is SHELF.

Table 5: Current Location of the Volumes

Volume	Volume EXPDT	Data set	Data set creation date	Current location
0n0000	today+1	TSOn.FINANCE.VITAL	today	SHELF
0n0001	today+1	TSOn.FINANCE.VITAL	today	SHELF
0n0002	today+1	TSOn.PAYROLL.DISASTER	today- 5	SHELF
0n0003	today+1	TSOn.PAYROLL.DISASTER	today- 4	SHELF
0n0004	today+1	TSOn.PAYROLL.DISASTER	today- 3	SHELF
0n0011	today+1	TSOn.GDG.G0001V00	today	SHELF
0n0012	today+1	TSOn.GDG.G0002V00	today	SHELF
0n0013	today+1	TSOn.GDG.G0003V00	today	SHELF
0n0014	today+1	TSOn.GDG.G0004V00	today	SHELF
0n0015	today+1	TSOn.GDG.G0005V00	today	SHELF

1. Logon to TSO as described in Exercise 1. Use the DFSMSrmm ISPF dialog or TSO subcommands to define the following vital record specifications:
2. Add ten bin numbers in each of the built-in storage locations in order to make the storage locations available to receive volumes during inventory management. Do not specify any specific bin number as a starting point for the ADDBIN subcommand. Specify an asterisk (*) for the bin number to start with.
3. Define a name type VRS that fulfills these requirements:
 - The name of the VRS is LOCAL0n
 - Store two units at the LOCAL location
4. Define a data set VRS that fulfills the following requirements:
 - Match data sets whose first qualifier is 'TSOn' and last qualifier is 'VITAL'
 - Keep every version of the data set
 - Keep the data sets at the REMOTE location
5. Define a data set VRS that fulfills the following requirements:

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- Match data sets whose first qualifier is 'TSOn' and last qualifier is 'DISASTER'
- Keep three versions of the data set
- Keep the current version of the data set at the DISTANT location
- Move the older versions of the data set to the location LOCAL by using the LOCAL0n VRS as the next VRS in a chain

___ 6. Define a data set VRS that fulfills the following requirements:

- Match GDGD data sets whose first two qualifiers are 'TSOn.GDG'
- Retain six versions of the data set
- Keep the first three versions of the data set at the home location
- Use the previously coded name type VRS LOCAL0n as the next VRS in the chain

___ 7. After having completed your definitions, you should see the following vital record specifications (Perform a VRS search for vital record specifications that show your userid as the owner):

Table 6: Vital Record Specifications

Data set name	Type	Location	Storenumber	Count	Nextvrs
TSOn.**.VITAL	DSN	Remote	99999	99999	-
TSOn.**.DISASTER	DSN	Distant	1	3	LOCAL0n
TSOn.GDG	GDG	Home	3	6	LOCAL0n
LOCAL0n	NAME	Local	2		-

Prediction of the housekeeping results

___ 1. Before doing a real housekeeping run, you should predict the destination of the pre-defined volumes after inventory management has been run. Use the information in the table below together with the matching VRSs. Please fill in the cells in the most right column of the table by writing the name of the target storage location.

Table 7: Prediction of the Destination after Inventory Management

Volume	Volume EXPDT	Data set	Data set creation date	Required location	Destination
0n0000	today+1	TSOn.FINANCE.VITAL	today	SHELF	
0n0001	today+1	TSOn.FINANCE.VITAL	today	SHELF	
0n0002	today+1	TSOn.PAYROLL.DISASTER	today- 5	SHELF	
0n0003	today+1	TSOn.PAYROLL.DISASTER	today- 4	SHELF	
0n0004	today+1	TSOn.PAYROLL.DISASTER	today- 3	SHELF	
0n0011	today+1	TSOn.GDG.G0001V00	today	SHELF	
0n0012	today+1	TSOn.GDG.G0002V00	today	SHELF	
0n0013	today+1	TSOn.GDG.G0003V00	today	SHELF	
0n0014	today+1	TSOn.GDG.G0004V00	today	SHELF	
0n0015	today+1	TSOn.GDG.G0005V00	today	SHELF	

- ___ 2. Wait until your instructor has run the inventory management. Now verify your predictions by displaying volume details of your volumes. To do this, perform a volume search. Alternatively, you can try the CONFIRM option on the librarian menu and select the lines that shows a status of PENDING and an action of MOVE.
- ___ 3. Check the content of the report file SS21.REPORT.FILE.L4 that was created during the inventory management run. Search for your VRSs and the matching data sets. If you don't find the expected results, try to find out what was wrong with your VRSs.
- ___ 4. Create movement reports by using the jobs provided in your TSOOn.EX5.JCL data set and by using report generator.
- ___ 5. Create a report of the contents of the activity file by using the job in your TSOOn.EX5.JCL(EDJACTP). Check for the appearance of your data sets and volumes in all the reports that are contained in the output of the job. The activity file was created during the trial run of inventory management.
- ___ 6. Now confirm the move of one of your volumes by using the line command CM in the volume action summary list (see 2.) CONFIRM). Watch the change of the contents of the fields **DESTINATION**, **LOCATION**, and **INTRANSIT**.

END OF LAB

Student Exercises

Exercise 6. Advanced DFSMSrmm Vital Record Specification Processing

What This Exercise is About

This lab covers how to merge several retention policies by combining vital record specifications for the retention management of tape volumes.

What You Should Be Able to Do

At the end of the lab, you should be able to:

- Chain vital record specifications by using the NEXTVRS and ANDVRS keywords of the ADDVRS subcommand
- Use a secondary VRS for a data set by using a management class

Introduction

You select the type of processing you want DFSMSrmm to perform by specifying the DFSMSrmm EDGRMMxx parmlib member OPTION VRSEL(OLD) or VRSEL(NEW) operand. If you use vital record specification retention and movement functions that are not supported by VRSEL(OLD), DFSMSrmm ignores the unsupported values, issues message EDG2227I, and continues to process the remaining policy. If you use VRSEL(NEW), DFSMSrmm processes any vital record specification functions that are available.

When you specify VRSEL(NEW), you can use the following additional functions not available when you use VRSEL(OLD).

You can include retention information on a name vital record specification. If you chain vital record specifications using the RMM ADDVRS NEXTVRS or ANDVRS subcommand operands, each vital record specification can have a different retention type. If you use the ANDVRS operand, all the retention criteria you specify must be true for DFSMSrmm to apply a policy. If you define subchains in a vital record specification chain, DFSMSrmm applies the policies defined in each subchain independent of the rest of the chain.

You can retain cycles of data sets by day, where a cycle is all copies of a data set created on the same day. This is an alternative

to retaining cycles of data sets where a cycle is a single occurrence of a data set.

You can use the current location where a data set resides instead of only considering the home location or destination when defining policies. This allows both automatic and command-driven movement to be used within an installation, particularly useful when the location to be managed needs to be managed to a threshold, such as for an Tape Library Dataserver.

You can manage data set retention and movement separately. Use a vital record specification based on management value or management class name to define retention criteria; use a vital record specification based on data set name mask for defining movement.

You can request that a volume expiration date be ignored when the data set is released from retention by the policy.

You can request that volumes are returned to scratch status in one expiration processing run when the only release action is return to scratch.

You can define vital record specification chains that have different retention types on each vital record specification in a chain.

You can define vital record specification management values and management classes and use the DFSMSrmm EDGUX100 exit or your management class ACS routine to assign these values so retention policies can be applied to data sets and volumes.

You run DFSMSrmm inventory management vital record processing for DFSMSrmm to apply retention and movement policies to data sets and volumes. To release a volume, you must run vital record processing and expiration processing. We recommend running inventory management vital record processing and expiration processing in a single job step.

You can use DFSMSrmm to release DFSMSHsm tapes that are requested to be purged by DFSMSHsm. By default, the expiration date protection for DFSMSHsm tapes is done by DFSMSHsm. DFSMSHsm uses 1999/365 as the expiration date for permanent retention. To enable extra days retention for purged DFSMSHsm tape volumes, you need to set up retention options in the vital record specifications that are used to retain the tape volumes.

You can control how DFSMSrmm returns volumes to scratch using the ADDVRS RELEASE(EXPIRYDATEIGNORE) operand and the ADDVRS RELEASE(SCRATCHIMMEDIATE) operand. If you specify the SCRATCHIMMEDIATE operand and the only release

action specified for volumes is SCRATCH, then you can return the volumes to scratch in a single run of inventory management expiration processing. If you specify the EXPIRYDATEIGNORE operand, DFSMSrmm ignores the volume expiration date when the volume last change date and time is earlier than the vital record processing run date and time. DFSMSrmm then sets the volume to pending release status.

In this lab, you can try out the processing of chains of VRSs by using NEXTVRS and ANDVRS as well as the processing of a secondary VRS for retention management of volumes.

Required Materials

SS21 Student Notebook

Exercise Instructions

Preface

- All exercises of this chapter depend on the availability of a terminal supporting 3270 emulation for each team of students and connectivity to the lab system. Your instructor will provide you with information how to connect with the lab system.

Defining Vital Record Specifications

We have pre-defined a set of volumes and data sets for each lab team as shown in the table below.

Table 8: Pre-defined Volumes and Data Sets

Volume	Volume EXPDT	Data set	Data set creation date	Data set last reference date
0n0000	today-1	TSOn.DAYS.EXP	today -10	today - 10
0n0001	today-1	TSOn.DAYS.NOEXP	today - 2	today - 2
0n0002	today-1	TSOn.LREF.EXP	today - 10	today - 10
0n0003	today-1	TSOn.LREF.NOEXP	today - 10	today - 10
0n0004	today-1	TSOn.GDG.G0001.V00	today - 7	today - 7
0n0005	today-1	TSOn.GDG.G0002.V00	today - 6	today - 6
0n0006	today-1	TSOn.C01V0001	today - 9	today - 9
0n0007	today-1	TSOn.C01V0002	today - 9	today - 9
0n0008	today-1	TSOn.CYC.C01V0001	today - 9	today - 9
0n0009	today-1	TSOn.CYC.C01V0002	today - 9	today - 9
0n0010	today-1	TSOn.SPEC.EXP.EXAMPLE	today - 9	today - 9
0n0011	today-1	TSOn.SPEC.NOEXP.EXAMPLE	today - 9	today - 9
0n0012	today-1	TSOn.CATALOG.NOT.EXAMPLE	today - 9	today - 9
0n0013	today-1	TSOn.CATALOG.EXAMPLE	today - 9	today - 9
0n0014	today + 7	TSOn.IGNORE.EXPDT	today	today

We have also pre-defined a set of VRSs for each lab team as shown in the table below.

Table 9: VRS Definitions

Data set name	Type	Retention type	Count	NEXTVRS ANDVRS Secondary VRS
TSOn.DAYS.**	DSN	Days	5	
TSOn.LREF.**	DSN	Lastreferencedays	5	
TSOn.CYC.C%%V%%%	DSN	Cycles	1	
TSOn.GDG	GDG	Cycles	1	
TSOn.C^V^	PGDG	Cycles	1	
TSOn.SPEC.EXP	DSN	Specific date	99999	
TSOn.SPEC.NOEXP	DSN	Specific date	99999	

Table 9: VRS Definitions

Data set name	Type	Retention type	Count	NEXTVRS ANDVRS Secondary VRS
TSOn.CATALOG.**	DSN	Cycles / Whilecatalog	99999	

- ___ 1. Logon as TSOn. Your instructor will provide you with your group number n and the logon password (see Exercise 1).
- ___ 2. Define a dsn type VRS for use as a secondary VRS that fulfills the following requirements:
 - The data set name should be 'Mn99000'
 - Keep the data sets as long as they are cataloged
- ___ 3. Now assign the management class Mn99000 to the data set TSOn.LREF.EXP by using the line command C in a data set list (you must perform a data set search before) to simulate the processing of the management class ACS routine.
- ___ 4. Define a name type VRS that fulfills the following requirements:
 - The name should be 'CATALOGn'
 - Keep the data sets as long as they are cataloged
- ___ 5. The VRSs TSOn.C^^V^^^^ should be connected with the new VRS that you have just defined by using ANDVRS. Use the line command C in a list of VRSs that show your userid as owner to change the VRS TSOn.C^^V^^^^ accordingly. Insert the name CATALOGn in the **NEXT VRS in chain** field and the type AND in the **Chain using** field of the change data set VRS panel.
- ___ 6. The VRSs TSOn.DAYS.** should be connected with the new VRS that you have just defined by using NEXTVRS. Use the line command C in a list of VRSs that show your userid as owner to change the VRS TSOn.DAYS.** accordingly. Insert the name CATALOGn in the **NEXT VRS in chain** field and the type NEXT in the **Chain using** field of the change data set VRS panel.
- ___ 7. Define a dsn type VRS that fulfills the following requirements:
 - Match data set whose first two qualifiers are 'TSOn.IGNORE'
 - When the data set is no longer protected by a VRS, the volume expiration date should be ignored
 - The data set should be retained for two days since its creation
- ___ 8. After having completed your definitions, you should see the following vital record specifications (Perform a VRS search for vital record specifications that show your userid as the owner):

Table 10: VRS Definitions

Data set name / name	Type	Retention type	Count	NEXTVRS ANDVRS Secondary VRS
TSOn.DAYS.**	DSN	Days	5	(NEXT) CATALOGn
TSOn.LREF.**	DSN	Lastreferencedays	5	(secondary) Mn99000
TSOn.CYC.C%%V%%%	DSN	Cycles	1	
TSOn.GDG	GDG	Cycles	1	
TSOn.C^V^	PGDG	Cycles	1	(AND) CATALOGn
TSOn.SPEC.EXP	DSN	Specific date	99999	
TSOn.SPEC.NOEXP	DSN	Specific date	99999	
TSOn.CATALOG.**	DSN	Cycles / Whilecatalog	99999	
CATALOGn	Name	WHILECATALOG	99999	
Mn99000	DSN	WHILECATALOG	99999	

Prediction of the housekeeping results

1. Before doing a real housekeeping run, you should predict which volumes will be retained by a VRS. Use the information in the table below together with the matching VRSs. The volumes that are not protected by a VRS will be updated by inventory management to show an availability of pending release because their expiration dates are being reached. Please fill in the cells in the most right column of the table by writing *RETAIN* or *RELEASE*.

Table 11: Retention of Volumes by VRSs

Volume	Volume EXPDT	Data set	Data set creation date	Data set last reference date	Volume is still retained or will be released
0n0000	today-1	TSOn.DAYS.EXP	today -10	today - 10	
0n0001	today-1	TSOn.DAYS.NOEXP	today - 2	today - 2	
0n0002	today-1	TSOn.LREF.EXP	today - 10	today - 10	
0n0003	today-1	TSOn.LREF.NOEXP	today - 10	today - 10	
0n0004	today-1	TSOn.GDG.G0001.V00	today - 7	today - 7	
0n0005	today-1	TSOn.GDG.G0002.V00	today - 6	today - 6	
0n0006	today-1	TSOn.C01V0001	today - 9	today - 9	
0n0007	today-1	TSOn.C01V0002	today - 9	today - 9	
0n0008	today-1	TSOn.CYC.C01V0001	today - 9	today - 9	
0n0009	today-1	TSOn.CYC.C01V0002	today - 9	today - 9	
0n0010	today-1	TSOn.SPEC.EXP.EXAMPLE	today - 9	today - 9	

Table 11: Retention of Volumes by VRSs

Volume	Volume EXPDT	Data set	Data set creation date	Data set last reference date	Volume is still retained or will be released
0n0011	today-1	TSOn.SPEC.NOEXP.EXAMPLE	today - 9	today - 9	
0n0012	today-1	TSOn.CATALOG.NOT.EXAMPLE	today - 9	today - 9	
0n0013	today-1	TSOn.CATALOG.EXAMPLE	today - 9	today - 9	
0n0014	today + 7	TSOn.IGNORE.EXPDT	today	today	

- ___ 2. Wait until your instructor has run the inventory management. Now verify your predictions by displaying volume details of your volumes. To do this, perform a volume search. Alternatively, you can try the CONFIRM option on the librarian menu and select the line that shows a status of PENDING.
- ___ 3. Check the content of the report file SS21.REPORT.FILE.L6 that was created during the inventory management run. Search for your VRSs and the matching data sets. If you don't find the expected results, try to find out what was wrong with your VRSs.
- ___ 4. Which subchain retains the volume 0n0000 currently?
- ___ 5. Which subchain retains the volume 0n0002 currently?
- ___ 6. Why is the data set TSOn.C^V^ not contained in the report file?
- ___ 7. Create a report of the contents of the activity file by using the job in your TSOn.EX6.JCL(EDJACTP). Check for the appearance of your data sets and volumes in all the reports that are contained in the output of the job. The activity file was created during the trial run of inventory management.
- ___ 8. Perform a trial inventory management that uses the date in five days as the actual date. Use member DHSPVY in your TSOn.EX6.JCL to perform the inventory management. Is the data set TSOn.IGNORE.** still contained in the report?

END OF LAB

Student Exercises

Appendix A. Exercise 1 Solution

Step 1

- ___ 1. Logon as TSO. Your instructor will provide you with your group number n and the logon password.
- ___ 2. Start the DFSMSrmm ISPF application using TSO RMMISPF from any command line. Note that the primary option menu appears as shown in chapter 1 of the student notes of the course material. You now have the choice of selecting a path like USER, LIBRARIAN, STORAGE ADMINISTRATOR, and so on.
- ___ 3. Finish the application. Now, please, start ISMF. In which mode does ISMF currently operate (*end user* or *storage administrator*)? Select DFSMSrmm from the primary option menu of ISMF. What is the look of the primary option menu of DFSMSrmm now? **If ISMF runs in end user mode, the DFSMSrmm ISPF application is presented by the end user menu.**
- ___ 4. Leave DFSMSrmm as well as ISMF again. Now start the DFSMSrmm application for a third time using command TSO RMMISPF LIBRARIAN. What is the look of DFSMSrmm now? Note that the primary option menu appears depending on the option used with the call of RMMISPF. For details see SC26-7404 *z/OS DFSMSrmm Guide and Reference*. **The DFSMSrmm ISPF application is presented by the librarian menu.**
- ___ 5. To perform a data set search, use the fast path command DATASET SEARCH from the command line of the current menu. Note that the data set search option is not offered in the librarian menu. We will perform a data set search later in this exercise. For details on using fast path commands see Appendix in SC26-7404 *z/OS DFSMSrmm Guide and Reference*.
- ___ 6. Now use another fast path command entering VOLUME SEARCH in the command line.

Step 2

- ___ 1. Perform a volume search for volumes that show your TSO userid as owner. Don't forget to use a fitting LIMIT (*) search argument because otherwise you will not see all of the 14 pre-defined volumes in the resulting volume list.
- ___ 2. Fill in the missing values in Table 1 using line commands in column 1 of the volume list. Note that some values are not recorded in the volume record but in the data set record. Line commands V, I, and L will help you to find the requested informations. Use PF1 for help if needed. The table just should help you to walk through the panels of volume details and data set details. You need not fill out all cells of the table to finish this part of the exercise successfully. The table should help you to take a guided walk through the panels that provide informations about volumes and data sets.

Table 12:

Volume	Expiration Date	Data Set Name	Volume Status	Volume Availability
0n0000	<i>not exceeded</i>	<i>TSOn.TAPEDSN. DATA1</i>	<i>MASTER</i>	<i>PENDING RELEASE</i>
0n0001	<i>not exceeded</i>	<i>TSOn.TAPEDSN. DATA2</i>	Creating Job <i>HAH9434</i>	Volume Location <i>SHELF</i>
0n0002	<i>exceeded</i>	<i>TSOn.BACKUP.T APEDAT1</i>	DS Creation Date <i>now-55 (now-56)</i>	DS Last Reference Date <i>now-10 (now-11)</i>
0n0003	<i>not exceeded</i>	<i>TSOn.BACKUP.T APEDAT2</i>	Volume Status <i>MASTER</i>	Media Type <i>HPCT</i>
0n0004	<i>exceeded</i>	<i>TSOn.GDG.G000 1V00</i>	Label Type <i>SL</i>	Assign Date <i>now-8 (now-9)</i>
0n0005	<i>not exceeded</i>	<i>TSOn.GDG.G000 2V00</i>	DS Sequence # <i>2</i>	Previous Volume <i>0n0010</i>
0n0006	<i>not exceeded</i>	<i>TSOn.KEYBOAR D.BACKUPV1</i>	Media Name <i>SQUARE</i>	Density <i>3480</i>
0n0007	<i>exceeded</i>	<i>TSOn.KEYBOAR D.BACKUPV2</i>	Block Count <i>23879</i>	Creating Job <i>B2CA995</i>
0n0008	<i>exceeded</i>	<i>TSOn.CYC.KEYB OARD.BACKUPV 1</i>	Recording Format <i>128TRACK</i>	Special Attributes <i>NONE</i>
0n0009	<i>not exceeded</i>	<i>TSOn.CYC.KEYB OARD.BACKUPV 2</i>	Compaction <i>NONE</i>	Abend while open <i>NO</i>
0n0010	<i>not exceeded</i>	<i>TSOn.SPEC.EXP. EXAMPLE</i> <i>TSOn.GDG.G000 2V00</i>	Number of Data Sets <i>2</i>	Next Volume <i>0n0005</i>
0n0011	<i>exceeded</i>	<i>TSOn.SPEC.NOE XP.EXAMPLE</i>	Volume Type <i>PHYSICAL</i>	File 1 system Id <i>XX21</i>
0n0012	<i>exceeded</i>	<i>TSOn.TMM.ARCH IVE.Y1998</i>	DS Block Size <i>32760</i>	Release Action <i>Return to scratch</i>
0n0013	<i>exceeded</i>	<i>TSOn.TMM.ARCH IVE.Y1999</i>	Vol. Availability <i>OPEN</i>	Volume Location <i>SHELF</i>

3. Leave the volume list now and make a new selection to perform a data set search.
Create a data set list containing all data sets residing on volumes that show your
TSO userid as owner. Don't forget to use a fitting LIMIT(*) search argument

because otherwise you will not see all the 15 data sets in the resulting data set list. Spend some minutes to try out the line commands available in the list.

- ___ 4. Why does the data set list contain 15 entries? Remember that the volume search you did before presented only 14 volumes that contain your user ID as the volume owner. **Data set *TSOn.GDG.G0002V00* appears twice because it is contained in two volumes.**

Step 3

- ___ 5. Take a look at the OPTIONS of the DFSMSrmm ISPF application. You may choose individual settings for processing, viewing, and sorting. See SC26-7404 *z/OS DFSMSrmm Guide and Reference* for details. Now, please, change processing options to B (background).
- ___ 6. Repeat the actions described under 1.) of step 2. A volume search will not be processed at once (in foreground), but the TSO RMM subcommand that is needed in order to perform the volume search is generated and stored in a data set for later processing (in background).
- ___ 7. Now leave the DFSMSrmm ISPF application. There will be presented an additional panel to allow a selection of processing options for the commands that were generated. Choose option **B** to browse the generated command. End browse and choose **S** to submit a job for processing of the volume search in batch TSO environment. Check the job output for results using SDSF. You will find the output of the job using the H option in SDSF. Select the job by using the line command S.

END OF LAB

Appendix B. Exercise 2 Solution

Step 1

- ___ 1. Logon as TSO_n. Your instructor will provide you with your group number *n* and the logon password (see Exercise 1).
- ___ 2. Start the DFSMS_{rmm} ISPF application using TSO RMMISPF from any command line. Perform a volume search for volumes that have your TSO userid as owner. Don't forget to use a fitting LIMIT (*) search argument because otherwise you will not see all of the 14 pre-defined volumes in the resulting volume list.
- ___ 3. Select appropriate values where offered in Table 2. Line command V will help you to find the requested informations. Use PF1 for help if needed. Predict the status and the availability of each volume after the end of the inventory management run and fill in your predictions in Table 2.

Table 13:

Volume	Expiration date	Status after end of housekeeping	Availability after end of housekeeping
0n0000	Not reached	Scratch	-
0n0001	Not reached	Master	-
0n0002	Reached	Master	Pending release
0n0003	Not reached	Master	-
0n0004	Reached	Master	Pending release
0n0005	Not reached	Master	-
0n0006	Not reached	Master	-
0n0007	Reached	Master	Pending release
0n0008	Reached	Master	Pending release
0n0009	Not reached	Master	-
0n0010	Not reached	Master	-
0n0011	Reached	Master	Pending release
0n0012	Reached	Master	Pending release
0n0013	Reached	Master	Pending release

- ___ 4. Wait until your instructor has run the inventory management. Now verify your predictions by displaying volume details of your volumes. To do this, repeat the volume search as described in 2.).

Step 2

- ___ 5. RPTTEXT processing created an extended extract file in data set SS21.XREPORT.EXTRACT.FILE.L1 during inventory management. This is the base for the reporting utilities you will now use. Submit the jobs that are contained in the data set TSO_n.EX1.JCL.

Student Exercises

- ___ 6. Have a look at the reports that were created by the jobs. Some of the reports are empty because in our current environment we have not covered all situations that are possible subjects of reporting. During the following exercises you will have the opportunity to repeat reporting to find an example of some of the reports that are currently empty.
- ___ 7. Now use report generator to produce some reports. You will find in report generator a number of pre-defined reports. These are the same reports as offered by extended reporting. Generate the JCL for some reports of your choice and submit them. Also, try to tailor a pre-defined report in report generator by adding or deleting a column. Follow the steps as described in Unit 2 of your student notes.
- ___ 8. Next, perform the volume search again as described in 2.). By using the line command C, you can change the characteristics of a volume. Switch the status of volume 0n0000 back to MASTER. This can be an important action to rescue a volume from being overwritten. If users want to read the volume again, they cannot request the volume when it is in status SCRATCH, because DFSMSrmm will reject specific requests for scratch volumes.
- ___ 9. Verify your change by re-displaying the volume details. What is the expiration date of the volume now? **The new expiration date is the current date plus five days (five days is the default retention period in our environment).**
- ___ 10. Repeat execution of EDGRPTD by submitting the job in TSO on EX1.JCL(EDGRPTD) again. Why does the scratch volume report still contain volume 0n0000? **The extended report data set wasn't refreshed after your change.**

END OF LAB

Appendix C. Exercise 4 Solution

- ___ 1. Logon to TSO as described in Exercise 1. Use DFSMSrmm ISPF dialog or TSO subcommands to define the following vital record specifications:
- ___ 2. Define a data set VRS that fulfills the following requirements:
 - Match data sets whose first two qualifiers are 'TSOn.DAYS'
 - Keep the data sets five days since creation

RMM ADDVRS DSN('TSOn.DAYS.') DAYS COUNT(5)**
- ___ 3. Define a data set VRS that fulfills the following requirements:
 - Match data sets whose first two qualifiers are 'TSOn.LREF'
 - Keep a data set as long as it has been referenced within the last five days

RMM ADDVRS DSN('TSOn.LREF.') LASTREFERENCEDAYS COUNT(5)**
- ___ 4. Define a data set VRS that fulfills the following requirements:
 - Match data sets whose first two qualifiers are 'TSOn.CYC'
 - Keep one version of the data set

RMM ADDVRS DSN('TSOn.CYC.C%%V%%') CYCLES COUNT(1)
- ___ 5. Define a data set VRS that fulfills the following requirements:
 - Match GDG data sets whose GDG base name is 'TSOn.GDG'
 - Keep one member of the GDG

RMM ADDVRS DSN('TSOn.GDG') GDG CYCLES COUNT(1)
- ___ 6. Define a data set VRS that fulfills the following requirements:
 - Match data sets whose first qualifier is 'TSOn' and the second qualifier is eight characters long with the first character being a 'C' and the fourth character being a 'V'. For other characters in the second qualifier use the ignore character as a masking character which, in fact, makes up a pseudo-GDG definition.
 - Keep one version of the pseudo-GDG

RMM ADDVRS DSN('TSOn.C^V^') CYCLES COUNT(1)
- ___ 7. Define a data set VRS that fulfills the following requirements:
 - Match data sets whose first three qualifiers are 'TSOn.SPEC.NOEXP'
 - Expire the vital record specification tomorrow

RMM ADDVRS DSN('TSOn.SPEC.NOEXP') CYCLES COUNT(99999)
- ___ 8. Define a data set VRS that fulfills the following requirements:

- Match data sets whose first two qualifiers are 'TSOn.CATALOG'
- Keep a data set as long as it is cataloged

RMM ADDVRS DSN('TSOn.CATALOG.**') WHILECATALOG

- ___ 9. After having completed your definitions, you should see the following vital record specifications (Perform a VRS search for vital record specifications that show your userid as the owner):

Table 14:

Data set name	Type	Retention type	Count	Delete date
ABEND	DSN	Days	8	1999/365
TSOn.DAYS.**	DSN	Days	5	1999/365
TSOn.LREF.**	DSN	Lastreferencedays	5	1999/365
TSOn.CYC.C%%V%%%	DSN	Cycles	1	1999/365
TSOn.GDG	GDG	Cycles	1	1999/365
TSOn.C^V^	PGDG	Cycles	1	1999/365
TSOn.SPEC.EXP	DSN	Specific date	99999	today - 1
TSOn.SPEC.NOEXP	DSN	Specific date	99999	today + 1
TSOn.CATALOG.**	DSN	Cycles / Whilecatalog	99999	1999/365

- ___ 10. Note that there is an additional VRS which was pre-defined for each team 'TSOn.SPEC.EXP' to show that if its delete date is exceeded, a VRS will be deleted during inventory management.
- ___ 11. Also, a VRS 'ABEND' was predefined. You are not the owner of this VRS, so you will not see this VRS in the resulting list of your VRS SEARCH action, but it might affect your predictions of the inventory management run. Keep in mind when you predict the result of the following inventory management run that, if any of your data sets was closed by abend, this VRS will supersede the VRS that matches the true data set name.

Prediction of the housekeeping results

- ___ 1. Before doing a real housekeeping run, you should predict which volumes will be retained by a VRS. Use the information in the table below together with the matching VRSs. The volumes that are not protected by a VRS will be updated by inventory management to show an availability of pending release because their expiration dates are being reached. Please fill in the cells in the most right column of the table by writing *RETAIN* or *RELEASE*.

Table 15:

Volume	Volume EXPDT	Data set	Data set creation date	Data set last reference date	Volume is still retained or will be released
On0000	today-1	TSOn.DAYS.EXP	today-10	today - 10	Release
On0001	today-1	TSOn.DAYS.NOEXP	today- 2	today - 2	Retain
On0002	today-1	TSOn.LREF.EXP	today- 10	today - 10	Release
On0003	today-1	TSOn.LREF.NOEXP	today- 10	today - 10	Retain
On0004	today-1	TSOn.GDG.G0001.V00	today- 7	today - 7	Release
On0005	today-1	TSOn.GDG.G0002.V00	today- 6	today - 6	Retain
On0006	today-1	TSOn.C01V0001	today- 9	today - 9	Release
On0007	today-1	TSOn.C01V0002	today- 9	today - 9	Retain
On0008	today-1	TSOn.CYC.C01V0001	today- 9	today - 9	Retain
On0009	today-1	TSOn.CYC.C01V0002	today- 9	today - 9	Retain
On0010	today-1	TSOn.SPEC.EXP.EXAMPLE	today- 9	today - 9	Retain
On0011	today-1	TSOn.SPEC.NOEXP.EXAMPLE	today- 9	today - 9	Retain
On0012	today-1	TSOn.CATALOG.NOT.EXAMPLE	today- 9	today - 9	Release
On0013	today-1	TSOn.CATALOG.EXAMPLE	today- 9	today - 9	Retain

- ___ 2. Wait until your instructor has run the inventory management. Now verify your predictions by displaying volume details of your volumes. To do this, perform a volume search. Alternatively, you can try the CONFIRM option on the librarian menu and select the line that shows a status of PENDING.
- ___ 3. Check the content of the report file SS21.REPORT.FILE.L4 that was created during the inventory management run. Search for your VRSs and the matching data sets. Try to find out what was wrong, if you don't find the expected results.
- ___ 4. Check for the VRS TSOn.SPEC.EXP.EXAMPLE, which should have been deleted during the inventory management.
- ___ 5. Why is volume On0010 still retained? **Because the data set name record of data set TSOn.SPEC.EXAMPLES shows that the last close of the data set was performed by abend processing, thus the VRS OPEN protects the volume.**
- ___ 6. Create a report of the contents of the activity file by using the job in your TSOn EX4.JCL(EDJACTP). Check for the appearance of your data sets and volumes in all the reports that are contained in the output of the job. The activity file was created during the trial run of inventory management.

END OF LAB

Student Exercises

Appendix D. Exercise 5 Solution

- ___ 1. Logon to TSO as described in Exercise 1. Use the DFSMSrmm ISPF dialog or TSO subcommands to define the following vital record specifications:
- ___ 2. Add ten bin numbers in each of the built-in storage locations in order to make the storage locations available to receive volumes during inventory management. Do not specify any specific bin number as a starting point for the ADDBIN subcommand. Specify an asterisk (*) for the bin number to start with.

RMM ADDBIN * LOCATION(LOCAL) COUNT(10)

RMM ADDBIN * LOCATION(DISTANT) COUNT(10)

RMM ADDBIN * LOCATION(REMOTE) COUNT(10)

- ___ 3. Define a name type VRS that fulfills these requirements:
 - The name of the VRS is LOCAL0n
 - Store two units at the LOCAL location

RMM ADDVRS NAME(LOCAL0n) LOCATION(LOCAL) STORENUMBER(2)

- ___ 4. Define a data set VRS that fulfills the following requirements:
 - Match data sets whose first qualifier is 'TSOn' and last qualifier is 'VITAL'
 - Keep every version of the data set
 - Keep the data sets at the REMOTE location

RMM ADDVRS DSN('TSOn..VITAL') COUNT(99999) LOCATION(REMOTE)**

- ___ 5. Define a data set VRS that fulfills the following requirements:
 - Match data sets whose first qualifier is 'TSOn' and last qualifier is 'DISASTER'
 - Keep three versions of the data set
 - Keep the current version of the data set at the DISTANT location
 - Move the older versions of the data set to the location LOCAL by using the LOCAL0n VRS as the next VRS in a chain

RMM ADDVRS DSN('TSOn..DISASTER') COUNT(3) LOCATION(DISTANT) -
STORENUMBER(1) NEXTVRS(LOCAL0n)**

- ___ 6. Define a data set VRS that fulfills the following requirements:
 - Match GDGD data sets whose first two qualifiers are 'TSOn.GDG'
 - Retain six versions of the data set
 - Keep the first three versions of the data set at the home location
 - Use the previously coded name type VRS LOCAL0n as the next VRS in the chain

**RMM ADDVRS DSN('TSOn.GDG') GDG COUNT(6) LOCATION(HOME) -
STORENUMBER(3) NEXTVRS(LOCAL0n)**

- ___ 7. After having completed your definitions, you should see the following vital record specifications (Perform a VRS search for vital record specifications that show your userid as the owner):

Table 16: Vital Record Specifications

Data set name	Type	Location	Storenumber	Count	Nextvrs
TSOn.**.VITAL	DSN	Remote	99999	99999	-
TSOn.**.DISASTER	DSN	Distant	1	3	LOCAL0n
TSOn.GDG	GDG	Home	3	6	LOCAL0n
LOCAL0n	NAME	Local	2		-

Prediction of the housekeeping results

- ___ 1. Before doing a real housekeeping run, you should predict the destination of the pre-defined volumes after inventory management has been run. Use the information in the table below together with the matching VRSs. Please fill in the cells in the most right column of the table by writing the name of the target storage location.

Table 17: Prediction of the Destination after Inventory Management

Volume	Volume EXPDT	Data set	Data set creation date	Required location	Destination
0n0000	today+1	TSOn.FINANCE.VITAL	today	SHELF	REMOTE
0n0001	today+1	TSOn.FINANCE.VITAL	today	SHELF	REMOTE
0n0002	today+1	TSOn.PAYROLL.DISASTER	today- 5	SHELF	LOCAL
0n0003	today+1	TSOn.PAYROLL.DISASTER	today- 4	SHELF	LOCAL
0n0004	today+1	TSOn.PAYROLL.DISASTER	today- 3	SHELF	DISTANT
0n0011	today+1	TSOn.GDG.G0001V00	today	SHELF	LOCAL
0n0012	today+1	TSOn.GDG.G0002V00	today	SHELF	LOCAL
0n0013	today+1	TSOn.GDG.G0003V00	today	SHELF	-
0n0014	today+1	TSOn.GDG.G0004V00	today	SHELF	-
0n0015	today+1	TSOn.GDG.G0005V00	today	SHELF	-

- ___ 2. Wait until your instructor has run the inventory management. Now verify your predictions by displaying volume details of your volumes. To do this, perform a volume search. Alternatively, you can try the CONFIRM option on the librarian menu and select the lines that shows a status of PENDING and an action of MOVE.
- ___ 3. Check the content of the report file SS21.REPORT.FILE.L4 that was created during the inventory management run. Search for your VRSs and the matching data sets. If you don't find the expected results, try to find out what was wrong with your VRSs.
- ___ 4. Create movement reports by using the jobs provided in your TSOn.EX5.JCL data set and by using report generator.

- ___ 5. Create a report of the contents of the activity file by using the job in your TSO_n.EX5.JCL(EDJACTP). Check for the appearance of your data sets and volumes in all the reports that are contained in the output of the job. The activity file was created during the trial run of inventory management.
- ___ 6. Now confirm the move of one of your volumes by using the line command CM in the volume action summary list (see 2.) CONFIRM). Watch the change of the contents of the fields **DESTINATION**, **LOCATION**, and **INTRANSIT**.

The content of the **DESTINATION** field moves to the **LOCATION** field. The **DESTINATION** field is cleared. The **INTRANSIT** field is set to **NO**.

Student Exercises

Appendix E. Exercise 6 Solution

- ___ 1. Logon as TSO_n. Your instructor will provide you with your group number *n* and the logon password (see Exercise 1).
- ___ 2. Define a dsn type VRS for use as a secondary VRS that fulfills the following requirements:

- The data set name should be 'Mn99000'
- Keep the data sets as long as they are cataloged

RMM ADDVRS DSN('Mn99000') WHILECATALOG

- ___ 3. Now assign the management class Mn99000 to the data set TSO_n.LREF.EXP by using the line command C in a data set list (you must perform a data set search before) to simulate the processing of the management class ACS routine.

RMM CHANGEDATASET 'TSO_n.LREF.EXP' MANAGEMENTCLASS(Mn99000)

- ___ 4. Define a name type VRS that fulfills the following requirements:

- The name should be 'CATALOG_n'
- Keep the data sets as long as they are cataloged

RMM ADDVRS NAME(CATALOG_n) WHILECATALOG

- ___ 5. The VRSs TSO_n.C^V^ should be connected with the new VRS that you have just defined by using ANDVRS. Use the line command C in a list of VRSs that show your userid as owner to change the VRS TSO_n.C^V^ accordingly. Insert the name CATALOG_n in the **NEXT VRS in chain** field and the type AND in the **Chain using** field of the change data set VRS panel.
- ___ 6. The VRSs TSO_n.DAYS.** should be connected with the new VRS that you have just defined by using NEXTVRS. Use the line command C in a list of VRSs that show your userid as owner to change the VRS TSO_n.DAYS.** accordingly. Insert the name CATALOG_n in the **NEXT VRS in chain** field and the type NEXT in the **Chain using** field of the change data set VRS panel.
- ___ 7. Define a dsn type VRS that fulfills the following requirements:
 - Match data set whose first two qualifiers are 'TSO_n.IGNORE'
 - When the data set is no longer protected by a VRS, the volume expiration date should be ignored
 - The data set should be retained for two days since its creation

RMM ADDVRS DSN('TSO_n.IGNORE.') DAYS COUNT(2) -
RELEASE(EXPIRYDATEIGNORE)**

- ___ 8. After having completed your definitions, you should see the following vital record specifications (Perform a VRS search for vital record specifications that show your userid as the owner):

Table 18: VRS Definitions

Data set name / name	Type	Retention type	Count	NEXTVRS ANDVRS Secondary VRS
TSOn.DAYS.**	DSN	Days	5	(NEXT) CATALOGn
TSOn.LREF.**	DSN	Lastreferencedays	5	(secondary) Mn99000
TSOn.CYC.C%%V%%%	DSN	Cycles	1	
TSOn.GDG	GDG	Cycles	1	
TSOn.C^^V^^^	PGDG	Cycles	1	(AND) CATALOGn
TSOn.SPEC.EXP	DSN	Specific date	99999	
TSOn.SPEC.NOEXP	DSN	Specific date	99999	
TSOn.CATALOG.**	DSN	Cycles / Whilecatalog	99999	
CATALOGn	Name	WHILECATALOG	99999	
Mn99000	DSN	WHILECATALOG	99999	

Prediction of the housekeeping results

1. Before doing a real housekeeping run, you should predict which volumes will be retained by a VRS. Use the information in the table below together with the matching VRSs. The volumes that are not protected by a VRS will be updated by inventory management to show an availability of pending release because their expiration dates are being reached. Please fill in the cells in the most right column of the table by writing *RETAIN* or *RELEASE*.

Table 19: Retention of Volumes by VRSs

Volume	Volume EXPDT	Data set	Data set creation date	Data set last reference date	Volume is still retained or will be released
0n0000	today-1	TSOn.DAYS.EXP	today -10	today - 10	RETAIN
0n0001	today-1	TSOn.DAYS.NOEXP	today - 2	today - 2	RETAIN
0n0002	today-1	TSOn.LREF.EXP	today - 10	today - 10	RETAIN
0n0003	today-1	TSOn.LREF.NOEXP	today - 10	today - 10	RETAIN
0n0004	today-1	TSOn.GDG.G0001.V00	today - 7	today - 7	RELEASE
0n0005	today-1	TSOn.GDG.G0002.V00	today - 6	today - 6	RETAIN
0n0006	today-1	TSOn.C01V0001	today - 9	today - 9	RELEASE
0n0007	today-1	TSOn.C01V0002	today - 9	today - 9	RELEASE
0n0008	today-1	TSOn.CYC.C01V0001	today - 9	today - 9	RETAIN
0n0009	today-1	TSOn.CYC.C01V0002	today - 9	today - 9	RETAIN
0n0010	today-1	TSOn.SPEC.EXP.EXAMPLE	today - 9	today - 9	RELEASE

Table 19: Retention of Volumes by VRsS

Volume	Volume EXPDT	Data set	Data set creation date	Data set last reference date	Volume is still retained or will be released
0n0011	today-1	TSOn.SPEC.NOEXP.EXAMPLE	today - 9	today - 9	RETAIN
0n0012	today-1	TSOn.CATALOG.NOT.EXAMPLE	today - 9	today - 9	RELEASE
0n0013	today-1	TSOn.CATALOG.EXAMPLE	today - 9	today - 9	RETAIN
0n0014	today + 7	TSOn.IGNORE.EXPDT	today	today	RETAIN

- ___ 2. Wait until your instructor has run the inventory management. Now verify your predictions by displaying volume details of your volumes. To do this, perform a volume search. Alternatively, you can try the CONFIRM option on the librarian menu and select the line that shows a status of PENDING.
- ___ 3. Check the content of the report file SS21.REPORT.FILE.L6 that was created during the inventory management run. Search for your VRsS and the matching data sets. If you don't find the expected results, try to find out what was wrong with your VRsS.
- ___ 4. Which subchain retains the volume 0n0000 currently? **CATALOGn**
- ___ 5. Which subchain retains the volume 0n0002 currently? **Mn99000**
- ___ 6. Why is the data set TSOn.C^^^V^^^ not contained in the report file? **The data set is not cataloged, therefore one of the conditions that are chained by ANDVRS is not true.**
- ___ 7. Create a report of the contents of the activity file by using the job in your TSOn.EX6.JCL(EDJACTP). Check for the appearance of your data sets and volumes in all the reports that are contained in the output of the job. The activity file was created during the trial run of inventory management.
- ___ 8. Perform a trial inventory management that uses the date in five days as the actual date. Use member DHSKPVY in your TSOn.EX6.JCL to perform the inventory management. Is the data set TSOn.IGNORE.** still contained in the report? **No**

Student Exercises
